

Second recirculation of SA ballot for Draft 3.2 of the
**IEC/IEEE 60802 Time-Sensitive Networking Profile for
Industrial Automation**

This is an unapproved draft prepared by the IEC/IEEE 60802 Joint Project.

NOTE – This page is not subject to ballot comments.

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TIME-SENSITIVE NETWORKING PROFILE FOR INDUSTRIAL AUTOMATION

FOREWORD

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International Standard IEC/IEEE 60802 was prepared by subcommittee 65C: Industrial networks, of IEC technical committee 65: Industrial-process measurement, control and automation, in cooperation with IEEE 802.1: Higher Layer LAN Protocols Working Group of IEEE 802: LAN/MAN Standards Committee of the IEEE computer society, under the IEC/IEEE Dual Logo Agreement between IEC and IEEE. It is an International Standard.

This document is published as an IEC/IEEE Dual Logo standard.

The text of this International Standard is based on the following IEC documents:

Draft	Report on voting
65C/XX/FDIS	65C/XX/RVD

Full information on the voting for its approval can be found in the report on voting indicated in the above table.

The language used for the development of this International Standard is English.

This document was drafted in accordance with the rules given in the ISO/IEC Directives, Part 2, available at www.iec.ch/members_experts/refdocs. The main document types developed by IEC are described in greater detail at www.iec.ch/publications/.

The IEC Technical Committee and IEEE Technical Committee have decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under webstore.iec.ch in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn, or
- revised.

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INTRODUCTION

This document defines time-sensitive networking profiles for industrial automation. The profile selects features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules for the digital data sheet and for remote procedure calls.

The profile meets the industrial automation market objective of converging Operations Technology (OT) and Information Technology (IT) networks by defining a common, standardized network infrastructure. This objective is accomplished by taking advantage of the improvements that Time-Sensitive Networking provides to IEEE 802.1 and IEEE 802.3 standard Ethernet networks by providing guaranteed data transport with bounded low latency, low latency variation, zero congestion loss for critical traffic, and high availability.

The profile helps the convergence of industrial communication networks by referring only to international standards to build the lower layers of the communication stack and their management.

Ethernet extended with Time-Sensitive Networking technology provides the features required in the area of industrial communication networks, such as:

- meeting low latency and latency variation requirements concerning data transmission;
- efficient exchange of data records on a frequent time period;
- reliable communications with calculable downtime;
- high availability meeting application requirements;
- efficient mechanisms for bandwidth utilization of exchanges of data records, with zero congestion loss;
- improved clock synchronization mechanisms, including support of multiple gPTP domains.

TIME-SENSITIVE NETWORKING PROFILE FOR INDUSTRIAL AUTOMATION

1 Scope

This document defines time-sensitive networking profiles for industrial automation. The profiles select features, options, configurations, defaults, protocols, and procedures of bridges, end stations, and LANs to build industrial automation networks. This document also specifies YANG modules defining read-only information available online and offline as a digital data sheet. This document also specifies YANG modules for remote procedure calls and actions to address requirements arising from industrial automation networks.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes the requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 9594-8:2020, *Information technology — Open systems interconnection — Part 8: The Directory: Public-key and attribute certificate frameworks*, available at: <https://www.iso.org/obp/ui/#iso:std:iso-iec:9594:-8:en>

IEEE Std 1588-2019, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*¹

IEEE Std 1588e-2024, *Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems Amendment: MIB and YANG Data Models*

IEEE Std 802.1AB-2016, *IEEE Standard for Local and metropolitan area networks – Station and Media Access Control Connectivity Discovery*

IEEE Std 802.1ABcu-2021, *IEEE Standard for Local and metropolitan networks – Station and Media Access Control Connectivity Discovery Amendment 1: YANG Data Model*

IEEE Std 802.1AR-2018, *IEEE Standard for Local and Metropolitan Area Networks – Secure Device Identity*

IEEE Std 802.1AS-2020, *IEEE Standard for Local and Metropolitan Area Networks – Timing and Synchronization for Time-Sensitive Applications*

IEEE Std 802.1ASdm-2024, *IEEE Standard for Local and Metropolitan Area Networks: Timing and Synchronization for Time-Sensitive Applications Amendment: Hot Standby*

IEEE Std 802.1ASdn-2024, *IEEE Standard for Local and Metropolitan Area Networks: Timing and Synchronization for Time-Sensitive Applications Amendment: YANG Data Model*

IEEE Std 802.1ASdr-2024, *IEEE Standard for Local and Metropolitan Area Networks – Timing and Synchronization for Time-Sensitive Applications Amendment: Inclusive Terminology*

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- 449 IEEE Std 802.1CB-2017, *IEEE Standard for Local and metropolitan area networks – Frame*
450 *Replication and Elimination for Reliability*
- 451 IEEE Std 802.1CBcv-2021, *IEEE Standard for Local and metropolitan area networks – Frame*
452 *Replication and Elimination for Reliability – Amendment 1: Information Model, YANG Data*
453 *Model and Management Information Base Module*
- 454 IEEE Std 802.1CBdb-2021, *IEEE Standard for Local and metropolitan area networks – Frame*
455 *Replication and Elimination for Reliability – Amendment 2: Extended Stream Identification*
456 *Functions*
- 457 IEEE Std 802.1Q-2022, *IEEE Standard for Local and Metropolitan Area Network – Bridges and*
458 *Bridged Networks*
- 459 IEEE Std 802.1Qcw-2023, *IEEE Standard for Local and Metropolitan Area Networks – Bridges*
460 *and Bridged Networks, Amendment: YANG Data Models for Scheduled Traffic, Frame*
461 *Preemption, and Per-Stream Filtering and Policing*
- 462 IEEE Std 802.1Qdj-2024, *Standard for Local and Metropolitan Area Networks – Bridges and*
463 *Bridged Networks, Amendment: Configuration Enhancements for Time-Sensitive Networking*
- 464 IEEE Std 802.1Qdx-2024, *Standard for Local and Metropolitan Area Networks – Bridges and*
465 *Bridged Networks, Amendment: YANG Data Models for the Credit-Based Shaper*
- 466 IEEE Std 802.1Qdy-2025, *Standard for Local and Metropolitan Area Networks – Bridges and*
467 *Bridged Networks, Amendment: YANG Data Models for the Multiple Spanning Tree Protocol*
- 468 IEEE Std 802.3-2022, *IEEE Standard for Ethernet*
- 469 IEEE Std 802.3.2-2019, *IEEE Standard for Ethernet – YANG Data Model Definitions*
- 470 IEEE Draft Std P802.3.2a, *Standard for Ethernet – YANG Data Model Definitions*
- 471 IEEE Std 802.3cx-2023, *IEEE Standard for Ethernet Amendment 6: Media Access Control*
472 *(MAC) Service Interface and Management Parameters to Support Improved Precision Time*
473 *Protocol (PTP) Timestamping Accuracy*
- 474 IEEE Std 802.3de-2022, *Standard for Ethernet Amendment 6: Enhancements to the MAC Merge*
475 *and Time Synchronization Service Interface for Point-to-Point 10 Mb/s Single Pair Ethernet*
- 476 IETF RFC 2131, Droms, R., *Dynamic Host Configuration Protocol*, March 1997, available at
477 <https://www.rfc-editor.org/info/rfc2131>
- 478 IETF RFC 2986, Nystrom, M. and Kaliski, B., *PKCS #10: Certification Request Syntax*
479 *Specification Version 1.7*, November 2000, available at <https://www.rfc-editor.org/info/rfc2986>
- 480 IETF RFC 3986, Berners-Lee, T., Fielding, R., and Masinter, L., *Uniform Resource Identifier*
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3 Terms, definitions, symbols, abbreviated terms and conventions

3.1 General

For the purposes of this document, the terms and definitions given in ITU-T G.8260, IEEE Std 802-2014, IEEE Std 802.3™-2022, IEEE Std 802.1Q™-2022, IEEE Std 802.1AS™-2020 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEEE Standards Dictionary Online: available at <https://dictionary.ieee.org>
- ITU-T Terms and Definitions database: available at https://www.itu.int/br_tsb_terms/#/

3.2 List of terms, abbreviated terms and definitions given in various standards

For the purposes of this document, the terms and definitions given in Table 1 apply.

For ease of understanding, the most important terms used within this document are listed in Table 1 but the definitions are not repeated.

Table 1 – List of terms

Term	Source
BTCA	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr™-2024
Bridge	IEEE Std 802.1Q-2022
Bridge Port	IEEE Std 802.1Q-2022
CFM	IEEE Std 802.1Q-2022
Clock	IEEE Std 802.1AS-2020
ClockTimeTransmitter	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockTimeReceiver	IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024
ClockSource	IEEE Std 802.1AS-2020
ClockTarget	IEEE Std 802.1AS-2020
CNC	IEEE Std 802.1Q-2022
Configuration Domain	IEEE Std 802.1Qdj™-2024
constant time error (cTE)	ITU-T G.8260

Term	Source
Customer Virtual Local Area Network (C-VLAN) component	IEEE Std 802.1Q-2022
CUC	IEEE Std 802.1Q-2022
device	IEEE Std 802.1AR™-2018
DLL	IEEE Std 802-2014
DTE	IEEE Std 802.3-2022
dynamic time error (dTE)	ITU-T G.8260
end entity (EE)	NIST Special Publication 800-57 Part 2, Revision 1
end station	IEEE Std 802-2014
Ethernet	IEEE Std 802.3-2022
FDB	IEEE Std 802.1Q-2022
FID	IEEE Std 802.1Q-2022
fingerprint	IETF RFC 7589
FQTSS	IEEE Std 802.1Q-2022
fractional frequency offset	IEEE Std 802.1AS-2020
frame	IEEE Std 802.1Q-2022
frame preemption	IEEE Std 802.1Q-2022
FRER	IEEE Std 802.1CB™-2017
gating cycle	IEEE Std 802.1Q-2022
gPTP communication path	IEEE Std 802.1AS-2020
gPTP domain	IEEE Std 802.1AS-2020
Grandmaster Clock	IEEE Std 802.1AS-2020
Grandmaster PTP Instance	IEEE Std 802.1AS-2020
IST	IEEE Std 802.1Q-2022
LAN	IEEE Std 802-2014
latency	IEEE Std 802.1Q-2022
Listener	IEEE Std 802.1Q-2022
LLDP	IEEE Std 802.1AB™-2016
LLDPDU	IEEE Std 802.1AB-2016
local clock	IEEE Std 802.1AS-2020
LocalClock	IEEE Std 802.1AS-2020
logical link	IEEE Std 802-2014
LPI	IEEE Std 802.3-2022
MAC	IEEE Std 802-2014
MMRP	IEEE Std 802.1Q-2022
MST	IEEE Std 802.1Q-2022
MVRP	IEEE Std 802.1Q-2022
NETCONF	IETF RFC 6241
PCP	IEEE Std 802.1Q-2022
PDU	IEEE Std 802.1Q-2022
PHY	IEEE Std 802.3-2022
PLS	IEEE Std 802.3-2022
Port	IEEE Std 802.1Q-2022
preciseOriginTimestamp	IEEE Std 802.1AS-2020

Term	Source
primary domain	IEEE Std 802.1ASdm™-2024
PTP End Instance	IEEE Std 802.1AS-2020
PTP Instance	IEEE Std 802.1AS-2020
PTP Link	IEEE Std 802.1AS-2020
PTP Port	IEEE Std 802.1AS-2020
PTP Relay Instance	IEEE Std 802.1AS-2020
PVID	IEEE Std 802.1Q-2022
redundancy	IEC 60050-192
residence time	IEEE Std 802.1AS-2020
secondary domain	IEEE Std 802.1ASdm-2024
signature suite	IEEE Std 802.1AR-2018
station	IEEE Std 802-2014
stream	IEEE Std 802.1Q-2022
synchronized time	IEEE Std 802.1AS-2020
Talker	IEEE Std 802.1Q-2022
time error	ITU-T G.8260
time-sensitive stream	IEEE Std 802.1Q-2022
traffic class	IEEE Std 802.1Q-2022
TLV	IEEE Std 802.3-2022
UNI	IEEE Std 802.1Q-2022
VID	IEEE Std 802.1Q-2022
VLAN	IEEE Std 802.1Q-2022
X.509	ISO/IEC 9594-8:2020
YANG	IETF RFC 6020

593

594 3.3 Terms defined in this document

595 3.3.1

596 application clock

597 clock used by the application to time events

598 Note 1 to entry: Events can be periodic or aperiodic.

599 3.3.2

600 Bridge component

601 Customer Virtual Local Area Network (C-VLAN) Bridge

602 3.3.3

603 control latency

604 time delay between the input to a sensor application and the output from an actuator application

605 Note 1 to entry: For the purposes of this document, control latency does not include latencies in the sensor,
606 actuator, or the physical system in a process.

607 3.3.4

608 deadline

609 application defined reference point that represents a time when data is required by the
610 application

3.3.5**digital data sheet**

information about the capabilities of an IA-station

3.3.6**end station component**

end station entity

3.3.7**Global Time**

synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale

3.3.8**IA-controller**

industrial automation function, consisting of a comparing element and a controlling element, that performs a specified control function

Note 1 to entry: An IA-controller exchanges data with several IA-devices or other IA-controllers for the purpose of control of a system.

Note 2 to entry: The primary categories of IA-controllers are distributed control systems (DCS), programmable logic controllers (PLCs), and programmable automation controllers (PACs).

3.3.9**IA-device**

industrial automation function, consisting of sensor or actuator elements to read or write process data

Note 1 to entry: An IA-device exchanges data with an IA-controller or other IA-devices for the purpose of control of a system.

3.3.10**IA-station**

material element or assembly of one or more end station components, and zero, one or more bridge components

Note 1 to entry: IA-controllers and IA-devices are industrial automation functions of IA-stations.

3.3.11**imprinting**

<security> equipping IA-stations with an LDevID credential, corresponding trust anchor and certificate-to-name mapping instructions

3.3.12**management entity**

IA-station function responsible for configuration of Bridge components, end station components and ports

Note 1 to entry: The management entity interacts with remote management.

3.3.13**network diameter**

number of links in the longest of all the calculated shortest paths between each pair of nodes in the network

3.3.14**network provisioning**

process of defining a consistent network configuration, which is applied to all stations

655 **3.3.15**
 656 **nominal frequency**
 657 ideal frequency with zero uncertainty

658 Note 1 to entry: The nominal frequency of the PTP timescale is further explained in IEEE Std 1588™-2019, 7.2.1,
 659 7.2.2, and Annex B.

660 **3.3.16**
 661 **non-proxied IA-station**
 662 IA-station which contains a ccA Bridge component or a ccA end station component

663 **3.3.17**
 664 **ppm**
 665 **μHz/Hz**

666 Note 1 to entry: The term "ppm" refers to a pure multiplier of 0,000 001 and is used in the context of this
 667 document as an SI unit term to allow readable terms conformant to various rules related to expressions.

668 **3.3.18**
 669 **proxy**
 670 IA-station whose Netconf server can be utilized by other IA-stations for the purpose of remote
 671 management

672 **3.3.19**
 673 **proxy target**
 674 IA-station which contains only ccB Bridge or end station components

675 **3.3.20**
 676 **Working Clock**
 677 synchronized time, derived from a gPTP domain, that is traceable to the PTP timescale, or to
 678 an ARB timescale that is continuous

679 Note 1 to entry: In general, the Working Clock is traceable to an ARB timescale; however, the Working Clock time
 680 can be correlated to a recognized timing standard.

681 **3.4 Abbreviated terms**

682	AEAD	Authenticated Encryption with Associated Data
683	AES	Advanced Encryption Standard
684	ARB	Arbitrary
685	ASCII	American Standard Code for Information Interchange
686	ASN	Abstract Syntax Notation
687	BTCA	Best timeTransmitter Clock Algorithm
688	CA	Certification Authority
689	CBC	Cipher Block Chaining
690	ccA	Conformance Class A
691	ccB	Conformance Class B
692	CFM	Connectivity Fault Management
693	CMLDS	Common Mean Link Delay Service
694	CMS	Cryptographic Message Syntax
695	CN	Common Name
696	CNC	Centralized Network Configuration
697	CRL	Certificate Revocation List
698	CRUDX	Create Read Update Delete eXecute
699	CSR	Certificate Signing Request

700	CUC	Centralized User Configuration
701	C-VLAN	Customer VLAN
702	DAC	Discretionary Access Control
703	DER	Distinguished Encoding Rules
704	DH	Diffie-Hellman
705	DHE	Diffie-Hellman Ephemeral
706	DLL	Data Link Layer
707	DMAC	Destination MAC Address
708	DNS	Domain Name Service
709	DSA	Digital Signature Algorithm
710	DTE	Data Terminal Equipment
711	EC	Elliptic Curve
712	ECC	Elliptic Curve Cryptography
713	ECDSA	Elliptic Curve Digital Signature Algorithm
714	EdDSA	Edwards-Curve Digital Signature Algorithm
715	EE	End Entity
716	FDB	Filtering Database
717	FID	Filtering Identifier
718	FQDN	Fully Qualified Domain Name
719	FQTSS	Forwarding and Queuing Enhancements for Time-Sensitive Streams
720	FRER	Frame Replication and Elimination for Reliability
721	GCM	Galois Counter Mode
722	gPTP	generalized Precision Time Protocol
723	HMAC	Keyed-Hashing for Message Authentication Code
724	HW	Hardware
725	IA	Industrial Automation
726	IDeVID	Initial Secure Device Identifier
727	IEC	International Electrotechnical Commission
728	IEEE	Institute of Electrical and Electronics Engineers
729	I-LAN	Internal Local Area Network
730	ISO	International Organization for Standardization
731	ISS	Internal Sublayer Service
732	IST	Internal Spanning Tree
733	IT	Information Technology
734	ITU	International Telecommunication Union
735	LDeVID	Locally Significant Secure Device Identifier
736	LLDP	Link Layer Discovery Protocol
737	LLDPDU	Link Layer Discovery Protocol Data Unit
738	LPI	Low Power Idle
739	LRP	Link-local Registration Protocol
740	MAC	Media Access Control
741	MD	Media-Dependent
742	MDI	Medium Dependent Interface

743	MMRP	Multiple MAC Registration Protocol
744	MST	Multiple Spanning Tree
745	MVRP	Multiple VLAN Registration Protocol
746	N/A	Not applicable
747	NACM	Network configuration Access Control Model
748	NETCONF	Network Configuration Protocol
749	NMDA	Network Management Datastore Architecture
750	NPE	Network Provisioning Entity
751	NRR	Neighbor Rate Ratio
752	OCSF	Online Certificate Status Protocol
753	OID	Object Identifier
754	OMG®	Object Management Group ²
755	OT	Operations Technology
756	OUI	Organizationally Unique Identifier
757	PCP	Priority Code Point
758	PCS	Profile Conformance Statement
759	PDU	Protocol Data Unit
760	PE	Path Entity
761	PEM	Privacy Enhanced Mail
762	PFS	Perfect Forward Secrecy
763	PHY	Physical Layer Device
764	PII	Personally Identifiable Information
765	PKCS	Public Key Cryptography Standards
766	PKI	Public Key Infrastructure
767	PKIX	Public Key Infrastructure X.509
768	PLS	Physical Signaling Sublayer
769	PPS	Pulse Per Second
770	PTP	Precision Time Protocol
771	PVID	Port VLAN Identifier
772	QoS	Quality of Service
773	RBAC	Role-Based Access Control
774	RFC	Request for Comments
775	RPC	Remote Procedure Call
776	RSA	Rivest-Shamir-Adleman
777	RAE	Resource Allocation Entity
778	SAN	Subject Alternative Name
779	SHA	Secure Hash Algorithm
780	STE	Sync Tree Entity
781	TDE	Topology Discovery Entity

² OMG®, UML® and Unified Modeling Language™ are either registered trademarks or trademarks of a product supplied by the Object Management Group, Inc. in the United States and/or other countries. This information is given for the convenience of users of this document and does not constitute an endorsement by IEEE or IEC of the product named. Equivalent products may be used if they can be shown to lead to the same results.

782	TLS	Transport Layer Security
783	TLV	Type, Length, Value
784	TOFU	Trust On First Use
785	TSN	Time-Sensitive Networking
786	TSN-IA	Time-Sensitive Networking for Industrial Automation
787	TTP	Trusted Third Party
788	UML®	Unified Modeling Language™
789	UNI	User/Network Interface
790	URL	Uniform Resource Locator
791	URN	Uniform Resource Name
792	VID	VLAN Identifier
793	VLAN	Virtual Local Area Network
794	YANG	Yet Another Next Generation data modeling language

795 3.5 Conventions

796 3.5.1 Convention for capitalizations

797 Capitalized terms are either based on the rules given in the ISO/IEC Directives Part 2 or
798 emphasize that these terms have a specific meaning throughout this document.

799 Throughout this document "bridge" can be used instead of "Bridge", except when

- 800 • it occurs at the beginning of a sentence or
- 801 • it is being used as (or part of) a specific term such as "VLAN Bridge" rather than being used
802 to identify bridges (potentially of any type) in general. If "VLAN Bridge" is meant where only
803 "Bridge" is written, a change to "VLAN Bridge" would be appropriate.

804 3.5.2 Unit conventions

805 This document uses:

- 806 • Gb/s for gigabits per second,
- 807 • Mb/s for megabits per second and,
- 808 • kb/s for kilobits per second.

809 3.5.3 Conventions for YANG contents

810 YANG modules and XML instance data for YANG shown in this document use the following
811 style:

812 Text style `higher-layer-if` text style

813 Contents of a YANG module use the following style:

```
814 <ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
815   <bridges>
816     <bridge> <!-- list -->
817       <name>functional-unit-x</name>
818     ...

```

819 3.5.4 Conventions for YANG selection and Digital Data Sheet

820 The digital data sheet expresses device capabilities and therefore, it is not required that all
821 nodes in a YANG module be included in the digital data sheet. YANG nodes in 6.4.9, marked
822 with [m], are mandatory nodes in the digital data sheet, nodes marked with [c] are conditional

mandatory if the IA-station supports the corresponding optional functionality. Nodes marked with [o] are optional nodes in the digital data sheet. These marking in no way affect whether the feature and associated YANG module is required for the IA-station. Please refer to Clause 5 for conformance criteria for the IA-station.

YANG node selections in 6.4.9 of parent nodes implicitly include all subsidiary child nodes.

4 Overview of TSN in industrial automation

4.1 Industrial application operation

Industrial network applications are based on three main types of building blocks, which can be combined in one IA-controller or provided as a combination of an IA-controller and IA-devices interconnected through a suitable communication network.

These basic building blocks are:

- IA-device Sensor subsystems, which provide input signals indicating the value of the parameter or state being monitored, such as temperature, pressure, or discrete input information;
- IA-controller subsystems, which operate on combinations of measurements and external demand settings to develop output requests, such as position corrections in a motion application;
- IA-device Actuator subsystems, which implement output requests that result in physical changes to the process or machine under control, such as a level in a storage tank, the speed of a printing press, or movement of a robot.

NOTE 1 In general, all subsystems have an internal state, based upon initial settings, and derived from execution; therefore, the application inputs are combined with the internal state to develop an updated internal state and associated outputs.

A control loop is formed when the process or machine responds to the IA-device Actuator output and produces a new measured value at the IA-device Sensor. The complete loop is shown in Figure 1 where an IA-controller and IA-devices are connected as end stations in the network.

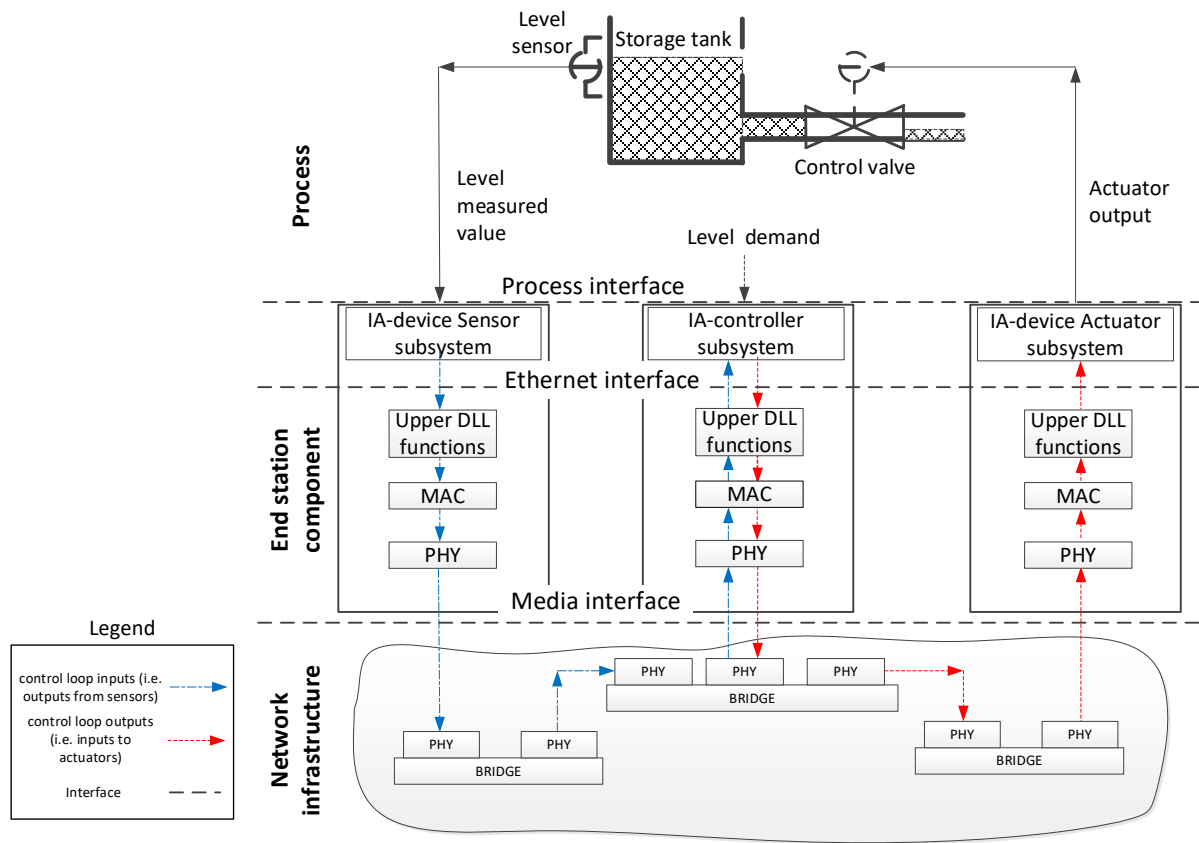


Figure 1 – Data flow in a control loop

In operation, the IA-device Sensor subsystem samples the measured value and the sampled values are transferred through the network as data packets for the IA-controller subsystem to compare with the demand value. After some computation time, the output is transferred from the IA-controller subsystem to the IA-device Actuator subsystem for implementation as a change in the external process.

This sequence repeats continuously as a regular operation using a Working Clock. The Working Clock is traceable to an ARB timescale or to the PTP timescale. Traceability to the PTP timescale is not required by all applications. For stability, the sequence repetition time (i.e., sampling time) is typically chosen to be between one fifth to one tenth of the time constant of the process response although shorter times can also be used.

NOTE 2 The actual ratio of the process response time constant to sampling time required for stability depends on the implementation.

Control latency is a critical factor in all types of control and is bounded. Components contributing to the control latency time (as shown in Figure 1) are:

- application time for sampling, computation, and processing within each IA-controller and IA-device. These are specific to the IA-device and IA-controller and known to the IA-device or IA-controller makers,

- the time for data transfer through the upper DLL functions, MAC and PHY layers within each IA-controller and IA-device. This time depends on the implementation of these components, their situation-dependent load and performance, and configuration elements related to QoS supported by these components, and
- the End Station and Bridge schedule and transfer time through the network. These are influenced by the configuration process, which allocates available bandwidth and priorities to various types of application messages.

Offline engineering of the network is possible, including the calculation of the control latency time. During system operation, management services are provided for diagnostics and checking the performance indicators of an installed network.

4.2 Industrial applications

4.2.1 General

Industrial applications can contain multiple tasks. These tasks are executed based upon time or other events. Thus, an industrial application can have multiple tasks executing on different cycles as shown in Figure 2 and Figure 3.

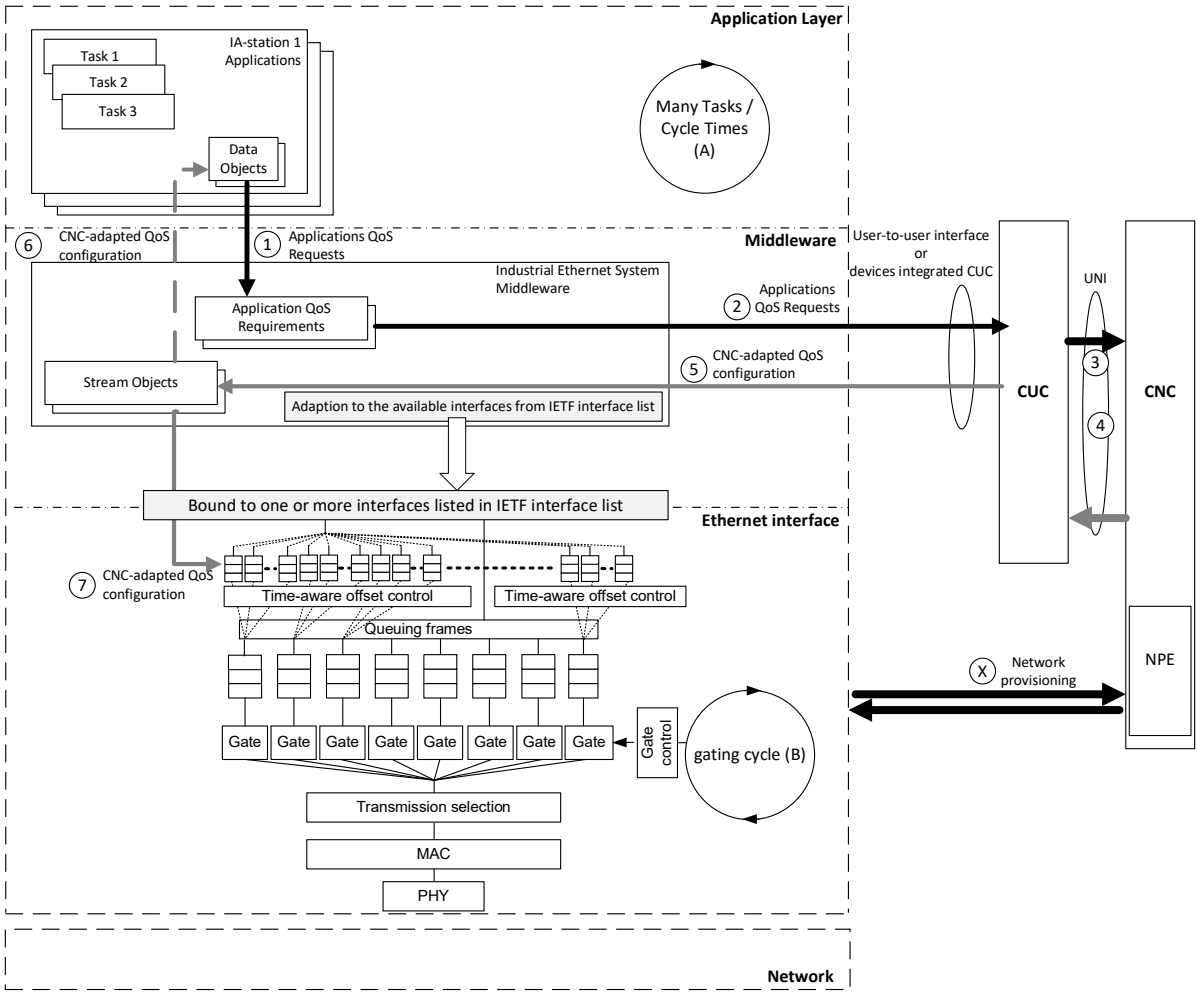
Examples of these tasks include the following.

- Background tasks, which are executed when no other task is running. There can be zero, one, or more such tasks in an industrial application.
- Main task which executes periodically. The start and execution of this task is often based upon the ARB timescale. There can be zero or one such task, in an industrial application.
- Global Time tasks. The start and execution of these tasks is often based upon Global Time (for example, at noon every day or at noon every Friday). There can be zero, one or more such tasks in an industrial application.
- Process driven tasks which are started by an event (for example, a sensor value reaches a defined point, or a process fault occurs). There can be zero, one or more such tasks in an industrial application.
- Control loop tasks which are bound to the Working Clock and started periodically. There can be zero, one or more such tasks in an industrial application.

A user defines the required automation tasks along with the data objects required as output and input for these tasks and the end station which hosts these tasks. Thus, these tasks are bound to data objects, which are exchanged between end stations per the user's definition. Many of these tasks have timing requirements, which are added as attributes to the assigned data objects. Examples of these attributes include:

- An update interval: [DataObject_Update_Interval] (time between two consecutive updates at the transmitting end station);
- A deadline: [DataObject_Deadline] (latest receive time at the end station, relative to the start of the DataObject Update Interval);
- The size of the DataObject: [DataObject_Data_Size];
- Other attributes to form a stream-list request according to IEEE Std 802.1Qdj-2024, 46.1.5.

NOTE These attributes are provided for illustration purposes. The list is not representative of all industrial applications. These are not network attributes.



- Legend:
- (X) IA-stations are configured by the CNC before any CUC to CNC action happens. For example, this configuration can include the gating cycle and transmission selection. Later accesses can happen concurrently with a CUC to CNC sequence.
 - (1) ... (7) Sequence of actions taken
- Time-aware offset control Queues are shared between multiple middleware components.

Figure 2 – IA-station interaction with CNC – Transmit path

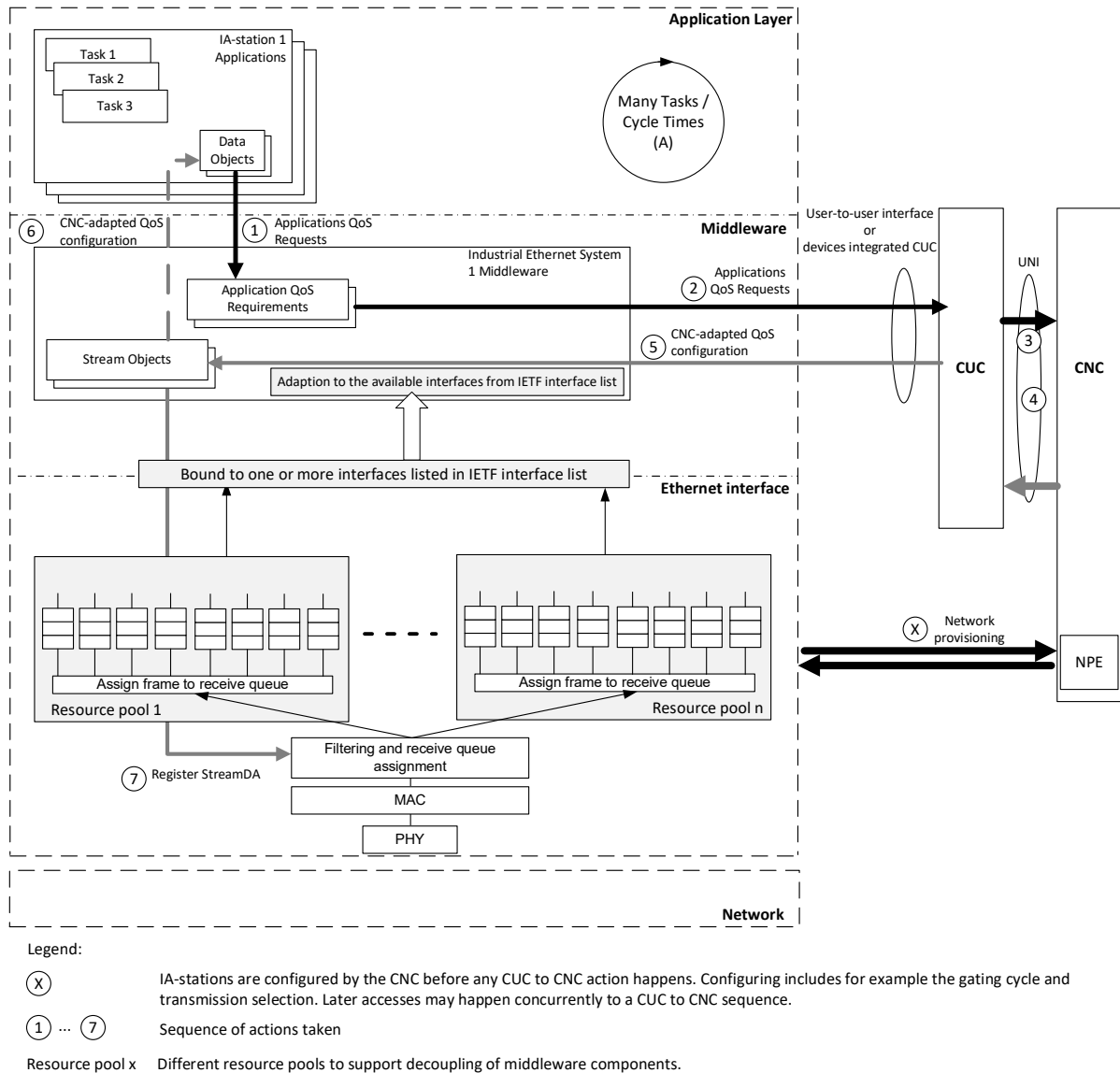


Figure 3 – IA-station interaction with CNC – Receive path

4.2.2 Control loop tasks

Multiple control loop tasks can be implemented and run in parallel in their automation devices. For example, this behavior can be implemented by using a common Working Clock, a common starting point relative to the Working Clock and a common duration for this control loop task at the involved IA-devices and IA-controllers. The data objects associated with the control loop share common values for some attributes (for example, the same values for DataObject_Update_Interval and DataObject_Deadline). Multiple control loop tasks can be implemented and run in parallel in their automation devices.

4.2.3 Start of control loop tasks

The calculation of the starting point for a control loop task is independent from the time when the device is powered up or connected to the Configuration Domain. The start of a control loop task, which is based on the Working Clock, can be calculated in the following manner: divide the Working Clock value, expressed as an integer, by the duration of the control loop task, expressed as an integer, whenever the Working Clock value increases by one. A remainder of zero provides the basis for the start of the control loop task.

NOTE The units of the Working Clock value and the units of the duration of the control loop task are the same.

Stations in the network associated with the control loop synchronize to a Working Clock using IEEE Std 802.1AS-2020.

4.3 IA-stations

An IA-station can be a simple end station acting as source or destination for control data traffic. In addition, an IA-station can be a combined functional unit that includes an end station component together with a Bridge component in one chassis. IA-stations, incorporating multiple functional units with several end station components and Bridge components within one chassis, can also be found in industrial automation. Within this kind of combined IA-station various components can be connected by internal ports and internal LANs. All components utilize a common management entity as shown in Figure 4.

Figure 4 shows an example IA-station incorporating four functional units in one chassis. Functional unit 1 and functional unit 2 each consist of a Bridge component and an end station component. The end station components are connected by internal ports via internal LANs to the Bridge components. The Bridge components include two external ports each. Functional unit 3 includes only a single end station component with one external port. Functional unit 4 includes a single end station component with two external ports.

IA-controllers and IA-devices as well as the management entity are IA-station functions acting as source of or destination for link layer data traffic. Thus, each IA-station incorporates at least one end station component where these functions can be located. Figure 4 shows that IA-station functions can either reside in a single end station component (IA-device 1, IA-controller 1, IA-device 2, IA-device 3, IA-controller 3) or in multiple end station components (IA-controller 2, management entity).

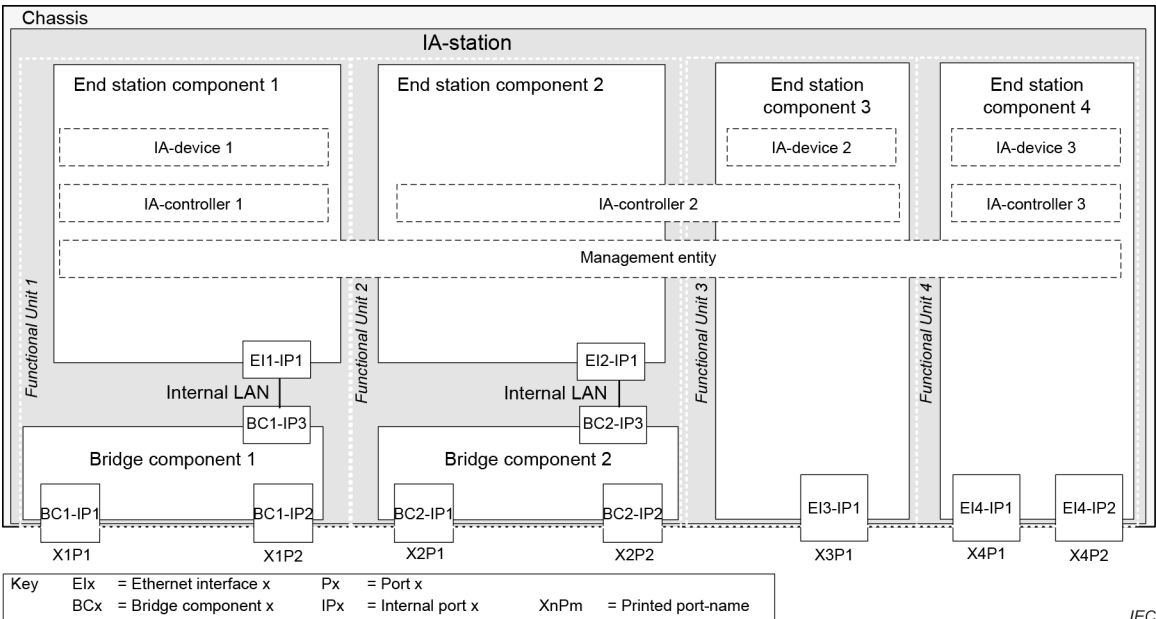


Figure 4 – IA-station example

4.4 Ethernet interface

One or more middleware components act as a layer between applications and the Ethernet interface. Figure 2 and Figure 3 show the relation between applications, middleware, Ethernet interface and the network. Various applications can run in parallel on an automation device. Data objects represent the information exchanged between applications running in different end stations. The application requirements contained in these data objects are translated by the

middleware into stream requirements for use by the CUC. This translation can be accomplished in one or both of the following ways.

- a) The user defines the data objects and translates them into stream requirements and end-station communication-configurations. A user-specific mechanism is used to configure the network components, establish paths, and the time-aware offset control.
- b) The user defines the data objects and associates them with QoS requirements for each stream (application QoS requirements). These can be forwarded as stream requirement requests by a CUC to a CNC. The CNC responds by providing a stream configuration response. The request and response are specified in IEEE Std 802.1Qdj-2024. This information is used to configure the time-aware offset control, which utilizes per-stream queues. The CUC can be integrated into the end station or can be accessed via a user-to-user protocol. The middleware uses this information for configuring Talkers and Listeners. This information is also used to add additional timing information to the data objects for application usage.

Time-aware offset control utilizes per-stream queues (see IEEE Std 802.1Q-2022, Figure 34-1) and the traffic specification of the streams, including transmission offsets, provided by the CNC to ensure the order of stream transmission.

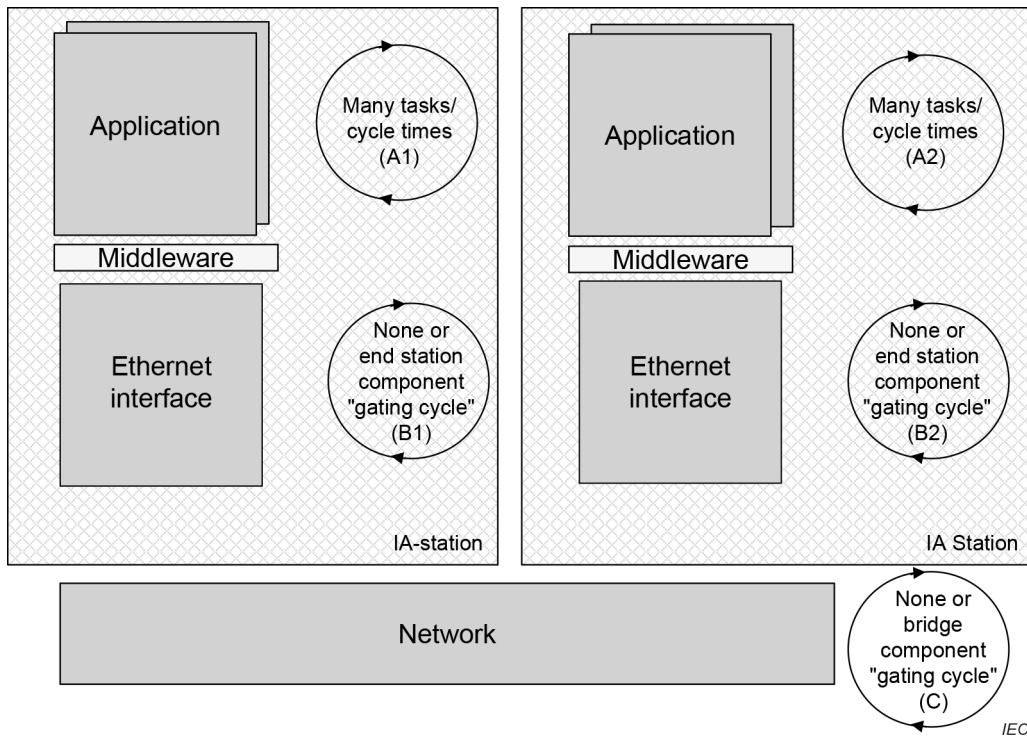


Figure 5 – Model for cycles

These automation systems, which are built from various end stations and connected via bridges, can share a common gating cycle or each station can have its own gating cycle. Alternatively, a bridge or end station can have no gating cycle (expressed as "none" in Figure 5).

4.5 Mechanisms to meet control loop latency requirements

Meeting latency requirements on a network can be accomplished using one or more combinations of the mechanisms enumerated below. The choice of a mechanism or a subset of the mechanisms listed below depends on the nature of the application(s) and the corresponding latency requirements:

- a) defining, testing, and simulating all possible application combinations and associated traffic patterns,

- b) overprovisioning the network,
- c) providing scheduled time slots for each application to transmit on the network,
- d) preempting lower priority traffic,
- e) providing scheduled time slots for certain traffic classes,
- f) time-aware offset control, and
- g) enforcing deterministic queuing delays in bridges.

NOTE This list is not comprehensive and not all mechanisms mentioned here are part of this specification. For specific mechanisms covered by this document please refer to Clause 5.

Frame preemption is specified in IEEE Std 802.1Q-2022 and IEEE Std 802.3-2022.

Reserving time on the network for certain traffic types can be done through enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 8.6.8.4. An aligned gating cycle is defined. Once a gating cycle is defined, portions of a cycle time can either be allocated to streams or classes of streams.

Multiple Talker-Listener pairs can be used for streams between end stations. Engineered time-triggered transmit can be used to coordinate transmission of all the traffic that shares a network to meet application requirements.

Creating a traffic load model in advance allows analysis of resulting traffic. It can be used to select and implement appropriate mechanisms to achieve latency requirements.

4.6 Translation between middleware and network provisioning

4.6.1 Mapping translation table

Application engineering can be done without knowledge of the network provisioning. Since the application is not aware of the network provisioning, it cannot directly map to the network configuration, for example, the use of PCP or VID as configured in the network. This problem is solved by providing a translation table, in the form of a YANG module definition, to the middleware. The IA-station's local YANG datastore contains this information.

Figure 6 shows examples of the translation models.

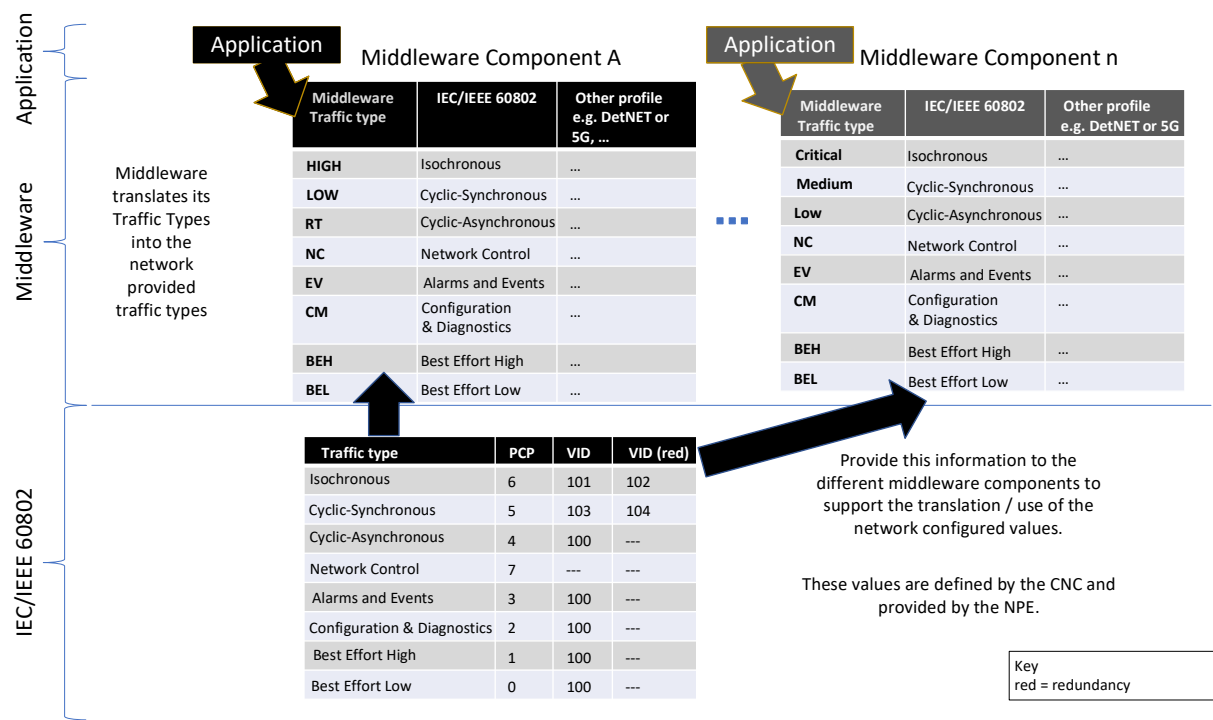


Figure 6 – Traffic type translation example

The YANG module `iecieee60802-tt-mapping.yang` is used to provide the required mapping information to all installed middleware and applications. The information in the YANG module provides the VLAN-ID, the assigned traffic types with their PCP values and redundancy information (6.4.10.8.7).

4.6.2 PTP Instances

PTP domain numbers are also configured during network provisioning. The middleware is informed about which PTP domain is assigned to which target clock by providing `descriptionDS.userDescription` names according to IEEE Std 1588-2019, 8.2.5.5, in a translation table.

`descriptionDS.userDescription` names allow the support of multiple middleware components at one IA-station using the same PTP Instances (see 6.2.13). An IA-station's local database stores this information.

Figure 7 shows examples of the translation models.

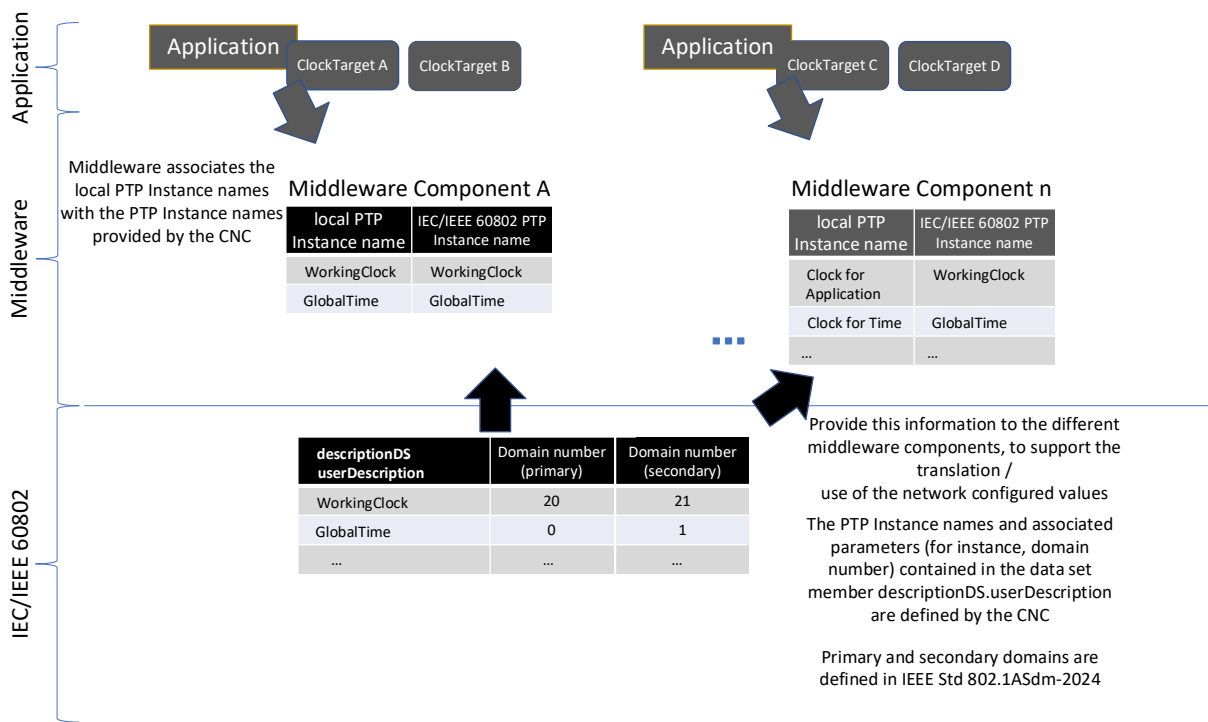


Figure 7 – PTP instance translation example

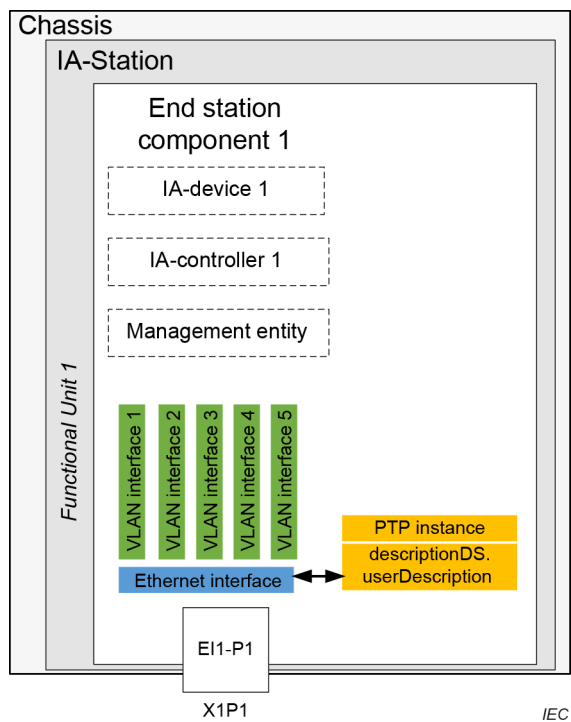


Figure 8 – descriptionDS.userDescription used for PTP instance translation

The userDescription contains the clock type (i.e., WorkingClock, GlobalTime, or both). This information is used by the middleware to align to the intended ClockTarget or ClockSource (see 6.2.13).

4.7 Industrial traffic types

4.7.1 General

Industrial automation applications make use of different traffic schemes/types for different functionalities (for example, parameterization, control, alarming). The various traffic patterns have different characteristics and thus impose different requirements on a network. To specify these traffic types, a two-step approach is used.

- a) First, define characteristics of generic traffic types (traffic type categories).
- b) Second, define instances of the generic traffic types (i.e. the traffic types).

4.7.2 Traffic type characteristics

The traffic type characteristics in Table 2 enable the identification of several distinct traffic types that are shared among sets of industrial applications.

Table 2 – Traffic type characteristics

Characteristic	Description
Cyclic	Traffic types consist of frames that can either be transmitted on a reoccurring time period (cyclic) or at no set period (acyclic). Available selections are: • required: traffic frames are transmitted cyclically, or • optional: implementation of cyclic traffic is at the discretion of the user.
Data delivery requirements	Denotes the delivery constraints for the traffic. Four options are specified. • Frame Latency: data delivery of a frame for a given Talker-Listener pair occurs within a bounded timespan. • Bandwidth: data delivery up to a certain number of frames or data size (including bursts of frames) occurs over a defined period. • Deadline: data delivery of a frame to a given Listener occurs at or before a specific point in time. • No: Denotes the case of traffic types with no special data delivery requirements
Time-triggered transmission	Talker data transmission occurs at a specific point in time based upon the Working Clock. Available selections are: • required, or • optional: implementation of time-triggered transmission is at the discretion of the user. Enhancements of scheduled traffic is only one means of achieving time-triggered transmission. Other, application-based, methods are possible
Stream-based	Indicates whether a traffic type requires registering a stream, i.e., allocating dedicated resources, for the traffic, before a given Talker can start transmitting data. • True: Data can only be transmitted after a Stream has been set up. • False: Data can be transmitted at any time.

4.7.3 Traffic type categories

4.7.3.1 General

The two-step approach described in 4.7.1 allows a clear differentiation between characteristics as seen from the "network" point of view and "application" point of view. Traffic type categories allow different IEEE 802 feature selections to achieve the goals of a specific network deployment. Four traffic type categories are identified in industrial automation systems:

- a) IA time-aware stream,
- b) IA stream,
- c) IA traffic engineered non-stream, and

d) IA non-stream.

4.7.3.2 IA time-aware stream

The characteristics of this traffic type category are shown in Table 3.

Table 3 – IA time-aware stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Deadline or Frame Latency
Time-triggered transmission	Required
Stream-based	True

4.7.3.3 IA stream

The characteristics of this traffic type category are shown in Table 4.

Table 4 – IA stream characteristics

Characteristics	
Cyclic	Required
Data delivery requirement	Frame Latency
Time-triggered transmission	Optional
Stream-based	True

4.7.3.4 IA traffic engineered non-stream

The characteristics of this traffic type category are shown in Table 5.

Table 5 – IA traffic engineered non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	Bandwidth
Time-triggered transmission	Optional
Stream-based	False

4.7.3.5 IA non-stream

The characteristics of this traffic type category are shown in Table 6.

Table 6 – IA non-stream characteristics

Characteristics	
Cyclic	Optional
Data delivery requirement	No
Time-triggered transmission	Optional
Stream-based	False

4.7.4 Traffic types

4.7.4.1 General

Table 7 summarizes relevant industrial automation traffic types and their associated characteristics. In an industrial automation system, other applications, such as audio or video, utilizes one of these traffic types.

Table 7 – Industrial automation traffic types summary

Traffic type name	Cyclic	Data delivery requirements	Time-triggered transmission	Stream based	Traffic type category
Isochronous	Required	Deadline	Required	True	IA time-aware stream
Cyclic-synchronous	Required	Frame Latency	Required	True	IA time-aware stream
Cyclic-asynchronous	Required	Frame Latency	Optional	True	IA stream
Alarms & events	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Configuration & Diagnostics	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Network Control	Optional	Bandwidth	Optional	False	IA traffic engineered non-stream
Best Effort High	Optional	No	Optional	False	IA non-stream
Best Effort Low	Optional	No	Optional	False	IA non-stream

4.7.4.2 Isochronous

Isochronous is a type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Listeners have individual deadline requirements. Cycle times are typically in the range of microseconds to tens of milliseconds. Frame size is typically below 500 octets. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.3 Cyclic-synchronous

Cyclic-synchronous is a type of IA time-aware stream traffic. This type of traffic is transmitted cyclically using time-triggered transmission. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of hundreds of microseconds to hundreds of milliseconds. Frame size is unconstrained except as indicated in 5.5.1. Talker-Listener pairs are synchronized to the Working Clock. The network is configured by the CNC to provide zero congestion loss for this traffic type. This type of traffic is normally used in control loop tasks.

4.7.4.4 Cyclic-asynchronous

Cyclic-asynchronous is a type of IA stream traffic. This type of traffic is transmitted cyclically with latency requirements bounded by the interval as specified in IEEE Std 802.1Q-2022, 46.2.3.5.1. Talker-Listener pairs have individual latency requirements. Cycle times are typically in the range of milliseconds to seconds. Frame size is unconstrained except as indicated in 5.5.1. Data exchanges between Talker-Listener pairs are typically not dependent on the Working Clock. This traffic type typically tolerates limited congestion loss. The network is

1103 configured by the CNC to handle this traffic type without loss, up to a certain number of frames
1104 or data size.

1105 **4.7.4.5 Alarms and events**

1106 Alarms and events are a type of IA traffic engineered non-stream. This type of traffic is
1107 transmitted cyclically or acyclically. This traffic expects bounded latency including time for
1108 retransmission in the range of milliseconds to hundreds of milliseconds. The source of the alarm
1109 or event typically limits the bandwidth allocated to this traffic. Frame size is unconstrained
1110 except as indicated in 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss
1111 is expected. The network is configured by the CNC to handle these frames, including bursts of
1112 frames, up to a certain number of frames or data size over a defined period.

1113 **4.7.4.6 Configuration and diagnostics**

1114 Configurations and diagnostics are a type of IA traffic engineered non-stream. This type of traffic
1115 is transmitted cyclically or acyclically. This traffic expects bounded latency, up to seconds,
1116 including time for retransmission. The source of configuration or diagnostics frames typically
1117 limits the bandwidth allocated to this traffic. Frame size is unconstrained except as indicated in
1118 5.5.1. Congestion loss can occur. Retransmission to mitigate frame loss is expected. The
1119 network is configured by the CNC to handle these frames, including bursts of frames, up to a
1120 certain number of frames or data size over a defined period.

1121 **4.7.4.7 Network control**

1122 Network control is a type of IA traffic engineered non-stream. This type of traffic can be
1123 transmitted cyclically or acyclically. This traffic expects bounded latency including time for
1124 retransmission. Frame size is unconstrained except as indicated in 5.5.1. The network is
1125 configured by the CNC to handle these frames, including bursts of frames, up to a certain
1126 number of frames or data size over a defined period. If these limits are exceeded congestion
1127 loss can occur. Network control is comprised of services required to maintain network operation.
1128 Examples include time synchronization, loop prevention, and topology detection.

1129 **4.7.4.8 Best effort**

1130 Best effort is a type of IA non-stream. The network is configured by the CNC so that these
1131 frames do not interfere with other traffic types. These frames are forwarded when resources
1132 are available. Congestion loss resulting in frame drop can occur. It is sometimes desirable to
1133 have more than one traffic class for best effort traffic (see Table 8).

1134 **4.7.4.9 Traffic class to traffic type mapping**

1135 Table 8 provides an example for the usage of traffic classes based on the traffic type:

1136 **Table 8 – Example traffic class to traffic type mapping**

Traffic class	Queue Assignment (8 queues)	Queue Assignment (4 queues)	Traffic type
7	6	2	Isochronous
6	5	1	Cyclic-Synchronous
5	4	1	Cyclic-Asynchronous
4	7	3	Network Control
3	3	0	Alarms and events
2	2	0	Configuration & Diagnostics
1	1	0	Best Effort High
0	0	0	Best Effort Low

NOTE An example of mapping of PCP and traffic type to an application is provided in Figure 6.

The traffic type categories definition allows different IEEE 802 feature selections to achieve specified goals. Moreover it helps in identification of the traffic protection mechanisms. Adherence to this example of a common mapping helps minimize potential conflicts between traffic types.

4.8 Security for TSN-IA

4.8.1 General

Subclause 4.8 describes selected aspects of TSN-IA security. Protecting the management of industrial communication is the main objective of TSN-IA security. The protection of communications that use industrial traffic types is not addressed by this document.

4.8.2 Security configuration model

Security configuration is a part of system engineering and configuration. The security configuration in this document does not encompass the supply of configuration objects for middleware and application security. Security configuration settles the prerequisites for protecting the establishment and management of communications that use industrial traffic types (see 4.7). It ensures that the security features of IA-stations (including CNCs) can be used for protecting message exchanges and authorizing the resource accesses during stream establishment and management. This security configuration supplies deployment-specific configuration objects to IA-stations. They encompass:

- instructions about cryptographic algorithms,
- credentials and trust anchors,
- instructions to interpret the outcome of peer entity authentication while enforcing resource access controls, and
- access control rules and permissions.

This security configuration uses NETCONF/YANG request/response exchanges, as follows.

- The to-be-configured IA-stations act in NETCONF server role with respect to their security configuration.
- A NETCONF client is responsible for setting-up IA-stations for security. This NETCONF client possesses information about the security relationship to be established during security configuration or about the expectations on the IA-stations in a Configuration Domain. It can be implemented as part of an interactive or automated process (for example an engineering tool, or CNC operation). As an implication, the security configuration includes options for interactive and automated setup, i.e. security configuration is done by human or non-human actors.

NOTE NETCONF notifications can also be used to recognize events such as a near-term end-of-life of certificate objects, especially EE certificate objects (see IETF RFC 4210, 3.1.1).

- The security configuration exchanges supply deployment-specific objects (trust anchors, credentials, etc.) to IA-stations and manages them. IA-stations that are in factory default state can only possess manufacturer-specific security objects (trust anchors, credentials, etc.) when booting initially. The protected NETCONF/YANG exchanges with IA-stations that are in factory default state are outlined in 4.8.3 to 4.8.6.

4.8.3 NETCONF/YANG processing

Securing NETCONF/YANG resources (for example, NETCONF sessions or managed objects) on NETCONF servers is specified by IETF RFC 6241 (NETCONF). Therefore, message exchange protection between NETCONF clients and servers as well as resource access authorization by NETCONF servers are utilized.

- IETF RFC 7589 and draft-ietf-netconf-over-tls13 (NETCONF-over-TLS) specify a solution to protect NETCONF message exchanges by TLS.
- IETF RFC 8341 (NACM) specifies three access control points, covering the request/response and notification model in NETCONF according to IETF RFC 8341, 2.1.

NETCONF servers enforce security as shown in Figure 9. The processing steps are executed upon the current configuration of the NETCONF server's YANG modules.

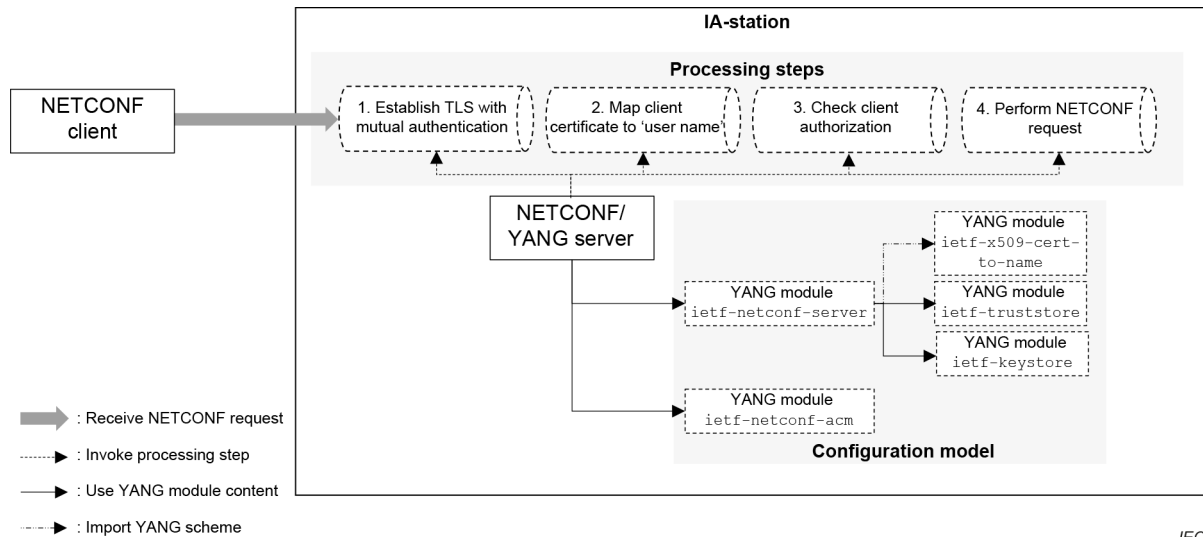


Figure 9 – NETCONF/YANG security processing steps

The processing steps on the side of NETCONF servers are as follows.

- 1) Establish a TLS connection with mutual authentication: the NETCONF server acts as TLS server and awaits connection requests of NETCONF clients (TLS clients). At the beginning of the TLS handshake, the TLS client and server negotiate the TLS protocol version to be used. During the TLS handshake the NETCONF server authenticates itself towards the NETCONF client by a credential from its ietf-keystore YANG module. In addition, the NETCONF server challenges the NETCONF client for authentication and verifies its authentication by trust anchors in its ietf-truststore YANG module according to 6.3.4. A successful mutual authentication is a prerequisite for proceeding to the next step.
- 2) Map the client certificate to a username: the NETCONF server maps the authenticated TLS client certificate to a "NETCONF username"³ by applying an ordered list of mapping instructions. These instructions are provided in its ietf-x509-cert-to-name YANG module. The applicable list item is identified by matching its configured fingerprint (according to IETF RFC 7589, Clause 7) against the certification path that was used for TLS client authentication (an end entity certificate or a CA certificate). According to the map type of the identified list item, the NETCONF server determines the "NETCONF username". This can be done by extracting information from the end entity certificate of the NETCONF client. A successful certificate-to-"NETCONF username" mapping is a prerequisite for proceeding to the next step.
- 3) Check client authorization: the NETCONF server checks if the NETCONF client has the permission to access the requested NETCONF/YANG resource based on its "NETCONF username" and the access control rules available in its ietf-netconf-acm YANG module. See 4.8.4 for more information about NETCONF/YANG access control. A successful authorization is a prerequisite for proceeding to the next step.
- 4) Perform NETCONF request: if all preceding steps succeeded, the NETCONF server performs the NETCONF request.

³ In this document and, in almost all cases, NETCONF username values do not represent human users.

1217 4.8.4 NETCONF/YANG access control

1218 NACM defines a YANG information model for describing permitted/denied access operations.
 1219 NETCONF servers are responsible for enforcing access control to their resources according to
 1220 the information in their ietf-netconf-acm YANG modules. NACM allows the description of
 1221 access-controlled resources in terms of NETCONF protocol operations, nodes in YANG
 1222 datastores or types of notification events. NACM uses character strings to represent the subject
 1223 actors, i.e. NETCONF clients. These character strings are known as "NETCONF username".
 1224 The NACM access control information of a NETCONF server is created, updated, and deleted
 1225 per IA-station. The management of this information happens along the IA-station lifecycle, for
 1226 example, manufacturing, bootstrapping, operation, maintaining, re-owning, destructing.
 1227 Moreover, the management of the NACM access control information itself is subject to NACM
 1228 access control.

1229 This document employs multiple YANG data models for fulfilling its purposes. This extends
 1230 beyond the above identified YANG modules (see 4.8.3). The NETCONF server on an IA-station
 1231 enforces access control for NETCONF/YANG resources. To meet this objective, the NETCONF
 1232 server on an IA-station is supplied with access control information for the used
 1233 NETCONF/YANG resources. NACM is employed for this purpose and profiles default access
 1234 control information for the NETCONF/YANG resources (see 6.3.2.2). This relieves other
 1235 organizations or individuals for example, manufacturers, integrators, operators, owners from
 1236 being responsible to create NACM access control information for the respective
 1237 NETCONF/YANG resources.

1238 NACM relies on character strings (known as "NETCONF username") to refer to clients. NACM
 1239 access control information as specified in this document, populates the "NETCONF username"
 1240 character strings in NACM with role names specified in 6.3.2.1.4 c). This allows to create default
 1241 NACM information without knowing actual names of individual entities. A role name can refer
 1242 to 0, 1 or more individual entities. It is the responsibility of users to assign role names to
 1243 individual entities. This happens by binding the assigned role names to the credentials of
 1244 individual entities. The current form to express this binding is a role extension in the identity
 1245 certificates of end entities defined in this document. These are NETCONF clients, i.e. these
 1246 role extensions appear in the end entity certificates of LDevID credentials for NETCONF clients.

1247 As initial step NACM maps the NETCONF username to a set of groups. The set of groups
 1248 determines the set of rules to be applied for access-controlled resources.

1249 4.8.5 Identity checking

1250 IETF RFC 7589 (NETCONF-over-TLS) specifies that NETCONF clients check the identity of
 1251 NETCONF servers (IETF RFC 7589, Clause 6) and that NETCONF servers verify the identity
 1252 of NETCONF clients (IETF RFC 7589, Clause 7).

1253 The NETCONF server identity check happens inside NETCONF clients. It matches an actual
 1254 against an expectation.

- 1255 • The actual server identity is established by the end entity certificate of the NETCONF server
 1256 (authenticated by means of TLS).
- 1257 • The expectations on server identity are established by the information that is used to
 1258 connect to the NETCONF server.

1259 IETF RFC 7589 refers to IETF RFC 6125, Clause 6, for the details of retrieving the actual and
 1260 comparing it against the expected.

1261 The NETCONF client identity check happens inside NETCONF servers. It also matches an
 1262 actual against an expectation.

- 1263 • The actual client identity is established by the end entity certificate of the NETCONF client
 1264 (authenticated by means of TLS).

- 1265 • The expectations on client identity are established by the contents of the ietf-netconf-acm
1266 and ietf-x509-cert-to-name YANG modules.

1267 The details of this check are subject to the requested NETCONF operation. IETF RFC 7589,
1268 Clause 7, specifies the mapping of an authenticated client certificate to a "NETCONF username"
1269 whose permissions are then enforced by IETF RFC 8341 (NACM). More information is provided
1270 in 4.8.3, steps 2 and 3.

1271 **4.8.6 Secure device identity**

1272 **4.8.6.1 Device identity**

1273 The term 'device' originates from IEEE Std 802.1AR-2018. It matches the term IA-station in this
1274 document.

1275 The device identity refers to a set of information items about a device that:

- 1276 • describes a device as a physical or virtual entity in a distributed system (identifier and
1277 attribute information);
- 1278 • is used by a device to describe itself as such entity (identifier and attribute information);
- 1279 • allows to interact with this device (addressing information, i.e. a specific identifier class).

1280 The targeted use case, for example application data exchanges, configuration exchanges,
1281 inventory, or ordering, determines the required amount of identity information about a device.

1282 The device identity of any single IA-station encompasses:

- 1283 • MAC addresses, IP addresses, TCP ports, DNS names;
- 1284 • ietf-hardware YANG module contents (IETF RFC 8348, Clause 3 and 7.1).

1285 **4.8.6.2 Verifiable device identity**

1286 Certain aspects of device identity are verified before relying on them during online interactions.
1287 These are examples.

- 1288 • DNS names or IP addresses are used to call the management entity of an IA-station, i.e. its
1289 NETCONF/YANG server. Their value represents the caller's expectation on the identity of
1290 their responder in network communications. Verification of the responder's identity helps
1291 defeat DNS spoofing, component impersonation and man-in-the-middle attacks. This is
1292 specified by IETF RFC 7589 and described in IETF RFC 6125, Clause 6. Passing this check
1293 is a prerequisite before NETCONF application exchanges can happen.
- 1294 • mfg-name values in instances of the ietf-hardware YANG module. These values make claims
1295 about the IA-station manufacturer. Their verification is a means to protect against
1296 counterfeiting.

1297 The verification of IA-station identity happens according to a model that is fully specified by this
1298 document. That verification can be done in a manufacturer-agnostic manner. This verification
1299 is important before supplying locally significant credentials especially LDevID to IA-stations that
1300 are in factory-default state.

1301 **4.8.6.3 Verification support mechanisms**

1302 **4.8.6.3.1 General**

1303 Subclause 4.8.6.3 considers mechanisms that support device identity verification during online
1304 interactions with IA-stations.

4.8.6.3.2 Secure transports

Sending information in plain form over a protected channel, e.g. ietf-hardware YANG module contents via NETCONF-over-TLS protects the transferred information during its transit through the network but does not vouch for the correctness of the received information, e.g. the mfg-name value.

4.8.6.3.3 Secure information

Protecting information objects by means of a cryptographic authentication code or digital signature enables verification of the authenticity and integrity of that information. These cryptographic authentication codes can use symmetric or asymmetric schemes. In case of asymmetric schemes, raw and self-signed public keys are distinguished from CA-signed public keys.

Asymmetric schemes with CA-signed public keys are preferable for the verifiable device identity use case: claimants and verifiers share a public key; the claimant possesses the corresponding private key. The establishment and storage of shared public keys uses public key certificates. For this approach self-signed CA certificates are to be established in an authentic manner. The number of self-signed CA certificates is independent from the number of verifiers (CNCs) as well as claimants (IA-stations).

4.8.6.3.4 IDevID and LDevID credentials

IDeVID and LDeVID credentials are specified by IEEE Std 802.1AR-2018. These objects are comprised of a certification path and a private key. The certification path encompasses an end entity certificate which contains verifiable device identity in a CA-signed form. The device identity verification happens after validating the certification path (IETF RFC 5280, Clause 6) and checking the proof-of-possession for the private key. The certification path validation demands trust anchors as input arguments (IETF RFC 5280, 6.1.1 input argument (d)).

Two types of credentials are distinguished by IEEE Std 802.1AR-2018:

- IDeVIDs are issued by device manufacturers. They represent an initial identity as it is known at device production-time. The initial device identity is not locally significant: it cannot contain deployment-specific information such as DNS names or IP addresses.
- LDeVIDs are issued by other actors, e.g. a device user. They represent a locally significant device identity: they can contain deployment-specific information, e.g. DNS names or IP addresses.

IEEE Std 802.1AR-2018, Clause 6, uses signature suites to describe the subject public key and the signature fields in IDeVID and LDeVID certification paths. This notion is different from TLS cipher suites.

NOTE IDeVID and LDeVID credentials also serve purposes beyond secure device identity, for instance the realization of secure transports. This facilitates the use case of NETCONF/YANG security setup from factory default state.

4.8.6.3.5 IDevID items beyond IEEE Std 802.1AR-2018

IEEE Std 802.1AR-2018 allows verification of the following identity items:

- certificate issuer (not necessarily: manufacturer) by issuer field (data type: ASN.1 Name), and
- if present: device instance by serialNumber value (data type: ASN.1 PrintableString).

NOTE 1 IEEE Std 802.1AR-2018 represents the initial device identity as an optional serialNumber attribute (OID 2.5.4.5) in the subject field of the EE certificate. This value is unique within the domain of significance of the EE certificate issuer.

NOTE 2 This verification can happen after certification path validation and the proof-of-possession checking for the private key.

The following bullet points describe options beyond IEEE Std 802.1AR-2018 for verifying the device identity of IA-stations in factory default state. It also identifies informational items for the corresponding checks:

- IA-station manufacturer check: using names that identify IA-station manufacturers, e.g. mfg-name in ietf-hardware YANG module,
- IA-station type check: using attributes that identify IA-station types, e.g. model-name, hw-revision, description in ietf-hardware YANG module, and
- IA-station instance check: using values that identify IA-station instances, e.g. serial-num in ietf-hardware YANG module.

The following model described in the bullet points applies to the verification of the initial device identity of IA-stations:

- the set of to-be-conducted checks is determined by IA-station and CNC users,
- an IA-station uses IDevID credentials to prove its device identity. The checking happens by means of online interactions in the operational network. It happens automatically and is done by CNCs. This does not depend on Configuration Domain external repositories, and
- other stakeholders, e.g. middleware/application consortia or individual manufactures are allowed to additionally express information items in IDevID credentials to reflect their device identity model. CNCs do not assess such additional information.

4.8.6.3.6 Device identity representation in IDevID and LDevID credentials

The best practices for representing verifiable device identity information in IDevID and LDevID credentials (see 6.3.3.2.2 for more information) are as follows.

- Corresponding information (actual values or references to them) appears in EE certificates:
 - IDevID EE certificates bind initial device identity items that are known by the device manufacturer at production time, e.g. mfg-name.
 - LDevID EE certificates bind locally significant device identity items that are known by other actors such as device users, e.g. DNS names or IP addresses. They can also bind initial device identity information.
 - Items that encode device naming information appear in the subjectAltName extension.
- NOTE This is specified in IETF RFC 5280, 4.2.1.6. It is further explained in IETF RFC 6125, 2.3.
- A binding can take one of following forms. Multiple forms can appear in one EE certificate:
 - by-value: the verifiable device identity information is represented by its value inside the IDevID resp. LDevID EE certificate. Examples are:
 - the product serialNumber in IDevID credentials (IEEE Std 802.1AR-2018), and
 - the hostname of the NETCONF/YANG server in LDevID credentials (IETF RFC 6125, Clause 6).
 - by-ref: the verifiable device identity information is represented by a reference inside the IDevID resp. LDevID EE certificate, not by its value.
 - The actual value can be provided by the device itself or by a device-external source.
 - If it is provided in form of an unprotected information object, then the reference object that is embedded to EE certificates includes a digest value.

5 Conformance

5.1 General

A claim of conformance to this document is a claim that the behavior of an implementation of an IA-station (see 5.5, 5.6) with its Bridge components (see 5.7, 5.8) and end station

1396 components (see 5.9, 5.10) meets the mandatory requirements of this document and may
1397 support options identified in this document.

1398 **5.2 Requirements terminology**

1399 The verbal forms for required expressions of provisions follow the conventions below.

- 1400 a) Requirements terminology is provided in the ISO/IEC Directives Part 2:2021, Clause 7. This
1401 document can be found at www.iec.ch/members_experts/refdocs.
- 1402 b) The Profile Conformance Statement (PCS) proformas (see Annex A) reflect the occurrences
1403 of the words "shall," "may," and "should" within this document.
- 1404 c) This document avoids needless repetition and apparent duplication of its formal
1405 requirements by using "is", "is not", "are", and "are not" for definitions and the logical
1406 consequences of conformant behavior. Behavior that is permitted but is neither always
1407 required nor directly controlled by an implementer or administrator, or whose conformance
1408 requirement is detailed elsewhere, is described by "can". Behavior that never occurs in a
1409 conformant implementation or system of conformant implementations is described by
1410 "cannot". The word "allow" is used as a replacement for the phrase "support the ability for",
1411 and the word "capability" means "can be configured to".

1412 **5.3 Profile conformance statement (PCS)⁴**

1413 The supplier of an implementation that is claimed to conform to this document shall provide the
1414 information necessary to identify both the supplier and the implementation and shall complete
1415 a copy of the PCS proforma provided in Annex A.

1416 **5.4 Conformance classes**

1417 This document includes conformance requirements and options that are related to an entire
1418 station, as well as conformance requirements and options that are related to single Bridge or
1419 end station components within an IA-station. Figure 10 illustrates this conformance model.

⁴ Copyright release for the PCS: Users of this document may freely reproduce the PCS contained in this document so that it can be used for its intended purpose.

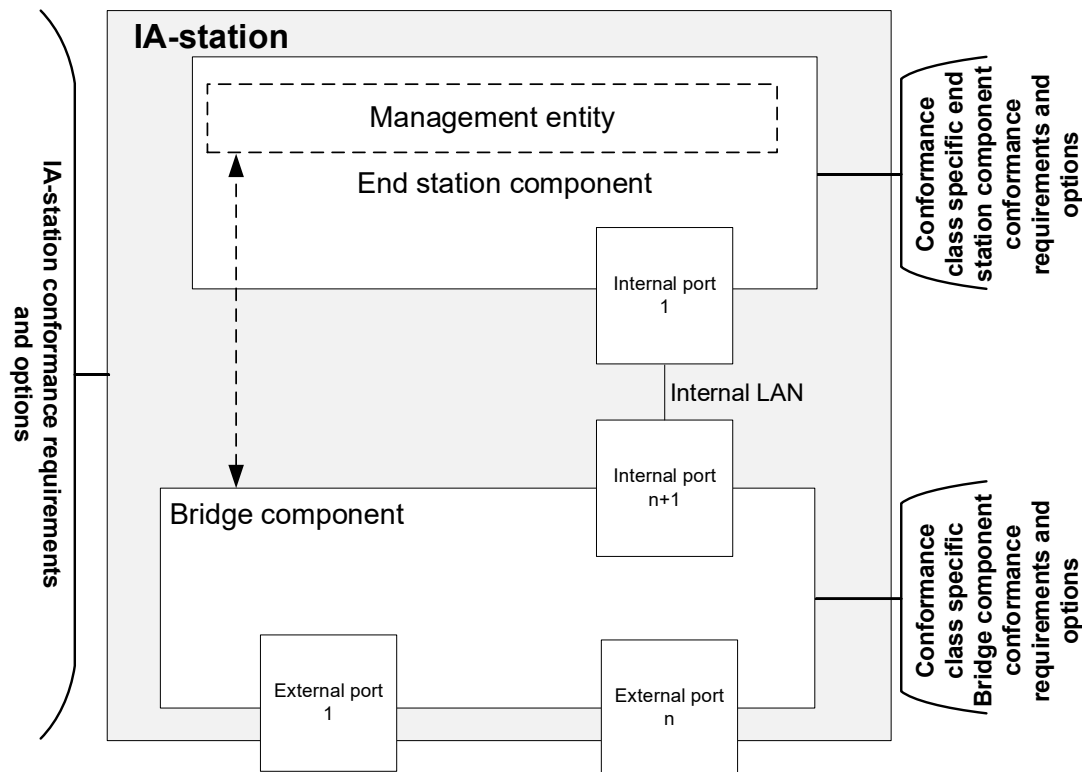


Figure 10 – IA-station conformance model

This document supports a variety of industrial use cases. In some of these use cases, support of certain TSN features can be mandatory, while in others, supporting these features can lead to non-optimal implementations. Therefore, this document defines two conformance classes that are applicable both to Bridge components and end station components. Conformance Class A (ccA) is feature-rich, i.e. tailored to use cases requiring support of many TSN-IA features. Conformance Class B (ccB) targets implementations that are more resource constrained. The details for the conformance classes are specified in 5.7 and 5.8 for Bridge components, and in 5.9 and 5.10 for end station components.

NOTE 1 It is the responsibility of the IA-station manufacturer to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single IA-station.

NOTE 2 It is the responsibility of the user to carefully consider the implications of mixing ccA and ccB Bridge components and end station components in a single Configuration Domain.

NOTE 3 Any Bridge compliant to this document is an IA-station. Any IA-station contains a management entity (i.e. an end station component).

5.5 IA-station requirements

5.5.1 IA-station PHY and MAC requirements for external ports

External ports of IA-stations that claim conformance to this document shall meet the following requirements.

- Support the Media Access Control (MAC) service specification according to IEEE Std 802.3-2022, Clause 2.
- Support the Media Access Control (MAC) frame and packet specifications according to IEEE Std 802.3-2022, Clause 3, especially the MAC Client Data field size according to IEEE Std 802.3-2022, 3.2.7, item c).
- Support Layer Management according to IEEE Std 802.3-2022, 5.2.4.

- d) Implement at least one IEEE Std 802.3-2022 MAC that shall operate in full-duplex mode, and associated IEEE Std 802.3-2022 PHY with a data rate of at least one of the following speeds: 10 Mb/s, 100 Mb/s, 1 000 Mb/s, 2.5 Gb/s, 5 Gb/s, or 10 Gb/s together with the corresponding managed objects:
 - 1) 10BASE-T1L MAU type according to IEEE Std 802.3-2022, Clauses 22 and 146,
 - 2) 100BASE-TX or 100BASE-FX MAU types according to IEEE Std 802.3-2022 Clauses 21, 22, 24, 25, 26 and IEEE Std 802.3-2022 23A, 28A, 28B, 28C, and 28D,
 - 3) 1000BASE-T or 1000BASE-SX MAU types according to IEEE Std 802.3-2022 Clauses 28, 34, 35, 36, 37, 38, and 40, and IEEE 802.3-2022 Annexes 28A, 28B, 28C, and 28D,
 - 4) 2.5GBASE-T or 5GBASE-T MAU types according to IEEE Std 802.3-2022, Clauses 28, 1458 125, and 126, and 802.3-2022 Annexes 28A, 28B, 28C, and 28D,
 - 5) 2.5GBASE-T1 or 5GBASE-T1 MAU types according to IEEE Std 802.3-2022, Clause 149,
 - 6) 10GBASE-T or 10GBASE-SR MAU types according to IEEE Std 802.3-2022, 1462 Clauses 44, 46, 49, 51, 52, 55, and IEEE Std 802.3-2022, Annexes 48A and 55A,
 - 7) 10GBASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 149,
 - 8) 100BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 96, and
 - 9) 1000BASE-T1 MAU type according to IEEE Std 802.3-2022, Clause 97.
- e) Support the YANG features and nodes of the ieee802-ethernet-interface module according to 6.4.9.2.1.
- f) Implement Ethernet support for time synchronization protocols according to IEEE Std 802.3-2022, Clause 90 as amended by IEEE Std 802.3cx™ -2023 and IEEE Std 802.3de™-2022 with the aTimeSyncSelectionDdmp set to 'symbol-after-sfd' according to IEEE Std 802.3cx-2023, 30.13.1.14.

Clauses and subclauses not mentioned can be implemented but are not obligatory.

5.5.2 IA-station topology discovery requirements

IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) The required capabilities according to IEEE Std 802.1AB-2016, 5.3 and IEEE Std 802.1ABcu™-2021, 5.3.
- b) Topology discovery and verification according to 6.5.
- c) The YANG features and nodes of the ieee802-dot1ab-lldp module according to 6.4.9.2.2.

5.5.3 IA-station requirements for time synchronization

These requirements are related to the entire IA-station with all its PTP Instances and PTP Ports. IA-stations for which a claim of conformance to this document is made shall support the following list of requirements.

- a) PTP Instance requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn™-2024, and IEEE Std 802.1ASdm-2024, 5.4.1 items a) through h).

NOTE A gPTP domain in a PTP End Instance can be used for Global Time, Working Clock, or both.

- b) PTP Instance defaults for BTCA according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2a.
- c) Timing and synchronization management according to IEEE Std 802.1AS-2020, 5.4.2 items j) and k).
- d) PTP Instance requirements according to 6.2.2.
- e) PTP Protocol requirements according to 6.2.3.

- f) Clock control system requirements according to 6.2.4.
- g) Error generation limits according to 6.2.5.
- h) PTP Instance synchronization status according to 6.2.6.
- i) The transmission of the Drift_Tracking TLV according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2 item n).
- j) External port configuration capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.4.2 f).
- k) MAC-specific timing and synchronization methods for IEEE Std 802.3 full-duplex links according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1AS-2020/Cor 1-2021, IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024, 5.5 items a) through e) and item j), and 11.2.17.
- l) The YANG features and nodes of the:
 - i) ieee1588-ptp-tt module according to 6.4.9.2.3.1,
 - ii) ieee802-dot1as-gtp module according to 6.4.9.2.3.2, and
 - iii) ieee802-dot1as-hs module according to 6.4.9.2.3.3.
- m) The message timestamp point according to IEEE Std 802.1AS-2020, 11.3.9.
- n) The Common Mean Link Delay Service (CMLDS) according to IEEE Std 802.1AS-2020, 11.2.17.
- o) The descriptionDS according to 6.2.13 and IEEE Std 1588-2019, 8.2.5.
- p) gtpCapableStateMachinesEnabled is FALSE according to IEEE Std 802.1ASdm-2024, 5.4.2a and 14.8.54a.

NOTE Setting gtpCapableStateMachinesEnabled to FALSE allows transmission of signaling message to be disabled by design.

5.5.4 Management requirements for non-proxied IA-stations

5.5.4.1 Secure management exchanges

IA-stations which contain a ccA Bridge component or a ccA end station component and for which a claim of conformance to this document is made shall support the following list of requirements.

- a) NETCONF server functionality according to IETF RFC 6241 including:
 - 1) candidate configuration capability as described in IETF RFC 6241, 8.3,
 - 2) rollback-on-error capability as described in IETF RFC 6241, 8.5, and
 - 3) validate capability as described in IETF RFC 6241, 8.6.
- b) NETCONF-over-TLS server supporting TLS version 1.2 according to 6.3.2.1.1 and 6.3.2.1.4.
- c) Secure Device Identity according to 6.3.3 and IEEE Std 802.1AR-2018, 5.3 a), using the signature suite in IEEE Std 802.1AR-2018 9.2, 5.3 d), and 5.3 i).
- d) PKIX according to 6.3.2.1.5 and IETF RFC 5280, 4.1, 4.2.1.1-3, 4.2.1.6, 6.1 and 6.2.
- e) NACM (IETF RFC 8341) supporting four different roles according to 6.3.2.1.5 c).
- f) The YANG features and nodes of the:
 - 1) ietf-keystore module according to 6.4.9.2.4.1,
 - 2) ietf-netconf-acm module according to 6.4.9.2.4.2 and,
 - 3) ietf-truststore according to 6.4.9.2.4.3.
- g) NETCONF Event Notifications according to IETF RFC 5277 including operations according to IETF RFC 5277, Clause 2.

- 1538 h) Dynamic Subscriptions to YANG Events and Datastores over NETCONF according to
1539 6.4.7.7.
 - 1540 i) NETCONF Extensions to support the Network Management Datastore Architecture (NMDA)
1541 as described in IETF RFC 8526.
 - 1542 j) DHCP client according to IETF RFC 2131, 4.1, 4.2, and 4.4.
 - 1543 k) At least one of the following asymmetric key pair generation methods:
1544 1) component-internal generation according to 6.3.4.3;
1545 2) component-external generation according to 6.3.4.3;
 - 1546 l) Storage of at least one IDevID credential according to 6.3.3.4.2.5 and one LDevID-
1547 NETCONF credential according to 6.3.4.1.
 - 1548 m) Support the 60802 device info TLV described in Annex E.
 - 1549 n) If the IA-station supports CNC functionality, that IA-station shall also support the following
1550 list of requirements.
1551 1) TSN CNC station requirements according to IEEE Std 802.1Q-2022, 5.29.
1552 2) NETCONF-over-TLS server and related client functionality 5.5.4.2.
1553 3) The common YANG modules, features, and nodes according to 6.4.9.2.
1554 4) The optional YANG modules, features, and nodes according to 6.4.9.3.
1555 5) Support the use of at least one customer VLAN Identifier for an isolation VLAN and one
1556 VLAN identifier for a default VLAN.
1557 6) Support CUC/CNC YANG modules, features and nodes according to 6.4.9.4.
 - 1558 o) If the IA-station supports CUC functionality, that IA-station shall also support the following
1559 list of requirements.
1560 1) Support NETCONF-over-TLS client functionality with client related security requirements
1561 according to 5.5.4.2.
1562 2) The TSN UNI YANG module, features, and nodes according to 6.4.9.4.1.
1563 3) The ietf-netconf-client module according to 6.4.9.4.1.
 - 1564 p) If the IA-station supports proxy functionality, that IA-station shall also support the
1565 iecIEEE60802-proxy YANG module according to 6.4.10.8.8.
- 1566 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1567 which a claim of conformance to this document is made should support internal key generation
1568 according to 6.3.4.3.2.

1569 **5.5.4.2 IA-station management YANG modules**

- 1570 IA-stations which contain a ccA Bridge component or a ccA end station component and for
1571 which a claim of conformance to this document is made shall support the YANG features and
1572 nodes for IA-station management of the:
- 1573 a) ietf-system-capabilities module according to 6.4.9.2.5.1,
 - 1574 b) ietf-yang-library module according to 6.4.9.2.5.2,
 - 1575 c) ietf-yang-push module according to 6.4.9.2.5.3,
 - 1576 d) ietf-notification-capabilities module according to 6.4.9.2.5.4,
 - 1577 e) ietf-subscribed-notifications module according to 6.4.9.2.5.5,
 - 1578 f) diagnostics using YANG-Push subscriptions according to 6.4.7,
 - 1579 g) ietf-netconf-monitoring module according to 6.4.9.2.5.6,
 - 1580 h) ietf-system module according to 6.4.9.2.5.7,
 - 1581 i) ietf-hardware module according to 6.4.9.2.5.8,
 - 1582 j) ietf-interfaces module according to 6.4.9.2.5.9,

- k) ieee802-dot1q-bridge module according to 6.4.9.2.5.10,
- l) iecieee60802-ethernet-interface module according to 6.4.9.2.5.11,
- m) ietf-netconf-server according to 6.4.9.2.5.12,
- n) iecieee60802-bridge according to 6.4.9.2.5.11,
- o) ietf-subscribed-notifications according to 6.4.9.2.5.13,
- p) iecieee60802-subscribed-notifications according to 6.4.9.2.5.13, and
- q) iecieee60802-ia-station according to 6.4.9.2.5.11.
- r) ieee802-dot1q-rstp-bridge according to 6.4.9.2.5.14.

5.5.4.3 Digital data sheet

IA-stations for which a claim of conformance to this document is made shall provide a 60802 instance data file according to 6.4.8. The instance data file shall contain at least the YANG nodes of 6.4.9 that are marked with [m]. Nodes marked with [c] shall be included if the associated feature is supported.

NOTE It is the user's responsibility to ensure that the filename is unique by using a standardized mechanism (for example, GUID, URL, or ReverseDomainName).

5.6 IA-station options

5.6.1 IA-station PHY and MAC options for external ports

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Power over Ethernet (PoE) over 2 Pairs according to IEEE Std 802.3-2022, Clause 33.
- b) Power Interfaces according to IEEE Std 802.3-2022, Clause 104.
- c) Power over Ethernet according to IEEE Std 802.3-2022 Clause 145.

5.6.2 IA-station options for time synchronization

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) The media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024.
- b) Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c).
- c) More than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d).
- d) Hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m).

5.6.3 IA-station options for management

IA-stations for which a claim of conformance to this document is made may support the following list of requirements.

- a) Writable-Running capability according to IETF RFC 6241, 8.2.
- b) Confirmed Commit capability according to IETF RFC 6241, 8.4.
- c) Distinct Startup capability according to IETF RFC 6241, 8.7.
- d) URL capability according to IETF RFC 6241, 8.8.
- e) XPath capability according to IETF RFC 6241, 8.9.
- f) NETCONF-over-TLS server supporting TLS version 1.3, according to 6.3.2.1.3.

g) PKIX according to IETF RFC 5280, 4.2.1.13, Clause 5 and 6.3.

5.6.4 Proxy target options

IA-stations which contain only ccB Bridge or end station components and for which a claim of conformance to this document is made may support the following list of requirements.

a) Secure management exchanges according to 5.5.4.1, a) through m) including support of IA-station management YANG modules according to 5.5.4.2.

b) Secure management exchanges using a management proxy according to 6.4.5.5 including the following list of requirements.

1) Provide a digital datasheet according to 5.5.4.3 including the YANG features and nodes of the `iecieeee60802-proxy-target` module according to 6.4.10.8.9.

2) Support the 60802 device info TLV described in Annex E.

5.7 Bridge component requirements

5.7.1 Common Bridge component requirements

A Bridge component implementation of any conformance class for which a claim of conformance to this document is made shall support the following list of requirements.

a) C-VLAN component requirements according to IEEE Std 802.1Q-2022, 5.5 and 5.4 except item o) in IEEE Std 802.1Q-2022, 5.4.

b) The use of Customer VLAN Identifiers (C-VID).

c) FDB to contain Static and Dynamic VLAN Registration Entries for a minimum of 10 VIDs according to IEEE Std 802.1Q-2022, 8.8.

NOTE 1 An example use case for 10 VIDs: 2 VIDs for IA time-aware stream or IA stream traffic, 2 VIDs for IA time-aware stream or IA stream redundancy, 4 VIDs for IA traffic engineered non-stream or IA non-stream traffic, 1 VID for the isolation VLAN, and 1 VID for the Default VLAN (see 6.4.5.2).

d) Translation of VIDs through support of the VID Translation Table or through support of both the VID Translation Table and Egress VID translation table on one or more Bridge Ports according to IEEE Std 802.1Q-2022, 6.9.

e) The strict priority algorithm for transmission selection on each port for each traffic class according to IEEE Std 802.1Q-2022, 8.6.8.1.

f) The capability to disable Priority-based flow control according to IEEE Std 802.1Q-2022, Clause 36, if it is implemented.

g) The Priority Regeneration requirements according to IEEE Std 802.1Q-2022, 5.4.1, item o).

h) MST according to IEEE Std 802.1Q-2022, 5.4.1.1 a) to i) and k) to o) and 6.4.2.4.

i) TE-MSTID according to IEEE Std 802.1Q-2022, 5.5.2, 8.6, and 8.8.

j) Spanning tree, VLAN, and TE-MSTID configuration according to 6.4.2.4.

k) YANG features and nodes of the `iecieeee60802-tt-mapping` module according to 6.4.10.8.7.

l) PSFP according to IEEE Std 802.1Q-2022, 8.6.5.2.1, with support for:

1) at least 3 stream filters per port according to IEEE Std 802.1Q-2022, 8.6.5.3,

2) at least one stream gate according to IEEE Std 802.1Q-2022, 8.6.5.4 with at least zero gate control list entries according to IEEE Std 802.1Q-2022, 12.31.1.4, and

3) at least 3 flow meters per port according to IEEE Std 802.1Q-2022, 8.6.5.5.

NOTE 2 The flow meter parameter values Excess Information Rate (EIR) = 0, Excess burst size (EBS) = 0 and Color Mode (CM) = `color_blind`, as configurable by the CNC, are applicable to many industrial automation use cases. When CM = `color_blind`, `DropOnYellow` (IEEE Std 802.1Q-2022, 8.6.5.5, item h), `MarkAllFramesRed` (IEEE Std 802.1Q-2022, 8.6.5.1.3, item j), and `MarkAllFramesRedEnable` (IEEE Std 802.1Q-2022, 8.6.5.5, item i) are not used.

NOTE 3 For example, an implementation could contain one flow meter for broadcast traffic, one flow meter for multicast traffic and one flow meter for unicast traffic. This example can be realized using 49 mask and match stream identification functions per Port and 49 associated stream filter instances (one-to-one relationship) that

1673 identify and map unicast (1 stream identification function for identifying all frame with the group address bit
 1674 cleared in the DA), broadcast (1 stream identification function for identifying all frames with all DA bits set) and
 1675 multicast (47 stream identification function with the DA group address bit set and one other DA bit cleared for
 1676 all 47 combinations).

1677 m) The YANG features and nodes for flow meter configuration according to 6.4.9.2.5.14.

1678 n) Stream identification component behaviors according to IEEE Std 802.1CB-2017, 5.3 and
 1679 IEEE Std 802.1CBdbTM-2021, 5.5 d).

1680 **5.7.2 ccA Bridge component requirements**

1681 A Bridge component implementation for which a claim of conformance to ccA of this document
 1682 is made shall support the following list of requirements.

1683 a) Common Bridge component requirements according to 5.7.1.

1684 b) At least 2 PTP Instances according to 5.5.3.

1685 c) Eight queues according to IEEE Std 802.1Q-2022, 8.6.6.

1686 d) Enhancements for scheduled traffic for data rates of 100 Mb/s and 1 Gb/s according to IEEE
 1687 Std 802.1Q-2022, 5.4.1 items ac) and ad) including:

1688 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022,
 1689 8.6.9.4.16 and Table 12-32,

1690 2) support of the YANG features and nodes of the ieee802-dot1q-sched module according
 1691 to 6.4.9.3.2, and

1692 3) support of the YANG features and nodes of the icieeee60802-sched-bridge module
 1693 according to 6.4.9.3.3.

1694 e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of
 1695 100 Mb/s and 1 Gb/s, including:

1696 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE
 1697 Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of
 1698 the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported
 1699 functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and

1700 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module
 1701 according to 6.4.9.3.4.

1702 **5.7.3 ccB Bridge component requirements**

1703 A Bridge component implementation for which a claim of conformance to ccB of this document
 1704 is made shall support the following list of requirements.

1705 a) Common Bridge component requirements according to 5.7.1.

1706 b) At least 1 PTP Instance according to 5.5.3.

1707 c) At least four queues according to IEEE Std 802.1Q-2022, 8.6.6.

1708 **5.8 Bridge component options**

1709 **5.8.1 Common Bridge component options**

1710 A Bridge component implementation of any conformance class for which a claim of conformance
 1711 to this document is made may support the operation of the credit-based shaper algorithm
 1712 according to IEEE Std 802.1Q-2022, 8.6.8.2 on any Port as the transmission selection algorithm
 1713 for at least 4 traffic classes including support of the YANG features and nodes of the ieee802-
 1714 dot1q-cbsa-bridge module according to 6.4.9.3.5.

1715 **5.8.2 ccA Bridge component options**

1716 A Bridge component implementation for which a claim of conformance to ccA of this document
 1717 is made may support the following list of requirements.

1718 a) Any or none of the common Bridge component options according to 5.8.1.

- b) More than 2 PTP Instances according to 5.5.3.
- c) Enhancements for scheduled traffic for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s according to IEEE Std 802.1Q-2022, 5.4.1 items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the ieee802-dot1q-sched module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the icieeee60802-sched-bridge module according to 6.4.9.3.3.
- d) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1 item ae), for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s, including:
 - NOTE IEEE Std 802.3de-2022, 99.1, comprises 10 Mb/s.
 - 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
 - 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module according to 6.4.9.3.4.

5.8.3 ccB Bridge component options

A Bridge component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

- a) Any or none of the common Bridge component options according to 5.8.1.
- b) Up to eight queues according to IEEE Std 802.1Q-2022, 8.6.6.
- c) More than 1 PTP Instance according to 5.5.3.
- d) Enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.4.1, items ac) and ad) including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the ieee802-dot1q-sched module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the icieeee60802-sched-bridge module according to 6.4.9.3.3.
- e) Frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, item ae), including:
 - 1) support of Interspersing Express Traffic with preemptable traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities for TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7, and
 - 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module according to 6.4.9.3.4.

5.9 End station component requirements

5.9.1 Common end station Component requirements

An end station component implementation of any conformance class for which a claim of conformance to this document is made shall support the following list of requirements.

- a) The use of at least one customer VID for IA traffic engineered non-stream or IA non-stream traffic.
- b) The use of an additional customer VID for IA time-aware stream traffic if that traffic type category is supported.

- c) The use of an additional customer VID for IA stream traffic if that traffic type category is supported.
- d) The use of an additional customer VID for IA time-aware stream traffic if redundancy for that traffic type category is supported.
- e) The use of an additional customer VID for IA stream traffic if redundancy for that traffic type category is supported.
- f) Participate in only a single Configuration Domain.
- g) The use of an additional customer VID for an isolation VLAN.
- h) The use of an additional customer VID for a default VLAN

5.9.2 ccA end station component requirements

An end station component implementation for which a claim of conformance to ccA of this document is made shall support the following list of requirements.

- a) Common end station component requirements according to 5.9.1.
- b) At least 2 PTP Instances according to 5.5.3.
- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, for data rates of 100 Mb/s and 1 Gb/s including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the ieee802-dot1q-sched module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the icieeee60802-sched-bridge module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, for data rates of 100 Mb/s, and 1 Gb/s, if the IA time-aware stream traffic or the IA stream traffic type categories are supported, including:
 - 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
 - 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module according to 6.4.9.3.4.

5.9.3 ccB end station component requirements

An end station component implementation for which a claim of conformance to ccB of this document is made shall support the following list of requirements:

- e) Common end station component requirements according to 5.9.1.
- f) At least 1 PTP Instance according to 5.5.3.

5.10 End station component options

5.10.1 Common end station component options

An end station component implementation of any conformance class for which a claim of conformance to this document is made may support the following list of requirements.

- a) The operation of the credit-based shaper algorithm according to IEEE Std 802.1Q-2022, 8.6.8.2 including support of the YANG features and nodes of the ieee802-dot1q-cbs module according to 6.4.9.3.5.
- b) Talker end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcvTM-2021, 5.6, and 5.7 c) on one or more ports

and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.

- c) Listener end system behaviors according to IEEE Std 802.1CB-2017, as amended by IEEE Std 802.1CBdb-2021 and IEEE Std 802.1CBcv-2021, 5.9 on one or more ports and for 1 or more Compound Streams, including support of the `ieee802-dot1cb-stream-identification` and `ieee802-dot1cb-frer` YANG modules according to 6.4.9.3.6.

5.10.2 ccA end station component options

An end station component implementation for which a claim of conformance to ccA of this document is made may support the following list of requirements.

- a) Common end station options according to 5.10.1
- b) More than 2 PTP Instances according to 5.5.3.
- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, for data rates of 10 Mb/s, 2,5 Gb/s, 5 Gb/s, and 10 Gb/s including:

NOTE IEEE Std 802.3de-2022, 99.1, comprises 10 Mb/s.

- 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and
- 2) support of the YANG features and nodes of the `ieee802-dot1q-preemption` module according to 6.4.9.3.4.

5.10.3 ccB end station component options

An end station component implementation for which a claim of conformance to ccB of this document is made may support the following list of requirements.

- a) Common end station component options according to 5.10.1.
- b) One or more PTP Instances according to 5.5.3.
- c) End station requirements for enhancements for scheduled traffic according to IEEE Std 802.1Q-2022, 5.25, including:
 - 1) TickGranularity of less than or equal to 10 ns according to IEEE Std 802.1Q-2022, 8.6.9.4.16 and Table 12-32,
 - 2) support of the YANG features and nodes of the `ieee802-dot1q-sched` module according to 6.4.9.3.2, and
 - 3) support of the YANG features and nodes of the `ieciieee60802-sched-bridge` module according to 6.4.9.3.3.
- d) End station requirements for frame preemption according to IEEE Std 802.1Q-2022, 5.26, including:
 - 1) support of Interspersing Express Traffic according to IEEE Std 802.3-2022, Clause 99 as amended by IEEE Std 802.3de-2022, including support of the Additional Ethernet Capabilities TLV in an LLDPDU to indicate supported functions of frame preemption according to IEEE Std 802.3-2022, 79.3.7 and Table 79-8, and

- 2) support of the YANG features and nodes of the ieee802-dot1q-preemption module according to 6.4.9.3.4.

6 Required functions for an industrial network

6.1 General

Clause 6 provides requirements specific to this document and the industrial use case.

6.2 Synchronization

6.2.1 General

An IA-station can contain more than one Grandmaster PTP Instance and PTP End Instance to support:

- a) hot-standby use cases, or
- b) Working Clock or Global Time.

For further explanation of the requirements for time synchronization, refer to Annex D.

6.2.2 PTP Instance requirements

A Grandmaster PTP Instance, a PTP Relay Instance and a PTP End Instance, and the Working Clock or Global Time clocks connected to them, shall meet the following requirements under their allowed working conditions and for their lifetime.

- a) The fractional frequency offset of the LocalClock relative to the nominal frequency shall be according to Table 9.
- b) The range of the rate of change of fractional frequency offset of the LocalClock shall be according to Table 9.
- c) During operation, the Working Clock and Global Time at Grandmaster PTP Instances and PTP End Instances shall increase monotonically, where monotonic means that for a time y that occurs after time x , the ClockTarget's timestamp of y is greater than or equal to the ClockTarget's timestamp of x .
- d) The Working Clock and Global Time at a PTP End Instance can be controlled by applying a frequency change over a period of time. The frequency applied can have a fine resolution to speed up or slow down the clock smoothly, and it has a total range of frequency adjustment.
- e) For the Global Time at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.
- f) For the Working Clock at a PTP End Instance, the maximum value of frequency adjustment shall be according to Table 9.

For Working Clock or Global Time, decoupled from a ClockTarget, a higher maximum value of frequency adjustment and maximum rate of change of fractional frequency offset are allowed. As soon as it is coupled (or coupled again), a) to f) apply.

Table 9 – Required values

Topic	Value
Local Clock at non-Grandmaster PTP Instance, range of fractional frequency offset relative to the nominal frequency	± 50 ppm
Local Clock at non-Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of fractional frequency offset with respect to the nominal frequency	± 25 ppm

Topic	Value
Working Clock and Global Time (acting as ClockSource) and Local Clock at Grandmaster PTP Instance, range of rate of change of fractional frequency offset with respect to the nominal frequency (steady state, see Annex D.2.3)	± 1 ppm/s
Working Clock and Global Time (acting as ClockSource) at Grandmaster PTP Instance, range of rate of change of fractional frequency offset (transient, see Annex D.2.3)	± 3 ppm/s
Working Clock and Global Time at PTP End Instance, maximum value of frequency adjustment	± 250 ppm over any observation interval of 1 ms
NOTE 1 If the Grandmaster PTP Instance implementation is such that its Working Clock and Local Clock are the same or otherwise locked to the same frequency, the normative requirements on the Working Clock take priority over those on the Local Clock. NOTE 2 The Maximum value of frequency adjustment represents an upper bound that limits how much a PTP End Instance can change the frequency of its Working Clock or Global Time during a given period. However, the adjustment is expected to be gradual over the defined interval rather than instantaneous. NOTE 3 The example algorithms that track clock drift use up to 4 seconds of historical data and can take that length of time to respond to changes in clock drift. The example algorithm has been used for simulating cases where no fast changes in the observed frequency drift rate were observed. In most of the real-life situations, this condition can be satisfied.	

1894

1895 6.2.3 PTP protocol requirements

1896 Table 10 shows the required protocol times.

1897

Table 10 – Protocol settings

Topic	Value
Nominal time between successive Announce messages (announce interval)	1 s
Nominal time between successive Pdelay_Req messages (Pdelay_Req message transmission interval)	125 ms
Range of allowed time between successive Pdelay_Req messages	119 ms to 131 ms
Nominal time between successive Sync messages at the Grandmaster (Sync message transmission interval)	125 ms
Range of allowed time between successive Sync messages at the Grandmaster	119 ms to 131 ms
Time between reception of a Sync message and transmission of the subsequent Sync message (i.e. residence time) at a PTP Relay instance	Maximum: 15 ms Measured Mean: ≤ 5 ms
Maximum time between transmission of a Sync message and transmission of the related Follow_Up message	2,5 ms
Time between reception of a Pdelay_Req message and transmission of the subsequent Pdelay_Resp message (i.e. Pdelay turnaround time).	Maximum: 15 ms
NOTE 1 A consequence of having a single allowed value of mean sync interval is that syncLocked mode is achieved. If the timeTransmitter port sync interval is the same as that of the timeReceiver port, syncLocked mode is achieved. NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

1898

1899 6.2.4 Clock control system requirements for PTP End Instances

1900 Table 11 shows the required clock control system characteristics at a PTP End Instance.

1901

Table 11 – Clock control system requirements

Topic	Value
Maximum Bandwidth	1,0 Hz
Minimum Bandwidth	0,9 Hz
Maximum Gain Peaking	2,2 dB
Minimum absolute value of Roll-off	20 dB/decade

Topic	Value
NOTE 1 For more information regarding the clock control system, see Annex C.	
NOTE 2 The values contained in this table apply to both the Working Clock and Global Time.	

6.2.5 Error generation limits

Table 12 shows the required limits on error generation at a Grandmaster PTP Instance. A limit on error generation for a Grandmaster PTP Instance is a limit on the amount of error it generates in the output Sync message compared to its Working Clock (acting as ClockSource) and Local Clock. See D.3.4.

Table 12 – Error generation limits for Grandmaster PTP Instance

Topic	Value
Accuracy of forwarded time information: (preciseOriginTimestamp + correctionField) in PTP timing message minus Working Clock at Grandmaster when Sync message is transmitted	Allowable range of the measured mean: - 10 ns to + 10 ns Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
Accuracy of forwarded Rate Ratio information: rateRatio field in Follow_Up information TLV minus true Rate Ratio between Working Clock at Grandmaster and Local Clock when Sync message is transmitted	Mean 0 ppm ± 0,1 ppm Standard deviation ≤ 0,1 ppm
Accuracy of syncEgressTimestamp: syncEgressTimestamp in Drift_Tracking TLV minus Local Clock when Sync message is transmitted	Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measure mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error.	
NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy.	

Table 13 shows the required limits on error generation at a PTP Relay Instance. A limit on error generation for a PTP Relay Instance is a limit on the amount of error it adds to the output Sync message compared to the input Sync message. These requirements are written for the case when errors due to change of fractional frequency offset of its Local Clock with respect to the nominal frequency and errors in the input Sync message are negligible with respect to the specified error generation limits. See D.3.5.

1916

Table 13 – Error generation limits for PTP Relay Instance

Topic	Value
Accuracy of residenceTime measurement (no clock drift): (preciseOriginTimestamp + correctionField) in the PTP timing message transmitted by PTP Relay Instance minus Working Clock at Grandmaster when the Sync message is transmitted while... • Working Clock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of the measured mean: - 2 ns to + 2 ns Range around the measured mean within which 90% of measurements fall: ± 10 ns Range around the measured mean within which 100% of measurements fall: ± 20 ns
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (no clock drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • Working Clock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of Neighbor Rate Ratio measurement and correct modification of Rate Ratio (Rate Ratio drift and Neighbor Rate Ratio drift): rateRatio field in the Follow_Up information TLV transmitted by PTP Relay Instance minus Rate Ratio from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (no clock drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) atGrandmaster is stable. • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,02 ppm
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable.	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm

Topic	Value
Accuracy of Neighbor Rate Ratio Drift measurement and correct modification of Rate Ratio Drift (Rate Ratio drift and Neighbor Rate Ratio drift) rateRatioDrift field in the Drift_Tracking TLV transmitted by PTP Relay Instance minus Rate Ratio Drift from the PTP Relay Instance's Local Clock to the Working Clock at the Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s	Allowable range of the measured mean: - 0,1 ppm to + 0,1 ppm Allowable measured standard deviation around the measured mean: 0,08 ppm
Accuracy of syncEgressTimestamp: syncEgressTimestamp in Drift_Tracking TLV minus Local Clock when Sync message is transmitted	Range around the measured mean within which 90% of measurements fall: ± 7 ns Range around the measured mean within which 100% of measurements fall: ± 10 ns
Accuracy of meanLinkDelay: meanLinkDelay measured by the PTP Relay Instance minus actual path delay	± 3 ns
Accuracy of Pdelay turnaround time measurement: (responseOriginTimestamp –requestReceiptTimestamp) for transmitted Pdelay_Resp and associated Pdelay_Resp_Follow_Up messages minus the actual interval between reception of the associated Pdelay_Req and transmission of the Pdelay_Resp as measured with respect to the Local Clock	Allowable range of the measured mean: - 2 ns to + 2 ns Range around the measured mean within which 90% of measurements fall: ± 10 ns Range around the measured mean within which 100% of measurements fall: ± 20 ns

Topic	Value
NOTE 1 “Allowable range of the measured mean” specifies limits on constant error. “Range around the measure mean” and “Standard deviation” specify limits on dynamic error. A limit on the constant error of syncEgressTimestamp is not specified because constant error in this characteristic is not a source of time synchronization error. NOTE 2 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy. NOTE 3 These requirements are for errors generated by the PTP Relay Instance. They assume input Sync messages with negligible errors. However, timestamps of Pdelay_Req reception and Pdelay_Resp transmission by the upstream PTP Instance should include expected timestamp errors when meanLinkDelay accuracy is assessed.	

1917

1918 Table 14 shows the required limits on error generation at a PTP End Instance. A limit on error
1919 generation for a PTP End Instance is a limit on the amount of error it adds to its Working Clock
1920 (acting as ClockTarget) compared to the input Sync message. These requirements are written
1921 for the case when errors due to change of fractional frequency offset of its Local Clock with
1922 respect to the nominal frequency and errors in the input Sync message are negligible with
1923 respect to the specified error generation limits. See D.3.6.

1924

Table 14 – Error generation limits for PTP End Instance

Topic	Value
Accuracy of recovered clock (no clock drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster is stable. • Local Clock at upstream PTP Instance is stable. • meanLinkDelay between upstream PTP Instance and PTP End Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 15 ns to + 15 ns
Accuracy of recovered clock (Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance is stable. • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 145 ns to + 15 ns

Topic	Value
Accuracy of recovered clock (Rate Ratio drift and Neighbor Rate Ratio drift): Working Clock (acting as ClockTarget) at PTP End Instance minus Working Clock (acting as Clock Source) at Grandmaster while... • WorkingClock (acting as ClockSource) at Grandmaster PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • Local Clock at upstream PTP Instance, fractional frequency offset with respect to the nominal frequency is increasing at 1 ppm/s • meanLinkDelay between upstream PTP Instance and PTP Relay Instance is negligible	Allowable range of cTE: - 10 ns to + 10 ns Allowable range of dTE: - 145 ns to + 15 ns
Accuracy of meanLinkDelay: meanLinkDelay measured by the PTP End Instance minus actual path delay	± 3 ns
NOTE 1 This document specifies statistical normative requirements without a test duration. It is the responsibility of a test plan to determine the test duration and pass/fail criteria, taking sampling error into consideration, and striking an acceptable balance between test duration and accuracy.	
NOTE 2 These requirements are for errors generated by the PTP End Instance. They assume input Sync messages with negligible errors. However, timestamps of Pdelay_Req reception and Pdelay_Resp transmission by the upstream PTP Instance should include expected timestamp errors when meanLinkDelay accuracy is assessed.	

1925

1926 **6.2.6 PTP Instance synchronization status**

1927 Industrial automation systems monitor the synchronization status of each PTP Instance to
1928 determine the viability of operations. This status is obtained from the isSynced global variable
1929 specified in IEEE Std 802.1ASdm-2024, 18.4.1.1.

1930 The PtpInstanceSyncStatus state machine according to IEEE Std 802.1ASdm-2024, 18.4, shall
1931 be supported by all PTP instances independent of whether hot standby is supported. Non-
1932 Grandmaster PTP Instances shall support one or more of the application interfaces of 9.3, 9.4,
1933 9.5, and 9.6 according to IEEE Std 802.1ASdm-2024. Grandmaster PTP Instances shall support
1934 the application interface of 9.2 according to IEEE Std 802.1ASdm-2024.

1935 **6.2.7 Application framework**

1936 Any step change in the time of a ClockSource or ClockTarget whose absolute value exceeds a
1937 user-defined threshold (for example 1 μs) leads to action being taken by the application or by
1938 a higher-layer entity.

1939 If the change is in Global Time, it is desirable that all consumers of that time be made aware of
1940 this change (i.e. a jump in Global Time from the value A to the value B), so that the actual time
1941 interval between the time corresponding to A and the time corresponding to B can be evaluated.

In the case of Working Clock, a time change that exceeds the user-defined threshold (for example 1 μ s) is avoided to protect assets and prevent damage. Thus, the ClockSource or ClockTarget can be decoupled (see Figure 13 and Figure 14) from the PTP-maintained clock when such a time change occurs.

In Figure 13, two ClockTargets are traceable to a reliable source of time, which should be synchronized to Global Time or Working Clock.

The status of a ClockSource, ClockTarget, ClockTimeTransmitter or ClockTimeReceiver is given by the status of the clock (see 6.2.6) as shown in Figure 11. When timestamps are provided to the application, the current ClockSource or ClockTarget state can also be provided to the application.

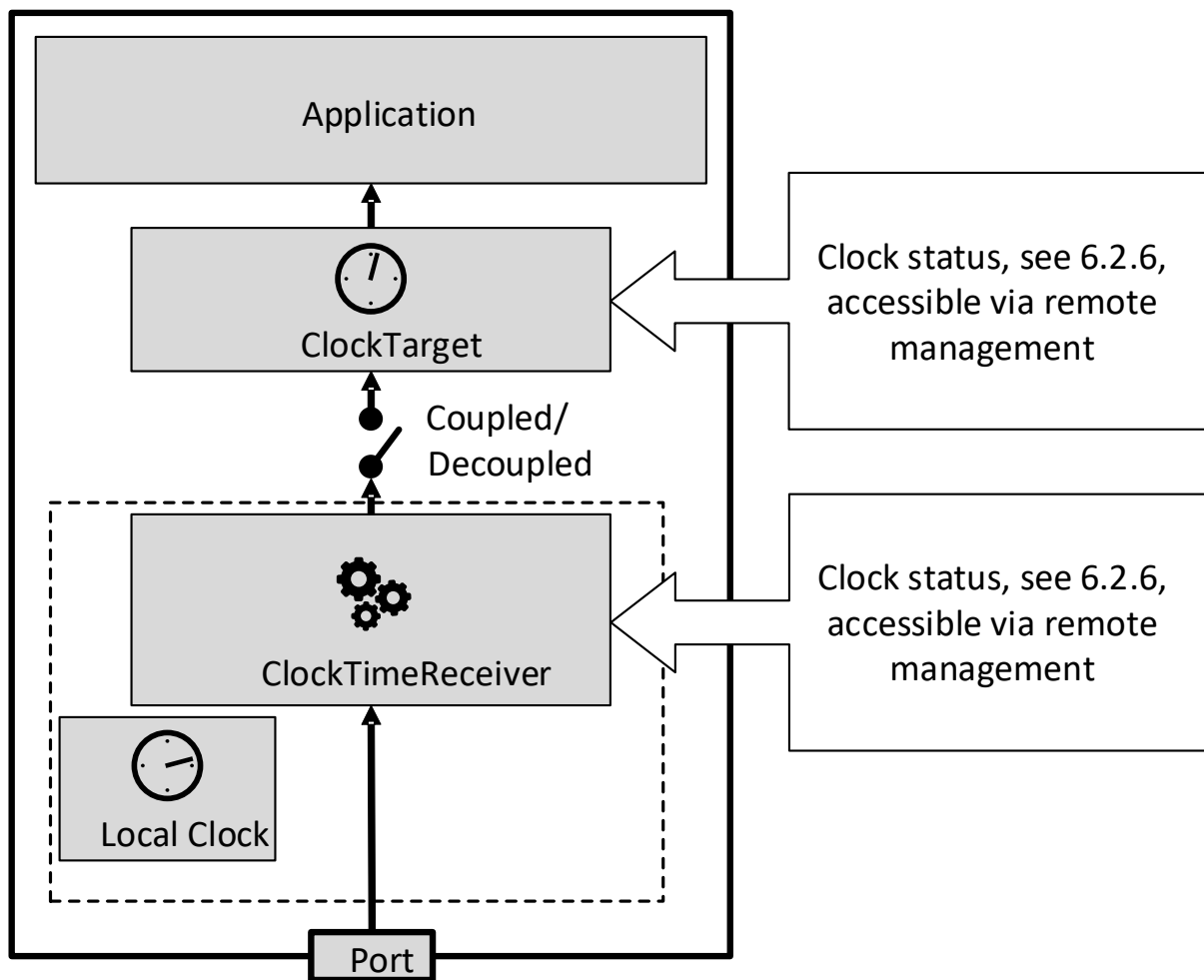


Figure 11 – Clock model

6.2.8 Working Clock domain framework

The gPTP domainNumber of a Working Clock domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, $\max|TE_R|$ of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 1 μ s (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

6.2.9 Global Time domain framework

The gPTP domainNumber of a Global Time domain is assigned by the CNC. In industrial applications, when the number of PTP Relay Instances between the Grandmaster PTP Instance and any PTP End Instance is less than or equal to 99, $\max|TE_R|$ of the synchronized time of any ClockTarget, relative to the Grandmaster ClockSource, is less than or equal to 100 μ s (see error budget A in Figure 14). Thus it is incumbent upon any PTP Instance to ensure that the requirements specified in 5.5.3, 6.2.2, 6.2.3, 6.2.4, and 6.2.5 are met.

6.2.10 IA-station model for clocks

Industrial automation applications, as described in 4.1, require synchronized time that is traceable to a known source (i.e. Global Time) and a source of time synchronized to the Working Clock. Figure 12 and Figure 13 show examples of the IA-station internal model for clocks with the two PTP Instances. It is possible for the ClockSource or ClockTarget to start decoupled or become decoupled from the ClockTimeTransmitter or ClockTimeReceiver, respectively, of a PTP Instance; the ClockSource or ClockTarget runs independently of the availability of the network or a Grandmaster. For example, if the PTP Instance enters a state where isSynced is FALSE, the application might choose to decouple its clock from the PTP Instance and continue to run on its internal clock. If isSynced for the PTP Instance changes to TRUE, the application can choose to again synchronize to the PTP Instance.

Figure 13 shows the IA-station internal model for clocks, with the two PTP Instances used as ClockTimeReceiver/ClockTarget.

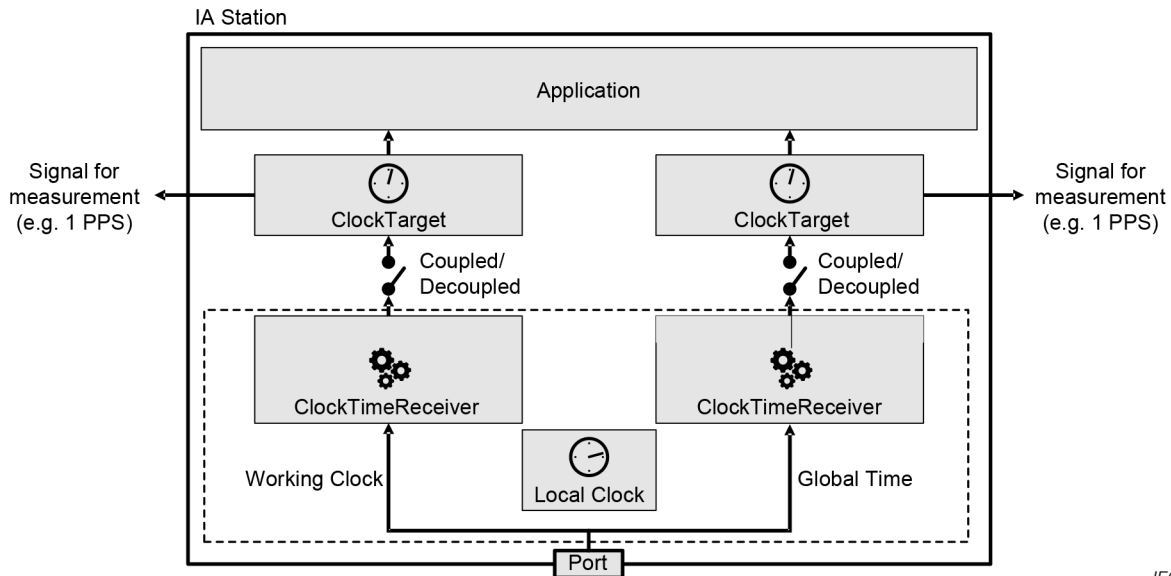


Figure 12 – Example clock usage principles for PTP End Instances

Figure 14 shows the IA-station internal model for clocks, with the two PTP Instances used as Grandmaster.

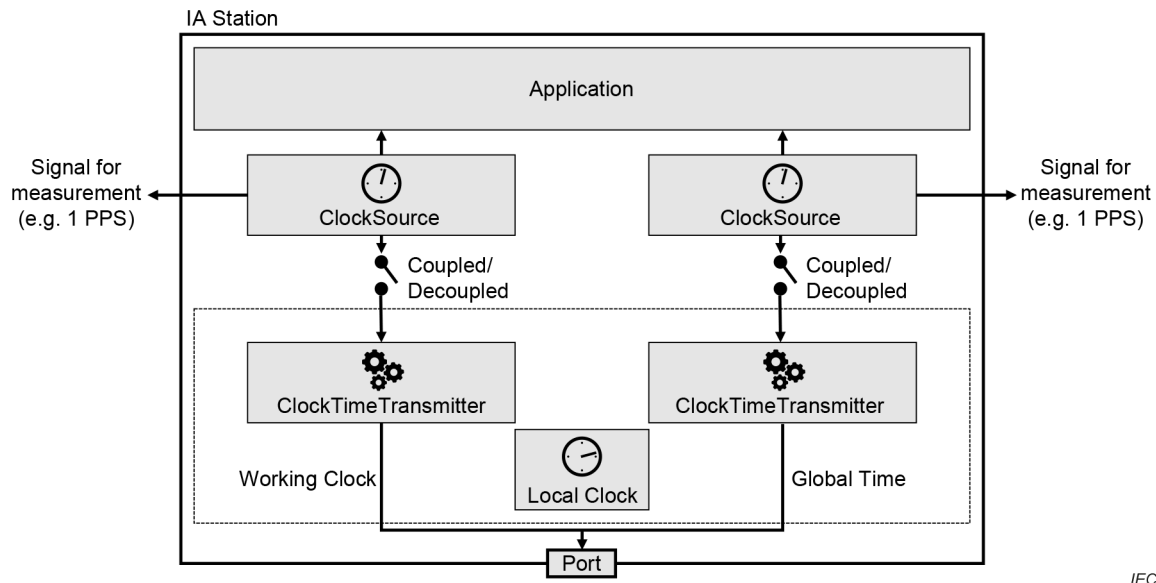


Figure 13 – Example clock usage principles for Grandmaster PTP Instances

6.2.11 Clock usage for the Ethernet interface

6.2.11.1 Time-aware offset control

Time-aware offset control (see 4.4), uses an assigned source of time and a definition when to start or to stop, which are dependent on the clock state.

The clock used is the ClockTarget or, in the case of a Grandmaster PTP Instance, the ClockSource.

IA time-aware streams are only transmitted while isSynced for the chosen ClockSource or ClockTarget is TRUE (see 6.2.6).

Thus, changes of the clock state directly influence the transmission of frames.

6.2.11.2 Gating cycle

To control the gating cycle, the gate control list uses an assigned source of time. Enabling and disabling the gate control list is dependent on the clock state.

The clock used is the ClockTarget or, in the case of a Grandmaster PTP Instance, the ClockSource.

The gating cycle is run using the chosen ClockSource or ClockTarget regardless of the value of isSynced (see 6.2.6).

6.2.12 Error model

Synchronization is transported over the entire path, from the Grandmaster PTP Instance to the PTP End Instance, through the intermediate PTP Relay Instances. All time errors, cTE and dTE, are accumulated during this process.

Time error can arise in the following processes:

- a) the transporting of time in PTP Instances and via PTP Links that connect PTP Instances,
- b) the providing of time to the Grandmaster PTP Instance, from the ClockSource entity via the ClockTimeTransmitter entity, and

c) the providing of time to a ClockTarget entity (end application) via the ClockTimeReceiver entity.

NOTE Item a) includes time error introduced in a PTP End Instance between the timeReceiver port and the ClockTimeReceiver entity, and between the ClockTimeTransmitter entity and a TimeTransmitter port.

An output synchronization signal (for example, 1 pulse per second (PPS)) synchronized to the Working Clock as shown in Figure 13 and Figure 14, at any PTP Instance, is used to measure the time error between the ClockSource of the Grandmaster and the ClockTarget of a PTP Instance that is not the Grandmaster. The additional error introduced by implementation of the output synchronization signal is in the range of –10 ns to +10 ns. Figure 14 shows the error budget principle used. These budgets do not include any deviation from the PTP timescale. Representative budgets are provided in Annex D.

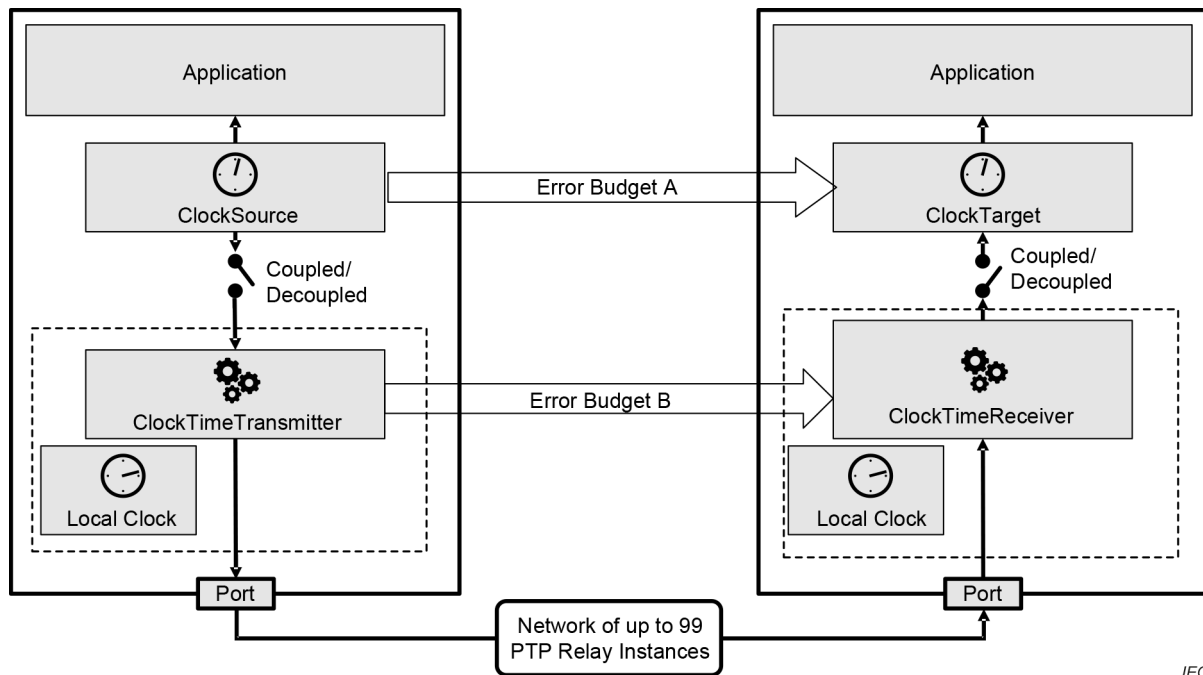


Figure 14 – Error budget scheme

Table 15 shows example values for the splitting of the available error budgets (see Figure 14).

Table 15 – Error budget

Domain	Error budget A	Error budget B
Working Clock	1 μ s	900 ns
Global Time	100 μ s	99,9 μ s

Global Time is often used for tracking events in industrial applications (i.e. sequence of events). Any usage of Global Time for time stamping of application events is allowed an error budget of 1 μ ms.

6.2.13 gPTP domains and PTP Instances

Any valid gPTP domain number as specified in IEEE Std 802.1AS-2020 can be used. The IEEE Std 1588-2019 attribute descriptionDS.userDescription shall be used according to Table 16 to support the translation of PTP Instances and middleware as described in 4.6.2. One gPTP domain can be used for both Working Clock and Global Time. If only one gPTP domain is used, then the requirements for the Working Clock apply (see 6.2.8).

2036

Table 16 – descriptionDS.userDescription of gPTP Domains

gPTP Domain	descriptionDS.userDescription
Working Clock (no hot standby configured)	"60802-WorkingClock"
Primary Working Clock (with configured hot standby)	"60802-Primary-WorkingClock"
Secondary Working Clock (with configured hot standby)	"60802-Secondary-WorkingClock"
Global Time (no hot standby configured)	"60802-GlobalTime"
Primary Global Time (with configured hot standby)	"60802-Primary-GlobalTime"
Secondary Global Time (with configured hot standby)	"60802-Secondary-GlobalTime"
GlobalTime and WorkingClock (no hot standby configured)	"60802-GlobalTime-WorkingClock"
Primary GlobalTime and WorkingClock (with configured hot standby)	"60802-Primary-GlobalTime-WorkingClock"
Secondary GlobalTime and WorkingClock (with hot standby configured)	"60802-Secondary-GlobalTime-WorkingClock"

2037

2038 The descriptionDS.userDescription attribute is represented in the ieee1588-ptp-tt YANG module
 2039 by the `user-description` leaf in the `description-ds` container of a PTP Instance.

2040 The linking between a gPTP domain and the IETF interfaces is provided by the `underlying-`
 2041 `interface`.

2042 6.3 Security model

2043 6.3.1 General

2044 Subclause 6.3 specifies the security model starting with NETCONF/YANG. It describes the
 2045 security functionality, the security objects in factory default state, the imprinting of Configuration
 2046 Domain-specific security objects and the secure configuration based on Configuration Domain-
 2047 specific security objects.

2048 6.3.2 Security functionality

2049 6.3.2.1 Message exchange protection

2050 6.3.2.1.1 NETCONF-over-TLSv1.2 profile

2051 Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to
 2052 IETF RFC 7589, where the to-be-configured IA-stations act in the NETCONF server role. TLS
 2053 protocol version 1.2 according to IETF RFC 5246, 6.2.3.3, 7.4.7.2 and 8.1.2 shall be supported
 2054 according to the following list of requirements.

2055 a) Mutual authentication in conjunction with the IDevID and LDevID credentials according to
 2056 6.3.4 and 6.3.5.

2057 b) The cipher suite TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 according to IETF
 2058 RFC 5289, 3.2 and Clause 5.

2059 NOTE IETF RFC 7589 implicitly mandates the cipher suite TLS_RSA_WITH_AES_128_CBC_SHA by referring to
 2060 IETF RFC 5246. This cipher suite is not used in this document because it requires excessive asymmetric key lengths,
 2061 it is not an Authenticated Encryption with Associated Data (AEAD) scheme, and it does not provide perfect forward
 2062 secrecy.

2063 c) Signature algorithm ECDSA with SHA-256 and Curve P-256 according to NIST FIPS 186-5
 2064 Digital Signature Standard (DSS).

2065 The TLS Certificate message from the TLS client and server contains the self-signed root
 2066 certificate. This approach allows simplifying and flattening the PKI hierarchy on base of the

current TLS client certificate to NETCONF username mapping algorithm in IETF RFC 7589. Implementations shall support TLS Certificate message with at least 2 certificate objects.

NETCONF-over-SSH according to IETF RFC 6242 is not used in this document.

NOTE This document selects TLS as a secure transport for NETCONF since TLS is the better match for the case of configuration clients that rely upon unattended or automated operation. This case is dominant in industrial automation.

6.3.2.1.2 NETCONF-over-TLSv1.2 options

The following list of TLS protocol version 1.2 requirements may be supported.

- a) The cipher suites TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 according to IETF RFC 5289, 3.2 and Clause 5, and TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 according to IETF RFC 7905, Clause 2.
- b) Signature algorithms ECDSA with SHA-512 and Curve P-521 according to NIST FIPS 186-5, Ed25519 according to IETF RFC 8032, 5.1, and Ed448 according to IETF RFC 8032, 5.2.

6.3.2.1.3 NETCONF-over-TLSv1.3 profile

Network configuration with NETCONF/YANG is protected by NETCONF-over-TLS according to IETF RFC 7589 and IETF draft-ietf-netconf-over-tls13. TLS protocol version 1.3 according to IETF RFC 8446 may be used with mutual authentication for NETCONF/YANG as follows.

- a) The cipher suites TLS_AES_128_GCM_SHA256, TLS_AES_256_GCM_SHA384 and TLS_CHACHA20_POLY1305_SHA256 according to IETF RFC 8446, 9.1.
- b) One or more of the following signature schemes:
 - ecdsa_secp256r1_sha256 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.3, ecdsa_secp521r1_sha512 according to NIST FIPS 186-5 and NIST SP 800-186, 3.2.1.5,
 - ed25519 according to IETF RFC 8032, 5.1, and
 - ed448 according to IETF RFC 8032, 5.2.

The TLS Certificate message from the TLS client and server contains the self-signed root certificate.

6.3.2.1.4 Certificate-to-name mapping

The IETF RFC 7589 based certificate-to-name mapping procedure is as follows.

NOTE IETF RFC 7589, Clause 7, specifies that NETCONF servers map client certificates to "NETCONF usernames" and specifies a concrete mapping procedure for this purpose. This mapping is represented by the YANG module ietf-x509-cert-to-name.

The list of mapping entries has a single element containing:

- fingerprint: the fingerprint of the trust anchor for the Configuration Domain, and
- map_type: ext-60802-roles.

The map-type ext-60802-roles maps the roles provided in the id-60802-pe-roles extension (defined in 6.3.2.1.5) of the end entity certificate presented by the NETCONF client to a NETCONF username. The UTF-8 string representation of each role is added to the NETCONF username in chronological order of the enumeration values, whereas multiple roles are separated by ':' character.

2107 6.3.2.1.5 Role extension

2108 The id-60802-pe-roles extension in LDevID-NETCONF end entity certificates shall be
2109 constructed as follows:

2110 a) Extension field extnID

2111 The extnID shall provide the following OBJECT IDENTIFIER to identify the id-60802-pe-roles
2112 extension:

```
2113 id-60802 OBJECT IDENTIFIER ::= { iso (1) iso-identified-organization (3) ieee (111)
2114 standards-association-numbered-series-standards (2) lan-man-stds (802) ieee802dot1(1)
2115 ieee802dot1stds(2) 1 }
```

```
2116
2117 id-60802-pe OBJECT IDENTIFIER ::= { id-60802 1 }
```

```
2118
2119 id-60802-pe-roles OBJECT IDENTIFIER ::= { id-60802-pe 1 }
```

2120

2121 b) Extension field critical

2122 The id-60802-pe-roles extension is marked as non-critical (critical:= FALSE).

2123 c) Extension field extnValue

```
2124 60802RoleNamesSyntax ::= SEQUENCE OF 60802RoleName
```

2125

```
2126 60802RoleName ::= ENUMERATED {
2127     SecurityAdminRole (0),
2128     ConfiguratorRole (1),
2129     StreamConfiguratorRole (2),
2130     SubscriberRole (3) }
```

2131 6.3.2.2 Resource access authorization

2132 Access control to NETCONF/YANG resources shall be protected by NACM according to IETF
2133 RFC 8341.

2134 NACM specifies a YANG data model (ietf-netconf-acm) for expressing rules to control access
2135 to NETCONF/YANG resources. This document profiles NACM to deliver role-based access
2136 control.

2137 NOTE 1 NACM does not natively deliver role-based access control but can be geared by profiling.

2138 This role-based model for security resources should be applied according to the following list
2139 of requirements.

- 2140 • The global switch enable-nacm is set to true.
- 2141 • The set of NETCONF/YANG resources of an IA-station is partitioned according to the YANG
2142 modules specified in 6.4.9 with a permission-to-role assignment as listed below. An access
2143 operation is allowed through the keyword "permitted" and not allowed through the keyword
2144 "denied".

2145 NOTE 2 NACM recognizes following "access-operations": create, read, update, delete, exec and uses the term write
2146 access for the access operations "create", "delete", and "update". This document uses the terms read, write and
2147 exec access.

- 2148 • All authenticated entities (default rules): All YANG modules: read access permitted, write
2149 access denied, exec-access denied.

2150 NOTE 3 The default rules apply for YANG modules that are listed in 6.4.9 but are not listed in the rules of the
2151 individual roles.

- 2152 • Rules for StreamConfiguratorRole: YANG module ieee802-dot1q-cnc-config: write and
2153 execute operations permitted.

- 2154 • Rules for SubscriberRole:
 - 2155 – YANG module ietf-subscribed-notifications: write and execute operations permitted, and
 - 2156 – YANG module ietf-yang-push: write and execute operations permitted.
- 2157 • Rules for ConfiguratorRole: All YANG modules except those listed below, write and execute
 2158 operations permitted:
 - 2159 – YANG modules for security configuration, i.e. ietf-truststore, ietf-keystore, path to cert-
 2160 to-name nodes of ietf-netconf-server, path to tls-server-parameters nodes of ietf-
 2161 netconf-server,
 - 2162 – YANG modules for stream configuration, i.e. ieee802-dot1q-cnc-config, and
 - 2163 – YANG modules for subscription configuration, i.e. ietf-subscribed-notifications, ietf-
 2164 yang-push.
- 2165 • Rules for SecurityAdminRole:
 - 2166 – YANG module ietf-truststore, path to certificate node of IDevID trust anchor: write and
 2167 execute operations denied, and
 - 2168 – YANG module ietf-truststore (besides path to certificate node of IDevID trust anchor):
 2169 write and execute operations permitted.
 - 2170 – YANG module ietf-keystore, path to asymmetric-key node of IDevID credential: write and
 2171 execute operations denied, and
 - 2172 – YANG module ietf-keystore (besides path to asymmetric-key node of IDevID credential):
 2173 write and execute operations permitted.
 - 2174 – YANG module ietf-netconf-server (besides path to cert-to-name nodes and path to tls-
 2175 server-parameters nodes): write and execute operations denied, and
 - 2176 – YANG module ietf-netconf-server, path to cert-to-name nodes: write and execute
 2177 operations permitted.
 - 2178 – YANG module ietf-netconf-server, path to tls-server-parameters nodes: write and
 2179 execute operations permitted.

2180 In addition, the following access control should be applied for NETCONF protocol operations:

- 2181 • <lock>, <unlock>: permitted for any role specified in this document,
- 2182 • <partial-lock>, <partial-unlock>: denied (not used in this document),
- 2183 • <get> and <get-config>: mapped to a "read" access operation to the target datastore,
- 2184 • <edit-config>: permitted for any role specified in this document,
- 2185 • <copy-config>: permitted for ConfiguratorRole,
- 2186 • <delete-config>: denied (not used in this document),
- 2187 • <commit>: permitted for any role specified in this document,
- 2188 • <discard-changes>: permitted for any role specified in this document,
- 2189 • <close-session>: permitted for any role specified in this document, and
- 2190 • <kill-session>: denied (not used in this document).

2191 This document does not specify the assignment of role names to actual system entities. This is
 2192 a duty of system owners or operators.

2193 **6.3.3 IDevID Profile**

2194 **6.3.3.1 General**

2195 IA-stations shall possess IDevID credentials according to 6.3.3. CNCs shall contain trust
 2196 anchors for validating IDevID credentials.

6.3.3.2 Object Contents

6.3.3.2.1 General

The IDevID credential contents shall comply to 6.3.3.2.2, 6.3.3.2.3, and IEEE Std 802.1AR-2018, Clause 6.

6.3.3.2.2 IA-station Identity

Any IDevID EE certificate of an IA-station shall take one of the following forms:

- raw form: the IDevID EE certificate complies to IEEE Std 802.1AR-2018, Clause 8, and
- extended form: the IDevID EE certificate complies to requirements provided IEEE Std 802.1AR-2018, Clause 8. The extended form of an IDevID EE certificate shall be constructed as follows:
 - the verifiable device identity shall appear as a URN in a GeneralName of type uniformResourceIdentifier in the subjectAltName extension,
 - the URN value shall be constructed according to IETF RFC 8141 and as follows:
 - namespace identifier: ieee (see IETF RFC 8069),
 - namespace-specific string: iec-ieee-60802#verifiable-device-identity, and
 - q-component (see IETF RFC 8141, 2.3.2) to parameterize the named resource: an ampersand-separated list of keyword=value tuples with following keywords and values. These tuples can appear in any order inside the q-component:
 - the keywords: hardware-rev, serial-num, mfg-name, model-name;
 - their corresponding values from the single "chassis" component list entry in the ietf-hardware YANG module (see 6.4.9.2.5.8) that represents the management entity of the IA-station respectively from its pre-material form in percent-encoding according to IETF RFC 3986, 2.1.

NOTE 1 These are the items with the YANG property config=false from the 'component' list entry that represents the management entity of the IA-station. The config=false items firmware-rev and software-rev are excluded to avoid IDevID credential updates in case of FW or SW updates.

NOTE 2 An object looks like urn:ieee:iec-ieee-60802#verifiable-device-identity?=mfg-name=<mfg-name>&model-name=<model-name>&hardware-rev=<hardware-rev>&serial-num=<serial-num>.

NOTE 3 One IDevID EE certificate can have one subjectAltName extension which can have one or more GeneralName entries. In particular, there can be one or more GeneralName entries of type uniformResourceIdentifier. This allows other organizations (e.g. middleware and application consortia or individual manufacturers) to also represent their perception of verifiable device identity in addition to the perception of this document.

6.3.3.2.3 Signature Suites

An IDevID shall utilize the signature suite: ECDSA P-256/SHA-256 according to NIST FIPS 186-5, NIST FIPS 180-4 and NIST SP 800-186 using the algorithm identifiers according to IETF RFC 5480.

An IDevID may utilize the following signature suites:

- ECDSA P-521/SHA-512 according to NIST FIPS 186-5, NIST FIPS 180-4 and NIST SP 800-186 using the algorithm identifiers according to IETF RFC 5480,
- EdDSA instance Ed25519 according to IETF RFC 8032 using Curve25519 according to IETF RFC 7748 and using the algorithm identifiers according to IETF RFC 8410, and
- EdDSA instance Ed448 according to IETF RFC 8032 using Curve448 according to IETF RFC 7748 and using the algorithm identifiers according to IETF RFC 8410.

2241 6.3.3.3 Information Model

2242 6.3.3.3.1 General

2243 The information model for IDevID credentials and trust anchors shall comply to YANG and
2244 NMDA, in particular the YANG modules ietf-keystore and ietf-truststore, as well as subsequent
2245 subclauses of 6.3.3.3.

2246 6.3.3.3.2 Entries

2247 IDevID credentials shall be provided in form of built-in keys of an IA-station by its manufacturer.
2248 In YANG, they are modeled as config-false nodes and are represented in the 'keystore'
2249 container that is instantiated by the YANG module ietf-keystore. The private key shall use the
2250 private-key-type choice hidden-private-key, i.e. the IDevID private key is not presented in
2251 NETCONF/YANG. The details of storing and protecting IDevID private keys as well as using
2252 them for signing purposes are implementation specific.

2253 Trust anchors for IDevID credentials are CNC user-configured data objects; these objects shall
2254 be available as applied configuration (IETF RFC 8342) upon CNCs. In YANG, they are modeled
2255 as config-true nodes and are represented in the 'truststore' container that is instantiated by the
2256 YANG module ietf-truststore.

2257 NOTE IA-station built-in trust anchors for use cases such as firmware/software update are not addressed in this
2258 document.

2259 6.3.3.3.3 Entry Manifolddness

2260 An IA-station shall possess one IDevID credential with a certification path plus trust anchor
2261 information issued under the required signature suite according to 6.3.3.2.3 as part of its factory
2262 default state.

2263 If an IA-station supports an optional signature suite according to 6.3.3.2.3, it shall possess in
2264 addition one IDevID credential with a certification path plus trust anchor information issued
2265 under the optional signature suite as part of its factory default state.

2266 An IA-station can have additional IDevID credential(s) with a certification path plus trust anchor
2267 information issued under a combination of any required or any supported optional DevID
2268 signature suites.

2269 If an IA-station possesses multiple IDevID credentials, then they shall be issued by the same
2270 organization (the IA-station manufacturer). Their EE certificates shall contain the same device
2271 identity information.

2272 A CNC shall support at least one trust anchor for IDevID credentials per supported IA-station
2273 manufacturer.

2274 6.3.3.3.4 Entry Naming

2275 IDevID credentials shall be present in an 'asymmetric-key' entry that is identified as: /ietf-
2276 keystore:keystore/asymmetric-keys/asymmetric-key/name=
2277 IDevID-<SignatureSuiteName>-<CertificateSerialNumberOfEECertificate>.

2278 IDevID trust anchors shall be present in 'certificate' entries that are identified as: /ietf-
2279 truststore:truststore/certificate-bags/certificate-bag/certificate/name=
2280 IDevID-<SignatureSuiteName>-<CertificateSerialNumberOfCACertificate>.

2281 Such entries shall be present underneath a 'certificate-bag' entry that is identified as: /ietf-
2282 truststore:truststore/certificate-bags/certificate-bag/name=IDevID.

2283 6.3.3.4 Processing model**2284 6.3.3.4.1 General**

2285 The processing model for IDevID credentials and trust anchors shall comply to IEEE Std
2286 802.1AR-2018 and 6.3.3.4.

2287 6.3.3.4.2 Credentials**2288 6.3.3.4.2.1 General**

2289 IDevID credentials are used in following use cases:

- 2290 • NETCONF/YANG security setup from factory default; the number of such events scales with
2291 the number of factory resets, i.e. this use case is performed sporadically. It is conducted by
2292 CNCs and encompasses a device identity verification, and
- 2293 • device identity verification happens as a subtask during NETCONF/YANG security setup
2294 from factory default. It can also happen at the discretion of the CNC user. The details of
2295 device identity verification are also subject to given policy.

2296 In these use cases, IA-stations act in claimant role and CNCs act in verifier role:

- 2297 • IA-stations shall present the certification path of and prove private key possession for an
2298 IDevID credential, and
- 2299 • CNCs shall validate the certification path, check the proof-of-possession for the private key,
2300 and verify the obtained device identity information.

2301 6.3.3.4.2.2 Creation

2302 IA-station manufacturers select the form factor for representing verifiable device identity in
2303 IDevID credentials: raw or extended form. The details of the IDevID credential issuance process
2304 are manufacturer-specific and not addressed in this document.

2305 IA-station manufacturers are not required to offer an update feature for IDevID credentials.

2306 6.3.3.4.2.3 Distribution

2307 IA-stations shall supply IDevID credentials in form of built-in keys, see 6.3.3.3.

2308 6.3.3.4.2.4 Use

2309 Verifiers (CNCs) shall perform the following checks when they challenge claimants (IA-stations)
2310 to authenticate themselves by means of an IDevID credential.

- 2311 • IDevID certification path validation according to IETF RFC 5280, Clause 6. Whether this
2312 validation happens with or without revocation checks is at the discretion of the CNC user.
2313 It is the responsibility of the CNC user to supply a trust anchor configuration (set of trusted
2314 certificates or trusted public keys), a revocation check instruction (Boolean) and optionally,
2315 X.509 CRL objects according to IETF RFC 5280, Clause 5, to CNCs. The certification path
2316 validation is passed if and only if the IDevID EE certificate is the leaf of a valid certification
2317 path that ends with a CA certificate which is signed by a configured trust anchor and which
2318 is not revoked (if revocation check is enabled).
- 2319 • Proof-of-possession checking for the private key. The proof-of-possession check is passed
2320 if and only if the IA-station possesses the private key which matches the public key in the
2321 IDevID EE certificate.

- It is the responsibility of the CNC user to establish and supply to CNCs: a device identity verification policy which determines the verifiable device identity subset that shall be checked by the CNC for the IA-stations in a Configuration Domain. This is a subset of {hardware-rev, serial-num, mfg-name, model-name}. The empty subset ("no-identity-check") as well as the whole set are allowed. The device identity verification for an IA-station instance shall behave according to the following list of requirements.
 - If this subset is empty, then the device identity check is passed. If the user chooses not to verify identity, information about the devices is considered unreliable. Tracking the unverified status of such devices is the responsibility of the user. It is the responsibility of the user to establish policies for the use of such devices.
 - If this subset is non-empty, then the CNC performs the following expected vs. actual check for each verifiable device identity item in this subset:
 - The check for any item in this subset is passed if the expected value (from ietf-hardware YANG module) matches the actual value (from the verifiable device identity URN value for this document in the subjectAltName extension of the IDevID EE certificate). This check fails if the IDevID has raw form.
 - The device identity check is passed if it is passed for all items in the subset.

IDevIDs in raw form (without verifiable device identity URN) can be used if the device identity verification setting option "no-identity-check" is employed. This allows to perform the NETCONF/YANG security setup from factory default for IA-stations with IDevID credentials in raw form. From CNC perspective these IA-stations remain anonymous.

NOTE This document does not specify a mechanism for device identity verification for IDevIDs in raw form. Whether and how device identity checks for such IA-stations are done in an offline mode is at the discretion of CNC users.

6.3.3.4.2.5 Storage

Credentials shall be stored persistently upon an IA-station. The details for implementing this persistent storage are IA-station manufacturer-specific and not addressed in this document.

IA-stations shall support storage of at least one IDevID credential.

6.3.3.4.2.6 Revocation

It is the responsibility of IA-station manufacturers to report revocation for the IDevID credentials issued by them in form of X.509 CRL objects according to IETF RFC 5280, Clause 5. These objects are made available in a form that allows relying parties, i.e. CNC users to retrieve them at their own discretion.

CNC users decide whether they support IDevID certification path validation with or without revocation:

- if revocation checks are disabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1, or
- if revocation checks are enabled, then certificate path validation shall be performed according to IETF RFC 5280, 6.1 and 6.3.

NOTE It is the responsibility of CNC users to obtain up-to-date X.509 CRL objects from manufactures and make them locally available for verifiers.

6.3.3.4.3 Trust anchors

6.3.3.4.3.1 General

Trust anchors are input arguments for certification path validation according to IETF RFC 5280, 6.1.1, input argument (d). Relying parties decide about these input arguments in a discretionary fashion, i.e. these objects are not created and distributed as literal trust anchor objects but in a pre-material form of, for example, self-signed certificate objects.

2368 NOTE The digital signature in self-signed certificates do not vouch for authenticity of this object: Actor X can issue
 2369 self-signed certificates featuring the name of actor A that cannot be distinguished from self-signed certificates issued
 2370 by A. The mechanisms to verify the authenticity of self-signed certificates are not addressed in this document.

2371 The trust anchors for use cases where IA-stations act in claimant role are determined by CNC
 2372 users.

2373 **6.3.3.4.3.2 Creation**

2374 The details of the issuance and update processes for trust anchors for validation of IDevID
 2375 credentials are not addressed by this document.

2376 **6.3.3.4.3.3 Distribution**

2377 With respect to use cases where IA-stations act in claimant role, e.g. NETCONF/YANG security
 2378 setup and device identity verification, the following model applies.

- 2379 • Issuers (IA-station manufacturers) create and distribute trust anchors. Issuers also provide
 2380 out-of-band means that allow relying parties to check the authenticity of these objects, and
- 2381 • Relying parties (CNC users) check the authenticity of trust anchors and decide about their
 2382 acceptance as trust anchors for certification path validation in a discretionary manner and
 2383 configure their verifiers (CNCs) accordingly.

2384 The details of distribution and validation of trust anchors are not addressed by this document.

2385 **6.3.3.4.3.4 Use**

2386 Trust anchors for IDevID credentials are used for certification path validation according to IETF
 2387 RFC 5280, 6.1.1 d). This concerns CNCs with respect to the use cases NETCONF/YANG
 2388 security setup from factory default, device identity verification.

2389 **6.3.3.4.3.5 Storage**

2390 Trust anchors for IDevID credentials shall be stored persistently upon CNCs. The details for
 2391 implementing this persistent storage are not addressed in this document.

2392 **6.3.3.4.3.6 Revocation**

2393 IA-station manufacturers are not required to support an authority revocation feature for IDevID
 2394 credential certification authorities.

2395 **6.3.4 Security setup based on IDevID**

2396 **6.3.4.1 General**

2397 IA-stations in factory default state shall conduct a security setup sequence for the Configuration
 2398 Domain. This sequence consists of the following steps, each step is described in 6.3.4.

- 2399 • imprintTrustAnchor: imprint of a Configuration Domain specific trust anchor to an IA-station
 2400 that allows to validate LDevID-NETCONF certificates presented by communication partners.
- 2401 • imprintCredential: imprint of a Configuration Domain specific credential to an IA-station, i.e.
 2402 a private key and the corresponding X.509 v3 end entity certificate according to ISO/IEC
 2403 9594-8 as profiled in IETF RFC 5280, Clause 4, (plus intermediate CA certificates, if
 2404 applicable) plus self-signed root CA certificate that serves as own LDevID credential.
- 2405 • imprintCertToNameMapping: imprint a Configuration Domain specific certificate-to-name
 2406 mapping to an IA-station.

2407 Credentials shall be stored persistently upon an IA-station. The details for implementing this
 2408 persistent storage are IA-station manufacturer-specific and not addressed in this document.

2409 IA-stations shall support storage of at least one LDevID-NETCONF credential.

2410 **6.3.4.2 imprintTrustAnchor**

2411 IA-stations in factory default state shall support the imprinting of a single Configuration Domain
2412 specific trust anchor according to IETF RFC 6024 via NETCONF-over-TLS according to a
2413 procedure called "provisional accept of client certificate", which uses an IDevID credential on
2414 NETCONF and TLS server side (IA-station) and a LDevID credential on NETCONF and TLS
2415 client side (for example, a CNC) and operates as follows at the NETCONF and TLS server.

- 2416 a) Challenge the client for TLS client authentication according to IETF RFC 7589 by sending
2417 a CertificateRequest message with an empty certificate_authorities entry.
- 2418 b) Perform certification path validation according to IETF RFC 5280, Clause 6, for the contents
2419 of the client's Certificate message. This certification path validation fails due to a missing
2420 trust anchor for the LDevID credential.
- 2421 c) Provisionally accept the failing certification path validation when the reason is "no matching
2422 trust anchor" (and only this reason) and proceed with the TLS exchange.
- 2423 d) Expect the client to send a trust anchor for LDevID over the provisionally accepted TLS
2424 session (no other object type).
- 2425 e) If the trust anchor in the NETCONF application payload was accepted, then redo the priorly
2426 failing certification path validation using this trust anchor, see step b).
- 2427 f) If this certification path revalidation is successful, then keep the TLS session alive and send
2428 an <rpc-reply> with success. The client then is expected to perform the NETCONF
2429 exchanges for imprintCredential (described in 6.3.4.3) and for imprintCertToNameMapping
2430 (described in 6.3.4.4) via the already established TLS session.
- 2431 g) If this certification path revalidation is not successful, then terminate the TLS session. The
2432 usual NETCONF/YANG hygiene applies. This is expected to remove the entry in the ietf-
2433 truststore that was created in step d).

2434 NOTE This "provisional accept of client certificate" is a mirrored version of the "provisional accept of server cert"
2435 in IETF RFC 8995.

2436 The "provisional accept of client cert" in factory default state shall skip the certificate-to-name
2437 mapping and shall use the NACM recovery session, i.e. skip permission checking. In this model
2438 all authenticated clients are accepted as authorized for doing the first imprinting of the LDevID
2439 credential and the corresponding trust anchor. Only contextual checks such as "once only when
2440 being in factory default state" are feasible. This model is also known as "trust on first use"
2441 (TOFU) and, e.g. also allows to read contents of the ietf-hardware module by the client for an
2442 extended identity check.

2443 The imprinting NETCONF client checks the actual server identity that is stated by the IA-station
2444 on TLS level.

2445 The NETCONF client checks the IDevID end entity certificate presented by the NETCONF
2446 server on TLS level for existence of subjectAltName extension with GeneralName entries of
2447 type uniformResourceIdentifier. If an entry contains the namespace identifier and the
2448 namespace-specific string as defined in 6.3.3.2.2, the presented server certificate is in
2449 extended form, otherwise it is in raw form.

2450 In case that the server certificate is in raw form, the following matching can be done:

- 2451 • Match a list of accepted (or blocked) manufacturers against the issuer or subject field entries
2452 of the certificate.
- 2453 • Match a list of accepted (or blocked) product instances against the product serial number
2454 from the subject field per accepted manufacturer.
- 2455 • Match the end entity certificate object as a whole against a list of pinned certificates.

2456 In case that the server certificate is in extended form, the following additional matching can be
 2457 done: match the q-components included in the verifiable device identity according to
 2458 6.3.3.2.2 against those that can be read out from the corresponding leaves of the YANG module
 2459 ietf-hardware or against reference values obtained by a method not addressed in this document.

2460 Details of how the matching happens depend on the implementation of the client that performs
 2461 this imprinting.

2462 The LDevID-NETCONF trust anchor certificate shall be imprinted using the truststore container
 2463 of the ietf-truststore module with:

- 2464 • /ts:truststore/ts:certificate-bags/ts:certificate-bag/ts:name = IEC60802,
- 2465 • /ts:truststore/ts:certificate-bags/ts:certificate-bag/[ts:name=IEC60802]/,
 - 2466 – ts:certificate/ts:name = IEC60802-LDevID,
 - 2467 – ts:certificate/ts:cert-data containing the IEC60802-LDevID trust anchor certificate data
 - 2468 object of type trust-anchor-cert-cms according to IETF RFC 9640, i.e., enveloped in
 - 2469 Base64-encoded CMS SignedData in degenerated form "certs-only" (no signature
 - 2470 value), and
 - 2471 – The imprintTrustAnchor step shall use the NETCONF operation <edit-config> according
 - 2472 to IETF RFC 6241 for the truststore container. The NETCONF operation <commit> is not
 - 2473 yet applied, but rather after successful completion of all security setup sequence steps.

2474 **6.3.4.3 imprintCredential**

2475 **6.3.4.3.1 General**

2476 The LDevID-NETCONF end entity certificate shall be provided as X.509 v3 public key certificate
 2477 according to ISO/IEC 9594-8 as profiled in IETF RFC 5280, Clause 4, with the following criteria:

- 2478 • contains the FQDN of the NETCONF server in its subjectAltName extension according to
- 2479 IETF RFC 7589, Clause 6, and IETF RFC 6125, 2.2 and B.7;
- 2480 • contains a public key and is signed by a signature suite according to 5.5.4.2 or 5.6.3;
- 2481 • contains a digitalSignature in its keyUsage extension;
- 2482 • has a finite validity period.

2483 NOTE The actual length of the validity period is at the discretion of the user of the Configuration Domain.

2484 Depending on the key generation capabilities, different steps are applied to this keystore
 2485 container.

2486 **6.3.4.3.2 Internal key generation**

2487 For IA-station with internal key generation capabilities, two NETCONF exchanges are
 2488 performed. Processing steps for the first NETCONF exchange shall be applied as follows at the
 2489 NETCONF server.

2490 a) Receive and process the NETCONF request message with action <generate-csr> and input
 2491 values as follows:

- 2492 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:
 2493 generate-csr/ks:input/ks:csr-format containing derived identity p10-csr-format according
 2494 to IETF RFC 9640, and
- 2495 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID_NETCONF]/ks:
 2496 generate-csr/ks:input/ks:csr-info containing a Base64-encoded PKCS#10
 2497 CertificationRequestInfo according to IETF RFC 2986, Clause 4.

- 2498 b) Base64-decode the <csr-info> value and parse it as a PKCS#10 CertificationRequestInfo
2499 object.
- 2500 c) Extract the algorithm information from the child element SubjectPublicKeyInfo of
2501 CertificationRequestInfo and randomly generate a key pair for the specified algorithm.
- 2502 d) Internally store the private key together with its metadata for example, algorithm information,
2503 <name> value in a secure manner.
- 2504 e) Put the public key into the (parsed) PKCS#10 CertificationRequestInfo.
- 2505 f) Serialize the PKCS#10 CertificationRequestInfo (including the public key).
- 2506 g) Use the private key to create signature value for the (serialized) PKCS#10
2507 CertificationRequestInfo (including the public key).
- 2508 h) Create a NETCONF reply message with /ks:keystore/ks:asymmetric-keys/ks:asymmetric-
2509 key/[ks:name=LDevID-NETCONF]/ks:generate-csr/ks:output/ks:p10-csr containing the data
2510 object of the previous step.

2511 In the second NETCONF exchange, the LDevID-NETCONF end entity certificate (plus
2512 intermediate CA certificates) shall be imprinted using the keystore container of the ietf-keystore
2513 module with:

- 2514 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
2515 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
2516 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
2517 – ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-
2518 NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus
2519 self-signed root CA certificate as data object of type end-entity-cert-cms according to
2520 IETF RFC 9640

2521 The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF
2522 RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but
2523 rather after successful completion of all security setup sequence steps.

2524 6.3.4.3.3 External key generation

2525 External key generation can be used for IA-stations without internal key generation capability.
2526 For external key generation, one NETCONF exchange is performed.

2527 The LDevID-NETCONF private key and end entity certificate (plus intermediate CA certificates)
2528 shall be imprinted using the keystore container of the ietf-keystore module with:

- 2529 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/ks:name = LDevID-NETCONF,
2530 • /ks:keystore/ks:asymmetric-keys/ks:asymmetric-key/[ks:name=LDevID-NETCONF]/,
2531 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF,
2532 – ks:certificates/ks:certificate/ks:public-key-format describing the encoding of the public
2533 key of the selected cryptographic algorithm according to IETF RFC 9640,
2534 – ks:certificates/ks:certificate/ks:public-key containing the public key value in the selected
2535 public-key-format,
2536 – ks:certificates/ks:certificate/ks:private-key-format describing the encoding of the private
2537 key of the selected cryptographic algorithm according to IETF RFC 9640,
2538 – ks:certificates/ks:certificate/ks:cleartext-private-key containing the private key value in
2539 the selected private-key-format,

2540 NOTE The private key is confidentiality protected by NETCONF-over-TLS even if the option <cleartext-private-key>
2541 is used.

- 2542 – ks:certificates/ks:certificate/ks:name = LDevID-NETCONF, and
2543 – ks:certificates/ks:certificate/ks:cert-data containing the certificate chain LDevID-
2544 NETCONF end entity certificate (plus intermediate CA certificates, if applicable) plus

2545 self-signed root CA certificate as data object of type end-entity-cert-cms according to
2546 IETF RFC 9640.

2547 The imprintCredential step shall use the NETCONF operation <edit-config> according to IETF
2548 RFC 6241 for the keystore container. The NETCONF operation <commit> is not yet applied, but
2549 rather after successful completion of all security setup sequence steps.

2550 External key generation can introduce security vulnerabilities during the generation and loading
2551 process. Ensuring those processes are secure is the responsibility of the user and not
2552 addressed in this document.

2553 **6.3.4.4 imprintCertToNameMapping**

2554 The Configuration Domain specific certificate-to-name mapping is imprinted in the ietf-netconf-
2555 server YANG module under the following node.

- 2556 • /ncs:netconf-server/ncs:listen/ncs:endpoint/ncs:tls/ncs:netconf-server-
2557 parameters/ncs:client-identity-mappings/ncs:cert-to-name, with the following leaves:
 - 2558 – id = 1,
 - 2559 – fingerprint = Configuration Domain specific fingerprint of the LDevID-NETCONF trust
2560 anchor using the hash algorithm sha256 according to IETF RFC 7589, Clause 7, and
 - 2561 – map-type = ext-60802-roles.

2562 The application of this map-type is described in 6.3.5, steps e) and f).

2563 The imprintCertToNameMapping step uses the NETCONF operation <edit-config> according to
2564 IETF RFC 6241 for the certificate-to-name mapping. Afterwards the NETCONF operation
2565 <commit> is applied to finalize the security setup sequence steps and to leave the factory
2566 default state.

2567 **6.3.5 Secure configuration based on LDevID-NETCONF**

2568 Configuration by NETCONF/YANG is protected by NETCONF-over-TLS as described in 6.3.2.1
2569 and NACM as described in 6.3.2.2. The NETCONF/YANG servers and clients shall use LDevID
2570 credentials for authentication.

2571 The procedure called "provisional accept of client certificate" as described in 6.3.4.2 is not
2572 applied anymore if the IA-station has left the factory default state. Instead, after successful
2573 establishment of a TLS session according to IETF RFC 7589 and IETF draft-ietf-netconf-over-
2574 tls13, the NETCONF server shall perform a certificate-to-name mapping and authorization
2575 check as follows.

- 2576 a) Compare the fingerprint of the trust anchor of the NETCONF client's certification path with
2577 the fingerprint contained in cert-to-name list entries of the x509c2n container for equal
2578 values.
- 2579 b) If no cert-name list entry match is found, then terminate the TLS session.
- 2580 c) If a cert-to-name list entry match is found, then verify if the map-type is equal to ext-60802-
2581 roles.
- 2582 d) If the map-type does not match, then terminate the TLS session.
- 2583 e) If the map-type value matches, then extract the role values from the id-60802-pe-roles
2584 certificate extension of the NETCONF client's TLS-authenticated end entity certificate. The
2585 output is a list of string values from the enumeration of specified role names according to
2586 this document.
- 2587 f) The list of role name string values is provided as input to NACM for permission checking.
2588 The access to the requested resource is checked according to the rules configured in the
2589 nacm container of the ietf-netconf-acm YANG module.

The NETCONF client checks if the expected identity to address the NETCONF server (IP address or DNS name) matches to the actual server identity that is stated by the IA-station on TLS level. This shall be done by comparing the expected identity with the subjectAltName extension of the TLS authenticated LDevID-NETCONF end entity certificate of the NETCONF server.

6.4 Management

6.4.1 General

Subclause 6.4 describes a model for configuration, deployment, and management of an industrial automation network.

6.4.2 IA-station management model

6.4.2.1 General

The management model of IA-stations covers simple end station IA-stations as well as combined IA-stations as described in 4.3. The IA-station management model is applied for topology discovery, network provisioning and stream establishment.

6.4.2.2 IEEE 802.1Q management model

In industrial automation both Bridge and end station components make use of IEEE 802.1Q specified functionality (for example, traffic classes, gate control). Thus, the IEEE 802.1Q management model is the basic management model to be applied to all IA-stations. Figure 15 shows the implementation of the IEEE Std 802.1Q Bridge model in YANG as specified in IEEE Std 802.1Q-2022, Clause 48. The IETF Interface Management YANG data model is specified in IETF RFC 8343.

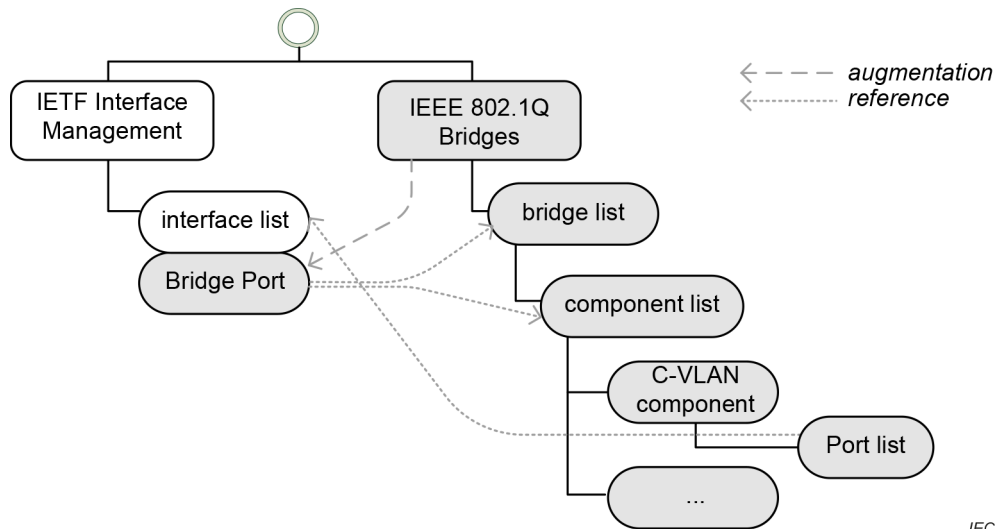


Figure 15 – Generic IEEE 802.1Q YANG Bridge management model

The IEEE 802.1Q Bridge model is organized as a bridge list where each bridge includes an underlying component list (for example, C-VLAN components). Each component has a Port list attached with references to the representation of the ports in the IETF interface list. The managed data of the ports is defined as Bridge Port augmentation to the IETF interface model. Each Bridge Port includes a reference to its bridge and component instances in the IEEE 802.1Q Bridge model.

The YANG data model for an IEEE 802.1Q Bridge is applied to IA-stations as follows.

- Each functional unit of an IA-station is modeled as bridge entry in the bridge list.

- Each Bridge and end station component of an IA-station is modeled as C-VLAN component.
- IA-station components belonging to the same functional unit are added to the component list of this functional unit's bridge entry.
- Each IA-station external or internal port is modeled as Bridge Port.

IA-station ports belonging to the same component are added to the Port list of the related component list entry.

Further YANG data models which are relevant for IA-stations are described in 6.4.9.

6.4.2.3 Internal LAN connection model

The modeling of internal connections between C-VLAN components within an IA-station is aligned to IEEE Std 802.1Q-2022, 17.3.2.2, which includes an I-LAN interface. As shown in Figure 16, the I-LAN interface is modeled as an ilan IETF interface object (see IETF RFC 7224) together with appropriate `higher-layer-if` and `lower-layer-if` reference objects to describe the internal connection.

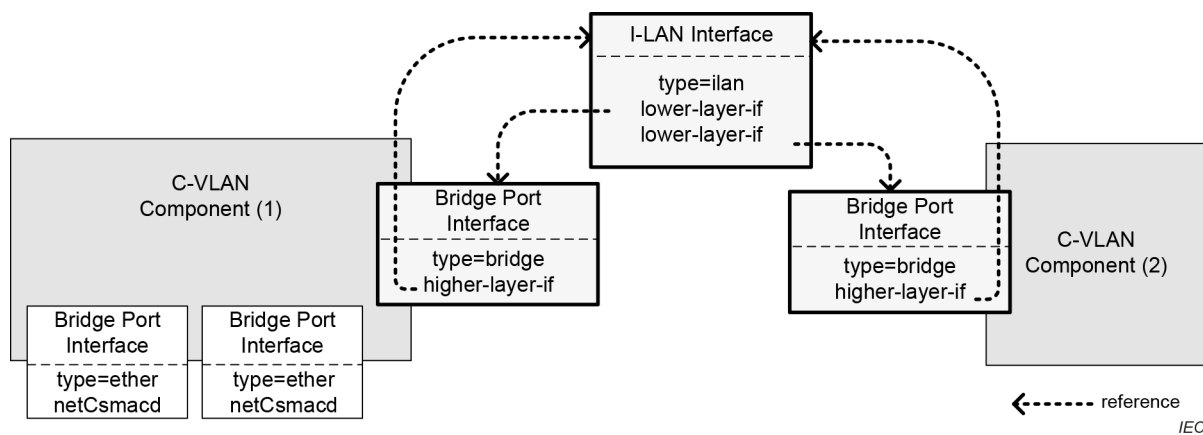


Figure 16 – Internal LAN connection management model

This internal LAN connection model comprises three configuration steps.

- The internal Ports of the C-VLAN components are modelled as IETF interfaces of type bridge with Bridge Port augmentation.
- An additional I-LAN interface of type ilan as described in IETF RFC 7224 is created.
- The I-LAN interface references the internal Bridge Port interfaces of the connected C-VLAN components as lower-layer-if, and the internal Bridge Port interfaces of the connected C-VLAN components reference the I-LAN interface as higher-layer-if.

Figure 17 shows the application of this model to the example IA-station of Figure 16.

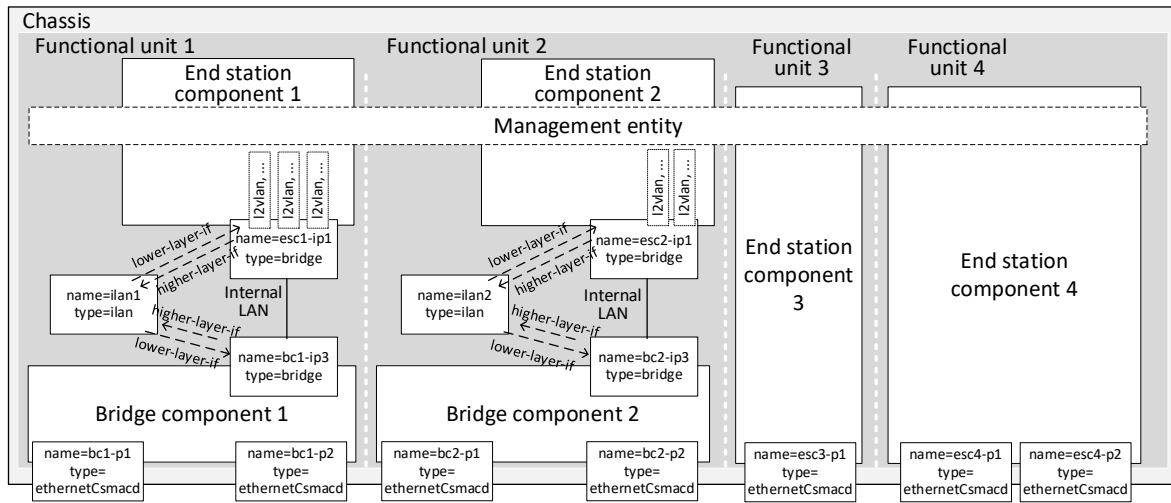


Figure 17 – IA-station example with IETF interfaces

NOTE Figure 17 represents an abstract model and is not intended to imply a particular implementation or partitioning.

Late binding of IA-station applications to the configured VLANs and priorities is performed using the `ieciieee60802-tt-mapping` YANG module in 6.4.10.8.7.

6.4.2.4 Spanning Tree, VLAN and TE-MSTID configuration

C-VLAN Bridge components of IA-stations shall support:

- the Common and Internal Spanning Tree (CIST) calculated by the Multiple Spanning Tree Algorithm and Protocol (MSTP), and
- the Traffic Engineering Multiple Spanning Tree Instance Identifier (TE-MSTID) as specified in IEEE Std 802.1Q-2022, 5.5.2.

The MSTP configuration is either default or accomplished by IA-station specific means.

CNCs configure VLANs in the `vlan` list in the `bridge-vlan` container of the `ieee802-dot1q-bridge` YANG module. Ports are assigned to a `vlan` as `static-filtering-entries` in a `filtering-database`.

NOTE `vlan`, in lowercase, refers to a YANG element.

VLANs are assigned to filtering databases in the `vid-to-fid` list of the `bridge-vlan` container. The filtering databases, and in consequence the VLANs, are by default assigned to the MSTP calculated Internal Spanning Tree and can be assigned to the TE-MSTID by management. IA-time-aware streams and IA-streams are assigned to the TE-MSTID.

TE-MSTID assignment is accomplished via the `bridge-mst` container of the `ieee802-dot1q-bridge` YANG module.

The following example shows the VID/FID/MSTID configuration of an IA-station's C-VLAN Bridge component, which supports three VLANs in three Forwarding Databases (VID 100 in FID 1, VID 101 in FID 2 and VID 102 in FID 3). FID 2 and FID 3 – and in consequence VID 101 and VID 102 – are assigned to the TE-MSTID. FID 1 – and in consequence VID 100 – is not assigned to a MSTID and thus, is implicitly assigned to the Internal Spanning Tree (IST).

Figure 18 shows the representation of this example configuration in the MST configuration table.

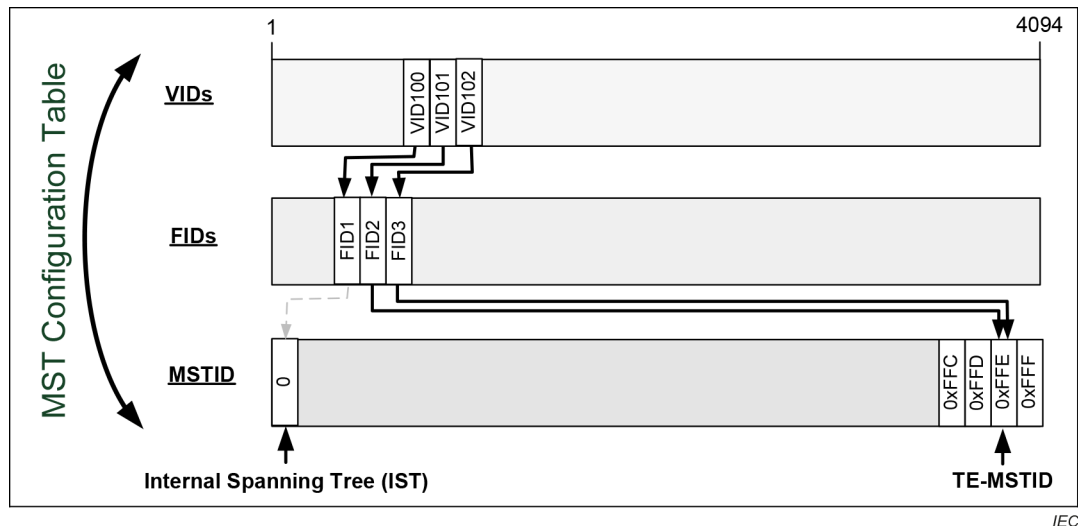


Figure 18 – VID/FID/MSTID example

The YANG-based configuration of this example is shown as YANG instance data snippet of the `ieee802-dot1q-bridge` YANG module. Herein the MST configuration table is included in component "bridge-component-x", which is part of bridge "functional-unit-x". It is the responsibility of the user to ensure that traffic type names are consistent with those provided in the YANG module `ieciieee60802-tt-mapping` (see 6.4.10.8.7).

```
<ieee802-dot1q-bridge xmlns:dot1q="urn:ieee:std:802.1Q:yang:ieee802-dot1q-bridge">
  <bridges>
    <bridge> <!-- list -->
      <name>functional-unit-x</name>
      ...
      <component> <!-- list -->
        <name>bridge-component-x</name>
        ...
        <bridge-vlan>
          <version>2</version> <!-- MST supported -->
          ...
          <vlan>
            <vid>100</vid>
            <name>60802-best-effort-hl-100</name> <!-- best effort high and
low -->
          </vlan>
          <vlan>
            <vid>101</vid>
            <name>60802-isochronous-101</name> <!-- isochronous -->
          </vlan>
          <vlan>
            <vid>102</vid>
            <name>60802-isochronous-102R</name> <!-- isochronous -->
          </vlan>
          ...
          <vid-to-fid>
            <vid>100</vid>
            <fid>1</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>101</vid>
            <fid>2</fid>
          </vid-to-fid>
          <vid-to-fid>
            <vid>102</vid>
            <fid>3</fid>
          </vid-to-fid>
        </bridge-vlan>
      ...
    </bridge>
  </bridges>
</ieee802-dot1q-bridge>
```

```

2719         <bridge-mst>
2720             ...
2721             <fid-to-mstid> <!-- list -->
2722                 <!-- fid 1 is implicitly assigned to mstid 0 -->
2723                 <fid>2</fid>
2724                 <mstid>4094</mstid> <!-- TE-MSTID -->
2725             </fid-to-mstid>
2726             <fid-to-mstid> <!-- list -->
2727                 <fid>3</fid>
2728                 <mstid>4094</mstid> <!-- TE-MSTID -->
2729             </fid-to-mstid>
2730         </bridge-mst>
2731         ...
2732     </component>
2733 </bridge>
2734 </bridges>
2735 </ieee802-dot1q-bridge>

```

2736 6.4.3 Discovery of IA-station internal structure

2737 LLDP provides information about the external connectivity of IA-stations. To identify the internal
2738 structure of complex IA-stations (see 4.3) the IEEE 802.1Q management model (see 6.4.2.2)
2739 and the IETF Interface management model are applied.

- 2740 • The functional units of an IA-station are represented as bridge entries in the bridge-list.
- 2741 • The components of a functional unit are represented as component entries in the associated
2742 bridge entry's component-list.
- 2743 • Internal LAN connections between components of a functional unit are identified by I-LAN
2744 entries in the IETF interface list (6.4.2.3).

2745 6.4.4 Network engineering model

2746 To understand the requirements for network configuration, deployment and management, an
2747 engineering model covering industrial use cases is required. The "fully centralized model"
2748 described in IEEE Std 802.1Q-2022, 46.1.3.3 includes two functional entities: the CUC and the
2749 CNC. The relationship between user and network configuration is described in IEEE Std
2750 802.1Q-2022, Clause 46. This document further elaborates this relationship to address use
2751 cases for industrial automation. A conceptual block diagram of a CNC is shown in Figure 19,
2752 which adds further details to the CNC specified in IEEE Std 802.1Q-2022 to serve the industrial
2753 automation use case. The following functional entities are introduced.

2754 a) The Topology Discovery Entity (TDE)

2755 The topology discovery entity is responsible for the topology discovery (i.e. Bridge
2756 component and end station component discovery). The TDE also performs a topology
2757 verification in cases where an expected topology is provided by the engineering tool. The
2758 resulting topology information is used by the CNC. The TDE detects added or removed IA-
2759 stations, including internal structure and connectivity. Thus, the CNC becomes aware of
2760 them. Overall, the TDE discovers and maintains an inventory of the devices, including their
2761 capabilities and the topology they form.

2762 b) The Path Entity (PE)

2763 The PE computes, establishes and maintains the forwarding paths for the IA time-aware
2764 stream and IA stream traffic type categories according to 4.7.3.

2765 c) The Sync Tree Entity (STE)

2766 The STE computes, establishes and maintains the sync trees. For example, for Working
2767 Clock and Global Time.

2768 d) The Resource Allocation Entity (RAE)

2769 The RAE is responsible for the allocation of the resources that are necessary for all traffic
2770 type categories, according to 4.7.3, to meet their requirements via their forwarding paths.
2771 For example, frame buffers at egress ports and FDB entries.

2772 e) The Network Provisioning Entity (NPE)

2773 The NPE applies a network policy provided by the Engineering Tool to the IA-stations within
 2774 the Configuration Domain. It uses the information discovered by the TDE to create a network
 2775 configuration based upon this policy which is then applied to all IA-stations. The CNC uses
 2776 the chosen network configuration together with the discovered IA-stations and their
 2777 capabilities as input for its stream calculation and deployment.

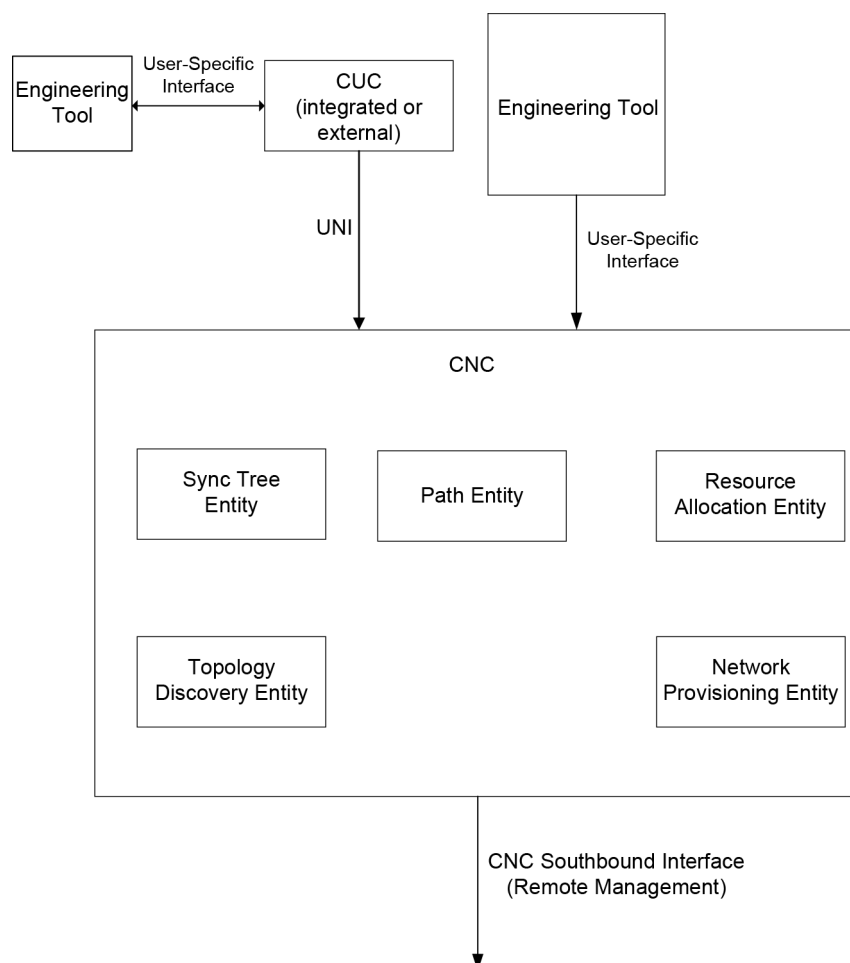
2778 A CNC includes these functional entities. The implementation of these functional entities and
 2779 the CNC can vary. The means of communication among these functional entities is
 2780 implementation dependent.

2781 If there are multiple CNCs in one Configuration Domain, then, by some means not addressed
 2782 by this document, only a single CNC is in charge at any time in the given Configuration Domain.

2783 The CNC can be in a dedicated station or integrated into any IA-controller or IA-device.
 2784 Generally, its engineering tool interface is user-specific and can only work with the compatible
 2785 engineering tools. The definition of this interface is not addressed in this document.

2786 The CUC can be in a dedicated station or integrated into any IA-controller or IA-device.
 2787 Generally, the CUC is user-specific. In industrial automation use cases, an IA-controller
 2788 integrated CUC is very likely.

2789 For stream establishment, the UNI of the CNC component is exposed.



IEC

Figure 19 – Structure and interfaces of a CNC

2792 Figure 20 shows an example of the structure of an IA-station which the CNC can discover and
 2793 manage.

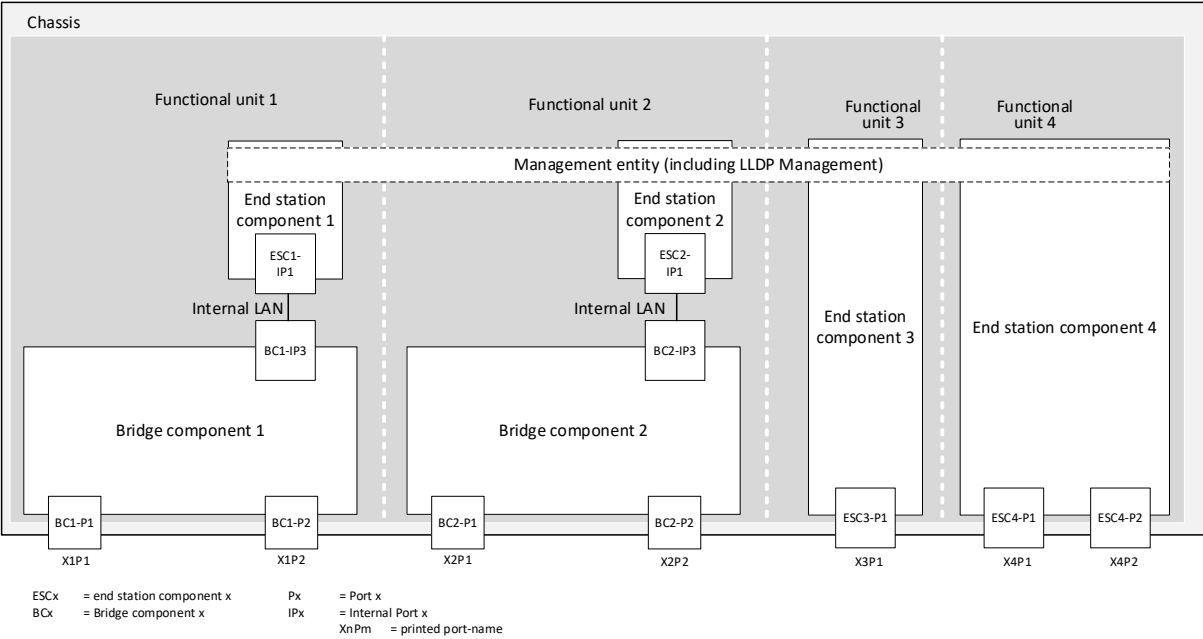


Figure 20 – IA-station structure example

Figure 21 shows the interaction of IA-stations with the CNC.

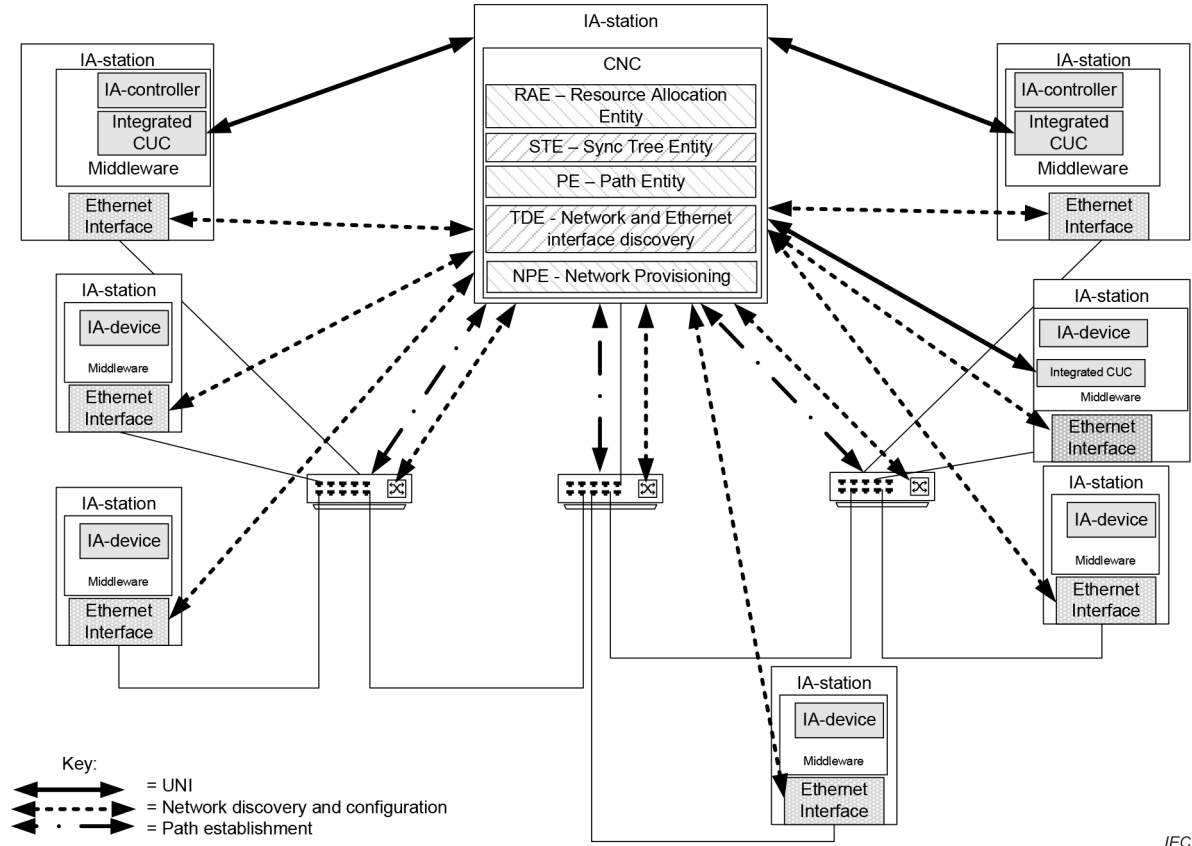


Figure 21 – CNC interaction

6.4.5 Operation

6.4.5.1 General

A representative model for network configuration is shown in Figure 22. This diagram maintains the traditional role of the IA-controller and the IA-device in an industrial automation network. IA-devices and IA-controllers require configuration from engineering tools (refer to engineering tools A, B, D, and E). These tools and associated interfaces are not addressed by this document. In this example, engineering tool C communicates directly with the CNC to provide traffic requirements for the network. The protocols that the engineering tool uses for communication with end stations are specific to the user application.

The UNI is the interface to the CNC which is serviced by NETCONF over TLS. The UNI service recognizes that industrial automation communications are typically connection oriented. There is a communication initiator, typically in an IA-controller, which is responsible for establishing those connections, determining what data is of interest and providing the required update rate. So, while an application/middleware of an IA-station (for example a Drive) understands what information it can produce and the maximum rate at which that information can be provided, until an IA-controller establishes a connection with that device, it does not know where that information goes and what update rate is required to close the control loop. The IA-controller gets this information from its engineering tool. There can be multiple IA-controllers in each Configuration Domain. The CNC uses the topology, the device capabilities, the device configuration, and the traffic specifications from the user to calculate a path for each Talker-Listener pair. The UNI then provides stream identification (VLAN, DMAC, etc.) to the middleware.

The operational management model, see Figure 22, reflects the model used in industrial automation. Figure 22 shows an active CNC managing multiple IA-stations. Each station can wholly incorporate a CUC and interact with the CNC directly.

Security requirements (see 6.3) are an important consideration for these networks and are integrated into the design, configuration, and deployment of any management model.

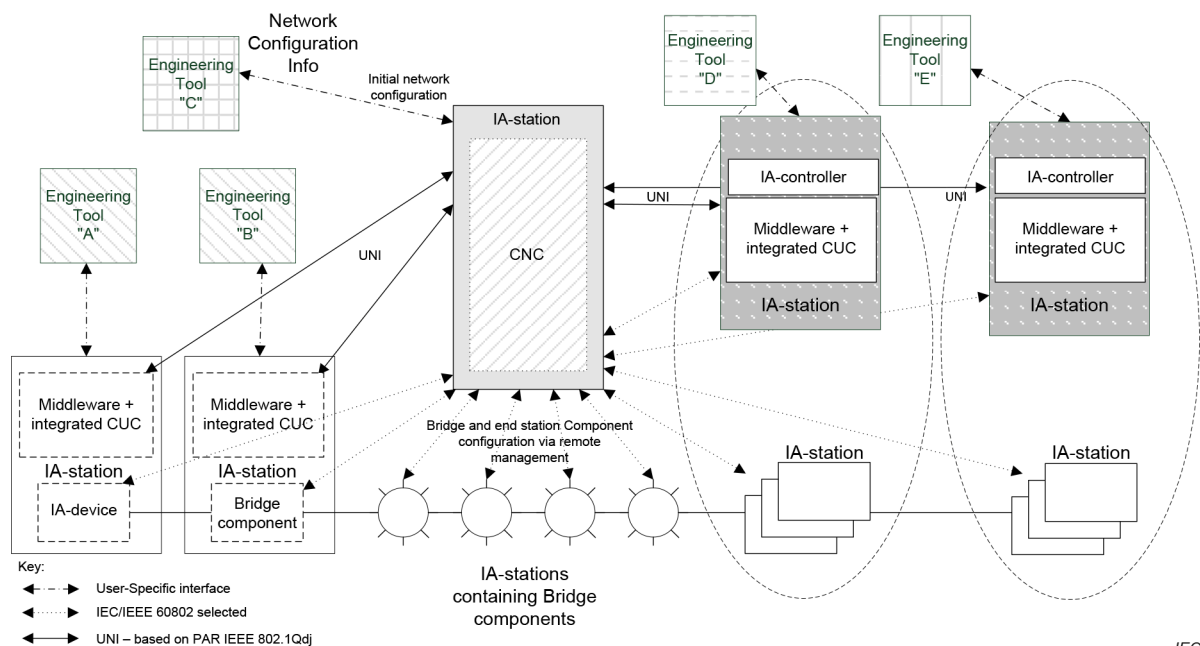


Figure 22 – Operational management model

Figure 23 shows the steps that are typically performed in the scope of the CUC-CNC interaction.

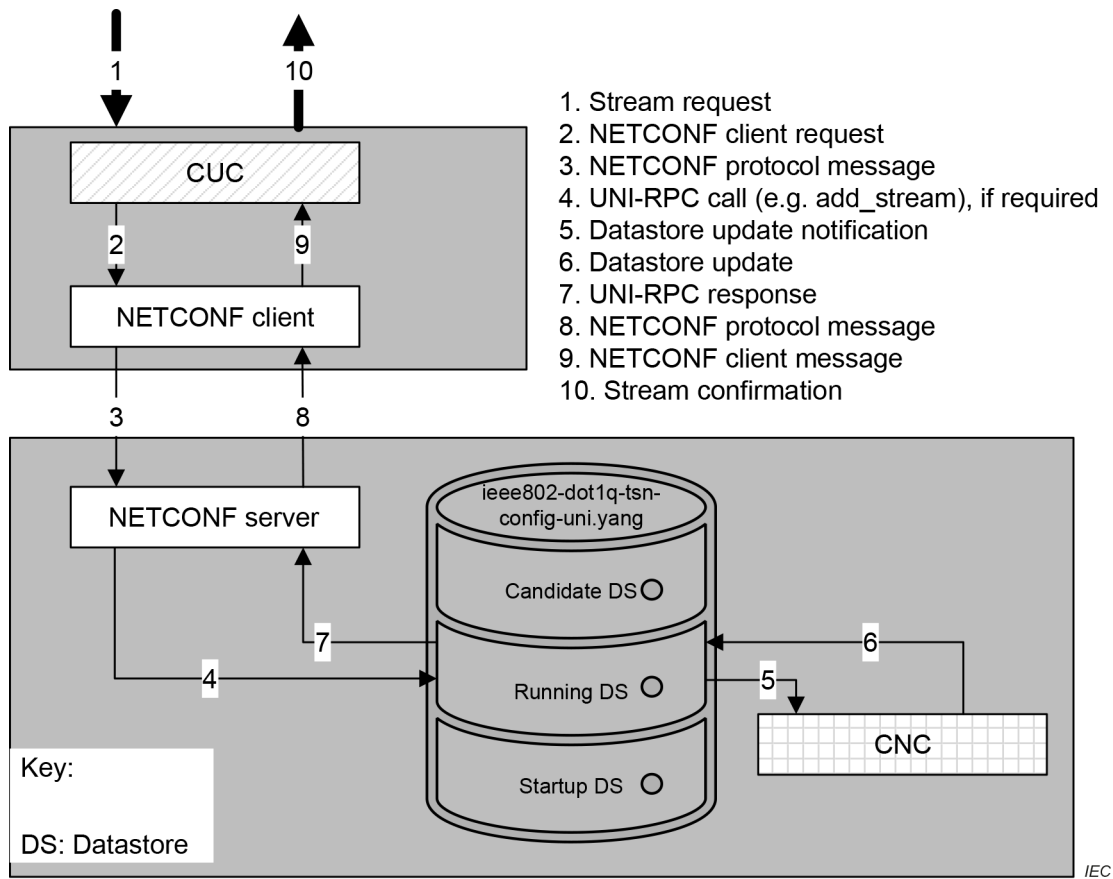


Figure 23 – UNI service model

After the computation of the paths and the scheduling or shaping configuration have been done, the CNC configures the IA-stations via NETCONF client. The typical steps that are performed in this process are shown in Figure 24 below.

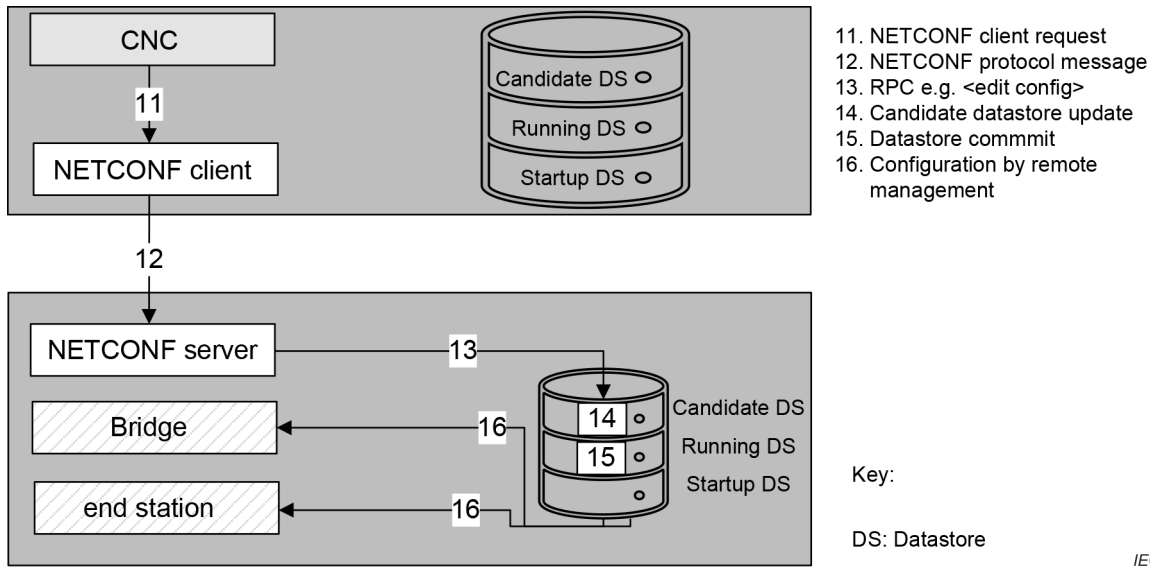


Figure 24 – CNC southbound

Instances of NETCONF servers and clients within a Configuration Domain are shown in Figure 25. IA-stations that contain a CNC or CUC entity contain both a NETCONF server and a NETCONF client. The CUC utilizes a NETCONF client for the UNI. The CNC utilizes a

NETCONF server to accommodate the UNI as well as remote network management of the end stations and bridges that are contained in the same chassis as the CNC entity. The CNC utilizes a NETCONF client for the southbound interface of the CNC, i.e. for the remote management of the bridges and end stations in the scope of stream configuration. All IA-stations have a NETCONF server to make remote management possible. The NETCONF server used by the CNC serves multiple NETCONF Clients (CUCs) within a single Configuration Domain whose requests clients can occur simultaneously.

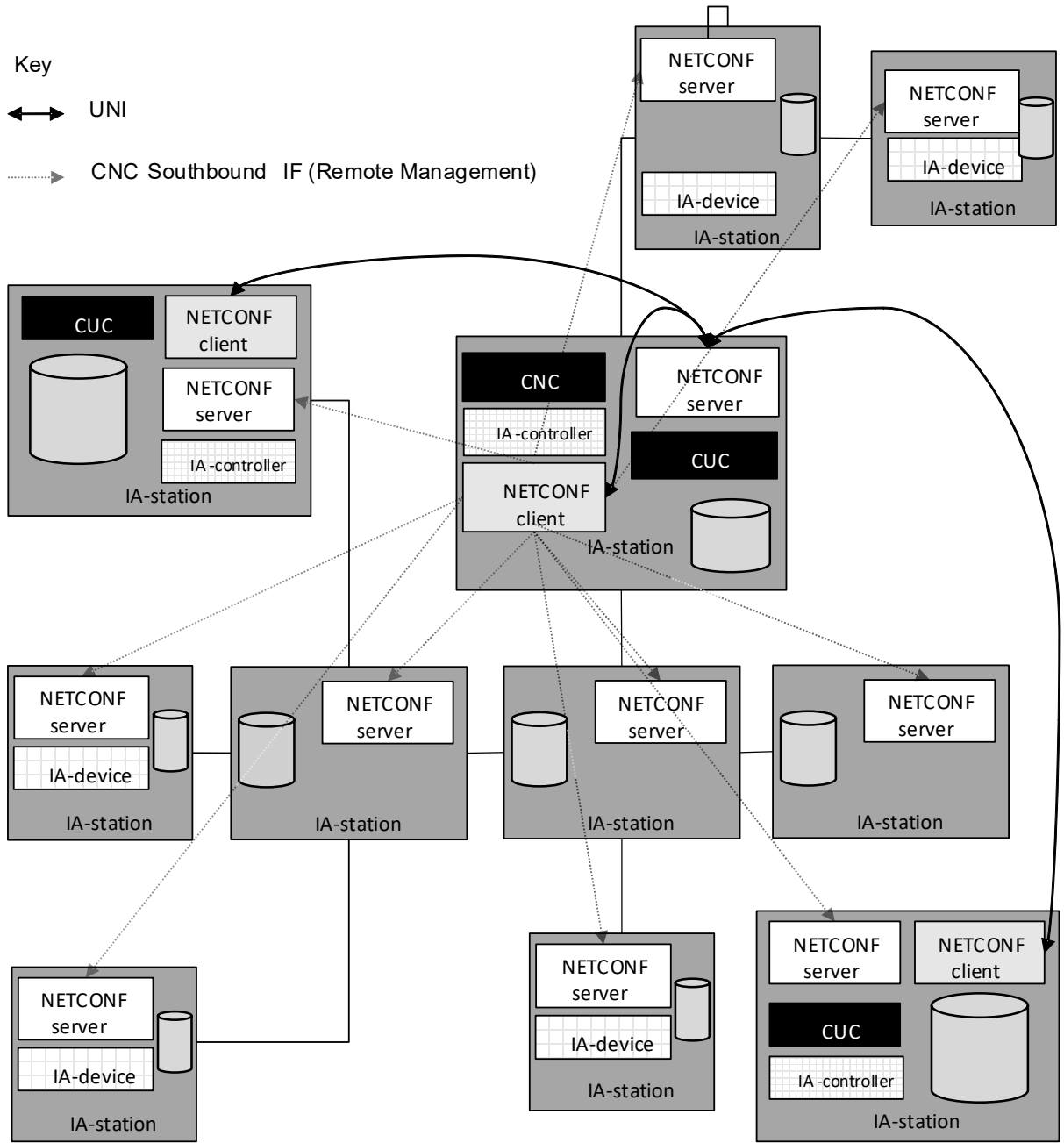


Figure 25 – NETCONF usage in a Configuration Domain

6.4.5.2 Domain port states

A CNC manages available network resources and assigns them to the IA-stations. Management of the network resources is only possible if the CNC owns these resources. Thus, no connected station is allowed to make use of network resources that are not granted by the CNC. The security configuration of a connected station allows remote access for the CNC.

2853 Protection of the network resources is done by managing the ports (see Figure 26) at the
2854 boundary of the Configuration Domain. The state of any newly connected station is unknown.
2855 The CNC is responsible for determining if the newly connected station is added to the
2856 Configuration Domain and configuring the IA-station appropriately.

2857 This port state model avoids any assumptions about configuration of added stations or network
2858 portions.

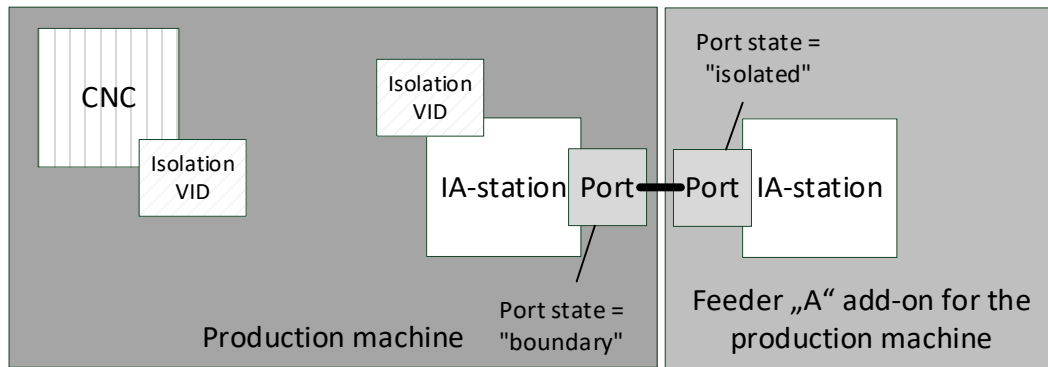


Figure 26 – Boundary port model

2861 Ports of an IA-station that is a member of a Configuration Domain have different states:

- 2862 • Isolated – a station connected via this port can only exchange information with a CNC. The
2863 CNC is responsible for establishing an isolation VID and for onboarding the station. In the
2864 isolated state:
 - 2865 – the port gets to or remains in isolated state in case of a link down event, e.g. when
2866 nothing is connected, or no link is established;
 - 2867 – the port gets to or remains in isolated state in case of a link up event;
 - 2868 – the port stays in isolated state as long as the neighbor is unknown, not able to enter
2869 Boundary state.
- 2870 • Boundary – a station connected via this port is not part of the Configuration Domain, but is
2871 allowed to access devices inside the Configuration Domain and to pass traffic through the
2872 Configuration Domain
- 2873 • Inside – a station connected via this port is part of the Configuration Domain

2874 The determination of whether a given port of an IA-station remains in the Isolated state or
2875 transitions to the Boundary or Inside state is performed by the CNC using remote management.
2876 A port acts as a domain boundary if it is in the Isolated or Boundary state.

2877 For example, a port could be configured as follows:

- 2878 • Isolated state:
 - 2879 – port is IST boundary,
 - 2880 – port is not part of a sync tree,
 - 2881 – port uses VLAN stripping for egress,
 - 2882 – port uses VLAN assignment and priority regeneration to assign all traffic to an isolated
2883 VLAN, and
 - 2884 – port uses an ingress rate limiter to control the amount of traffic for the Configuration
2885 Domain.
- 2886 • Boundary state:
 - 2887 – port is part of IST,

- 2888 – port is part of a sync tree,
- 2889 – port uses VLAN stripping for egress,
- 2890 – port uses VLAN assignment and priority regeneration to assign all traffic to a default
- 2891 VLAN, and
- 2892 – port uses an ingress rate limiter to control the amount of traffic for the Configuration
- 2893 Domain.
- 2894 • Inside state:
- 2895 – port is part of IST,
- 2896 – port is part of a sync tree, and
- 2897 – port is part of the active topology for stream and non-stream traffic.

2898 An example workflow includes the following steps executed by the CNC:

2899 a) Topology discovery:

2900 1) Case A: Link down / Port not connected:

2901 i) set port to isolated state;

2902 ii) configure a NETCONF subscription "on data change" to the port state leaf

2903 2) Case B: Neighbor is not a Configuration Domain member:

2904 i) set port to boundary state;

2905 ii) configure a NETCONF subscription "on data change" to the port state leaf.

2906 3) Case C: Neighbor is not a Configuration Domain member – but part of expected topology:

2907 i) set port to boundary state;

2908 ii) configure the neighbor station as Configuration Domain member;

2909 iii) set port to inside state.

2910 b) NETCONF subscription trigger:

2911 Issued to the CNC upon change of subscribed YANG data.

2912 **6.4.5.3 Engineered network**

2913 For an offline engineered (based on the available digital data sheets of the used IA-stations)
2914 centralized approach with fixed topology, fixed stations and fixed paths, the user provides traffic
2915 requirements, path information, topology information and expected network configuration to the
2916 CNC. The CNC then uses the TDE, RAE and the NPE to perform the calculation of paths,
2917 resources, and stream schedules necessary to meet the specified traffic requirements and
2918 deploys the result of these calculations via remote management. The CNC also provides the
2919 relevant results to the CUC via the UNI. The CUC then configures the end stations using the
2920 user-to-user interface (see Figure 3).

2921 The workflow for this example consists of the following steps:

2922 a) The user determines:

2923 1) the expected network topology,

2924 2) the expected stations and its capabilities, value ranges and quantities,

2925 3) the expected paths and resources,

2926 4) the required streams, and

2927 5) the requirements for IA non-stream traffic.

2928 This step focuses on network capabilities including the Ethernet interface of the end stations.
2929 For example, if the end station is a sensor, the user considers the Ethernet interface capabilities
2930 of the sensor as they apply to the physical world.

2931 b) Engineering Tool provides this information to the CNC via a user-specific interface.

2932 Although the communication between the CNC and any Engineering Tool is user-specific, the
2933 CNC obtains all necessary information via the integrated TDE and NPE.

2934 c) The CNC uses the TDE to discover the topology and checks it against the expected
2935 topology. The NPE is used to configure the IA-stations of the Configuration Domain.

2936 d) The CNC uses STE and NPE to setup, validate, and monitor synchronization configuration
2937 in the Configuration Domain.

2938 e) The CNC uses the information from engineering item a), steps 1 to 5, above to respond to
2939 requests from middleware (with integrated CUC) using UNI. These requests are handled
2940 using the already established communication paths received from the user.

2941 If the CNC is not required after commissioning, then the CNC can be removed after setting up
2942 the IA-stations. That requires that all IA-stations have a persistent storage for the data provided
2943 by the CNC.

2944 **6.4.5.4 Dynamic topology**

2945 **6.4.5.4.1 General**

2946 For a centralized approach with a dynamic topology and dynamic paths, the user provides the
2947 network policy to the CNC. The TDE performs topology discovery including IA-station
2948 capabilities (YANG representation of the digital data sheet) and the NPE performs network
2949 configuration for the CNC. IA-stations then provide traffic requirements via the middleware to
2950 the CNC via the UNI. The CNC then uses the TDE, RAE, and NPE to perform the calculation of
2951 paths, resources, and stream schedules necessary to meet the specified traffic requirements
2952 and deploys the result of these calculations via remote management. The CNC also provides
2953 the relevant results to the CUC via the UNI. The CUC then configures the end stations using
2954 the user-to-user interface (see Figure 3).

2955 The workflow for this example consists of the following steps.

2956 a) The user determines the network policy and provides it to the CNC.

2957 b) The TDE continuously discovers the physical network topology and station capabilities of
2958 each station using remote management.

2959 c) The NPE uses the information gathered in steps a) to b) to configure the IA-stations in the
2960 Configuration Domain.

2961 d) The CNC uses STE and NPE to setup, validate and monitor time synchronization
2962 configuration in the Configuration Domain.

2963 The CNC uses the information from steps a) to d) to respond to requests from middleware using
2964 UNI. The CNC establishes streams in the bridges via a remote management protocol.

2965 **6.4.5.4.2 Adding an IA-station**

2966 Each IA-station added to the Configuration Domain is discovered by the TDE and receives the
2967 initial network configuration from the NPE. After this, the middleware can request stream
2968 establishment.

2969 When an IA-station is added to the network, it is isolated until the CNC determines that its traffic
2970 requirements can be accommodated without disrupting other traffic (see 6.4.5.2).

2971 **6.4.5.4.3 Removing an IA-station**

2972 Each IA-station removed from the Configuration Domain is discovered by the TDE. A
2973 neighboring station can receive an updated network configuration by the NPE. After this, the
2974 removed IA-station is no longer part of the Configuration Domain.

6.4.5.4.4 Replacing an IA-station

In the simplest case, replacing an IA-station is simply the sequence of removing an IA-station (6.4.5.4.3) and adding an IA-station (6.4.5.4.2). In more complex cases, the user can take other precautions or actions following deployment.

6.4.5.5 Management proxies

6.4.5.5.1 General

Implementing NETCONF server functionality can be burdensome for resource constrained IA-stations. IA-stations which contain only ccB Bridge or end station components may utilize the NETCONF server of another IA-station as a management proxy. IA-stations supporting proxy functionality (proxies), maintain a list of the types of proxy functionality supported as well as the IA-stations for which they act as proxy (proxy targets). Proxy types are described in E.1.3. The information is maintained in the YANG module `ieciieee60802-proxy` as described in 6.4.10.8.8.

6.4.5.5.2 Proxy operation

When onboarding a new proxy target, a different process is followed. Proxies communicate with proxy targets via the isolation VLAN. When a new IA station is connected, the following sequence is followed.

- a) The IA-station to which the proxy target is connected (neighbor), which is already part of the configuration domain, reports a change in link status to the CNC.
- b) The CNC reads the stored LLDP information related to the proxy target from the neighbor via NETCONF. The neighbor can be a proxied device in which case the LLDP information about the proxy target would be retrieved using the neighbor's proxy.
- c) Each proxy target identifies itself using the 60802 device info TLV described in Annex E.
- d) Based on this TLV data, the CNC determines whether the connected station:
 - natively speaks NETCONF/YANG (no information regarding proxy operation is included), or
 - requires a proxy, and if so, which one.
- e) The proxy onboards the new device via a user-specific protocol. This user-specific protocol is not described in this document. Ensuring the security of this user-specific mechanism is not described in this document. If security equivalent to the secure management exchanges as described in 5.5.4.1 is not implemented in the user-specific protocol, security of the network can be compromised.
- f) The proxy communicates the result of the Authentication to the CNC. For instance, the CNC can utilize dynamic subscriptions and the on-change feature of the `ietf-yang-push` module to automatically be notified when the new proxy target is added to the proxy's list of proxy targets. Based upon the results of the authentication, the CNC sets the boundary port state appropriately
- g) The CNC communicates with the appropriate proxy and configures the device accordingly. The proxy handles the NETCONF parameterization and translates it into the required organization-defined or vendor-specific protocol and configures the proxy target.

6.4.5.6 Engineered network extended by dynamic topology

The engineered and dynamic topology workflows can be used together. For instance, modular machines, robot tool changers or more general plug & produce can add or remove modules. The basic machine is handled as an engineered network. Additional modules or removed modules are handled dynamically.

6.4.6 Engineered time-synchronization spanning tree

6.4.6.1 General

Engineered time-synchronization spanning tree (sync tree) for a given gPTP domain refers to the usage of external port configuration instead of BTCA for the construction of a desired sync tree with the Grandmaster PTP Instance as the root (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 10.3.1). The Grandmaster PTP Instance can reside in a dedicated grandmaster-capable IA-station.

One of the advantages of engineered sync trees is to enable a planned, deterministic, and stable configuration of the IEEE Std 802.1AS-2020 sync tree for a given gPTP domain. For example, this approach prevents sync tree changes in case of IA-station addition or removal from the network. Hot standby (see IEEE Std 802.1ASdm-2024) is a use case of an engineered sync tree.

6.4.6.2 Sync tree requirements

If an engineered synchronization spanning tree is used, the sync tree requirements for all participating PTP Instances in a gPTP domain are specified in 5.5.3, h).

6.4.6.3 STE phases

6.4.6.3.1 General

The STE follows the logical sequence described in 6.4.6.3 if an engineered sync tree is utilized in a gPTP domain. Each STE phase describes an externally observable behavior of the participating PTP Instances in a gPTP domain.

6.4.6.3.2 Discovery phase

In discovery phase, STE utilizes the topology discovered by the TDE to verify the capabilities and status of participating IA-stations via a diagnostics entity (see 6.4.7.1) by reading the following managed objects.

- The status of oper-status parameter is up (see IETF RFC 8343) for all participating Ethernet links.
- The status of isMeasuringDelay (see IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 14.16.4) is TRUE for all PTP Ports.
- The status of asCapable (see IEEE Std 802.1AS-2020, 14.8.7) is TRUE for all PTP Ports.
- The status of asCapableAcrossDomains (see IEEE Std 802.1AS-2020, 14.16.5) is TRUE for all LinkPorts.
- The status of gmCapable (see IEEE Std 802.1AS-2020, 14.2.7) is TRUE, only applicable to the Grandmaster PTP Instance.

STE should use the information collected via managed objects and the discovered topology to verify the constraints on the gPTP domain, for example:

- Verify that the number of PTP Relay Instances (hops) between the Grandmaster PTP Instance and any given timeReceiver PTP End Instance is within the limit prescribed by for example, CNC.

6.4.6.3.3 Provisioning phase

In provisioning phase, STE should apply the desired configuration to all participating PTP Instances, for example:

- externalPortConfigurationEnabled is set to true,

- 3062 • the desiredState of all PTP ports of the Grandmaster PTP Instance is set to
3063 TimeTransmitterPort,
- 3064 • the desiredState of exactly one PTP port of all the other PTP Instances is set to
3065 TimeReceiverPort,
- 3066 • the desiredState of remaining PTP ports that are part of sync tree in non-Grandmaster PTP
3067 Relay Instances is set to TimeTransmitterPort, and
- 3068 • the desiredState of all other PTP ports is set to PassivePort.

3069 Then STE should validate, for example, the syncLocked (see IEEE Std 802.1AS-2020, 14.8.52)
3070 parameter is TRUE for all PTP ports of PTP Relay Instances that are in TimeTransmitterPort
3071 state.

3072 **6.4.6.4 Adding an IA-station**

3073 Each IA-station added to the gPTP domain is discovered by STE via TDE. It is the responsibility
3074 of the CNC to on-board this newly added station. IA-stations can receive an updated gPTP
3075 configuration via STE.

3076 A newly installed IA-station can disrupt the operation of a gPTP domain. The extent of disruption
3077 is dependent on the location of the IA-station in the gPTP domain and the type of PTP Instance
3078 running on that IA-station. For example, if PTP Instances are arranged in a daisy-chain
3079 formation and an IA-station with a non-Grandmaster Relay Instance is installed in the middle of
3080 a daisy-chain, then this change can disrupt the operation of downstream Instances.

3081 **6.4.6.5 Removing an IA-station**

3082 The removal of a station from the gPTP domain is detected by STE via TDE. IA-stations can
3083 receive an updated gPTP configuration via STE.

3084 **6.4.6.6 Replacing an IA-station**

3085 An IA-station replacement follows the sequence of removing an IA-station according to 6.4.6.5
3086 and adding an IA-station according to 6.4.6.4.

3087 **6.4.7 Diagnostics**

3088 **6.4.7.1 General**

3089 Diagnosis for an IA-station is done by monitoring YANG representation of the IA-station's local
3090 database.

3091 A vendor can implement an observer in a diagnostics entity, which could reside in the CNC.
3092 This diagnostics entity uses the information provided by remote management to define the
3093 monitored objects and set up fitting notifications.

3094 **6.4.7.2 Observer model**

3095 A diagnostic entity can select any objects described via YANG and observe them via NETCONF.
3096 The NETCONF binding is specified in IETF RFC 8640, and the subscription model in IETF RFC
3097 8641. NETCONF messages can be pipelined, i.e. a client can invoke multiple RPCs without
3098 having to wait for RPC result messages first. RPC messages are specified in IETF RFC 6241,
3099 and notification messages are specified in IETF RFC 5277. To reduce the load on the diagnostic
3100 entity when many stations are providing notifications, the diagnostic objects can be monitored
3101 and notifications can be retrieved from individual IA-stations.

3102 Figure 27 shows the model of a diagnostic observer.

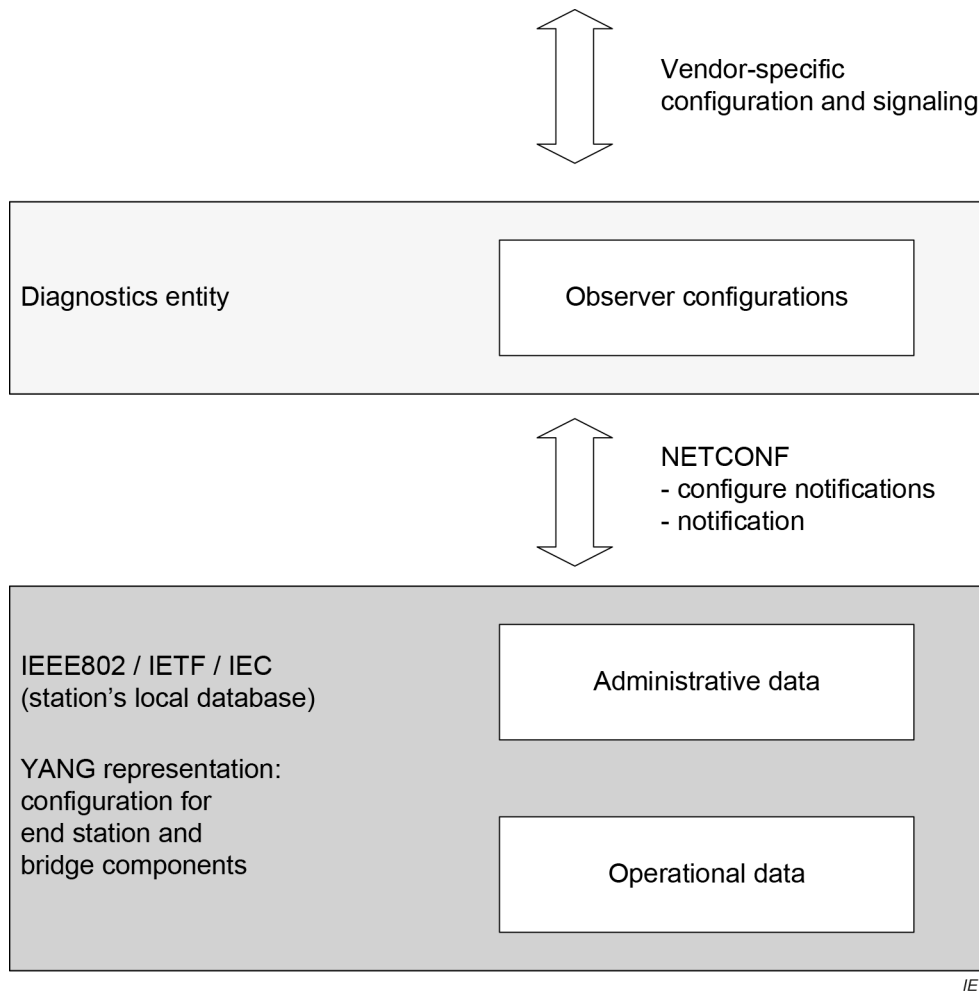


Figure 27 – Observer model

6.4.7.3 Usage of YANG Push

For diagnostics, an IA-station shall support YANG-Push subscriptions according to IETF RFC 8641 (YANG Push) and IETF RFC 8639 (Subscribed Notifications).

IA-stations shall support the "subtree" selection filter as specified in IETF RFC 8641, 3.6

6.4.7.4 Mandatory RPCs

An IA-station shall support following RPCs as specified in IETF RFC 8641:

- a) establish-subscription,
- b) modify-subscription,
- c) delete-subscription, and
- d) kill-subscription.

6.4.7.5 Mandatory notifications

An IA-station shall support following notifications as specified in IETF RFC 8641:

- a) subscription-resumed,
- b) subscription-modified,
- c) subscription-terminated,

- 3120 d) subscription-suspended,
- 3121 e) push-update, and
- 3122 f) push-change-update.

3123 **6.4.7.6 Mandatory diagnostics data nodes**

3124 An IA-station shall provide the following data nodes for NETCONF subscriptions.

- 3125 a) Change of link-status per Ethernet port:

3126 /ietf-interfaces/interface/oper-status

- 3127 b) Change of MAU-type per Ethernet port:

3128 /ieee802-ethernet-lldp/lldp/port/ operational-mau-type

- 3129 c) Change of sync-status:

- 3130 1) per PTP Instance:

3131 for every application-clock: /ieciieee60802-
3132 bridge/bridges/bridge/component/clock/is-synced

- 3133 2) per hot standby Instance:

3134 /dotlas-hs/ptp/common-services/hss/hot-standby-system-
3135 list/hot-standby-system-ds/hot-standby-system-state

- 3136 d) Data to be provided as periodic time-aligned subscriptions:

- 3137 1) Dropped frames statistic counters per Ethernet interface

- 3138 – /ietf-interfaces/interface/statistics/in-octets
- 3139 – /ietf-interfaces/interface/statistics/in-discards
- 3140 – /ietf-interfaces/interface/statistics/in-errors
- 3141 – /ietf-interfaces/interface/statistics/out-octets
- 3142 – /ietf-interfaces/interface/statistics/out-discards
- 3143 – /ietf-interfaces/interface/statistics/out-errors

- 3144 2) VLAN specific counters per Ethernet Interface and VLAN ID

- 3145 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3146 port/statistics/octets-rx
- 3147 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3148 port/statistics/octets-tx
- 3149 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3150 port/statistics/forward-outbound
- 3151 – /ieee802-dot1q-bridge/interfaces/interface/bridge-
3152 port/statistics/discard-inbound

3153 **6.4.7.7 Usage of NETCONF notifications**

3154 IA-stations shall implement the binding of a stream of events according to IETF RFC 8640
3155 (NETCONF Notifications) using the "encode-xml" feature and the "NETCONF" event stream of
3156 IETF RFC 8639.

3157 An IA-station shall support dynamic subscriptions according to 6.4.7.6, IETF RFC 8640 Clauses
3158 5, 6 and 7. The number of dynamic subscriptions shall be reported.

3159 **6.4.8 Data sheet**

3160 **6.4.8.1 General**

3161 Data sheets containing the capabilities, value ranges and quantities of IA-stations allow a user
3162 to select appropriate IA-stations and enable users to configure a system using online and offline
3163 engineering. See Annex B for quantities in a representative Configuration Domain.

3164 Online data sheets are modeled using YANG. YANG modeling is used for the offline data sheet
3165 to keep the offline (6.4.5.3) and online (6.4.5.4) format the same.

3166 **6.4.8.2 Digital data sheet of an IA-station**

3167 Both engineering models, offline via an engineering tool and online with plug & produce by the
3168 CNC, require information about the capabilities of an IA-station, for example, states,
3169 configurations, or supported features. An example depicting the creation of a digital data sheet
3170 is provided in Figure 28.

3171 This data is extracted from the implemented YANG modules, which are available in the local
3172 database of the IA-station.

3173 The data from the implemented YANG modules is also available offline in the form of a digital
3174 data sheet of an IA-station as a digital data sheet file.

3175 The digital data sheet of an IA-station provides a collection of all instantiated data nodes of all
3176 YANG modules that are required by this document (see 6.4.9). A manufacturer may reduce the
3177 instance data set by removing statistical config-false YANG nodes from the digital data sheet.

3178 The digital data sheet does not contain any additional information that is not modeled by the
3179 YANG modules that exist in the local database of the IA-station.

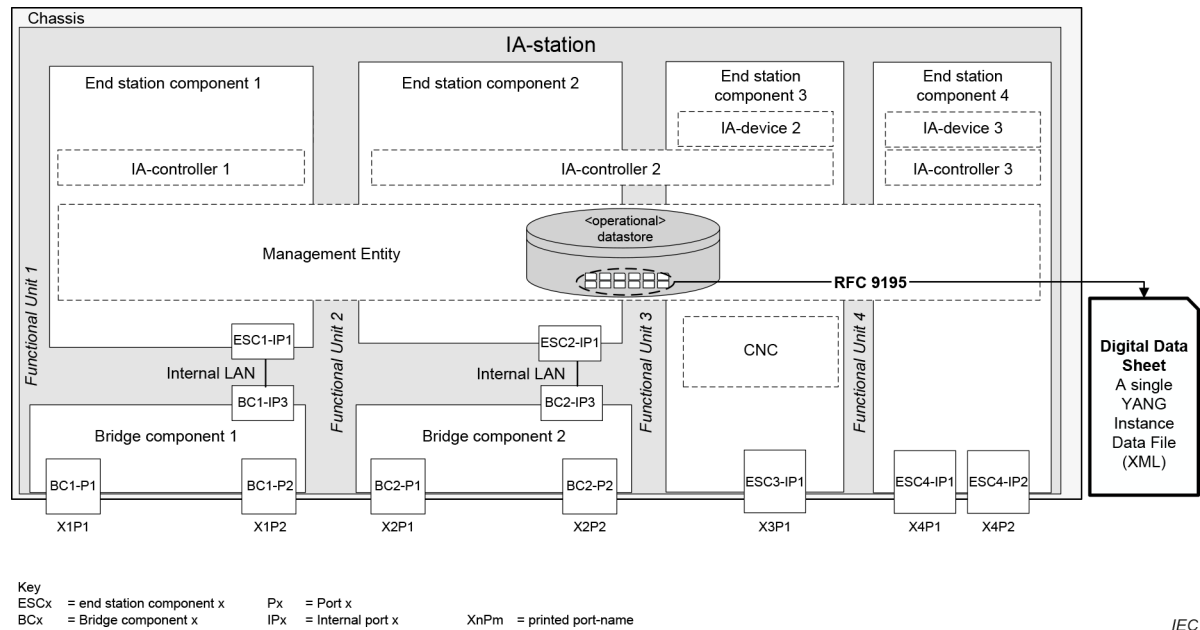
3180 The data sheet contains a single instance data set. It carries complete configuration and state
3181 data of each YANG module that is present in the local database of the IA-station.

3182 The identity of the datastore with which the instance data set is associated and the format of
3183 the YANG instance data set shall be reported according to IETF RFC 9195. The file format is
3184 based on the XML encoding. It is created by applying the respective XML encoding rules for
3185 the YANG structure of all YANG modules included in the digital data sheet.

3186 A user uses the information from the digital data sheet to understand the quantities and
3187 capabilities of an IA-station, which is required for successful offline engineering of the network.

3188 The features of a CNC are used for offline and online engineering or diagnostics. For this
3189 purpose, YANG modules are used that allow structured access to the local database of the CNC
3190 according to 6.4.9.2.5.11.

3191 Any IA-station can include a CNC entity in which case the collection of YANG modules of such
3192 IA-station includes all CNC specific YANG modules for example, the ieee802-dot1q-cnc-config
3193 YANG module. Since all IA-stations meet the requirements from 5.5.4, the CNC related YANG
3194 instance data is automatically included in the digital data sheet of the IA-station that hosts the
3195 CNC as described in 6.4.9.2.



IEC

Figure 28 – Creation of the digital data sheet of an IA-station

6.4.9 YANG representation of managed objects and nodes

6.4.9.1 General

All managed objects shall be represented in the YANG 1.1 format as described in IETF RFC 7950. The markings (i.e. [m], [o], [c]) indicate whether the node is included in the digital data sheet (see 3.5.4). These markings are independent of the conformance criteria for an IA-station (see 5.2).

6.4.9.2 Common YANG modules, features, and nodes

6.4.9.2.1 IEEE standard for Ethernet

IA-stations shall support the `ieee802-ethernet-interface` YANG module according to IEEE Std 802.3.2™-2019 as modified by IEEE Draft Std P802.3.2a with the following nodes:

[o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/duplex`

[o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/speed`

[o] `/ieee802-ethernet-interface/interfaces/interface/ethernet/flow-control/pause/direction` (if the feature "ethernet-pause" is supported)

[o] `ieee802-ethernet-lldp/lldp/port/operational-mau-type`

[o] `ieee802-ethernet-lldp/lldp/port/preemption-supported`

[o] `ieee802-ethernet-lldp/lldp/port/preemption-enabled`

[o] `ieee802-ethernet-lldp/lldp/port/preemption-active`

[o] `ieee802-ethernet-lldp/lldp/port/additional-fragment-size`

[o] `ieee802-ethernet-lldp/lldp/port/local-max-frame-size`

3218 **6.4.9.2.2 Station and media access control connectivity discovery**

3219 IA-stations shall support the following nodes from the ieee802-dot1ab-lldp YANG module
3220 according to IEEE Std 802.1ABcu-2021 with values and value ranges according to 6.5.

- 3221 [o] /ieee802-dot1ab-lldp/lldp/message-fast-tx
- 3222 [o] /ieee802-dot1ab-lldp/lldp/message-tx-hold-multiplier
- 3223 [o] /ieee802-dot1ab-lldp/lldp/message-tx-interval
- 3224 [o] /ieee802-dot1ab-lldp/lldp/reinit-delay
- 3225 [o] /ieee802-dot1ab-lldp/lldp/tx-credit-max
- 3226 [o] /ieee802-dot1ab-lldp/lldp/tx-fast-init
- 3227 [o] /ieee802-dot1ab-lldp/lldp/notification-interval
- 3228 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics
- 3229 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/last-change-time
- 3230 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-inserts
- 3231 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-deletes
- 3232 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-drops
- 3233 [o] /ieee802-dot1ab-lldp/lldp/remote-statistics/remote-ageouts
- 3234 [m] /ieee802-dot1ab-lldp/lldp/local-system-data
- 3235 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id-subtype
- 3236 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/chassis-id
- 3237 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-name
- 3238 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-description
- 3239 [m] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-
3240 supported
- 3241 [o] /ieee802-dot1ab-lldp/lldp/local-system-data/system-capabilities-
3242 enabled
- 3243 [o] /ieee802-dot1ab-lldp/lldp/port
- 3244 [o] /ieee802-dot1ab-lldp/lldp/port/name
- 3245 [o] /ieee802-dot1ab-lldp/lldp/port/dest-mac-address
- 3246 [o] /ieee802-dot1ab-lldp/lldp/port/admin-status

3247 [o] /ieee802-dot1ab-lldp/lldp/port/notification-enable

3248 [o] /ieee802-dot1ab-lldp/lldp/port/tlvs-tx-enable

3249 [o] /ieee802-dot1ab-lldp/lldp/port/message-fast-tx

3250 [o] /ieee802-dot1ab-lldp/lldp/port/message-tx-hold-multiplier

3251 [o] /ieee802-dot1ab-lldp/lldp/port/message-tx-interval

3252 [o] /ieee802-dot1ab-lldp/lldp/port/reinit-delay

3253 [o] /ieee802-dot1ab-lldp/lldp/port/tx-credit-max

3254 [o] /ieee802-dot1ab-lldp/lldp/port/tx-fast-init

3255 [o] /ieee802-dot1ab-lldp/lldp/port/notification-interval

3256 [o] /ieee802-dot1ab-lldp/lldp/port/management-address-tx-port

3257 [o] /ieee802-dot1ab-lldp/lldp/port/port-id-subtype

3258 [o] /ieee802-dot1ab-lldp/lldp/port/port-id

3259 [o] /ieee802-dot1ab-lldp/lldp/port/port-desc

3260 [o] /ieee802-dot1ab-lldp/lldp/port/remote-systems-data

3261 **6.4.9.2.3 Synchronization**

3262 **6.4.9.2.3.1 Timesync**

3263 IA-stations shall support the ieee1588-ptp-tt YANG module according to IEEE Std 1588e™-
3264 2024 with the following features:

- 3265 • cmls (Common Mean Link Delay Service), and
- 3266 • external-port-config.

3267 IA-stations shall support the ieee1588-ptp-tt YANG module according to IEEE Std 1588e-2024
3268 with the following nodes:

3269 [o] /ieee1588-ptp-tt/ptp/instances/instance/instance-index

3270 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/clock-identity

3271 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/number-ports

3272 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/priority1

3273 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/domain-number

3274 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/time-receiver-
3275 only

3276 [o] /ieee1588-ptp-tt/ptp/instances/instance/default-ds/sdo-id

3277 [o] /ieee1588-ntp-tt/ntp/instances/instance/default-ds/instance-enable

3278 [o] /ieee1588-ntp-tt/ntp/instances/instance/default-ds/external-port-
3279 config-enable

3280 [o] /ieee1588-ntp-tt/ntp/instances/instance/default-ds/instance-type

3281 [o] /ieee1588-ntp-tt/ntp/instances/instance/description-ds/user-
3282 description

3283 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-index

3284 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/underlying-interface

3285 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/port-state

3286 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/delay-mechanism

3287 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/port-ds/port-enable

3288 [o] /ieee1588-ntp-tt/ntp/instances/ports/port/external-port-config-
3289 port-ds/desired-state

3290 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/default-ds/clock-
3291 identity

3292 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/default-ds/number-link-
3293 ports

3294 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/port-index

3295 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/underlying-
3296 interface

3297 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3298 ds/port-identity/clock-identity

3299 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3300 ds/port-identity/port-number

3301 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3302 ds/domain-number

3303 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3304 ds/service-measurement-valid

3305 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3306 ds/mean-link-delay

3307 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3308 ds/scaled-neighbor-rate-ratio

3309 [o] /ieee1588-ntp-tt/ntp/common-services/cmlds/ports/port/link-port-
3310 ds/log-min-pdelay-req-interval

3311 [m] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3312 ds/version-number

3313 [m] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3314 ds/minor-version-number

3315 [o] /ieee1588-ptp-tt/ptp/common-services/cmlsds/ports/port/link-port-
3316 ds/delay-asymmetry

3317 **6.4.9.2.3.2 Timesync (ieee802-dot1as-gptp)**

3318 IA-stations shall support the ieee802-dot1as-gptp YANG module according to IEEE Std
3319 802.1ASdn-2024™ with the following nodes:

3320 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/gm-capable

3321 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/current-
3322 utc-offset-valid

3323 [o] /ieee802-dot1as-gptp/ptp/instances/instance/default-ds/ptp-
3324 timescale

3325 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-
3326 receipt-timeout

3327 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/current-one-
3328 step-tx-oper

3329 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/use-mgt-one-
3330 step-tx-oper

3331 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/mgt-one-
3332 step-tx-oper

3333 [o] /ieee802-dot1as-gptp/ptp/instances/ports/port/port-ds/sync-locked

3334 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3335 port-ds/cmlsds-link-port-enabled

3336 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3337 port-ds/is-measuring-delay

3338 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3339 port-ds/as-capable-across-domains

3340 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3341 port-ds/mean-link-delay-thresh

3342 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3343 port-ds/current-log-pdelay-req-interval

3344 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3345 port-ds/use-mgt-log-pdelay-req-interval

3346 [o] /ieee802-dot1as-gptp/ptp/common-services/cmlsds/ports/port/link-
3347 port-ds/mgt-log-pdelay-req-interval

3348 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3349 port-ds/current-compute-rate-ratio

3350 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3351 port-ds/use-mgt-compute-rate-ratio

3352 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3353 port-ds/mgt-compute-rate-ratio

3354 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3355 port-ds/current-compute-mean-link-delay

3356 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3357 port-ds/use-mgt-compute-mean-link-delay

3358 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3359 port-ds/mgt-compute-mean-link-delay

3360 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3361 port-ds/allowed-lost-responses

3362 [o] /ieee802-dot1as-gptp/ptp/common-services/cmls/ports/port/link-
3363 port-ds/allowed-faults

3364 **6.4.9.2.3 Timesync (ieee802-dot1as-hs)**

3365 IA-stations shall support the ieee802-dot1as-hs YANG module according to IEEE Std
3366 802.1ASdm-2024 with the following nodes:

3367 [o] /ieee802-dot1as-hs/ptp/instances/instance/parent-ds/gmPresent

3368 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/is-
3369 synced

3370 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-
3371 pdelay

3372 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-sync

3373 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/nrr-
3374 comp-method

3375 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/as-
3376 capable-across-domains

3377 [o] /ieee802-dot1as-hs/ptp/instances/instance/ptp-instance-ds/gptp-
3378 capable-state-machines-enabled

3379 **6.4.9.2.4 Security configuration modules**

3380 **6.4.9.2.4.1 YANG module for a keystore**

3381 IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the
3382 following features:

- 3383 • central-truststore-supported, and
- 3384 • asymmetric-keys.

3385 IA-stations shall support the ietf-keystore YANG module according to IETF RFC 9642 with the
3386 following nodes:

- 3387 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/name
- 3388 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key-
3389 format
- 3390 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/public-key
- 3391 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/private-
3392 key-format
- 3393 [o] /ietf-keystore/keystore/asymmetric-keys/asymmetric-key/hidden-
3394 private-key
- 3395 [o] /ietf-keystore/certificates/certificate/name
- 3396 [o] /ietf-keystore/certificates/certificate/cert-data
- 3397 [o] /ietf-keystore/certificates/certificate/expiration-date
- 3398 [o] /ietf-keystore/certificates/certificate/csr-info
- 3399 [o] /ietf-keystore/certificates/certificate/certificate-signing-
3400 request

3401 **6.4.9.2.4.2 Network configuration access control**

3402 IA-stations shall support the ietf-netconf-acm YANG module according to IETF RFC 8341 with
3403 the following nodes:

- 3404 [o] /ietf-netconf-acm/nacm/enable-nacm
- 3405 [o] /ietf-netconf-acm/nacm/read-default
- 3406 [o] /ietf-netconf-acm/nacm/write-default
- 3407 [o] /ietf-netconf-acm/nacm/exec-default
- 3408 [o] /ietf-netconf-acm/nacm/enable-external-groups
- 3409 [o] /ietf-netconf-acm/nacm/groups
- 3410 [o] /ietf-netconf-acm/nacm/rule-list

3411 **6.4.9.2.4.3 A YANG data module for a truststore**

3412 IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the
3413 following features:

- 3414 • central-keystore-supported, and
- 3415 • certificates.

3416 IA-stations shall support the ietf-truststore YANG module according to IETF RFC 9641 with the
3417 following nodes:

3418 [o] /ietf-truststore/truststore/certificate-bags/certificate-bag/name

3419 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3420 bag/certificate/name

3421 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3422 bag/certificate/cert-data

3423 [o] /ietf-truststore/truststore/certificate-bags/certificate-
3424 bag/certificate/expiration-date

3425 **6.4.9.2.5 IA-station management**

3426 **6.4.9.2.5.1 System capabilities**

3427 IA-stations shall support the ietf-system-capabilities YANG modules according to IETF RFC
3428 9196 with the following nodes:

3429 [m] /ietf-system-capabilities/system-capabilities/datastore-
3430 capabilities/datastore

3431 [m] /ietf-system-capabilities/system-capabilities/datastore-
3432 capabilities/per-node-capabilities

3433 [m] /ietf-system-capabilities/system-capabilities/datastore-
3434 capabilities/on-change-supported

3435 **6.4.9.2.5.2 YANG library**

3436 IA-stations shall support the ietf-yang-library YANG module according to IETF RFC 8525 with
3437 the following nodes:

3438 [m] /ietf-yang-library/yang-library/module-set

3439 [m] /ietf-yang-library/yang-library/schema

3440 [m] /ietf-yang-library/yang-library/datastore

3441 [m] /ietf-yang-library/yang-library/content-id

3442 **6.4.9.2.5.3 YANG push**

3443 IA-stations shall support the ietf-yang-push YANG module according to IETF RFC 8641, 4.1,
3444 with the on-change feature.

3445 IA-stations shall support the ietf-yang-push YANG module according to IETF RFC 8641, 4.1,
3446 with the following nodes:

3447 [o] /ietf-subscribed-notifications/filters/selection-filter

3448 [o] /ietf-subscribed-
3449 notifications/subscriptions/subscription/target/datastore

3450 [o] /ietf-subscribed-notifications/subscriptions/subscription/update-
3451 trigger

3452 **6.4.9.2.5.4 YANG notification capabilities**

3453 IA-stations shall support the ietf-notification-capabilities YANG module according to IETF RFC
3454 9196 with the following nodes:

3455 [m] /ietf-notification-capabilities/system-capabilities/subscription-
3456 capabilities

3457 [m] /ietf-notification-capabilities/system-capabilities/datastore-
3458 capabilities/per-node-capabilities/subscription-capabilities

3459 **6.4.9.2.5.5 YANG notifications**

3460 IA-stations shall support the ietf-subscribed-notifications YANG module according to IETF RFC
3461 8639 with the following features:

- 3462 • Configured,
- 3463 • encode-xml, and
- 3464 • subtree.

3465 IA-stations shall support the ietf-subscribed-notifications YANG module according to IETF RFC
3466 8639 with the following nodes:

3467 [o] /ietf-subscribed-notifications/streams/stream/name

3468 [o] /ietf-subscribed-notifications/streams/stream/description

3469 [o] /ietf-subscribed-notifications/streams/stream/replay-support

3470 [o] /ietf-subscribed-notifications/streams/stream/replay-log-creation-
3471 time

3472 [o] /ietf-subscribed-notifications/streams/stream/replay-log-aged-time

3473 [o] /ietf-subscribed-notifications/filters/stream-filter/name

3474 [o] /ietf-subscribed-notifications/filters/stream-filter/filter-spec

3475 [o] /ietf-subscribed-notifications/subscriptions/subscription/id

3476 [o] /ietf-subscribed-notifications/subscriptions/subscription/target

3477 [o] /ietf-subscribed-notifications/subscriptions/subscription/stop-
3478 time

3479 [o] /ietf-subscribed-notifications/subscriptions/subscription/dscp

3480 [o] /ietf-subscribed-
3481 notifications/subscriptions/subscription/weighting

3482 [o] /ietf-subscribed-
3483 notifications/subscriptions/subscription/dependency

3484 [o] /ietf-subscribed-
3485 notifications/subscriptions/subscription/transport

3486 [o] /ietf-subscribed-notifications/subscriptions/subscription/encoding

3487 [o] /ietf-subscribed-notifications/subscriptions/subscription/purpose

3488 [o] /ietf-subscribed-
3489 notifications/subscriptions/subscription/notification-message-origin

3490 [o] /ietf-subscribed-
3491 notifications/subscriptions/subscription/configured-subscription-state

3492 [o] /ietf-subscribed-
3493 notifications/subscriptions/subscription/receivers

3494 **6.4.9.2.5.6 NETCONF monitoring**

3495 IA-stations shall support the ietf-netconf-monitoring YANG module according to IETF RFC 6022
3496 with the following nodes:

3497 [m] /ietf-netconf-monitoring/netconf-state/capabilities

3498 [m] /ietf-netconf-monitoring/netconf-state/datastores

3499 [m] /ietf-netconf-monitoring/netconf-state/schemas

3500 **6.4.9.2.5.7 System management**

3501 IA-stations shall support the ietf-system YANG module according to IETF RFC 7317 with the
3502 following nodes:

3503 [o] /ietf-system/system/contact

3504 [o] /ietf-system/system/hostname

3505 [o] /ietf-system/system/location

3506 **6.4.9.2.5.8 Hardware management**

3507 IA-stations shall support the ietf-hardware YANG module according to IETF RFC 8348 with the
3508 following nodes:

3509 [m] /ietf-hardware/hardware/component/name

3510 [m] /ietf-hardware/hardware/component/class

3511 [m] /ietf-hardware/hardware/component/description

3512 [m] /ietf-hardware/hardware/component/hardware-rev

3513 [m] /ietf-hardware/hardware/component/software-rev

3514 [o] /ietf-hardware/hardware/component/serial-num

3515 [m] /ietf-hardware/hardware/component/mfg-name

3516 [m] /ietf-hardware/hardware/component/model-name

3517 An IA-station shall provide exactly one /ietf-hardware/component with class "chassis" and may
3518 provide further components with other classes.

3519 **6.4.9.2.5.9 Interface management**

3520 IA-stations shall support the ietf-interfaces YANG module according to IETF RFC 8343 with the
3521 following nodes:

3522 [m] /ietf-interfaces/interfaces/interface/name

3523 [m] /ietf-interfaces/interfaces/interface/description

3524 [m] /ietf-interfaces/interfaces/interface/type

3525 [o] /ietf-interfaces/interfaces/interface/enabled

3526 [o] /ietf-interfaces/interfaces/interface/oper-status

3527 [o] /ietf-interfaces/interfaces/interface/phys-address

3528 [o] /ietf-interfaces/interfaces/interface/higher-layer-if

3529 [o] /ietf-interfaces/interfaces/interface/lower-layer-if

3530 [o] /ietf-interfaces/interfaces/interface/speed

3531 [o] /ietf-interfaces/interfaces/interface/statistics/discontinuity-
3532 time

3533 [o] /ietf-interfaces/interfaces/interface/statistics/in-octets

3534 [o] /ietf-interfaces/interfaces/interface/statistics/in-discards

3535 [o] /ietf-interfaces/interfaces/interface/statistics/in-errors

3536 [o] /ietf-interfaces/interfaces/interface/statistics/out-octets

3537 [o] /ietf-interfaces/interfaces/interface/statistics/out-discards

3538 [o] /ietf-interfaces/interfaces/interface/statistics/out-errors

3539 **6.4.9.2.5.10 Bridge and end station component management**

3540 **6.4.9.2.5.10.1 General**

3541 IA-stations shall support the ieee802-dot1q-bridge YANG module according to
3542 IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw™-2023 with the
3543 following feature: ingress-filtering.

3544 IA-stations shall support the ieee802-dot1q-bridge YANG module according to
3545 IEEE Std 802.1Q-2022, Clause 48, as amended by IEEE Std 802.1Qcw-2023 with the following
3546 nodes. A distinction is made between nodes that shall be supported by bridge and end station
3547 components, or by bridge components only.

3548 **6.4.9.2.5.10.2 Bridge nodes to be supported by bridge and end station components**

- 3549 [m] /ieee802-dot1q-bridge/bridges/bridge/name
- 3550 [o] /ieee802-dot1q-bridge/bridges/bridge/address
- 3551 [m] /ieee802-dot1q-bridge/bridges/bridge/bridge-type
- 3552 [m] /ieee802-dot1q-bridge/bridges/bridge/ports
- 3553 [m] /ieee802-dot1q-bridge/bridges/bridge/components
- 3554 [m] /ieee802-dot1q-bridge/bridges/bridge/component/name
- 3555 [o] /ieee802-dot1q-bridge/bridges/bridge/component/id
- 3556 [m] /ieee802-dot1q-bridge/bridges/bridge/component/type
- 3557 [o] /ieee802-dot1q-bridge/bridges/bridge/component/traffic-class-
3558 enabled
- 3559 [m] /ieee802-dot1q-bridge/bridges/bridge/component/ports
- 3560 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-port
- 3561 [m] /ieee802-dot1q-bridge/bridges/bridge/component/capabilities
- 3562 [m] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3563 database/size
- 3564 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3565 database/static-vlan-registration-entries
- 3566 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3567 database/vlan-registration-entry

3568 **6.4.9.2.5.10.3 Filtering-database nodes to be supported by bridge components**

- 3569 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3570 database/aging-time
- 3571 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3572 database/static-entries
- 3573 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3574 database/dynamic-entries
- 3575 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3576 database/dynamic-vlan-registration-entries
- 3577 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3578 database/mac-address-registration-entries
- 3579 [o] /ieee802-dot1q-bridge/bridges/bridge/component/filtering-
3580 database/filtering-entry

3581 6.4.9.2.5.10.4 Permanent-database nodes to be supported by bridge components

3582 [m] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3583 database/size

3584 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3585 database/static-entries

3586 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3587 database/static-vlan-registration-entries

3588 [o] /ieee802-dot1q-bridge/bridges/bridge/component/permanent-
3589 database/filtering-entry

3590 6.4.9.2.5.10.5 Bridge-vlan nodes to be supported by bridge and end station components

3591 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/version

3592 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-
3593 vids

3594 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-
3595 vlan/override-default-pvid

3596 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vlan

3597 6.4.9.2.5.10.6 Bridge-vlan nodes to be supported by bridge components

3598 [m] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/max-
3599 msti

3600 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-
3601 fid-allocation

3602 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/fid-to-
3603 vid-allocation

3604 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-vlan/vid-to-
3605 fid

3606 6.4.9.2.5.10.7 Bridge-mst nodes to be supported by bridge components

3607 [o] /ieee802-dot1q-bridge/bridges/bridge/component/bridge-mst

3608 6.4.9.2.5.10.8 Bridge-port nodes to be supported by bridge and end station components

3609 [m] /ietf-interfaces/interfaces/interface/bridge-port/bridge-name

3610 [m] /ietf-interfaces/interfaces/interface/bridge-port/component-name

3611 [m] /ietf-interfaces/interfaces/interface/bridge-port/port-type

3612 [o] /ietf-interfaces/interfaces/interface/bridge-port/pvid

3613 [o] /ietf-interfaces/interfaces/interface/bridge-port/default-priority

3614 [m] /ietf-interfaces/interfaces/interface/bridge-port/traffic-class

3615 [o] /ietf-interfaces/interfaces/interface/bridge-port/statistics
 3616 [m] /ietf-interfaces/interfaces/interface/bridge-port/capabilities
 3617 [m] /ietf-interfaces/interfaces/interface/bridge-port/type-capabilities
 3618 [o] /ietf-interfaces/interfaces/interface/bridge-port/transmission-
 3619 selection-algorithm-table

3620 **6.4.9.2.5.10.9 Bridge-port nodes to be supported by bridge component ports**

3621 [o] /ietf-interfaces/interfaces/interface/bridge-port/priority-
 3622 regeneration
 3623 [o] /ietf-interfaces/interfaces/interface/bridge-port/acceptable-frame
 3624 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-ingress-
 3625 filtering
 3626 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-vid-
 3627 translation-table
 3628 [o] /ietf-interfaces/interfaces/interface/bridge-port/vid-translations
 3629 [o] /ietf-interfaces/interfaces/interface/bridge-port/enable-egress-
 3630 vid-translation-table
 3631 [o] /ietf-interfaces/interfaces/interface/bridge-port/egress-vid-
 3632 translations

3633 **6.4.9.2.5.11 IEC/IEEE 60802 YANG modules**

3634 IA-stations shall support the `ieciieee60802-ethernet-interface` YANG module according to this
 3635 document with the following nodes:

3636 [m] /ieciieee60802-ethernet-
 3637 interface/interfaces/interface/ethernet/supported-mau-types/mau-type
 3638 [m] /ieciieee60802-ethernet-
 3639 interface/interfaces/interface/ethernet/supported-mau-
 3640 types/preemption-supported

3641 IA-stations shall support the `ieciieee60802-bridge` YANG module according to this document
 3642 with the following nodes:

3643 [m] /ieciieee60802-bridge/interfaces/interface/bridge-port/max-burst-
 3644 params
 3645 [m] /ieciieee60802-bridge/interfaces/interface/bridge-port/committed-
 3646 data-rates
 3647 [m] /ieciieee60802-bridge/interfaces/interface/bridge-
 3648 port/transmission-selection-algorithm
 3649 [m] /ieciieee60802/interfaces/interface/bridge-port/supported-resource-
 3650 pools

- 3651 [m] /ieciieee60802-bridge/bridges/bridge/component/frer-supported
- 3652 [m] /ieciieee60802-bridge/bridges/bridge/component/max-redundant-
3653 streams
- 3654 [m] /ieciieee60802-bridge/bridges/bridge/component/max-fids
- 3655 [m] /ieciieee60802-bridge/bridges/bridge/component/max-fdb-entries
- 3656 [m] /ieciieee60802-bridge/bridges/bridge/component/delay-variance
- 3657 [m] /ieciieee60802-bridge/bridges/bridge/component/max-ptp-instances
- 3658 [m] /ieciieee60802-bridge/bridges/bridge/component/max-hot-standby-
3659 systems
- 3660 [m] /ieciieee60802-bridge/bridges/bridge/component/clock
- 3661 [m] /ieciieee60802/interfaces/interface/bridge-port/phy-latencies/mau-
3662 type
- 3663 [m] /ieciieee60802/interfaces/interface/bridge-port/phy-
3664 latencies/latency
- 3665 [m] /ieciieee60802/interfaces/interface/bridge-port/phy-
3666 latencies/latency-variation
- 3667 **IA-stations shall support the ieciieee60802-ia-station YANG module according to this document**
3668 **with the following nodes:**
- 3669 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-lldp
- 3670 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-timesync
- 3671 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-keystore
- 3672 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-truststore
- 3673 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-nacm
- 3674 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-yang-library
- 3675 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-yang-push
- 3676 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-yang-
3677 notifications
- 3678 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-netconf-
3679 monitoring
- 3680 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-netconf-
3681 client
- 3682 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-cnc-config

3683 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-sched-
3684 traffic

3685 [m] /ieciieee60802-ia-station/ia-station-capabilities/capability-frame-
3686 preemption

3687 IA-stations shall support the ieciieee60802-subscribed-notifications YANG module according to
3688 this document with the following nodes:

3689 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-
3690 subscriptions

3691 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-on-
3692 change-subscription-leaves

3693 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-periodic-
3694 subscription-leaves

3695 [m] /ieciieee60802-subscribed-notifications/subscriptions/max-periodic-
3696 subscription-interval

3697 IA-stations shall support the ieciieee60802-proxy YANG module according to this document with
3698 the following nodes:

3699 [c] /ieciieee60802-proxy/proxy/max-num-proxy-targets

3700 [c] /ieciieee60802-proxy/proxy/proxy-types

3701 [o] /ieciieee60802-proxy/proxy/proxy-targets

3702 IA-stations shall support the ieciieee60802-proxy-target YANG module according to this
3703 document with the following nodes:

3704 [c] /ieciieee60802-proxy-target/proxy-target/required-proxy-type

3705 **6.4.9.2.5.12 NETCONF server**

3706 IA-stations shall support the ietf-netconf-server YANG module according to draft-ietf-netconf-
3707 netconf-client-server, 3.1.1, with the following features:

- 3708 • tls-call-home, and
- 3709 • central-netconf-server-supported.

3710 IA-stations shall support the ietf-netconf-server YANG module according to draft-ietf-netconf-
3711 netconf-client-server, 3.3, with the following nodes:

3712 [o] /ietf-netconf-server/netconf-server/listen/idle-timeout

3713 [o] /ietf-netconf-server/netconf-server/listen/endpoint/name

3714 [o] /ietf-netconf-server/netconf-
3715 server/listen/endpoint/transport/tls/netconf-server-parameters

3716 [o] /ietf-netconf-server/netconf-
3717 server/listen/endpoint/transport/tls/tls-server-parameters

3718 [o] /ietf-netconf-server/netconf-server/call-home/netconf-client/name

3719 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3720 client/endpoints/endpoint/name

3721 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3722 client/endpoints/endpoint/transport/tls/netconf-server-parameters

3723 [o] /ietf-netconf-server/netconf-server/call-home/netconf-
3724 client/endpoints/endpoint/transport/tls/tls-server-parameters

3725 **6.4.9.2.5.13 Flow Meter Management**

3726 IA-stations which incorporate a bridge component shall support the ieee802-dot1q-stream-
3727 filters-gates YANG module according to IEEE Std 802.1Qcz-2023 as amended by IEEE Std
3728 802.1Qcw-2023 with the following nodes:

3729 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3730 filters/stream-filter-instance-table/stream-filter-instance-id

3731 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3732 filters/stream-filter-instance-table/stream-handle

3733 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3734 filters/stream-filter-instance-table/flow-meter-ref

3735 [o] /ieee802-dot1q-bridge/bridges/bridge/component/stream-
3736 filters/stream-filter-instance-table/flow-meter-enable

3737 [m] /ieee802-dot1q-bridge/bridges/bridge/component/stream-filters/max-
3738 stream-filter-instances

3739 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3740 gates/stream-gate-instance-table

3741 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3742 gates/max-stream-gate-instances

3743 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/stream-
3744 gates/supported-list-max

3745 IA-stations which incorporate a bridge component shall support the ieee802-dot1cb-stream-
3746 identification YANG module according to IEEE Std 802.1CBcv-2021 as amended by IEEE Std
3747 802.1CBdb-2021 with the following nodes:

3748 [o] /ieee802-dot1cb-stream-identification/stream-identity/index

3749 [o] /ieee802-dot1cb-stream-identification/stream-identity/handle

3750 [o] /ieee802-dot1cb-stream-identification/stream-identity/out-
3751 facing/input-port

3752 [o] /ieee802-dot1cb-stream-identification/stream-
3753 identity/parameters/mask-and-match-stream-identification/destination-
3754 mac-mask

3755 [o] /ieee802-dot1cb-stream-identification/stream-
3756 identity/parameters/mask-and-match-stream-identification/destination-
3757 mac-match

3758 NOTE For example, an implementation could contain per out-facing/input-port one mask and match stream
3759 identification for broadcast traffic, one mask and match stream identification for multicast traffic and one mask and
3760 match stream identification for unicast traffic.

3761 IA-stations which incorporate a bridge component shall support the ieee802-dot1q-psfp-bridge
3762 YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

3763 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3764 meters/flow-meter-instance-table/flow-meter-instance-id

3765 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3766 meters/flow-meter-instance-table/committed-information-rate

3767 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3768 meters/flow-meter-instance-table/committed-burst-size

3769 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3770 meters/flow-meter-instance-table/excess-information-rate

3771 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3772 meters/flow-meter-instance-table/excess-burst-size

3773 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3774 meters/flow-meter-instance-table/coupling-flag

3775 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3776 meters/flow-meter-instance-table/color-mode

3777 [o] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3778 meters/flow-meter-instance-table/drop-on-yellow

3779 [m] /ieee802-dot1q-psfp-bridge/bridges/bridge/component/flow-
3780 meters/max-flow-meter-instances

3781 **6.4.9.2.5.14 RSTP/MSTP**

3782 IA-stations shall support the ieee802-dot1q-rstp-bridge YANG module according to IEEE Std
3783 802.1Qdy™-2025 with the following nodes.

3784 [o] /ieee802-dot1q-rstp-bridge/bridges/bridge/rstp

3785 [o] /ieee802-dot1q-rstp-bridge/interfaces/interface/bridge-port/rstp

3786 IA-stations shall support the ieee802-dot1q-mstp-bridge YANG module according to IEEE Std
3787 802.1Qdy-2025 with the following nodes.

3788 [o] /ieee802-dot1q-mstp-bridge/bridges/bridge/component/bridge-
3789 mst/bridge-mstp

3790 [o] /ieee802-dot1q-mstp-bridge/interfaces/interface/bridge-port/port-
3791 mstp

3792 **6.4.9.3 Optional YANG data models, features, and nodes**

3793 **6.4.9.3.1 General**

3794 The following YANG modules, features and nodes shall be supported by IA-stations if the
3795 functionality they describe is included.

3796 **6.4.9.3.2 Scheduled traffic**

3797 IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-
3798 sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following feature:
3799 `scheduled-traffic`.

3800 IA-stations supporting the enhancements for scheduled traffic shall support the ieee802-dot1q-
3801 sched-bridge YANG module according to IEEE Std 802.1Qcw-2023 with the following nodes:

3802 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3803 `parameter-table/queue-max-sdu-table`

3804 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3805 `parameter-table/gate-enabled`

3806 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3807 `parameter-table/admin-gate-states`

3808 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3809 `parameter-table/oper-gate-states`

3810 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3811 `parameter-table/admin-control-list`

3812 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3813 `parameter-table/oper-control-list`

3814 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3815 `parameter-table/admin-cycle-time`

3816 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3817 `parameter-table/oper-cycle-time`

3818 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3819 `parameter-table/admin-cycle-time-extension`

3820 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3821 `parameter-table/oper-cycle-time-extension`

3822 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3823 `parameter-table/admin-base-time`

3824 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3825 `parameter-table/oper-base-time`

3826 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3827 `parameter-table/config-change`

3828 [o] `ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-`
3829 `parameter-table/config-change-time`

3830 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3831 parameter-table/tick-granularity

3832 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3833 parameter-table/current-time

3834 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3835 parameter-table/config-pending

3836 [o] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3837 parameter-table/config-change-error

3838 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3839 parameter-table/supported-list-max

3840 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3841 parameter-table/supported-cycle-max

3842 [c] ieee802-dot1q-sched-bridge/interfaces/interface/bridge-port/gate-
3843 parameter-table/supported-interval-max

3844 **6.4.9.3.3 IEC/IEEE 60802 YANG modules**

3845 IA-stations that support enhancements for scheduled traffic shall support the iecieee60802-
3846 sched-bridge YANG module according to this document with the following nodes:

3847 [c] /ieciieee60802-sched-bridge/interfaces/interface/bridge-port/gate-
3848 parameter-table/min-gating-times

3849 **6.4.9.3.4 Frame preemption**

3850 IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall
3851 support the ieee802-dot1q-preemption-bridge YANG module according to IEEE Std
3852 802.1Qcw-2023 with the following feature: frame-preemption.

3853 IA-stations supporting frame preemption according to IEEE Std 802.1Q-2022, 5.4.1, ae), shall
3854 support the ieee802-dot1q-preemption-bridge YANG module according to IEEE Std
3855 802.1Qcw-2023 with the following nodes:

3856 [o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
3857 port/frame-preemption-parameters/frame-preemption-status-table

3858 [o] ieee802-dot1q-preemption-bridge/interfaces/interface/bridge-
3859 port/frame-preemption-parameters/preemption-active

3860 **6.4.9.3.5 Credit-based shaper**

3861 IA-stations supporting the credit-based shaper according to IEEE Std 802.1Q-2022, 8.6.8.2,
3862 shall support the ieee802-dot1q-cbsa-bridge YANG module according to IEEE Std 802.1Qdx™-
3863 2024 with the following nodes.

3864 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3865 port/cbsa/cbsa-parameter-table/traffic-class

3866 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3867 port/cbsa/cbsa-parameter-table/admin-idle-slope

3868 [c] /ieee802-dot1q-cbsa-bridge/interfaces/interface/bridge-
3869 port/cbsa/cbsa-parameter-table/oper-idle-slope

3870 **6.4.9.3.6 FRER**

3871 IA-stations supporting FRER according to 5.10.1 item b) or item c), shall support the
3872 ieee802-dot1cb-stream-identification and ieee802-dot1cb-frer YANG modules according to
3873 IEEE Std 802.1CBcv-2021 with the following nodes:

3874 [o] /ieee802-dot1cb-stream-identification/stream-identity/index
3875 [o] /ieee802-dot1cb-stream-identification/stream-identity/handle
3876 [o] /ieee802-dot1cb-stream-identification /stream-identity/out-
3877 facing/input-port
3878 [o] /ieee802-dot1cb-stream-identification /stream-identity/out-
3879 facing/output-port
3880 [o] /ieee802-dot1cb-stream-identification /stream-
3881 identity/parameters/null-stream-identification
3882 [o] /ieee802-dot1cb-frer/frer/sequence-generation/index
3883 [o] /ieee802-dot1cb-frer/frer/sequence-generation/stream
3884 [o] /ieee802-dot1cb-frer/frer/sequence-generation/direction-out-facing
3885 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/index
3886 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/stream
3887 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/port
3888 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/direction-out-facing
3889 [o] /ieee802-dot1cb-frer/frer/sequence-recovery/algorithm/vector
3890 [o] /ieee802-dot1cb-frer/frer/sequence-identification/port
3891 [o] /ieee802-dot1cb-frer/frer/sequence-identification/direction-out-
3892 facing
3893 [o] /ieee802-dot1cb-frer/frer/sequence-identification/stream
3894 [o] /ieee802-dot1cb-frer/frer/sequence-identification/encapsulation/r-
3895 tag
3896 [o] /ieee802-dot1cb-frer/frer/stream-split

3897 **6.4.9.4 CUC/CNC YANG**

3898 **6.4.9.4.1 NETCONF Client**

3899 IA-stations with CNC or CUC functionality shall support the ietf-netconf-client YANG module
3900 according to draft-ietf-netconf-netconf-client-server, 2.1.1, with the following features:

- 3901 • `tls-initiate`,
- 3902 • `tls-listen`, and
- 3903 • `central-netconf-client-supported`.

3904 **6.4.9.4.2 YANG module for CNC-Config**

3905 IA-stations with CNC or CUC functionality shall support the ieee802-dot1q-cnc-config YANG
3906 module according to IEEE Std 802.1Qdj-2024 with the node:
3907 `[o] /ieee802-dot1q-cnc-config/cnc-config`

3908 **6.4.10 YANG Data Model**

3909 **6.4.10.1 General**

3910 Subclause 6.4.10 specifies the YANG data model for IA-stations. YANG (IETF RFC 7950) is a
3911 data modelling language used to model configuration data and state data for remote network
3912 management protocols. The selected YANG-based remote network management protocol is
3913 NETCONF (IETF RFC 6241). A YANG module specifies the organization and rules for the
3914 management data, and a mapping from YANG to the specific encoding enables the data to be
3915 understood correctly by both client (e.g. network manager) and server (e.g. IA-stations).

3916 **6.4.10.2 YANG framework**

3917 The core of the YANG module for IEC/IEEE 60802 IA-stations consists of YANG "augment"
3918 statements, used to add members to the tree of existing YANG modules plus one new module
3919 for IEC/IEEE 60802 specific objects.

3920 **6.4.10.3 IEC/IEEE 60802 Specific managed objects**

3921 **6.4.10.3.1 General**

3922 Subclause 6.4.10.3 defines the set of managed objects, and their functionality, that provides
3923 additional information about an IA-station that is required by a CNC to calculate network
3924 configurations.

3925 IEC/IEEE 60802 specific managed objects are specified:

- 3926 • per Ethernet interface, i.e. external port, in 6.4.10.3.2,
- 3927 • per end station component internal or external port in 6.4.10.3.3 and 6.4.10.3.4,
- 3928 • per bridge component internal or external port in 6.4.10.3.4,
- 3929 • per end station component in 6.4.10.3.5 and 6.4.10.3.7,
- 3930 • per bridge component in 6.4.10.3.6 and 6.4.10.3.7, and
- 3931 • per IA-station in 6.4.10.3.8.

3932 IEC/IEEE 60802 specific managed objects for CNC entities are specified in 6.4.10.3.9.

3933 **6.4.10.3.2 IEC/IEEE 60802 managed objects per Ethernet interface**

3934 **6.4.10.3.2.1 supportedMauTypes**

3935 The list of supported MAU Types including the data:

3936 a) mauType

3937 The value is the supported MAU Type derived from the list position of the corresponding
3938 dot3MauType as listed in IETF RFC 4836, Clause 5.

3939 b) preemptionSupported

3940 The Boolean value indicates if preemption is supported by the MAU Type.

3941 NOTE The operational MAU Type of an Ethernet interface is provided as leaf operational-mau-type of the
3942 ieee802-ethernet-lldp YANG module. The operational MAU Type is included in the supportedMauTypes list.

3943 **6.4.10.3.3 IEC/IEEE 60802 managed objects per end station component port**

3944 **6.4.10.3.3.1 worstCasePacketGap**

3945 The value is the worst case maximum inter-packet gap between consecutive frames in a traffic
3946 burst expressed in bit-times.

3947 NOTE Minimum interPacketGap is defined in IEEE Std 802.3-2022, 1.4.362. The worst-case-packet-gap is never
3948 less than the minimum interPacketGap.

3949 **6.4.10.3.3.2 maxBurstFrames**

3950 The value is the maximum number of frames that can be sent with minimal inter packet gap.

3951 **6.4.10.3.3.3 maxBurstBytes**

3952 The value is the maximum number of octets that can be sent with minimal inter packet gap.

3953 **6.4.10.3.3.4 committedDataRates**

3954 The list of committed data rates per traffic class and supported line speed including the data:

3955 a) committedInformationRate

3956 The value is the bandwidth limit in kbit/s.

3957 b) committedBurstSize

3958 The value is the burst size limit in octets.

3959 **6.4.10.3.4 IEC/IEEE 60802 managed objects per bridge or end station component port**

3960 **6.4.10.3.4.1 transmissionSelectionAlgorithm**

3961 The list of supported transmission selection algorithms according to IEEE Std 802.1Q-2022,
3962 8.6.8, per traffic class.

3963 **6.4.10.3.4.2 supportedResourcePools**

3964 The list of supported buffer resource pools including the data:

3965 a) resourcePoolName

3966 The value is the name of a resource pool.

3967 b) coveredTimeInterval

3968 The value specifies the covered buffering time given as rational number of seconds for the
3969 highest supported link speed.

3970 c) resourcePoolTrafficClasses

3971 The list of the traffic classes to be served by the resource pool.

3972 **6.4.10.3.4.3 minGatingTimes**

3973 The list of minimum gating times per supported line speed including the data:

3974 a) minCycleTime

3975 The value is the minimum value supported by this port of the AdminCycleTime and
3976 OperCycleTime parameters given as rational number of seconds.

3977 b) minIntervalTime

3978 The value is the minimum value supported by this port of the TimeIntervalValue parameter
3979 in nanoseconds.

3980 **6.4.10.3.5 IEC/IEEE 60802 managed objects per end station component**

3981 **6.4.10.3.5.1 frerSupported**

3982 The value indicates if FRER is supported.

3983 **6.4.10.3.5.2 maxRedundantStreams**

3984 The value is the maximum number of supported redundant streams.

3985 **6.4.10.3.6 IEC/IEEE 60802 managed objects per bridge component**

3986 **6.4.10.3.6.1 delayVariance**

3987 The value indicates variance in delay depending upon the use of a singleValue or
3988 multipleValues (see 6.4.10.3.6.2).

3989 **6.4.10.3.6.2 delayTimes**

3990 The list of minimum and maximum frame length independent and frame length dependent delay
3991 time values of frames as they pass through a bridge component. These values are given:

- 3992 • per supported MAU Type pair and traffic class, if delayVariance is singleValue, or
- 3993 • per port pair with supported MAU Types and traffic class, if delayVariance is multipleValues.

3994 The list includes the data:

3995 a) independentDelayMin

3996 The value is the minimum delay portion that is independent of frame length according to
3997 IEEE Std 802.1Q-2022, 12.32.1.1.

3998 b) independentDelayMax

3999 The value is the maximum delay portion that is independent of frame length according to
4000 IEEE Std 802.1Q-2022, 12.32.1.1.

4001 c) dependentDelayMin

4002 The value is the minimum delay portion that is dependent on frame length according to IEEE
4003 Std 802.1Q-2022, 12.32.1.2.

4004 d) dependentDelayMax

4005 The value is the maximum delay portion that is dependent on frame length according to
4006 IEEE Std 802.1Q-2022, 12.32.1.2.

4007 **6.4.10.3.6.3 latency**

4008 The PHY latency in ns.

4009 **6.4.10.3.6.4 latencyVariation**

4010 The PHY latency variation in ns.

4011 **6.4.10.3.7 IEC/IEEE 60802 managed objects per bridge or end station component**

4012 **6.4.10.3.7.1 maxFids**

4013 The value is the maximum number of supported FIDs.

4014 **6.4.10.3.7.2 maxFdbEntries**

4015 The list of the maximum number of static (6.4.10.3.7.3) and dynamic (6.4.10.3.7.4) FDB entries
4016 per FDB.

4017 **6.4.10.3.7.3 maxStaticFdbEntries**

4018 The value is the maximum number of static FDB entries.

4019 **6.4.10.3.7.4 maxDynamicFdbEntries**

4020 The value is the maximum number of dynamic FDB entries.

4021 **6.4.10.3.7.5 maxPtpInstances**

4022 The value is the maximum number of supported PTP Instances.

4023 **6.4.10.3.7.6 maxHotStandbySystems**

4024 The value is the maximum number of supported HotStandbySystem entities (see IEEE Std
4025 802.1ASdm-2024).

4026 **6.4.10.3.7.7 clockList**

4027 The list of supported application clock entities including the data:

4028 a) clockIdentity

4029 The clock identity of the application clock.

4030 b) clockTarget

4031 The Boolean value indicates if the application clock is a clock target (TRUE) or clock source
4032 (FALSE).

4033 c) arbSupported

4034 The Boolean value indicates if the application clock supports the ARB timescale.

4035 d) ptpSupported

4036 The Boolean value indicates if the application clock supports the PTP timescale.

4037 e) hotStandbySupported

4038 The Boolean value indicates if the application clock supports hot standby.

4039 f) attachedPtpInstance

4040 The value is a reference to the PTP or hot standby Instance, that is attached to the
4041 application clock.

4042 g) isSynced

4043 The Boolean value indicates if the application clock is either synchronized to the attached
4044 PTP Instance (TRUE) or to an internal/external ClockSource (FALSE).

4045 **6.4.10.3.8 IEC/IEEE 60802 managed objects per IA-station**

4046 **6.4.10.3.8.1 maxSubscriptions**

4047 The value is the maximum number of supported NETCONF Server subscriptions.

4048 **6.4.10.3.8.2 maxOnChangeSubscriptionLeaves**

4049 The value is the maximum number of supported leaves for NETCONF Server on-change
4050 subscriptions according to IETF RFC 8641.

4051 **6.4.10.3.8.3 maxPeriodicSubscriptionLeaves**

4052 The value is the maximum number of supported leaves for NETCONF Server periodic
4053 subscriptions according to IETF RFC 8641.

4054 **6.4.10.3.8.4 minPeriodicSubscriptionInterval**

4055 The value is the minimum periodic subscription interval in centiseconds (0,01 seconds) for
4056 NETCONF Server periodic subscriptions according to IETF RFC 8641.

4057 **6.4.10.3.8.5 capabilityLLDP**

4058 This Boolean value indicates if LLDP is supported.

4059 **6.4.10.3.8.6 capabilityTimesync**

4060 This Boolean value indicates if Timesync is supported.

4061 **6.4.10.3.8.7 capabilityKeystore**

4062 This Boolean value indicates if Keystore is supported.

4063 **6.4.10.3.8.8 capabilityNACM**

4064 This Boolean value indicates if NACM is supported.

4065 **6.4.10.3.8.9 capabilityTruststore**

4066 This Boolean value indicates if Truststore is supported.

4067 **6.4.10.3.8.10 capabilityYangLibrary**

4068 This Boolean value indicates if YANG library is supported.

4069 **6.4.10.3.8.11 capabilityYangPush**

4070 This Boolean value indicates if Yang Push is supported.

4071 **6.4.10.3.8.12 capabilityYangNotifications**

4072 This Boolean value indicates if YANG notifications is supported.

4073 **6.4.10.3.8.13 capabilityNetconfMonitoring**

4074 This Boolean value indicates if NETCONF Monitoring is supported.

4075 **6.4.10.3.8.14 capabilityNetconfClient**

4076 This Boolean value indicates if NETCONF client is supported.

4077 **6.4.10.3.8.15 capabilityCncConfig**

4078 This Boolean value indicates if CNC configuration is supported.

4079 **6.4.10.3.8.16 capabilitySchedTraffic**

4080 This Boolean value indicates if scheduled traffic is supported.

4081 **6.4.10.3.8.17 capabilityFramePreemption**

4082 This Boolean value indicates if frame preemption is supported.

4083 **6.4.10.3.9 IEC/IEEE 60802 managed objects per proxy station**

4084 **6.4.10.3.9.1 maxNumProxyTargets**

4085 The value indicates how many proxy targets can be handled by this proxy.

4086 **6.4.10.3.9.2 proxyType**

4087 OID of proxy targets handled by this proxy.

4088 **6.4.10.3.9.3 device**

4089 The identification of the device.

4090 **6.4.10.3.9.4 deviceType**

4091 The type of the device.

4092 **6.4.10.3.9.5 proxyAddr**

4093 The ip-address that is used by the CNC to configure the proxy target.

4094 **6.4.10.3.9.6 port**

4095 The port-number that is used by the CNC to configure the proxy target.

4096 **6.4.10.3.9.7 deviceAddr**

4097 The ip-address that is used by the proxy to address the proxy target.

4098 **6.4.10.3.9.8 macAddr**

4099 The mac-address that is used by the proxy to address the proxy target.

4100 **6.4.10.3.10 IEC/IEEE 60802 managed objects per proxy target**

4101 **6.4.10.3.10.1 proxyType**

4102 OID of proxy that this proxy target requires.

4103 **6.4.10.3.11 IEC/IEEE 60802 managed objects for CNC entities**

4104 **6.4.10.3.11.1 maxConfigurationDomains**

4105 The value is the maximum number of supported Configuration Domains.

4106 **6.4.10.3.11.2 maxCUCs**

4107 The value is the maximum number of supported CUC entities.

4108 **6.4.10.3.11.3 maxIAstations**

4109 The value is the maximum number of supported IA-stations.

4110 **6.4.10.3.11.4 maxNetworkDiameter**

4111 The value is the maximum supported network diameter.

4112 **6.4.10.3.11.5 maxStreams**

4113 The value is the maximum number of supported streams.

4114 **6.4.10.3.11.6 maxNumSeamlessTrees**

4115 The value is the maximum number of trees supported for seamless redundancy of a stream.

4116 **6.4.10.3.11.7 hotStandbySupported**

4117 The Boolean value indicates if hot standby is supported.

4118 **6.4.10.4 RPCs and actions specific to this document**

4119 **6.4.10.4.1 RPC ia-factory-reset**

4120 **6.4.10.4.1.1 General**

4121 In contrast to the original factory-reset RPC in IETF RFC 8808, this RPC puts the device into a
4122 state where a subsequent configuration by a CNC component results in a functioning IA-station
4123 according to this document. Depending on the factory default configuration, after being reset,
4124 the device can become unreachable on the network.

4125 **6.4.10.4.1.2 Input**

4126 There is no input.

4127 **6.4.10.4.1.3 Output**

4128 There is no output.

4129 **6.4.10.4.2 Action add-streams**

4130 **6.4.10.4.2.1 General**

4131 This Action requests a CNC to add a list of streams.

4132 **6.4.10.4.2.2 Input**

4133 The RPC requires the following input parameters.

- 4134 a) Cuid – The ID of the CUC for which the streams are to be added.
- 4135 b) StreamId – The Stream ID is a unique identifier of a Stream request and corresponding
4136 configuration.
- 4137 c) Container Talker – The Talker container contains:
 - 4138 – Talker's behavior for Stream (how/when transmitted),
 - 4139 – Talker's requirements from the network, and

- 4140 – TSN capabilities of the Talker's interface(s).
- 4141 d) List Listener – Each Listener list entry contains:
- 4142 – Listener's requirements from the network, and
- 4143 – TSN capabilities of the Listener's interface(s).

4144 **6.4.10.4.2.3 Output**

4145 Result – Status information indicating if Stream addition has been successful.

4146 **6.4.10.4.3 Action remove-listener**

4147 **6.4.10.4.3.1 General**

4148 This Action removes listeners from a stream.

4149 **6.4.10.4.3.2 Input**

4150 List Listener – A list of indices of listeners to be removed from a stream.

4151 **6.4.10.4.3.3 Output**

4152 Result – Status information indicating if Stream addition has been successful.

4153 **6.4.10.5 IEC/IEEE 60802 YANG data models**

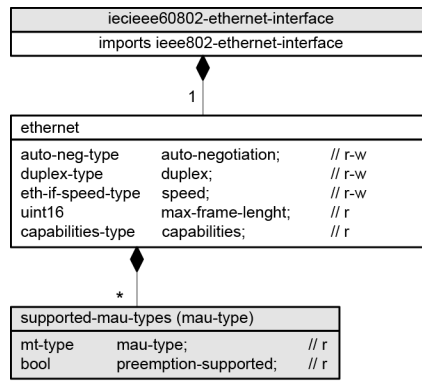
4154 A representation similar to UML® is used to provide an overview of the hierarchy of the
4155 IEC/IEEE 60802 YANG data model.

4156 A UML-like representation of the management model is provided in Figure 29, Figure 30,
4157 Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and Figure 37. The purpose
4158 of a UML-like diagram is to express the model design in a concise manner. The structure of the
4159 UML-like representation shows the name of the object followed by a list of properties for the
4160 object. The properties indicate their type and accessibility. It should be noted that UML-like
4161 representation is meant to express simplified semantics for the properties. It is not meant to
4162 provide the specific datatype used to encode the object in either MIB or YANG.

4163 NOTE – OMG® UML® 2.5 conventions together with C++ language constructs are used as a representation to convey
4164 model structure and relationships.

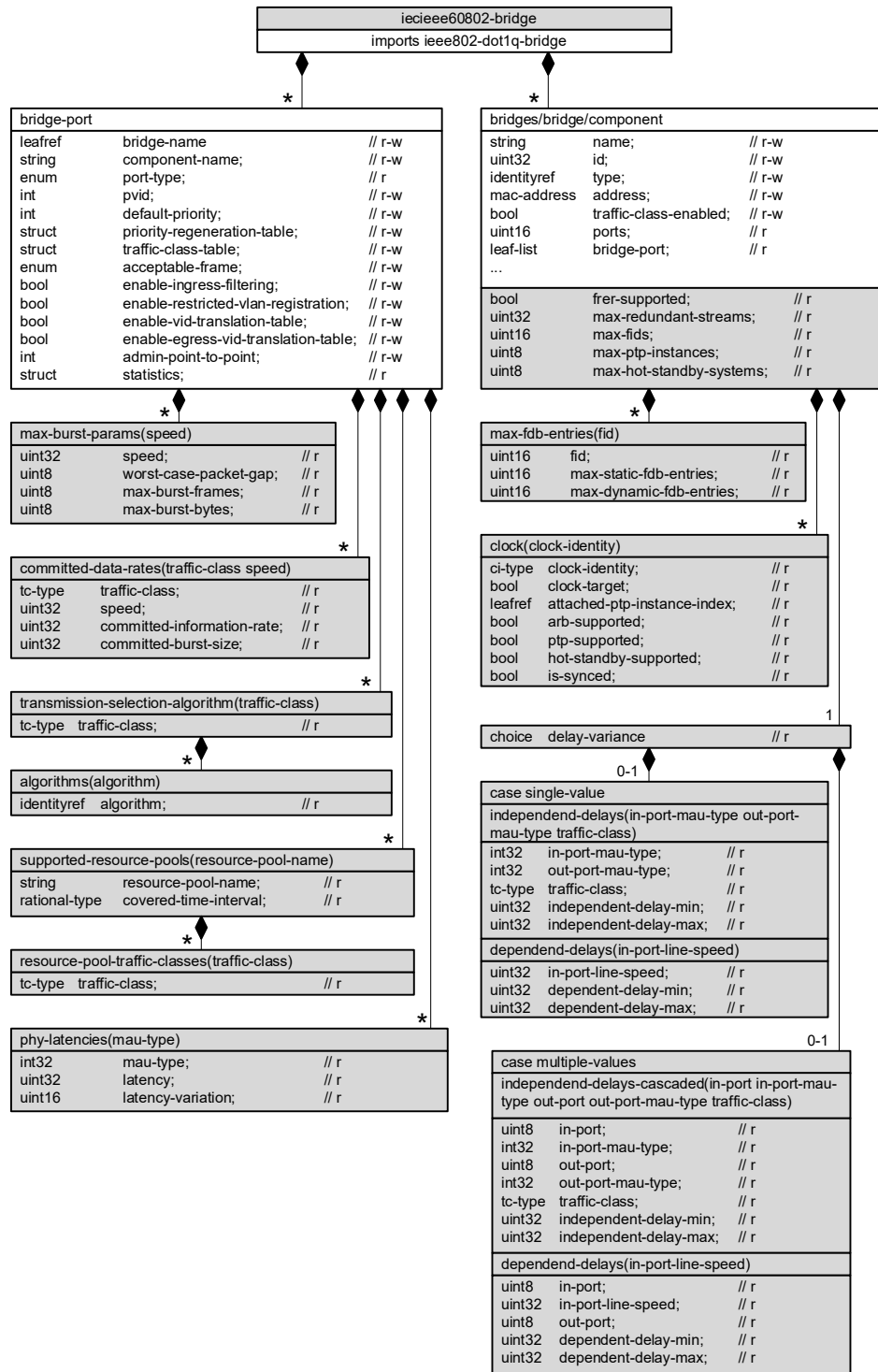
4165 For all UML® figures, data that is imported from original modules is shown in white, and data
4166 that is augmented according to IEC/IEEE 60802 is shown in grey.

4167 Figure 29, Figure 30, Figure 31, Figure 32, Figure 33, Figure 34, Figure 35, Figure 36, and
4168 Figure 37 provide an overview of the IEC/IEEE 60802 augmentations.



IEC

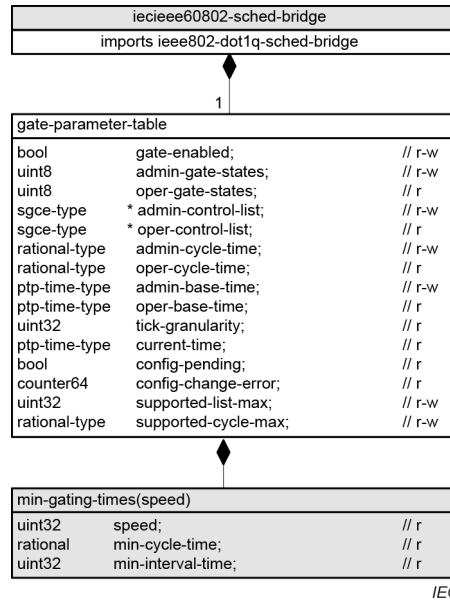
Figure 29 – Module iecieee60802-ethernet-interface



4171

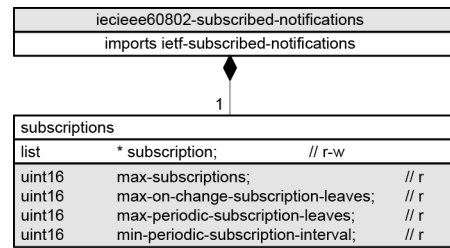
4172

Figure 30 – Module iecieee60802-bridge



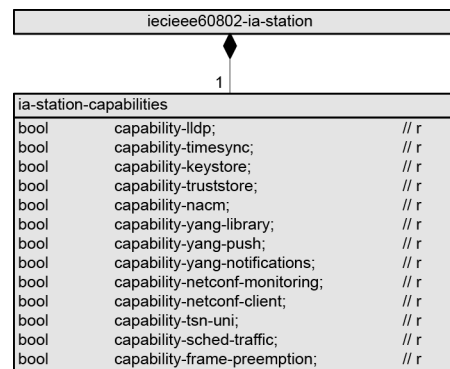
IEC

Figure 31 – Module icieeee60802-dot1-sched-bridge



IEC

Figure 32 – Module icieeee60802-subscribed-notifications



IEC

Figure 33 – Module icieeee60802-ia-station

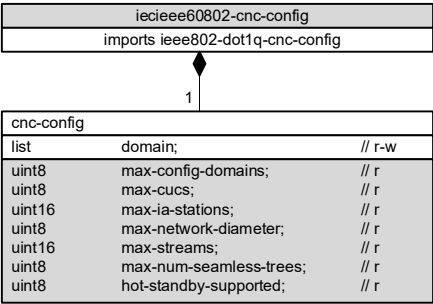


Figure 34 – Module icieeee60802-cnc-config

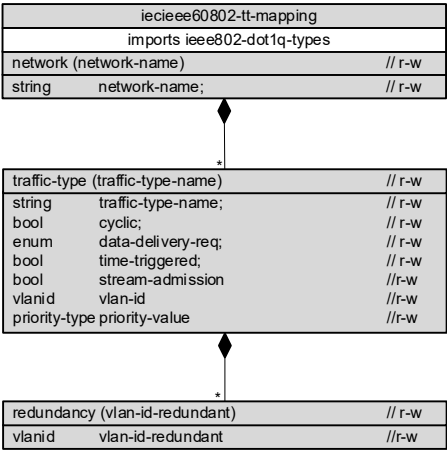


Figure 35 – Module icieeee60802-tt-mapping

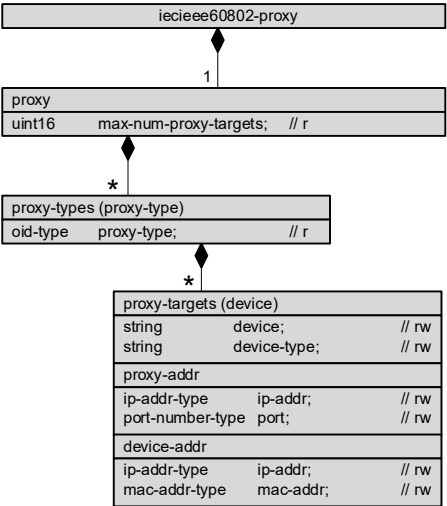


Figure 36 – Module icieeee60802-proxy

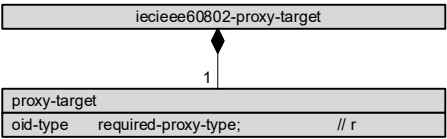


Figure 37 – icieeee60802-proxy-target

6.4.10.6 Structure of IEC/IEEE 60802 YANG data models

The YANG data models specified by this standard use the YANG modules summarized in Table 17.

In the YANG module definitions, if any discrepancy between the "description" text and the corresponding definition in any other part of this standard occurs, the definitions outside Clause 6 take precedence.

Table 17 – Summary of the YANG modules

Module	Description
ieee802-ethernet-interface	This module contains YANG definitions for configuring IEEE Std 802.3 Ethernet Interfaces.
ietf-interfaces	This module contains a collection of YANG definitions for managing network interfaces.
ieciieee60802-ethernet-interface	This module augments ieee802-ethernet-interface.
ieee802-types	This module contains a collection of generally useful derived data types for IEEE YANG data models.
ieee802-dot1q-bridge	This module describes the bridge configuration model for IEEE 802.1Q Bridges.
ieee802-dot1q-types	This module contains common types used within dot1q-bridge modules.
ieciieee60802-bridge	This module augments ieee802-dot1q-bridge.
ieee802-dot1q-sched-bridge	This module provides for management of IEEE Std 802.1Q Bridges that support Scheduled Traffic Enhancements.
ieciieee60802-dot1q-sched-bridge	This module augments ieee802-dot1q-sched-bridge.
ieee802-dot1cb-frer	This module provides management objects that control the frame replication and elimination from IEEE Std 802.1CB-2017.
ieee1588-ptp-tt	This module defines a data model for the configuration and state of IEEE Std 1588 clocks.
ietf-netconf-acm	This module provides management for the Network Configuration Access Control Model.
ieee802-dot1q-cnc-config	This module provides the Time-Sensitive Networking (TSN) User/Network Interface (UNI) for the exchange of information between CUC and CNC that are required to configure TSN Streams in a TSN network.
ieciieee60802-cnc-config	This module augments ieee802-dot1q-cnc-config.
ieciieee60802-ia-station	This module provides read-only information about the capabilities and RPCs for IEC/IEEE 60802 IA-stations.
ietf-subscribed-notifications	This module defines a YANG data model for subscribing to event records and receiving matching content in notification messages.
ieciieee60802-subscribed-notifications	This module augments ietf-subscribed-notifications.
ieciieee60802-tt-mapping	This module provides information on characteristics of traffic classes configured in a network to allow applications to map application traffic classes to them
ieciieee60802-proxy.yang	This module contains configuration and capability information for IEC/IEEE 60802 proxy stations
ieciieee60802-proxy-target.yang	This module contains capability information for IEC/IEEE 60802 proxy targets

6.4.10.7 YANG schema tree definitions

6.4.10.7.1 General

The schema tree is provided as an overview of the YANG modules. The symbols and their meaning are specified in YANG Tree Diagrams (IETF RFC 8340).

6.4.10.7.2 Module iecieee60802-ethernet-interface

```

4200 module: iecieee60802-ethernet-interface
4201
4202   augment /if:interfaces/if:interface/eth-if:ethernet:
4203     +-ro supported-mau-types* [mau-type]
4204     +-ro mau-type                uint32
4205     +-ro preemption-supported?   boolean
4206

```

6.4.10.7.3 Module iecieee60802-bridge

```

4207 module: iecieee60802-bridge
4208
4209   augment /if:interfaces/if:interface/bridge:bridge-port:
4210     +-ro max-burst-params* [speed]
4211     | +-ro speed                uint32
4212     | +-ro worst-case-packet-gap? uint8
4213     | +-ro max-burst-frames?     uint8
4214     | +-ro max-burst-bytes?      uint8
4215     +-ro committed-data-rates* [traffic-class speed]
4216     | +-ro traffic-class         dot1q-types:traffic-class-type
4217     | +-ro speed                 uint32
4218     | +-ro committed-information-rate? uint32
4219     | +-ro committed-burst-size?   uint32
4220     +-ro transmission-selection-algorithm* [traffic-class]
4221     | +-ro traffic-class         dot1q-types:traffic-class-type
4222     | +-ro algorithms* [algorithm]
4223     | +-ro algorithm            identityref
4224     +-ro supported-resource-pools* [resource-pool-name]
4225     | +-ro resource-pool-name    string
4226     | +-ro covered-time-interval
4227     | | +-ro numerator?         uint32
4228     | | +-ro denominator?      uint32
4229     | +-ro resource-pool-traffic-classes* [traffic-class]
4230     | +-ro traffic-class         dot1q-types:traffic-class-type
4231     +-ro phy-latencies* [mau-type]
4232     +-ro mau-type                uint32
4233     +-ro latency?                uint32
4234     +-ro latency-variation?      uint16
4235   augment /bridge:bridges/bridge:bridge/bridge:component:
4236     +-ro frer-supported?          boolean
4237     +-ro max-redundant-streams?   uint32
4238     +-ro max-fids?               uint16
4239     +-ro max-fdb-entries* [fid]
4240     | +-ro fid                   uint16
4241     | +-ro max-static-fdb-entries? uint16
4242     | +-ro max-dynamic-fdb-entries? uint16
4243     +-ro (delay-variance)?
4244     | +-: (single-value)
4245     | | +-ro independent-delays* [in-port-mau-type out-port-mau-type
4246   traffic-class]
4247     | | | +-ro in-port-mau-type    int32
4248     | | | +-ro out-port-mau-type   int32
4249     | | | +-ro traffic-class       dot1q-types:traffic-class-type
4250     | | | +-ro independent-delay-min? uint32
4251     | | | +-ro independent-delay-max? uint32
4252     | | +-ro dependent-delays* [in-port-line-speed]
4253     | | +-ro in-port-line-speed    uint32
4254     | | +-ro dependent-delay-min?  uint32
4255     | | +-ro dependent-delay-max?  uint32
4256     | +-: (multiple-values)
4257     | +-ro independent-delays-cascaded* [in-port in-port-mau-type out-
4258   port out-port-mau-type traffic-class]
4259

```

```

4260 | | +-ro in-port uint8
4261 | | +-ro in-port-mau-type int32
4262 | | +-ro out-port uint8
4263 | | +-ro out-port-mau-type int32
4264 | | +-ro traffic-class dot1q-types:traffic-class-type
4265 | | +-ro independent-delay-min? uint32
4266 | | +-ro independent-delay-max? uint32
4267 | +-ro dependent-delays-cascaded* [in-port in-port-line-speed out-
4268 port]
4269 | +-ro in-port uint8
4270 | +-ro in-port-line-speed uint32
4271 | +-ro out-port uint8
4272 | +-ro dependent-delay-min? uint32
4273 | +-ro dependent-delay-max? uint32
4274 +-ro max-ptp-instances? uint8
4275 +-ro max-hot-standby-systems? uint8
4276 +-ro clock* [clock-identity]
4277 | +-ro clock-identity ptp-tt:clock-identity
4278 | +-ro clock-target? boolean
4279 | +-ro attached-ptp-instance-index? -> /ptp-
4280 tt:ptp/instances/instance/instance-index
4281 | +-ro arb-supported? boolean
4282 | +-ro ptp-supported? boolean
4283 | +-ro hot-standby-supported? boolean
4284 | +-ro is-synced? boolean

```

6.4.10.7.4 Module iecieee60802-sched-bridge

```

4286 module: iecieee60802-sched-bridge
4287
4288 augment /if:interfaces/if:interface/bridge:bridge-port/sched-bridge:gate-
4289 parameter-table:
4290 +-ro min-gating-times* [speed]
4291 | +-ro speed uint32
4292 | +-ro min-cycle-time
4293 | | +-u ieee802:rational-grouping
4294 | +-ro min-interval-time? uint32

```

6.4.10.7.5 Module iecieee60802-cnc-config

```

4296 module: iecieee60802-cnc-config
4297
4298 augment /dot1q-cnc-config:cnc-config:
4299 +-ro max-config-domains? uint8
4300 +-ro max-cucs? uint8
4301 +-ro max-ia-stations? uint16
4302 +-ro max-network-diameter? uint8
4303 +-ro max-streams? uint16
4304 +-ro max-num-seamless-trees? uint8
4305 +-ro hot-standby-supported? uint8
4306 +-x add_streams
4307 | +-w input
4308 | | +-w cuc-id? string
4309 | | +-w stream-list* [stream-id]
4310 | | +-w stream-id tsn-types:stream-id-type
4311 | | +-w talker
4312 | | | +-w stream-rank
4313 | | | | +-w rank? uint8
4314 | | | +-w end-station-interfaces* [mac-address interface-name]
4315 | | | | +-w mac-address string
4316 | | | | +-w interface-name string
4317 | | | +-w data-frame-specification* [index]
4318 | | | | +-w index uint8
4319 | | | +-w (field)?

```

```

4320         |         |         |         +-:(ieee802-mac-addresses)
4321         |         |         |         |   +--w ieee802-mac-addresses
4322         |         |         |         |       +--w destination-mac-address?   string
4323         |         |         |         |       +--w source-mac-address?       string
4324         |         |         |         +-:(ieee802-vlan-tag)
4325         |         |         |         |   +--w ieee802-vlan-tag
4326         |         |         |         |       +--w priority-code-point?   uint8
4327         |         |         |         |       +--w vlan-id?                 uint16
4328         |         |         |         +-:(ipv4-tuple)
4329         |         |         |         |   +--w ipv4-tuple
4330         |         |         |         |       +--w source-ip-address?         inet:ipv4-
4331 address    |         |         |         |
4332         |         |         |         |       +--w destination-ip-address?   inet:ipv4-
4333 address    |         |         |         |
4334         |         |         |         |       +--w dscp?                     uint8
4335         |         |         |         |       +--w protocol?                 uint16
4336         |         |         |         |       +--w source-port?              uint16
4337         |         |         |         |       +--w destination-port?         uint16
4338         |         |         |         +-:(ipv6-tuple)
4339         |         |         |         |   +--w ipv6-tuple
4340         |         |         |         |       +--w source-ip-address?         inet:ipv6-
4341 address    |         |         |         |
4342         |         |         |         |       +--w destination-ip-address?   inet:ipv6-
4343 address    |         |         |         |
4344         |         |         |         |       +--w dscp?                     uint8
4345         |         |         |         |       +--w protocol?                 uint16
4346         |         |         |         |       +--w source-port?              uint16
4347         |         |         |         |       +--w destination-port?         uint16
4348         |         |         |         +--w traffic-specification
4349         |         |         |         |   +--w interval
4350         |         |         |         | |   +--w numerator?                 uint32
4351         |         |         |         | |   +--w denominator?               uint32
4352         |         |         |         | |   +--w max-frames-per-interval?   uint16
4353         |         |         |         | |   +--w max-frame-size?             uint16
4354         |         |         |         | |   +--w transmission-selection?     uint8
4355         |         |         |         | |   +--w time-aware!
4356         |         |         |         | |       +--w earliest-transmit-offset? uint32
4357         |         |         |         | |       +--w latest-transmit-offset?  uint32
4358         |         |         |         | |       +--w jitter?                 uint32
4359         |         |         |         |   +--w user-to-network-requirements
4360         |         |         |         | |   +--w num-seamless-trees?          uint8
4361         |         |         |         | |   +--w max-latency?                 uint32
4362         |         |         |         |   +--w interface-capabilities
4363         |         |         |         | |   +--w vlan-tag-capable?             boolean
4364         |         |         |         | |   +--w cb-stream-iden-type-list*     uint32
4365         |         |         |         | |   +--w cb-sequence-type-list*       uint32
4366         |         |         |         | +--w listener* [index]
4367         |         |         |         | |   +--w index                       uint32
4368         |         |         |         | |   +--w end-station-interfaces* [mac-address interface-name]
4369         |         |         |         | | |   +--w mac-address                 string
4370         |         |         |         | | |   +--w interface-name              string
4371         |         |         |         | |   +--w user-to-network-requirements
4372         |         |         |         | | |   +--w num-seamless-trees?          uint8
4373         |         |         |         | | |   +--w max-latency?                 uint32
4374         |         |         |         | |   +--w interface-capabilities
4375         |         |         |         | | |   +--w vlan-tag-capable?             boolean
4376         |         |         |         | | |   +--w cb-stream-iden-type-list*     uint32
4377         |         |         |         | | |   +--w cb-sequence-type-list*       uint32
4378         +-rw output
4379         +-rw result?   boolean
4380     augment /dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
4381     config:cuc/dot1q-cnc-config:stream:
4382         +--x remove_listener

```

```

4383     +--w input
4384     |   +--w listener* [index]
4385     |       +--w index      uint32
4386     +--rw output
4387     +--rw result?    boolean

```

4388 6.4.10.7.6 Module **iecieeee60802-ia-station**

```

4389 module: iecieeee60802-ia-station
4390   +--ro ia-station-capabilities
4391       +--ro capability-lldp?                boolean
4392       +--ro capability-timesync?            boolean
4393       +--ro capability-keystore?            boolean
4394       +--ro capability-truststore?          boolean
4395       +--ro capability-nacm?                boolean
4396       +--ro capability-yang-library?        boolean
4397       +--ro capability-yang-push?          boolean
4398       +--ro capability-yang-notifications?  boolean
4399       +--ro capability-netconf-monitoring?  boolean
4400       +--ro capability-netconf-client?      boolean
4401       +--ro capability-cnc-config?          boolean
4402       +--ro capability-sched-traffic?       boolean
4403       +--ro capability-frame-preemption?    boolean
4404
4405   rpcs:
4406       +--x ia-factory-reset

```

4407 6.4.10.7.7 Module **iecieeee60802-subscribed-notifications**

```

4408 module: iecieeee60802-subscribed-notifications
4409
4410   augment /sn:subscriptions:
4411       +--ro max-subscriptions?                uint16
4412       +--ro max-on-change-subscription-leaves? uint16
4413       +--ro max-periodic-subscription-leaves? uint16
4414       +--ro min-periodic-subscription-interval? uint16

```

4415 6.4.10.7.8 Module **iecieeee-tt-mapping**

```

4416 module: iecieeee60802-tt-mapping
4417   +--rw tt-mapping
4418       +--rw network* [network-name]
4419           +--rw network-name    string
4420           +--rw traffic-type* [traffic-type-name]
4421               +--rw traffic-type-name    string
4422               +--rw cyclic?              boolean
4423               +--rw data-delivery-req?    enumeration
4424               +--rw time-triggered?       boolean
4425               +--rw stream-based?         boolean
4426               +--rw vlan-id?              dot1q-types:vlanid
4427               +--rw redundancy* [vlan-id-redundant]
4428                   | +--rw vlan-id-redundant    dot1q-types:vlanid
4429               +--rw priority-value?       dot1q-types:priority-type

```

4430 6.4.10.7.9 Module **iecieeee60802-proxy**

```

4431 module: iecieeee60802-proxy
4432   +--rw proxy!
4433       +--ro max-num-proxy-targets?    uint16
4434       +--ro proxy-types* [proxy-type]
4435           +--ro proxy-type            string
4436       +--ro proxy-targets* [device]
4437           +--ro device                string
4438           +--ro device-type?          string
4439       +--ro proxy-addr
4440           | +--ro ip-addr?            inet:ip-address

```



```

4441         |  +--ro port?          inet:port-number
4442         +--ro device-addr
4443         +--ro ip-addr?         inet:ip-address
4444         +--ro mac-addr?       ieee:mac-address

```

4445 **6.4.10.7.10 Module iecieee60802-proxy-target**

```

4446 module: iecieee60802-proxy-target
4447     +--rw proxy-target!
4448     +--ro required-proxy-type?  string

```

4449 **6.4.10.8 YANG modules^{5,6}**

4450 **6.4.10.8.1 Module iecieee60802-ethernet-interface**

```

4451 module iecieee60802-ethernet-interface {
4452     yang-version 1.1;
4453     namespace
4454         "urn:ieee:std:60802:yang:ieciieee60802-ethernet-interface";
4455     prefix ia-eth-if;
4456
4457     import ieee802-ethernet-interface {
4458         prefix eth-if;
4459     }
4460     import ietf-interfaces {
4461         prefix if;
4462     }
4463
4464     organization
4465         "IEEE 802.1 Working Group and IEC subcommittee 65C:
4466         Industrial networks, of IEC technical committee 65:
4467         Industrial-process measurement, control and automation";
4468     contact
4469         "WG-URL: http://ieee802.org/1/
4470         WG-EMail: stds-802-1-1@ieee.org
4471
4472         Contact: IEEE 802.1 Working Group Chair
4473                 Postal: C/O IEEE 802.1 Working Group
4474                 IEEE Standards Association
4475                 445 Hoes Lane
4476                 Piscataway, NJ 08854
4477                 USA
4478
4479         E-mail: stds-802-1-chairs@ieee.org";
4480     description
4481         "Management objects that provide information about IEC/IEEE 60802
4482         IA-Stations as specified in IEC/IEEE 60802.
4483
4484         Copyright (C) IEC/IEEE (2025).
4485         This version of this YANG module is part of IEC/IEEE 60802;
4486         see the standard itself for full legal notices.";
4487
4488     revision 2025-03-12 {
4489         description "Published as part of IEC/IEEE 60802-2025.
4490             The following reference statement identifies each referenced
4491             IEEE Standard as updated by applicable amendments.";
4492         reference
4493             "IEC/IEEE 60802 TSN profile for industrial automation:

```

⁵ Copyright release for YANG: Users of this document may freely reproduce the YANG modules contained in this document so that they can be used for their intended purpose.

⁶ An ASCII version of each YANG module defined in this document is attached to the PDF of this document and can also be obtained from the IEEE 802.Website at <https://1.ieee802.org/yang-modules/>.

```

4494     IEC/IEEE 60802-2025.
4495     IEEE Std 802.1Q Bridges and Bridged Networks:
4496     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
4497     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
4498     IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
4499 }
4500
4501 augment "/if:interfaces/if:interface/eth-if:ethernet" {
4502     description
4503         "Augment IEEE Std 802.3 ethernet.";
4504     list supported-mau-types {
4505         key "mau-type";
4506         config false;
4507         description
4508             "Contains a list of supported MAU parameters.";
4509         leaf mau-type {
4510             type int32;
4511             config false;
4512             description
4513                 "The value is the supported MAU Type derived from the list
4514                 position of the corresponding dot3MauType as listed in
4515                 Clause 5 of IETF RFC 4836.";
4516             reference
4517                 "Item a) in 6.4.10.3.2.1 of IEC/IEEE 60802";
4518         }
4519         leaf preemption-supported {
4520             type boolean;
4521             config false;
4522             description
4523                 "The Boolean value indicates if preemption is supported by
4524                 the MAU Type.";
4525             reference
4526                 "Item b) in 6.4.10.3.2.1 of IEC/IEEE 60802";
4527         }
4528     }
4529 }
4530 }
4531

```

6.4.10.8.2 Module `ieciieee60802-bridge`

```

4532 module iecieee60802-bridge {
4533     yang-version 1.1;
4534     namespace "urn:ieee:std:60802:yang:ieciieee60802-bridge";
4535     prefix ia-bridge;
4536
4537     import ieee802-types {
4538         prefix ieee802;
4539     }
4540     import ieee802-dot1q-bridge {
4541         prefix bridge;
4542     }
4543     import ietf-interfaces {
4544         prefix if;
4545     }
4546     import ieee802-dot1q-types {
4547         prefix dot1q-types;
4548     }
4549     import ieee1588-ptp-tt {
4550         prefix ptp-tt;
4551     }
4552
4553     organization
4554         "IEEE 802.1 Working Group and IEC subcommittee 65C:
4555

```

```
4556     Industrial networks, of IEC technical committee 65:
4557     Industrial-process measurement, control and automation";
4558     contact
4559     "WG-URL: http://ieee802.org/1/
4560     WG-EMail: stds-802-1-1@ieee.org
4561
4562     Contact: IEEE 802.1 Working Group Chair
4563             Postal: C/O IEEE 802.1 Working Group
4564             IEEE Standards Association
4565             445 Hoes Lane
4566             Piscataway, NJ 08854
4567             USA
4568
4569     E-mail: stds-802-1-chairs@ieee.org";
4570     description
4571     "Management objects that provide information about
4572     IEC/IEEE 60802 IA-Stations as specified in IEC/IEEE 60802.
4573
4574     Copyright (C) IEC/IEEE (2025).
4575     This version of this YANG module is part of IEC/IEEE 60802;
4576     see the standard itself for full legal notices.";
4577
4578     revision 2025-03-12 {
4579         description
4580         "Published as part of IEC/IEEE 60802-2025.
4581         The following reference statement identifies each referenced
4582         IEEE Standard as updated by applicable amendments.";
4583         reference
4584         "IEC/IEEE 60802 TSN profile for industrial automation:
4585         IEC/IEEE 60802-2025.
4586         IEEE Std 802.1Q Bridges and Bridged Networks:
4587         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
4588         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
4589         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
4590     }
4591
4592     augment "/if:interfaces/if:interface/bridge:bridge-port" {
4593         description
4594         "Augment IEEE Std 802.1 bridge.";
4595         list max-burst-params {
4596             key "speed";
4597             config false;
4598             description
4599             "The list of maximum burst parameters per supported line
4600             speed.";
4601             leaf speed {
4602                 type uint32;
4603                 description
4604                 "This value is the line speed in Mbps.";
4605             }
4606             leaf worst-case-packet-gap {
4607                 type uint8;
4608                 config false;
4609                 description
4610                 "The value is the worst case maximum inter-packet gap
4611                 between consecutive frames in a traffic burst expressed
4612                 in bit-times.";
4613                 reference
4614                 "Item a) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4615             }
4616             leaf max-burst-frames {
4617                 type uint8;
4618                 config false;
```

```

4619         description
4620             "The value is the maximum number of frames that can be sent
4621             with minimal inter packet gap.";
4622         reference
4623             "Item b) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4624     }
4625     leaf max-burst-bytes {
4626         type uint8;
4627         config false;
4628         description
4629             "The value is the maximum number of octets that can be sent
4630             with minimal inter packet gap.";
4631         reference
4632             "Item c) in 6.4.10.3.3.1 of IEC/IEEE 60802";
4633     }
4634 }
4635 list committed-data-rates {
4636     key "traffic-class speed";
4637     config false;
4638     description
4639         "The list of committed data rates per traffic class and
4640         supported line speed.";
4641     leaf traffic-class {
4642         type dot1q-types:traffic-class-type;
4643         description
4644             "The traffic class of the entry (0..7).";
4645         reference
4646             "8.6.6 of IEEE Std 802.1Q";
4647     }
4648     leaf speed {
4649         type uint32;
4650         description
4651             "This value is the line speed in Mbps.";
4652     }
4653     leaf committed-information-rate {
4654         type uint32;
4655         config false;
4656         description
4657             "The value is the bandwidth limit in kbit/s.";
4658         reference
4659             "Item a) in 6.4.10.3.3.2 of IEC/IEEE 60802";
4660     }
4661     leaf committed-burst-size {
4662         type uint32;
4663         config false;
4664         description
4665             "The value is the burst size limit in bytes.";
4666         reference
4667             "Item b) in 6.4.10.3.3.2 of IEC/IEEE 60802";
4668     }
4669 }
4670 list transmission-selection-algorithm {
4671     key "traffic-class";
4672     config false;
4673     description
4674         "The list of supported transmission selection algorithms
4675         according to 8.6.8 of IEEE Std 802.1Q per traffic class.";
4676     leaf traffic-class {
4677         type dot1q-types:traffic-class-type;
4678         config false;
4679         description
4680             "Traffic class. (0..7)";
4681         reference

```

```
4682         "8.6.6 of IEEE Std 802.1Q";
4683     }
4684     list algorithms {
4685         key "algorithm";
4686         config false;
4687         description
4688             "The list of supported transmission selection algorithms
4689             according to 8.6.8 of IEEE Std 802.1Q for this traffic
4690             class.";
4691         leaf algorithm {
4692             type identityref {
4693                 base dot1q-types:transmission-selection-algorithm;
4694             }
4695             config false;
4696             description
4697                 "Transmission selection algorithm";
4698             reference
4699                 "8.6.8 of IEEE Std 802.1Q";
4700         }
4701     }
4702 }
4703 list supported-resource-pools {
4704     key "resource-pool-name";
4705     config false;
4706     description
4707         "The list of supported buffer resource pools.";
4708     leaf resource-pool-name {
4709         type string;
4710         config false;
4711         description
4712             "The value is the name of a resource pool.";
4713         reference
4714             "Item a) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4715     }
4716     container covered-time-interval {
4717         config false;
4718         uses ieee802:rational-grouping;
4719         description
4720             "The value is the covered buffering time given as rational
4721             number of seconds for the highest supported link speed.";
4722         reference
4723             "Item b) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4724     }
4725     list resource-pool-traffic-classes {
4726         key "traffic-class";
4727         config false;
4728         description
4729             "The list of the traffic classes to be served by the
4730             resource pool.";
4731         reference
4732             "Item c) in 6.4.10.3.4.2 of IEC/IEEE 60802";
4733         leaf traffic-class {
4734             type dot1q-types:traffic-class-type;
4735             description
4736                 "The traffic class of the entry.";
4737             reference
4738                 "8.6.6 of IEEE Std 802.1Q";
4739         }
4740     }
4741 }
4742 list phy-latencies {
4743     key "mau-type";
4744     config false;
```

```

4745     description
4746         "Latency information for each MAU type";
4747     leaf mau-type {
4748         type int32;
4749         config false;
4750         description
4751             "The MAU type";
4752     }
4753     leaf latency {
4754         type uint32;
4755         config false;
4756         description
4757             "The PHY latency in ns";
4758         reference
4759             "6.4.10.3.6.3 of IEC/IEEE 60802";
4760     }
4761     leaf latency-variation {
4762         type uint16;
4763         config false;
4764         description
4765             "The PHY latency variation in ns";
4766         reference
4767             "6.4.10.3.6.4 of IEC/IEEE 60802";
4768     }
4769 }
4770 }
4771
4772 augment "/bridge:bridges/bridge:bridge/bridge:component" {
4773     description
4774         "Augment IEEE Std 802.1 bridge component.";
4775     leaf frer-supported {
4776         type boolean;
4777         config false;
4778         description
4779             "The Boolean value indicates if FRER is supported.";
4780         reference
4781             "6.4.10.3.5.1 of IEC/IEEE 60802";
4782     }
4783     leaf max-redundant-streams {
4784         type uint32;
4785         config false;
4786         description
4787             "The value is the maximum number of supported redundant
4788             streams.";
4789         reference
4790             "6.4.10.3.5.2 of IEC/IEEE 60802";
4791     }
4792     leaf max-fids {
4793         type uint16;
4794         config false;
4795         description
4796             "The value is the maximum number of supported FIDs.";
4797         reference
4798             "6.4.10.3.7.1 of IEC/IEEE 60802";
4799     }
4800     list max-fdb-entries {
4801         key "fid";
4802         config false;
4803         description
4804             "The list of the maximum number of static and dynamic
4805             FDB entries per FID.";
4806         reference
4807             "6.4.10.3.7.2 of IEC/IEEE 60802";

```

```
4808     leaf fid {
4809         type uint16;
4810         config false;
4811         description
4812             "The FID number";
4813     }
4814     leaf max-static-fdb-entries {
4815         type uint16;
4816         config false;
4817         description
4818             "The value is the maximum number of static FDB
4819             entries.";
4820         reference
4821             "6.4.10.3.7.3 of IEC/IEEE 60802";
4822     }
4823     leaf max-dynamic-fdb-entries {
4824         type uint16;
4825         config false;
4826         description
4827             "The value is the maximum number of dynamic FDB entries.";
4828         reference
4829             "6.4.10.3.7.4 of IEC/IEEE 60802";
4830     }
4831 }
4832 choice delay-variance {
4833     config false;
4834     description
4835         "The value indicates variance in delay depending upon the use
4836         of a singleValue or multipleValues.";
4837     reference
4838         "6.4.10.3.6.1 of IEC/IEEE 60802";
4839     case single-value {
4840         list independent-delays {
4841             key "in-port-mau-type out-port-mau-type traffic-class";
4842             config false;
4843             description
4844                 "The list of minimum and maximum frame length
4845                 independent delay time values of frames as they pass
4846                 through a bridge component.";
4847             reference
4848                 "6.4.10.3.6.2 of IEC/IEEE 60802";
4849             leaf in-port-mau-type {
4850                 type int32;
4851                 config false;
4852                 description
4853                     "The MAU type of the input port";
4854             }
4855             leaf out-port-mau-type {
4856                 type int32;
4857                 config false;
4858                 description
4859                     "The MAU type of the input port";
4860             }
4861             leaf traffic-class {
4862                 type dot1q-types:traffic-class-type;
4863                 config false;
4864                 description
4865                     "The traffic class of the entry.";
4866                 reference
4867                     "8.6.6 of IEEE Std 802.1Q";
4868             }
4869             leaf independent-delay-min {
4870                 type uint32;
```

```

4871         config false;
4872         description
4873             "The value is the minimum delay portion that is
4874             independent of frame length according to 12.32.1.1.
4875             of IEEE 802.1Q";
4876         reference
4877             "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4878     }
4879     leaf independent-delay-max {
4880         type uint32;
4881         config false;
4882         description
4883             "The value is the maximum delay portion that is
4884             independent of frame length according to 12.32.1.1.
4885             of IEEE 802.1Q";
4886         reference
4887             "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4888     }
4889 }
4890 list dependent-delays {
4891     key "in-port-line-speed";
4892     config false;
4893     description
4894         "The list of minimum and maximum frame length dependent
4895         delay time values of frames as they pass through a
4896         bridge component";
4897     reference
4898         "6.4.10.3.6.2 of IEC/IEEE 60802";
4899     leaf in-port-line-speed {
4900         type uint32;
4901         config false;
4902         description
4903             "This value is the line speed in Mbps.";
4904     }
4905     leaf dependent-delay-min {
4906         type uint32;
4907         config false;
4908         description
4909             "The value is the minimum delay portion that is
4910             dependent on frame length according to 12.32.1.2.
4911             of IEEE 802.1Q";
4912         reference
4913             "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4914     }
4915     leaf dependent-delay-max {
4916         type uint32;
4917         config false;
4918         description
4919             "The value is the maximum delay portion that is
4920             dependent on frame length according to 12.32.1.2.
4921             of IEEE 802.1Q";
4922         reference
4923             "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4924     }
4925 }
4926 }
4927 case multiple-values {
4928     list independent-delays-cascaded {
4929         key "in-port in-port-mau-type out-port out-port-mau-type
4930             traffic-class";
4931         config false;
4932         description
4933             "The list of minimum and maximum frame length

```



```
4934         independent delay time values of frames as they pass
4935         through a bridge component.";
4936     reference
4937         "6.4.10.3.6.2 of IEC/IEEE 60802";
4938     leaf in-port {
4939         type uint8;
4940         config false;
4941         description
4942             "The port number of the input port";
4943     }
4944     leaf in-port-mau-type {
4945         type int32;
4946         config false;
4947         description
4948             "The MAU type of the input port";
4949     }
4950     leaf out-port {
4951         type uint8;
4952         config false;
4953         description
4954             "The port number of the output port";
4955     }
4956     leaf out-port-mau-type {
4957         type int32;
4958         config false;
4959         description
4960             "The MAU type of the input port";
4961     }
4962     leaf traffic-class {
4963         type dot1q-types:traffic-class-type;
4964         config false;
4965         description
4966             "The traffic class of the entry.";
4967         reference
4968             "8.6.6 of IEEE Std 802.1Q";
4969     }
4970     leaf independent-delay-min {
4971         type uint32;
4972         config false;
4973         description
4974             "The value is the minimum delay portion that is
4975             independent of frame length according to 12.32.1.1.
4976             of IEEE 802.1Q";
4977         reference
4978             "Item a) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4979     }
4980     leaf independent-delay-max {
4981         type uint32;
4982         config false;
4983         description
4984             "The value is the maximum delay portion that is
4985             independent of frame length according to 12.32.1.1.
4986             of IEEE 802.1Q";
4987         reference
4988             "Item b) in 6.4.10.3.6.2 of IEC/IEEE 60802";
4989     }
4990 }
4991 list dependent-delays-cascaded {
4992     key "in-port in-port-line-speed out-port";
4993     config false;
4994     description
4995         "The list of minimum and maximum frame length dependent
4996         delay time values of frames as they pass through a
```

```

4997         bridge component";
4998     reference
4999         "6.4.10.3.6.2 of IEC/IEEE 60802";
5000     leaf in-port {
5001         type uint8;
5002         config false;
5003         description
5004             "The port number of the input port";
5005     }
5006     leaf in-port-line-speed {
5007         type uint32;
5008         config false;
5009         description
5010             "This value is the line speed in Mbps.";
5011     }
5012     leaf out-port {
5013         type uint8;
5014         config false;
5015         description
5016             "The port number of the output port";
5017     }
5018     leaf dependent-delay-min {
5019         type uint32;
5020         config false;
5021         description
5022             "The value is the minimum delay portion that is
5023             dependent on frame length according to 12.32.1.2.
5024             of IEEE 802.1Q";
5025         reference
5026             "Item c) in 6.4.10.3.6.2 of IEC/IEEE 60802";
5027     }
5028     leaf dependent-delay-max {
5029         type uint32;
5030         config false;
5031         description
5032             "The value is the maximum delay portion that is
5033             dependent on frame length according to 12.32.1.2.
5034             of IEEE 802.1Q";
5035         reference
5036             "Item d) in 6.4.10.3.6.2 of IEC/IEEE 60802";
5037     }
5038 }
5039 }
5040 }
5041 leaf max-ptp-instances {
5042     type uint8;
5043     config false;
5044     description
5045         "The value is the maximum number of supported PTP
5046         Instances.";
5047     reference
5048         "6.4.10.3.7.5 of IEC/IEEE 60802";
5049 }
5050 leaf max-hot-standby-systems {
5051     type uint8;
5052     config false;
5053     description
5054         "The value is the maximum number of supported
5055         HotStandbySystem entities";
5056     reference
5057         "6.4.10.3.7.6 of IEC/IEEE 60802";
5058 }
5059 list clock {

```

```
5060     key "clock-identity";
5061     config false;
5062     description
5063         "The list of supported application clock entities.";
5064     reference
5065         "6.4.10.3.7.7 of IEC/IEEE 60802";
5066     leaf clock-identity {
5067         type ptp-tt:clock-identity;
5068         config false;
5069         description
5070             "The clock identity of the application clock.";
5071         reference
5072             "Item a) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5073     }
5074     leaf clock-target {
5075         type boolean;
5076         config false;
5077         description
5078             "The Boolean value indicates if the application clock is a
5079             clock target (TRUE) or clock source (FALSE).";
5080         reference
5081             "Item b) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5082     }
5083     leaf attached-ptp-instance-index {
5084         type leafref {
5085             path "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-
5086 tt:instance-index";
5087         }
5088         config false;
5089         description
5090             "The value is a reference to the index of the PTP or hot
5091             standby Instance, that is attached to the application
5092             clock.";
5093         reference
5094             "Item f) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5095     }
5096     leaf arb-supported {
5097         type boolean;
5098         config false;
5099         description
5100             "The Boolean value indicates if the application clock
5101             supports the ARB timescale.";
5102         reference
5103             "Item c) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5104     }
5105     leaf ptp-supported {
5106         type boolean;
5107         config false;
5108         description
5109             "The Boolean value indicates if the application clock
5110             supports the PTP timescale.";
5111         reference
5112             "Item d) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5113     }
5114     leaf hot-standby-supported {
5115         type boolean;
5116         config false;
5117         description
5118             "The Boolean value indicates if the application clock
5119             supports the hot standby.";
5120         reference
5121             "Item e) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5122     }
```

```

5123     leaf is-synced {
5124         type boolean;
5125         config false;
5126         description
5127             "The Boolean value indicates if the application clock is
5128             either synchronized to the attached PTP Instance (TRUE)
5129             or to an internal/external ClockSource (FALSE).";
5130         reference
5131             "Item g) in 6.4.10.3.7.7 of IEC/IEEE 60802";
5132     }
5133 }
5134 }
5135 }

```

5136 **6.4.10.8.3 Module iecieee60802-sched-bridge**

```

5137 module iecieee60802-sched-bridge {
5138     yang-version 1.1;
5139     namespace "urn:ieee:std:60802:yang:ieciieee60802-sched-bridge";
5140     prefix ia-sched-bridge;
5141
5142     import ieee802-types {
5143         prefix ieee802;
5144     }
5145     import ieee802-dot1q-bridge {
5146         prefix bridge;
5147     }
5148     import ieee802-dot1q-sched-bridge {
5149         prefix sched-bridge;
5150     }
5151     import ietf-interfaces {
5152         prefix if;
5153     }
5154
5155     organization
5156         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5157         Industrial networks, of IEC technical committee 65:
5158         Industrial-process measurement, control and automation";
5159     contact
5160         "WG-URL: http://ieee802.org/1/
5161         WG-EMail: stds-802-1-1@ieee.org
5162
5163         Contact: IEEE 802.1 Working Group Chair
5164         Postal: C/O IEEE 802.1 Working Group
5165         IEEE Standards Association
5166         445 Hoes Lane
5167         Piscataway, NJ 08854
5168         USA
5169
5170         E-mail: stds-802-1-chairs@ieee.org";
5171     description
5172         "Management objects that provide information about IEC/IEEE 60802
5173         IA-Stations as specified in IEC/IEEE 60802.
5174
5175         Copyright (C) IEC/IEEE (2025).
5176         This version of this YANG module is part of IEC/IEEE 60802;
5177         see the standard itself for full legal notices.";
5178
5179     revision 2025-03-12 {
5180         description "Published as part of IEC/IEEE 60802-2025.
5181         The following reference statement identifies each referenced
5182         IEEE Standard as updated by applicable amendments.";
5183         reference
5184             "IEC/IEEE 60802 TSN profile for industrial automation:

```

```

5185     IEC/IEEE 60802-2025.
5186     IEEE Std 802.1Q Bridges and Bridged Networks:
5187     IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5188     IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5189     IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5190 }
5191
5192 augment "/if:interfaces/if:interface/bridge:bridge-port/sched-
5193 bridge:gate-parameter-table" {
5194     description
5195         "Augment IEEE Std 802.1 bridge/gate-parameter-table.";
5196     list min-gating-times {
5197         key "speed";
5198         config false;
5199         description
5200             "The list of minimum gating times per supported line speed.";
5201         reference
5202             "6.4.10.3.4.3 of IEC/IEEE 60802";
5203         leaf speed {
5204             type uint32;
5205             config false;
5206             description
5207                 "This value is the line speed in Mbps.";
5208         }
5209         container min-cycle-time {
5210             uses ieee802:rational-grouping;
5211             description
5212                 "The value is the minimum value supported by this port of
5213                 the AdminCycleTime and OperCycleTime parameters given as
5214                 rational number of seconds.";
5215             reference
5216                 "Item a) in 6.4.10.3.4.3 of IEC/IEEE 60802";
5217         }
5218         leaf min-interval-time {
5219             type uint32;
5220             description
5221                 "The value is the minimum value supported by this port of
5222                 the TimeIntervalValue parameter in nanoseconds.";
5223             reference
5224                 "Item b) in 6.4.10.3.4.3 of IEC/IEEE 60802";
5225         }
5226     }
5227 }

```

6.4.10.8.4 Module iecieee60802-cnc-config

```

5229 module iecieee60802-cnc-config {
5230     yang-version 1.1;
5231     namespace "urn:ieee:std:60802:yang:iecieee60802-cnc-config";
5232     prefix ia-cnc-config;
5233
5234     import ieee802-dot1q-cnc-config {
5235         prefix dot1q-cnc-config;
5236     }
5237     import ieee802-dot1q-tsn-types {
5238         prefix tsn-types;
5239     }
5240
5241     organization
5242         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5243         Industrial networks, of IEC technical committee 65:
5244         Industrial-process measurement, control and automation";
5245     contact
5246         "WG-URL: http://ieee802.org/1/

```

```

5247     WG-EMail: stds-802-1-1@ieee.org
5248
5249     Contact: IEEE 802.1 Working Group Chair
5250             Postal: C/O IEEE 802.1 Working Group
5251             IEEE Standards Association
5252             445 Hoes Lane
5253             Piscataway, NJ 08854
5254             USA
5255
5256     E-mail: stds-802-1-chairs@ieee.org";
5257 description
5258     "Management objects that provide information about IEC/IEEE 60802
5259     IA-Stations as specified in IEC/IEEE 60802.
5260
5261     Copyright (C) IEC/IEEE (2025).
5262     This version of this YANG module is part of IEC/IEEE 60802;
5263     see the standard itself for full legal notices.";
5264
5265 revision 2025-01-10 {
5266     description "Published as part of IEC/IEEE 60802-2025.
5267     The following reference statement identifies each referenced
5268     IEEE Standard as updated by applicable amendments.";
5269     reference
5270         "IEC/IEEE 60802 TSN profile for industrial automation:
5271         IEC/IEEE 60802-2025.
5272         IEEE Std 802.1Q Bridges and Bridged Networks:
5273         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5274         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5275         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5276 }
5277
5278 augment "/dot1q-cnc-config:cnc-config" {
5279     description
5280         "Augment main container in cnc-config.";
5281     leaf max-config-domains {
5282         type uint8;
5283         config false;
5284         description
5285             "The value is the maximum number of supported configuration
5286             domains.";
5287         reference
5288             "6.4.10.3.9.1 of IEC/IEEE 60802";
5289     }
5290     leaf max-cucs {
5291         type uint8;
5292         config false;
5293         description
5294             "The value is the maximum number of supported CUC entities.";
5295         reference
5296             "6.4.10.3.9.2 of IEC/IEEE 60802";
5297     }
5298     leaf max-ia-stations {
5299         type uint16;
5300         config false;
5301         description
5302             "The value is the maximum number of supported IA-stations.";
5303         reference
5304             "6.4.10.3.9.3 of IEC/IEEE 60802";
5305     }
5306     leaf max-network-diameter {
5307         type uint8;
5308         config false;
5309         description

```

```
5310         "The value is the maximum supported network diameter.";
5311     reference
5312         "6.4.10.3.9.4 of IEC/IEEE 60802";
5313 }
5314 leaf max-streams {
5315     type uint16;
5316     config false;
5317     description
5318         "The value is the maximum number of supported streams.";
5319     reference
5320         "6.4.10.3.9.5 of IEC/IEEE 60802";
5321 }
5322 leaf max-num-seamless-trees {
5323     type uint8;
5324     config false;
5325     description
5326         "The value is the maximum number of trees supported for
5327         seamless redundancy of a stream.";
5328     reference
5329         "6.4.10.3.9.6 of IEC/IEEE 60802";
5330 }
5331 leaf hot-standby-supported {
5332     type uint8;
5333     config false;
5334     description
5335         "The Boolean value indicates if PTP hot standby is
5336         supported.";
5337     reference
5338         "6.4.10.3.9.7 of IEC/IEEE 60802";
5339 }
5340 action add_streams {
5341     description
5342         "This Action requests a CNC to add a list of streams.";
5343     input {
5344         leaf cuc-id {
5345             type string;
5346             description
5347                 "The CUC ID where the streams are to be added";
5348         }
5349         list stream-list {
5350             key "stream-id";
5351             description
5352                 "List of Streams that should be added.";
5353             leaf stream-id {
5354                 type tsn-types:stream-id-type;
5355                 description
5356                     "The Stream ID is a unique identifier of a Stream
5357                     request and corresponding configuration. It is used to
5358                     associate a CUC's Stream request with a CNC's
5359                     corresponding response.";
5360             }
5361             container talker {
5362                 description
5363                     "The Talker container contains: - Talker's behavior for
5364                     Stream (how/when transmitted) - Talker's requirements
5365                     from the network - TSN capabilities of the Talker's
5366                     interface(s).";
5367                 uses tsn-types:group-talker;
5368             }
5369             list listener {
5370                 key "index";
5371                 description
5372                     "Each Listener list entry contains: - Listener's
```

```

5373         requirements from the network - TSN capabilities of
5374         the Listener's interface(s).";
5375     leaf index {
5376         type uint32;
5377         description
5378             "This index is provided in order to provide a unique
5379             key per list entry.";
5380     }
5381     uses tsn-types:group-listener;
5382 }
5383 }
5384 }
5385 output {
5386     leaf result {
5387         type boolean;
5388         description
5389             "Returns status information indicating if Stream addition
5390             has been successful.";
5391     }
5392 }
5393 }
5394 }
5395
5396 augment "/dot1q-cnc-config:cnc-config/dot1q-cnc-config:domain/dot1q-cnc-
5397 config:cuc/dot1q-cnc-config:stream" {
5398     description
5399         "Augment stream list in tsc-config-uni.";
5400     action remove_listener {
5401         description
5402             "This Action removes listeners from a stream.";
5403         input {
5404             list listener {
5405                 key "index";
5406                 description
5407                     "Each Listener list entry contains: - Listener's
5408                     requirements from the network - TSN capabilities of the
5409                     Listener's interface(s).";
5410                 leaf index {
5411                     type uint32;
5412                     description
5413                         "This index is provided in order to provide a unique
5414                         key per list entry.";
5415                 }
5416             }
5417         }
5418         output {
5419             leaf result {
5420                 type boolean;
5421                 description
5422                     "Returns status information indicating if listene removal
5423                     has been successful.";
5424             }
5425         }
5426     }
5427 }
5428 }

```

6.4.10.8.5 Module iecieee60802-ia-station

```

5430 module iecieee60802-ia-station {
5431     yang-version 1.1;
5432     namespace "urn:ieee:std:60802:yang:ieciieee60802-ia-station";
5433     prefix ias;
5434 }

```



```
5435     import ietf-datastores {
5436         prefix ds;
5437         reference
5438             "IETF RFC 8342: Network Management Datastore Architecture
5439             (NMDA)";
5440     }
5441     import ietf-netconf-acm {
5442         prefix nacm;
5443         reference
5444             "IETF RFC 8341: Network Configuration Access Control Model";
5445     }
5446
5447     organization
5448         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5449         Industrial networks, of IEC technical committee 65:
5450         Industrial-process measurement, control and automation";
5451     contact
5452         "WG-URL: http://ieee802.org/1/
5453         WG-EMail: stds-802-1-1@ieee.org
5454
5455         Contact: IEEE 802.1 Working Group Chair
5456                 Postal: C/O IEEE 802.1 Working Group
5457                 IEEE Standards Association
5458                 445 Hoes Lane
5459                 Piscataway, NJ 08854
5460                 USA
5461
5462         E-mail: stds-802-1-chairs@ieee.org";
5463     description
5464         "Capability information and reset to factory defaults
5465         functionality for IEC/IEEE 60802 IA-Stations as specified in
5466         IEC/IEEE 60802.
5467
5468         Copyright (C) IEC/IEEE (2025).
5469         This version of this YANG module is part of IEC/IEEE 60802;
5470         see the standard itself for full legal notices.";
5471
5472     revision 2025-03-12 {
5473         description "Published as part of IEC/IEEE 60802-2025.
5474         The following reference statement identifies each referenced
5475         IEEE Standard as updated by applicable amendments.";
5476         reference
5477             "IEC/IEEE 60802 TSN profile for industrial automation:
5478             IEC/IEEE 60802-2025.
5479             IEEE Std 802.1Q Bridges and Bridged Networks:
5480             IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5481             IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5482             IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5483     }
5484
5485     feature ia-factory-default-datastore {
5486         description
5487             "Indicates that the factory default configuration is
5488             available as a datastore.";
5489     }
5490
5491     identity ia-factory-default {
5492         if-feature "ia-factory-default-datastore";
5493         base ds:datastore;
5494         description
5495             "This read-only datastore contains the factory default
5496             configuration for the device that will be used to replace
5497             the contents of the read-write conventional configuration
```

```

5498         datastores during a 'ia-factory-reset' RPC operation.";
5499     }
5500
5501     container ia-station-capabilities {
5502         config false;
5503         description
5504             "This container provides read only information about an
5505             ia-station's capabilities.";
5506         reference
5507             "IEC/IEEE 60802 - YANG Data Model";
5508         leaf capability-lldp {
5509             type boolean;
5510             config false;
5511             description
5512                 "The value is true if the device supports LLDP.";
5513             reference
5514                 "6.4.10.3.8.5 of IEC/IEEE 60802";
5515         }
5516         leaf capability-timesync {
5517             type boolean;
5518             config false;
5519             description
5520                 "The value is true if the device supports Timesync.";
5521             reference
5522                 "6.4.10.3.8.6 of IEC/IEEE 60802";
5523         }
5524         leaf capability-keystore {
5525             type boolean;
5526             config false;
5527             description
5528                 "The value is true if the device supports Keystore.";
5529             reference
5530                 "6.4.10.3.8.7 of IEC/IEEE 60802";
5531         }
5532         leaf capability-truststore {
5533             type boolean;
5534             config false;
5535             description
5536                 "The value is true if the device supports Truststore.";
5537             reference
5538                 "6.4.10.3.8.9 of IEC/IEEE 60802";
5539         }
5540         leaf capability-nacm {
5541             type boolean;
5542             config false;
5543             description
5544                 "The value is true if the device supports NACM.";
5545             reference
5546                 "6.4.10.3.8.8 of IEC/IEEE 60802";
5547         }
5548         leaf capability-yang-library {
5549             type boolean;
5550             config false;
5551             description
5552                 "The value is true if the device supports YANG library.";
5553             reference
5554                 "6.4.10.3.8.10 of IEC/IEEE 60802";
5555         }
5556         leaf capability-yang-push {
5557             type boolean;
5558             config false;
5559             description
5560                 "The value is true if the device supports YANG push.";

```

```
5561         reference
5562             "6.4.10.3.8.11 of IEC/IEEE 60802";
5563     }
5564     leaf capability-yang-notifications {
5565         type boolean;
5566         config false;
5567         description
5568             "The value is true if the device supports YANG
5569             notifications.";
5570         reference
5571             "6.4.10.3.8.12 of IEC/IEEE 60802";
5572     }
5573     leaf capability-netconf-monitoring {
5574         type boolean;
5575         config false;
5576         description
5577             "The value is true if the device supports NETCONF
5578             monitoring.";
5579         reference
5580             "6.4.10.3.8.13 of IEC/IEEE 60802";
5581     }
5582     leaf capability-netconf-client {
5583         type boolean;
5584         config false;
5585         description
5586             "The value is true if the device supports NETCONF client.";
5587         reference
5588             "6.4.10.3.8.14 of IEC/IEEE 60802";
5589     }
5590     leaf capability-cnc-config {
5591         type boolean;
5592         config false;
5593         description
5594             "The value is true if the device supports the CNC configuration.";
5595         reference
5596             "6.4.10.3.8.15 of IEC/IEEE 60802";
5597     }
5598     leaf capability-sched-traffic {
5599         type boolean;
5600         config false;
5601         description
5602             "The value is true if the device supports scheduled
5603             traffic.";
5604         reference
5605             "6.4.10.3.8.16 of IEC/IEEE 60802";
5606     }
5607     leaf capability-frame-preemption {
5608         type boolean;
5609         config false;
5610         description
5611             "The value is true if the device supports frame preemption.";
5612         reference
5613             "6.4.10.3.8.17 of IEC/IEEE 60802";
5614     }
5615 }
5616
5617 rpc ia-factory-reset {
5618     nacm:default-deny-all;
5619     description
5620         "The server resets all datastores to their factory
5621         default contents and any nonvolatile storage back to
5622         factory condition, deleting all dynamically
5623         generated files, including those containing keys,
```

certificates, logs, and other temporary files.

Depending on the factory default configuration, after being reset, the device may become unreachable on the network.

In contrast to the original factory-reset RPC in IETF RFC 8808, this RPC puts the device into a state where a subsequent configuration by a CNC component results in a functioning 60802 IA-station";

```
}
}
```

6.4.10.8.6 Module `iecieeee60802-subscribed-notifications`

```
module iecieeee60802-subscribed-notifications {
  yang-version 1.1;
  namespace
    "urn:ieee:std:60802:yang:iecieeee60802-subscribed-notifications";
  prefix ia-sn;

  import ietf-subscribed-notifications {
    prefix sn;
  }

  organization
    "IEEE 802.1 Working Group and IEC subcommittee 65C:
     Industrial networks, of IEC technical committee 65:
     Industrial-process measurement, control and automation";
  contact
    "WG-URL: http://ieee802.org/1/
     WG-EMail: stds-802-1-1@ieee.org

     Contact: IEEE 802.1 Working Group Chair
              Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              445 Hoes Lane
              Piscataway, NJ 08854
              USA

     E-mail: stds-802-1-chairs@ieee.org";
  description
    "Management objects that provide information about IEC/IEEE 60802
     IA-Stations as specified in IEC/IEEE 60802.

     Copyright (C) IEC/IEEE (2025).
     This version of this YANG module is part of IEC/IEEE 60802;
     see the standard itself for full legal notices.";

  revision 2024-02-19 {
    description "Published as part of IEC/IEEE 60802-2025.
     The following reference statement identifies each referenced
     IEEE Standard as updated by applicable amendments.";
    reference
      "IEC/IEEE 60802 TSN profile for industrial automation:
       IEC/IEEE 60802-2025.
       IEEE Std 802.1Q Bridges and Bridged Networks:
       IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
       IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
       IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
  }

  augment "/sn:subscriptions" {
```

```
5685     description
5686         "Augment subscriptions in ietf-subscribed-notifications.";
5687     leaf max-subscriptions {
5688         type uint16;
5689         config false;
5690         description
5691             "The value is the maximum number of supported NETCONF Server
5692             subscriptions.";
5693         reference
5694             "6.4.10.3.8.1 of IEC/IEEE 60802";
5695     }
5696     leaf max-on-change-subscription-leaves {
5697         type uint16;
5698         config false;
5699         description
5700             "The value is the maximum number of supported leaves for
5701             NETCONF Server on-change subscriptions according to IETF
5702             RFC 8641.";
5703         reference
5704             "6.4.10.3.8.2 of IEC/IEEE 60802";
5705     }
5706     leaf max-periodic-subscription-leaves {
5707         type uint16;
5708         config false;
5709         description
5710             "The value is the maximum number of supported leaves for
5711             NETCONF Server periodic subscriptions according to IETF
5712             RFC 8641.";
5713         reference
5714             "6.4.10.3.8.3 of IEC/IEEE 60802";
5715     }
5716     leaf min-periodic-subscription-interval {
5717         type uint16;
5718         config false;
5719         description
5720             "The value is the minimum periodic subscription interval in
5721             centiseconds (0.01 seconds) for NETCONF Server periodic
5722             subscriptions according to IETF RFC 8641.";
5723         reference
5724             "6.4.10.3.8.4 of IEC/IEEE 60802";
5725     }
5726 }
5727 }
```

5728 **6.4.10.8.7 Module iecieee60802-tt-mapping**

```
5729 module iecieee60802-tt-mapping {
5730     yang-version "1.1";
5731     namespace "urn:ieee:std:60802:yang:iecieee60802-tt-mapping";
5732     prefix tt-mapping;
5733
5734     import ieee802-dot1q-types {
5735         prefix dot1q-types;
5736     }
5737
5738     organization
5739         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5740         Industrial networks, of IEC technical committee 65:
5741         Industrial-process measurement, control and automation";
5742     contact
5743         "WG-URL: http://ieee802.org/1/
5744         WG-EMail: stds-802-1-1@ieee.org
5745 }
```

```

5746     Contact: IEEE 802.1 Working Group Chair
5747     Postal: C/O IEEE 802.1 Working Group
5748             IEEE Standards Association
5749             445 Hoes Lane
5750             Piscataway, NJ 08854
5751             USA
5752
5753     E-mail: stds-802-1-chairs@ieee.org";
5754     description
5755         "This module provides information on characteristics of traffic classes
5756         configured in a network to allow applications to map application
5757 traffic
5758         classes to them.
5759
5760     Copyright (C) IEC/IEEE (2025).
5761
5762     This version of this YANG module is part of IEC/IEEE 60802;
5763     see the standard itself for full legal notices.";
5764     revision 2024-12-13 {
5765         description
5766             "Published as part of IEC/IEEE 60802-2025.
5767
5768             The following reference statement identifies each referenced
5769             IEEE Standard as updated by applicable amendments.";
5770         reference
5771             "IEC/IEEE 60802 TSN profile for industrial automation:
5772             IEC/IEEE 60802-2025.
5773             IEEE Std 802.1Q Bridges and Bridged Networks:
5774             IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5775             IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5776             IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5777     }
5778
5779     container tt-mapping {
5780         description
5781             "Top-level container for the traffic class module.";
5782         list network {
5783             key "network-name";
5784             description
5785                 "A list of networks, each containing one or more sets of traffic
5786 class
5787                 characteristics.";
5788             leaf network-name {
5789                 type string;
5790                 description
5791                     "The network-name uniquely identifies a network.";
5792             }
5793             list traffic-type {
5794                 key "traffic-type-name";
5795                 description
5796                     "A list of traffic types in the network, each with its associated
5797                 traffic type characteristics.";
5798                 leaf traffic-type-name {
5799                     type string;
5800                     description
5801                         "The traffic-type-name uniquely identifies the traffic type.";
5802                 }
5803                 leaf cyclic {
5804                     type boolean;
5805                     description
5806                         "Indicates if cyclic transmission is mandatory (TRUE) or
5807 optional (FALSE)
5808                     for this traffic type.";

```

```

5809         reference
5810         "4.7.2 of IEC/IEEE 60802-2025";
5811     }
5812     leaf data-delivery-req {
5813         type enumeration {
5814             enum frame-latency {
5815                 value 0;
5816                 description
5817                 "Data delivery of a frame for a given Talker-Listener pair
5818                 occurs within a bounded timespan.";
5819             }
5820             enum flow-latency {
5821                 value 1;
5822                 description
5823                 "Data delivery up to a certain number of frames or data
5824 size          (including bursts of frames) occurring over a defined
5825                 period.";
5826             }
5827             enum deadline {
5828                 value 2;
5829                 description
5830                 "Data delivery of a frame to a given Listener occurs at or
5831                 before a specific point in time.";
5832             }
5833             enum no {
5834                 value 3;
5835                 description
5836                 "No special data delivery requirements.";
5837             }
5838         }
5839     }
5840     description
5841     "The data-delivery-req provides information on the data
5842     delivery requirements for this traffic type.";
5843     reference
5844     "4.7.2 of IEC/IEEE 60802-2025";
5845 }
5846 leaf time-triggered {
5847     type boolean;
5848     description
5849     "Indicates if it is required (TRUE) or optional (FALSE) that
5850     transmission occurs at a specific point in time based upon
5851     the Working Clock.";
5852     reference
5853     "4.7.2 of IEC/IEEE 60802-2025";
5854 }
5855 leaf stream-based {
5856     type boolean;
5857     default true;
5858     description
5859     "Indicates if traffic of this traffic type requires
5860     registering a stream, i.e., allocating dedicated
5861     resources, for the traffic, before a given Talker
5862     can start transmitting data (TRUE), or if registering
5863     a stream is not required (FALSE).";
5864     reference
5865     "4.7.2 of IEC/IEEE 60802-2025";
5866 }
5867 leaf vlan-id {
5868     type dot1q-types:vlanid;
5869     description
5870     "The VID this traffic type is mapped to.";
5871 }

```

```

5872     list redundancy {
5873         key "vlan-id-redundant";
5874         description
5875             "A list of VIDs this traffic type is mapped to for redundant
5876             transmission. There is one entry in the list per redundant
5877             transmission ";
5878         leaf vlan-id-redundant {
5879             type dot1q-types:vlanid;
5880             description
5881                 "The VID this traffic type is mapped for a redundant
5882 transmission.";
5883         }
5884     }
5885     leaf priority-value {
5886         type dot1q-types:priority-type;
5887         description
5888             "The PCP this traffic type is mapped to.";
5889     }
5890 }
5891 }
5892 }
5893 }

```

6.4.10.8.8 Module iecieee60802-proxy

```

5895 module iecieee60802-proxy {
5896     yang-version 1.1;
5897     namespace "urn:ieee:std:60802:yang:ieciieee60802-proxy";
5898     prefix pr;
5899
5900     import ieee802-types {
5901         prefix ieee;
5902     }
5903     import ietf-inet-types {
5904         prefix inet;
5905     }
5906
5907     organization
5908         "IEEE 802.1 Working Group and IEC subcommittee 65C:
5909         Industrial networks, of IEC technical committee 65:
5910         Industrial-process measurement, control and automation";
5911     contact
5912         "WG-URL: http://ieee802.org/1/
5913         WG-EMail: stds-802-1-1@ieee.org
5914
5915         Contact: IEEE 802.1 Working Group Chair
5916                 Postal: C/O IEEE 802.1 Working Group
5917                 IEEE Standards Association
5918                 445 Hoes Lane
5919                 Piscataway, NJ 08854
5920                 USA
5921
5922         E-mail: stds-802-1-chairs@ieee.org";
5923     description
5924         "Configuration and capability information for IEC/IEEE 60802 proxy
5925         targets as specified in IEC/IEEE 60802.
5926
5927         Copyright (C) IEC/IEEE (2025).
5928         This version of this YANG module is part of IEC/IEEE 60802;
5929         see the standard itself for full legal notices.";
5930
5931     revision 2025-03-12 {
5932         description
5933             "Published as part of IEC/IEEE 60802-2025.

```



```
5934         The following reference statement identifies each referenced
5935         IEEE Standard as updated by applicable amendments.";
5936     reference
5937         "IEC/IEEE 60802 TSN profile for industrial automation:
5938         IEC/IEEE 60802-2025.
5939         IEEE Std 802.1Q Bridges and Bridged Networks:
5940         IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
5941         IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
5942         IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
5943     }
5944
5945     container proxy {
5946         presence "The presence of this container indicates that this IA station
5947         provides proxy functionality for proxy targets";
5948         description
5949             "This container provides information about a
5950             60802 IA station with proxy functionality.";
5951         reference
5952             "IEC/IEEE 60802 - YANG Data Model";
5953         leaf max-num-proxy-targets {
5954             type uint16;
5955             config false;
5956             description
5957                 "The value indicates how many proxy targets can be handled by this
5958                 proxy component.";
5959             reference
5960                 "6.4.10.3.9.1 of IEC/IEEE 60802";
5961         }
5962         list proxy-types {
5963             key "proxy-type";
5964             config false;
5965             description
5966                 "Supported proxy types";
5967             leaf proxy-type {
5968                 type string {
5969                     pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
5970                 }
5971                 config false;
5972                 description
5973                     "OID of proxy targets handled by this proxy.";
5974                 reference
5975                     "6.4.10.3.9.2 of IEC/IEEE 60802";
5976             }
5977             list proxy-targets {
5978                 key "device";
5979                 description
5980                     "List of proxy targets that are connected to this proxy";
5981                 leaf device {
5982                     type string;
5983                     description
5984                         "The identification of the device.";
5985                     reference
5986                         "6.4.10.3.9.3 of IEC/IEEE 60802";
5987                 }
5988                 leaf device-type {
5989                     type string;
5990                     description
5991                         "The type of the device.";
5992                     reference
5993                         "6.4.10.3.9.4 of IEC/IEEE 60802";
5994                 }
5995             }
5996             container proxy-addr {
5997                 description
```

```

5997         "This container provides the address information
5998         that is used by the CNC to configure the proxy target.";
5999     leaf ip-addr {
6000         type inet:ip-address;
6001         description
6002             "The ip-address that is used by the CNC to configure
6003             the proxy target.";
6004         reference
6005             "6.4.10.3.9.5 of IEC/IEEE 60802";
6006     }
6007     leaf port {
6008         type inet:port-number;
6009         description
6010             "The port-number that is used by the CNC to configure
6011             the proxy target.";
6012         reference
6013             "6.4.10.3.9.6 of IEC/IEEE 60802";
6014     }
6015 }
6016 container device-addr {
6017     description
6018         "This container provides the address information
6019         that is used by the proxy to address the proxy target.";
6020     leaf ip-addr {
6021         type inet:ip-address;
6022         description
6023             "The ip-address that is used by the proxy to address
6024             the proxy target.";
6025         reference
6026             "6.4.10.3.9.7 of IEC/IEEE 60802";
6027     }
6028     leaf mac-addr {
6029         type ieee:mac-address;
6030         description
6031             "The mac-address that is used by the proxy to address
6032             the proxy target.";
6033         reference
6034             "6.4.10.3.9.8 of IEC/IEEE 60802";
6035     }
6036 }
6037 }
6038 }
6039 }
6040 }

```

6.4.10.8.9 Module iecieee60802-proxy-target

```

6042 module iecieee60802-proxy-target {
6043     yang-version 1.1;
6044     namespace "urn:ieee:std:60802:yang:ieciieee60802-proxy-target";
6045     prefix cd;
6046
6047     organization
6048         "IEEE 802.1 Working Group and IEC subcommittee 65C:
6049         Industrial networks, of IEC technical committee 65:
6050         Industrial-process measurement, control and automation";
6051     contact
6052         "WG-URL: http://ieee802.org/1/
6053         WG-EMail: stds-802-1-1@ieee.org
6054
6055         Contact: IEEE 802.1 Working Group Chair
6056         Postal: C/O IEEE 802.1 Working Group
6057         IEEE Standards Association

```

```
6058         445 Hoes Lane
6059         Piscataway, NJ 08854
6060         USA
6061
6062         E-mail: stds-802-1-chairs@ieee.org";
6063     description
6064         "Capability information for IEC/IEEE 60802 proxy targets as
6065         specified in IEC/IEEE 60802.
6066
6067         Copyright (C) IEC/IEEE (2025).
6068         This version of this YANG module is part of IEC/IEEE 60802;
6069         see the standard itself for full legal notices.";
6070
6071     revision 2025-01-08 {
6072         description
6073             "Published as part of IEC/IEEE 60802-2025.
6074             The following reference statement identifies each referenced
6075             IEEE Standard as updated by applicable amendments.";
6076         reference
6077             "IEC/IEEE 60802 TSN profile for industrial automation:
6078             IEC/IEEE 60802-2025.
6079             IEEE Std 802.1Q Bridges and Bridged Networks:
6080             IEEE Std 802.1Q-2022, IEEE Std 802.1Qcz-2023,
6081             IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qdj-2024,
6082             IEEE Std 802.1Qdx-2024, IEEE Std 802.1Qdy-2025.";
6083     }
6084
6085     container proxy-target {
6086         presence "The presence of this container indicates that an IA station
6087             is a proxy target that requires a proxy device";
6088         description
6089             "This container provides information about a
6090             60802 proxy target device.";
6091         reference
6092             "IEC/IEEE 60802 - YANG Data Model";
6093         leaf required-proxy-type {
6094             type string {
6095                 pattern '[0-9A-F]{2}(-[0-9A-F]{2}){2}';
6096             }
6097             config false;
6098             description
6099                 "OID of proxy that this proxy target requires.";
6100             reference
6101                 "6.4.10.3.10.1 of IEC/IEEE 60802";
6102         }
6103     }
6104 }
```

6105 6.5 Topology discovery and verification

6106 6.5.1 Topology discovery and verification requirements

6107 Electrical engineering of machines with multiple IA-stations includes the definition of the
6108 machine internal network topology (i.e. the engineered topology).

6109 The machine internal network topology includes type specific data of IA-stations (for example
6110 model name or manufacturer name) as well as instance specific data (for example IP addresses
6111 or DNS names).

The electrical engineering data of the network topology is used:

- during commissioning so that machine planning and installation are identical;
- by the TDE during operation to verify that the actual topology of the Configuration Domain matches the engineered topology;
- by maintenance staff during repair to easily identify failed IA-stations, ports, or links to be replaced.

Repair and replacement of an IA-station do not require verification of the updated engineered topology so that the TDE does not produce a verification error.

IA-stations are necessarily pre-configured when they are repaired or replaced. IA-stations report type and instance data as described in 6.5.3.

6.5.2 Topology discovery overview

6.5.2.1 General

LLDP enables the discovery of IA-stations, their external ports, and their external connectivity. A Topology Discovery Entity can query LLDP data by remote management to derive the physical network topology.

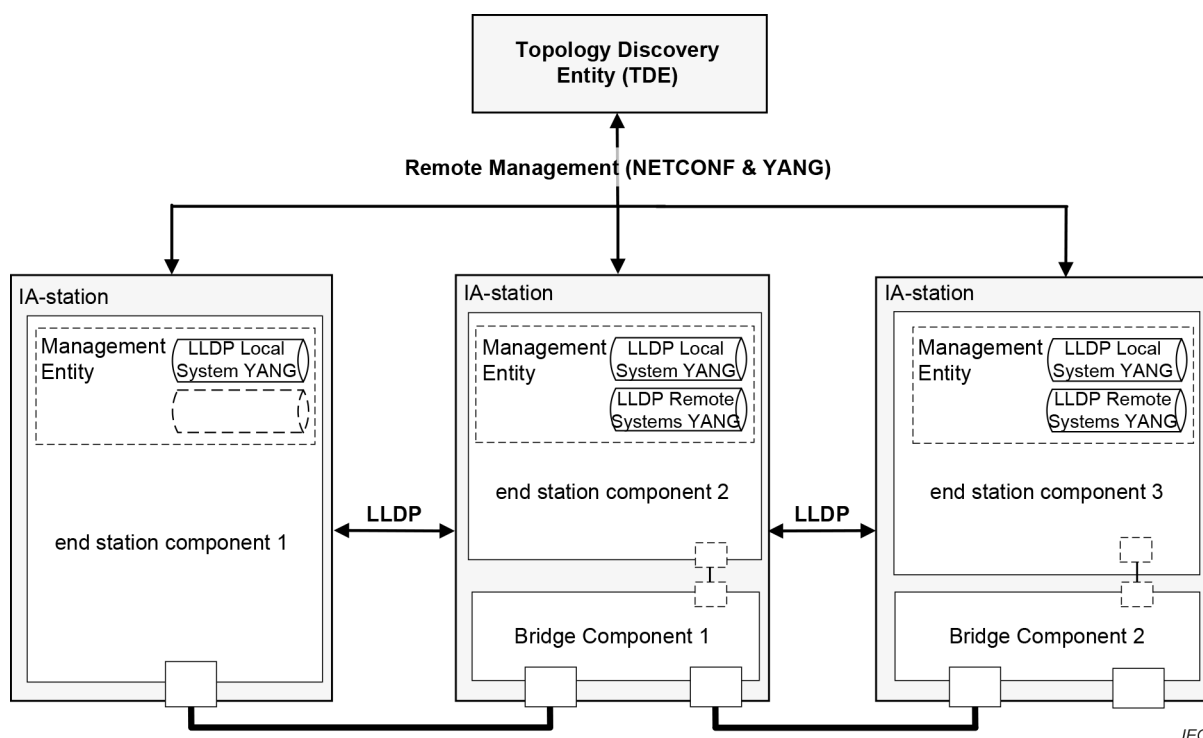


Figure 38 – Usage example of LLDP

Figure 38 illustrates a network showing the LLDP agent implementations in an IA-station consisting of a single end station component and two IA-stations with end station and Bridge components (see 4.3). The LLDP protocol is used to convey neighborhood information among peers, and NETCONF is used between the TDE and the IA-stations to query this neighborhood information from the IA-stations. This information allows the TDE to discover IA-stations and the physical network topology.

NOTE A Topology Discovery Entity (TDE) can be run from anywhere in the network with reachability to the to-be-discovered devices.

6137 IA-stations announce themselves via LLDP to support discovery by the TDE. Announcements
6138 contain the management address (see 6.5.2.4.6) and system capabilities (see 6.5.2.4.5) for the
6139 discovery operation. The announced system capabilities information enables the TDE to identify
6140 IA-stations with multiple end station and Bridge components. The TDE can use the definitions
6141 in 6.4.3 for the discovery of the internal structure of such IA-stations.

6142 To allow for operational behavior and exchanged information, IA-stations support the local
6143 system YANG (see 6.4.9.2.2). IA-stations that include a Bridge component additionally support
6144 the processing of received LLDP messages and support the remote systems YANG (see
6145 6.4.9.2.2).

6146 **6.5.2.2 LLDP operational control parameters**

6147 LLDP defines several operational parameters that control the protocol behavior (see IEEE Std
6148 802.1AB-2016, 10.5.1). These parameter definitions apply to all external ports of an IA-station.

6149 NOTE According to IEEE Std 802.1AB-2016, 9.1.1 c), changes to the local system that impact information
6150 exchanged via LLDP immediately trigger the transmission of an LLDPDU to communicate the local changes as quickly
6151 as possible to any neighboring systems.

6152 An IA-station shall support LLDP transmit mode (adminStatus enabledTxOnly) on an external
6153 end station component port and may support transmit and receive mode (adminStatus
6154 enabledRxTx) on that port (see IEEE Std 802.1AB-2016, 10.5.1).

6155 An IA-station shall support LLDP transmit and receive mode (adminStatus enabledRxTx) on an
6156 external Bridge component port (see IEEE Std 802.1AB-2016, 10.5.1).

6157 **6.5.2.3 LLDPDU transmission, reception, and addressing**

6158 The destination address to be used for LLDPDU transmission (dest-mac-address) shall be the
6159 nearest bridge group MAC address, i.e., 01-80-C2-00-00-0E, on all ports to limit the scope of
6160 LLDPDU propagation to a single physical link (see IEEE Std 802.1AB-2016, 7.1 a).

6161 NOTE IEEE Std 802.1AB-2016 defines LLDPDUs to be transmitted untagged, i.e. frames do not carry priority
6162 information for traffic class selection. At the same time, IEEE Std 802.1AB-2016 neither specifies a well-defined
6163 device-internal priority nor management capabilities for the configuration of the traffic class to be used for the
6164 transmission of LLDPDUs. It is the user's responsibility to prevent LLDPDUs from interfering with the transmission
6165 of time-critical control data.

6166 **6.5.2.4 LLDP TLV selection**

6167 **6.5.2.4.1 General**

6168 An IA-station transmitting LLDPDUs shall include the LLDP TLVs selected in 6.5.2.4 and may
6169 include additional TLVs (tlvs-tx-enable). An IA-station receiving LLDPDUs shall process
6170 LLDPDUs.

6171 Each LLDPDU shall contain the following LLDP TLVs specified in IEEE Std 802.1AB-2016, 8.5:

- 6172 • exactly one Chassis ID TLV according to 6.5.2.4.2,
- 6173 • exactly one Port ID TLV according to 6.5.2.4.3,
- 6174 • exactly one Time To Live TLV according to 6.5.2.4.4,
- 6175 • exactly one System Capabilities TLV according to 6.5.2.4.5, and
- 6176 • one or more Management Address TLVs according to 6.5.2.4.6.

6177 NOTE The concatenation of the Chassis ID and Port ID fields enables the recipient of an LLDPDU to identify the
6178 sending LLDP agent/port.

6179 **6.5.2.4.2 Chassis ID TLV**

6180 The Chassis ID field shall contain the same value for all transmitted LLDPDUs independent
6181 from the transmitting port of the IA-station, i.e. be a non-volatile identifier which is unique within
6182 the context of the administrative domain.

6183 The Chassis ID subtype field (chassis-id-subtype) should contain subtype 4, indicating that the
6184 Chassis ID field (chassis-id) contains a MAC address to achieve the Chassis ID's desired
6185 uniqueness. For IA-stations with multiple unique MAC addresses, any one of the IA-station's
6186 MAC addresses may be used and shall be the same for all external ports of that IA-station.

6187 **6.5.2.4.3 Port ID TLV**

6188 The Port ID field shall contain the same value for all transmitted LLDPDUs for a given external
6189 port, i.e. be a non-volatile, IA-station-unique identifier of the LLDPDU-transmitting port.

6190 The Port ID subtype field (port-id-subtype) should contain subtype 5, indicating that the Port ID
6191 field contains the port interface name (name) according to IETF RFC 8343.

6192 IA-stations should restrict the system-defined port ID to read-only access and a maximum name
6193 length of 255 characters. The names should match the port names printed on the chassis.

6194 **6.5.2.4.4 Time To Live TLV**

6195 The Time To Live value shall be set according to IEEE Std 802.1AB-2016, 8.5.4.

6196 **6.5.2.4.5 System capabilities TLV**

6197 An IA-station consisting of a single end station component shall set the system capabilities and
6198 enabled capabilities fields (system-capabilities-supported, system-capabilities-enabled) to
6199 Station Only (i.e. bit 8 set to 1) for all transmitted LLDPDUs.

6200 An IA-station consisting of at least one end station component and at least one Bridge
6201 component shall set the system capabilities and enabled capabilities fields to Station Only (i.e.
6202 bit 8 set to 1) and C-VLAN component (i.e. bit 9 set to 1) for all transmitted LLDPDUs.

6203 NOTE The combination of the Station Only and C-VLAN component flags is used as a marker indicating to the TDE
6204 that the internal structure of the IA-station consists of multiple components. This is a deliberate deviation from IEEE
6205 Std 802.1AB-2016, Table 8-4, which states in a footnote: "The Station Only capability is intended for devices that
6206 implement only an end station capability, and for which none of the other capabilities in the table apply. Bit 8 therefore
6207 is not set in conjunction with any other bits."

6208 **6.5.2.4.6 Management address TLV**

6209 An IA-station shall announce at least one IPv4 address by which its Management entity (see
6210 4.3) can be reached (management-address-tx-port).

6211 **6.5.2.5 LLDP remote systems data**

6212 An IA-station supporting the remote systems YANG shall be able to store information from at
6213 least one neighbor per external port.

6214 Receiving LLDPDUs from more neighbors than supported on a given port shall result in the last
6215 one received being saved to the remote systems YANG as described in
6216 IEEE Std 802.1AB-2016, 9.2.7.7.5.

6.5.3 Topology verification overview

Topology verification checks discovered topologies against engineered topologies. Topology verification data includes for every IA-station:

- model name,
- manufacturer name, and
- management address.

Topology verification data includes for every external port of an IA-station:

- port name, and
- remote connection (i.e. management address and port name of connected IA-station).

To support topology verification IA-stations shall support LLDP YANG data as specified in 6.4.9.2.2 and Hardware Management YANG data as specified in 6.4.9.2.5.8.

IA-station hardware instance specific data like MAC addresses or serial numbers are not considered for topology verification. This kind of data changes after a repair and replacement operation and thus, induces a topology verification error.

6.6 CNC

6.6.1 General

Subclause 6.6 describes stream destination MAC address handling at the CNC.

6.6.2 Stream destination MAC address range

A CNC manages the destination MAC address for requested streams. This destination MAC address together with the VID identifies the path used for these streams. Thus, a stream destination MAC address is unique together with the VID in a Configuration Domain.

Figure 39 shows the possible selections of a CNC for a contiguous address range. The CNC selects an OUI and an offset of the address range for the stream destination MAC addresses.

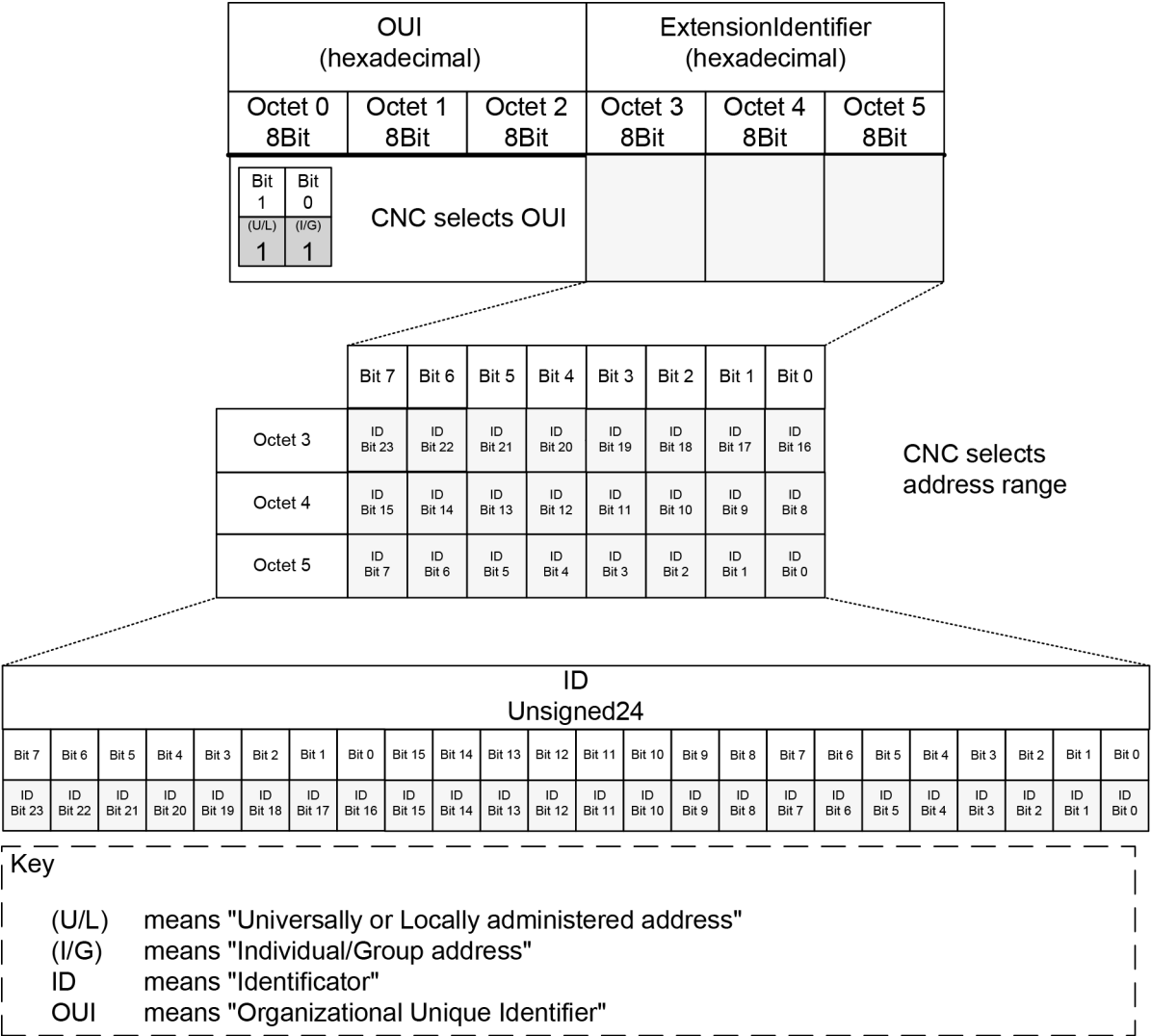
An address range of 2 048 stream destination MAC addresses allows together with a VID the usage of 2 048 streams. Each additional VID used for streams allows an additional 2 048 streams.

EXAMPLE

CNC selected OUI:= 00-80-C2

CNC selected address range:= 0.2047

CNC selected VID:= 101



6247

IEC

6248

Figure 39 – Stream destination MAC Address

Annex A (normative)

PCS proforma – Time-sensitive networking profile for industrial automation

A.1 General⁷

The supplier of an implementation that is claimed to conform to the profile specified in this document shall complete the corresponding Profile Conformance Statement (PCS) proforma, which is presented in a tabular format based on the format used for Protocol Implementation Conformance Statement (PICS) proformas.

The tables do not contain an exhaustive list of all requirements that are stated in the referenced standards; for example, if a row in a table asks whether the implementation is conformant to Standard X, and the answer "Yes" is chosen, then it is assumed that it is possible, for that implementation, to fill out the PCS proforma specified in Standard X to show that the implementation is conformant; however, the tables in this document only further refine those elements of conformance to Standard X where particular answers are required for the profiles specified here.

A completed PCS proforma is the PCS for the implementation in question. The PCS is a statement of which capabilities and options of the protocol have been implemented. The PCS can have several uses, including use by the following.

- a) Protocol implementer, as a checklist to reduce the risk of failure to conform to the document through oversight.
 - b) Supplier and acquirer, or potential acquirer, of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PCS proforma.
 - c) User, or potential user, of the implementation, as a basis for initially checking the possibility of interworking with another implementation.
- NOTE While interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PCS.
- d) Protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.
 - e) The user, to verify whether the IA-station, as described by the PCS, fulfills use-case requirements.

A.2 Abbreviated terms and special symbols

A.2.1 Status symbols

- | | |
|------|---|
| M | mandatory |
| O | optional |
| O.n | optional, but support of at least one of the group of options labeled by the same numeral n is required |
| X | prohibited |
| pred | conditional-item symbol, including predicate identification (see A.3.4) |
| ¬ | logical negation, applied to a conditional item's predicate |

⁷ Copyright release for the PCS: Users of this document may freely reproduce the PCS contained in this document so that they can be used for their intended purpose.

6291 **A.2.2 General abbreviated terms**

6292 N/A not applicable

6293 PCS Profile Conformance Statement

6294 **A.3 Instructions for completing the PCS proforma**

6295 **A.3.1 General structure of the PCS proforma**

6296 The first part of the PCS proforma, implementation identification and protocol summary, is to
6297 be completed as indicated with the information necessary to identify fully both the supplier and
6298 the implementation.

6299 The main part of the PCS proforma is a fixed-format questionnaire, divided into several
6300 subclauses, each containing a number of individual items. Answers to the questionnaire items
6301 are to be provided in the rightmost column, either by simply marking an answer to indicate a
6302 restricted choice (usually Yes or No) or by entering a value or a set or range of values. There
6303 are some items where two or more choices from a set of possible answers can apply; all relevant
6304 choices are to be marked. Each item is identified by an item reference in the first column. The
6305 second column contains the question to be answered; the third column records the status of
6306 the item – whether support is mandatory, optional, or conditional (see also A.3.4). The fourth
6307 column contains the reference or references to the material that specifies the item in the main
6308 body of this document, and the fifth column provides the space for the answers.

6309 The PCS indicates support of one of the conformance classes, ccA or ccB, per bridge and end-
6310 station component, specified in this profile.

6311 A single IA-station can incorporate the functionality of one or more of the functions listed in this
6312 PCS. For example, an IA-station could have both an end station component and a Bridge
6313 component.

6314 A supplier can also provide (or be required to provide) further information, categorized as either
6315 additional information (see A.3.2) or exception information (see A.3.3). When present, each
6316 kind of further information is to be provided in a further subclause of items labeled Ai or Xi,
6317 respectively, for cross-referencing purposes, where (i) is any unambiguous identification for the
6318 item (for example, simply a numeral). There are no other restrictions on its format and
6319 presentation.

6320 A completed PCS proforma, including any Additional Information and Exception Information, is
6321 the Protocol Implementation Conformation Statement for the implementation in question.

6322 NOTE Where an implementation is capable of being configured in more than one way, a single PCS can be used
6323 to describe all such configurations. However, the supplier has the choice of providing more than one PCS, each
6324 covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer
6325 presentation of the information.

6326 **A.3.2 Additional information**

6327 Items of Additional Information allow a supplier to provide further information intended to assist
6328 the interpretation of the PCS. It is not intended or expected that a large quantity be supplied,
6329 and a PCS can be considered complete without any such information. Examples can be an
6330 outline of the ways in which a (single) implementation can be set up to operate in a variety of
6331 environments and configurations, or information about aspects of the implementation that are
6332 outside the scope of this document but that have a bearing on the answers to some items.

6333 References to items of Additional Information can be entered next to any answer in the
6334 questionnaire and can be included in items of Exception Information.

A.3.3 Exception information

It can occasionally happen that a supplier wishes to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this item. Instead, the supplier shall write the missing answer into the Support column, together with an Xi reference to an item of Exception Information and shall provide the appropriate rationale in the Exception item itself.

An implementation for which an Exception item is required in this way does not conform to this document.

NOTE A possible reason for the situation described previously is that a defect in this document has been reported, a correction for which is expected to change the requirement not met by the implementation.

A.3.4 Conditional status

A.3.4.1 Conditional items

The PCS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply (mandatory or optional) are dependent on whether certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the "Not Applicable" (N/A) answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "pred: S" where pred is a predicate as described in A.3.4.2, and S is a status symbol, M or O.

If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate: the answer column is to be marked in the usual way. If the value of the predicate is false, the "Not Applicable" (N/A) answer is to be marked.

A.3.4.2 Predicates

A predicate is one of the following:

- a) an item-reference for an item in the PCS proforma: the value of the predicate is true if the item is marked as supported and is false otherwise,
- b) a predicate-name, for a predicate defined as a Boolean expression constructed by combining item-references using the Boolean operator OR: the value of the predicate is true if one or more of the items is marked as supported, or
- c) the logical negation symbol "¬" prefixed to an item-reference or predicate-name: the value of the predicate is true if the value of the predicate formed by omitting the "¬" symbol is false, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

A.4 Common requirements

A.4.1 Instructions

One instance of Clause A.4 shall be filled out per IA-station.

A.4.2 Implementation identification

The entire PCS proforma is a form that shall be filled out by a supplier according to Table A.1.

Table A.1 – Implementation identification template

Supplier	
Contact point for queries about the PCS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification, for example, name(s) and version(s) of machines and operating system names	

Only the first three items are required for all implementations; other information can be completed as appropriate in meeting the requirement for full identification. The terms "Name" and "Version" should be interpreted appropriately to correspond with a supplier's terminology (for example, Type, Series, Model).

A.4.3 Profile summary

Table A.2 shows the profile summary template.

Table A.2 – Profile summary template

Identification of profile specification	IEC/IEEE 60802 – Time-Sensitive Networking profile for industrial automation			
Identification of amendments (Amd) and corrigenda (Corr) to the PCS proforma that have been completed as part of the PCS	Amd	:	Corr	:
	Amd	:	Corr	:
Have any Exception items been required? (See A.3.3: the answer "Yes" means that the implementation does not conform to IEC/IEEE 60802)	No	[]	Yes	[]
Date of Statement				

A.4.4 Implementation summary

The form in Table A.3 is used to indicate the type of system that the PCS describes.

Table A.3 – Implementation type

Item	Feature	Status	References	Support
BC-CCA-N	State the number of Conformance Class A bridge components implemented by the IA-station.	O	5.7.2, 5.8.2	Number _____
BC-CCB-N	State the number of Conformance Class B bridge components implemented by the IA-station.	O	5.7.3, 5.8.3	Number _____
ESC-CCA-N	State the number of Conformance Class A end station components implemented by the IA-station.	O.1	5.9.2, 5.10.2	Number _____
ESC-CCB-N	State the number of Conformance Class B end station components implemented by the IA-station.	O.1	5.9.3, 5.10.3	Number _____
CNC	Does the IA-station include a CNC?	O	5.5.4.1 m)	Yes [] No []
CUC	Does the IA-station include a CUC?	O	5.5.4.1 n)	Yes [] No []
PRX	Does the IA-station support proxy functionality?	O	5.5.4.1 o)	Yes [] No []

A.5 IA-station requirements and options

A.5.1 Instructions

One instance of Clause A.5 shall be filled out for an IA-station.

A.5.2 IA-station requirements

The form in Table A.4 is used to indicate the IA-station requirements.

Table A.4 – IA-station requirements

Item	Feature	Status	References	Support
IASTA-1	Does the IA-station support PHY and MAC requirements for external ports?	M	5.5.1	Yes []
IASTA-2	Does the IA-station support topology discovery requirements?	M	5.5.2	Yes []
IASTA-3	Does the IA-station support requirements for time synchronization?	M	5.5.3	Yes []
IASTA-4	Does the IA-station provide a digital data sheet?	M	5.5.4.3	Yes []

A.5.3 Management Requirements for non-proxied IA-stations

The form in Table A.5 is used to indicate management requirements for IA-stations containing conformance class A components

Table A.5 – Additional requirements for non-proxied IA-stations

Item	Feature	Status	References	Support
IASTA-5	Does the IA-station support requirements for Secure management exchanges?	M	5.5.4.1 a) through l)	Yes []
IASTA-6	Number of of Dynamic Subscriptions to YANG Events and Datastores over NETCONF	M	5.5.4.1 h)	Number _____
IASTA-7	Does the IA-station support management YANG modules?	M	5.5.4.2	Yes []

A.5.4 IA-station PHY and MAC options for external ports

The form in Table A.6 is used to indicate PHY and MAC options for external ports.

Table A.6 – IA-station PHY and MAC options

Item	Feature	Status	References	Support
DOT3-1	Does the IA-station support PoE over 2 pairs?	O	5.6.1 a)	Yes [] No [] N/A []
DOT3-2	Does the IA-station support Power Interfaces?	O	5.6.1 b)	Yes [] No [] N/A []
DOT3-3	Does the IA-station support PoE?	O	5.6.1 c)	Yes [] No [] N/A []

A.5.5 IA-station options for time synchronization

The form in Table A.7 is used to indicate options for time synchronization.

6408

Table A.7 – IA-station time synchronization options

Item	Feature	Status	References	Support
PTP-1	Does the IA-station support media-independent TimeTransmitter capability according to IEEE Std 802.1AS-2020, 5.4.2 item b) as amended by IEEE Std 802.1ASdr-2024?	O	5.6.2 a)	Yes [] No []
PTP-2	Does the IA-station support Grandmaster PTP Instance capability Grandmaster PTP Instance capability according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdr-2024 and IEEE Std 802.1ASdm-2024, 5.4.2 item c)?	O	5.6.2 b)	Yes [] No []
PTP-3	Does the IA-station support more than one PTP port as a PTP Relay Instance according to IEEE Std 802.1AS-2020, 5.4.2 item d)?	O	5.6.2 c)	Yes [] No []
PTP-4	Does the IA-station support hot standby redundancy requirements according to IEEE Std 802.1AS-2020 as amended by IEEE Std 802.1ASdm-2024, 5.4.2, item m)?	O	5.6.2 d)	Yes [] No []

6409

6410 **A.5.6 IA-station secure management exchange options**

6411 The form in Table A.8 is used to indicate options for secure management exchange.

6412

Table A.8 – IA-station secure management exchange options

Item	Feature	Status	References	Support
SECMGMT-5	Does the IA-station support Writable-Running capability?	O	5.6.3 a)	Yes [] No []
SECMGMT-6	Does the IA-station support Confirmed Commit capability?	O	5.6.3 b)	Yes [] No []
SECMGMT-7	Does the IA-station support Distinct Startup capability?	O	5.6.3 c)	Yes [] No []
SECMGMT-8	Does the IA-station support URL capability?	O	5.6.3 d)	Yes [] No []
SECMGMT-9	Does the IA-station support XPath capability?	O	5.6.3 e)	Yes [] No []
SECMGMT-10	Does the IA-station support NETCONF-over-TLS server supporting TLS version 1.3	O	5.6.3 f)	Yes [] No []

6413

6414 **A.5.7 Proxy target requirements**

6415 The form in Table A.9 is used to indicate requirements for proxy targets.

6416

Table A.9 – Proxy target requirements

Item	Feature	Status	References	Support
PT-1	Does the IA-station support Secure management exchanges?	O	5.6.4 a)	Yes [] No []
PT-1	Does the IA-station support Secure management exchanges using a management proxy?	O	5.6.4 b)	Yes [] No []

6417

6418 **A.6 Bridge component**

6419 **A.6.1 Bridge component requirements**

6420 The form in Table A.10 is used to indicate bridge component requirements.

Table A.10 – Bridge component requirements

Item	Feature	Status	References	Support
BC-1	Does the bridge component support the common bridge component requirements?	M	5.7.1	Yes []
BC-2	Does the bridge component support ccA bridge component requirements?	O.2	5.7.2	Yes [] No []
BC-3	Does the bridge component support ccB bridge component requirements?	O.2	5.7.3	Yes [] No []

A.6.2 Instructions

One instance of Clause A.6 shall be filled out per bridge component implemented by an IA-station.

A.6.3 Bridge Common bridge component options

The form in Table A.11 is used to indicate bridge component options.

Table A.11 – Common bridge component options

Item	Feature	Status	References	Support
BC-4	Does the bridge component support the operation of the credit-based shaper algorithm?	O	5.8.1	Yes [] No []

A.6.4 Conformance class A bridge component options

The form in Table A.12 is used to indicate options for bridge components conforming to conformance class A.

Table A.12 – Conformance class A bridge component options

Item	Feature	Status	References	Support
CCA-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.2 a)	Yes [] No [] N/A []
CCA-BC-2	Does the bridge component support more than 2 PTP instances?	O	5.8.2 b)	Yes [] No [] N/A []
CCA-BC-3	State the number of PTP instances supported by the bridge component.	CCA-BC-2:M	5.8.2 b)	Number ____
CCA-BC-4	Does the bridge component support enhancements for scheduled traffic for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 c)	Yes [] No [] N/A []
CCA-BC-5	Does the bridge component support frame preemption for the 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s data rates?	O	5.8.2 d)	Yes [] No [] N/A []

A.6.5 Conformance class B bridge component options

The form in Table A.13 is used to indicate options for bridge components conforming to conformance class B.

Table A.13 – Conformance class B bridge component options

Item	Feature	Status	References	Support
CCB-BC-1	Does the bridge component support any of the common bridge component options?	O	5.8.3 a)	Yes [] No [] N/A []
CCB-BC-2	Does the bridge component support more than 4 but not more than 8 egress queues?	O	5.8.3 b)	Yes [] No [] N/A []

CCB-BC-3	State the number of egress queues supported by the bridge component.	CCB-BC-2:M	5.8.3 b)	Number ____
CCB-BC-4	Does the bridge component support more than 1 PTP Instance?	O	5.8.3 c)	Yes [] No [] N/A []
CCB-BC-5	State the number of PTP Instances supported by the bridge component.	CCB-BC-4:M	5.8.3 c)	Number ____
CCB-BC-6	Does the bridge component support enhancements for scheduled traffic?	O	5.8.3 d)	Yes [] No [] N/A []
CCB-BC-7	Does the bridge component support frame preemption?	O	5.8.3 e)	Yes [] No [] N/A []

6439

6440 **A.7 End station component**

6441 **A.7.1 Instructions**

6442 One instance of Clause A.7 shall be filled out per end station component implemented by an
6443 IA-station.

6444 **A.7.2 Common end station component requirements**

6445 The form in Table A.14 is used to indicate common requirements for end station components.

6446 **Table A.14 – Common end station component requirements**

Item	Feature	Status	References	Support
ESC-1	Does the end station component support the common end station component requirements?	M	5.9.1	Yes []
ESC-2	Does the end station component support the ccA end station component requirements?	O.3	5.9.2	Yes [] No []
ESC-3	Does the end station component support the ccB end station component requirements?	O.3	5.9.3	Yes [] No []

6447

6448 **A.7.3 Common end station component options**

6449 The form in Table A.15 is used to indicate options for end station components.

6450 **Table A.15 – Common end station component options**

Item	Feature	Status	References	Support
ESC-4	Does the end station component support the operation of the credit-based shaper?	O	5.10.1 a)	Yes [] No []
ESC-5	Does the end station component support talker end system behaviors?	O	5.10.1 b)	Yes [] No []
ESC-6	Does the end station component support listener end system behaviors?	O	5.10.1 c)	Yes [] No []

6451

6452 **A.7.4 Conformance class A end station component options**

6453 The form in Table A.16 is used to indicate options for end station components conforming to
6454 conformance class A.

6455 **Table A.16 – Conformance class A end station component options**

Item	Feature	Status	References	Support
CCA-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.2 a)	Yes [] No [] N/A []

CCA-ESC-2	Does the end station component support more than 2 PTP Instances?	O	5.10.2 b)	Yes [] No [] N/A []
CCA-ESC-3	State the number of PTP Instances supported by the end-station component.	CCA-ESC-2:M	5.10.2 b)	Number ____
CCA-ESC-4	Does the end station component support enhancements for scheduled traffic for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 c)	Yes [] No [] N/A []
CCA-ESC-5	Does the end station component support requirements for frame preemption for data rates 10 Mb/s, 2,5 Gb/s, 5 Gb/s, or 10 Gb/s?	O	5.10.2 d)	Yes [] No [] N/A []

6456

6457 **A.7.5 Conformance class B end station component options**

6458 The form in Table A.17 is used to indicate options for end station components conforming to
 6459 conformance class B.

6460 **Table A.17 – Conformance class B end station component options**

Item	Feature	Status	References	Support
CCB-ESC-1	Does the end station component support any of the common end station component options?	O	5.10.3 a)	Yes [] No [] N/A []
CCB-ESC-2	Does the end station component support one or more PTP Instances?	O	5.10.3 b)	Yes [] No [] N/A []
CCB-ESC-3	State the number of PTP Instances supported by the end-station component.	CCB-ESC-2:M	5.10.3 b)	Number ____
CCB-ESC-4	Does the end station component support enhancements for scheduled traffic?	O	5.10.3 c)	Yes [] No [] N/A []
CCB-ESC-5	Does the end station component support requirements for frame preemption?	O	5.10.3 d)	Yes [] No [] N/A []

6461

Annex B (informative)

Representative Configuration Domain

The following quantities are representative of what can be supported in a single Configuration Domain.

- IA-stations: 1 024.
- Network diameter: 64.
- Streams per IA-controller for IA-controller to IA-device (C2D) communication:
 - 512 Talker and ≥ 512 Listener streams, and
 - 1 024 Talker and $\geq 1\,024$ Listener streams in case of seamless redundancy.
- Streams per IA-controller for IA-controller to IA-controller (C2C) communication:
 - 64 Talker and ≥ 64 Listener streams, and
 - 128 Talker and ≥ 128 Listener streams in case of seamless redundancy.
- Streams per IA-device for IA-device-to-IA-device (D2D) communication:
 - 2 Talker and 2 Listener streams, and
 - 4 Talker and 4 Listener streams in case of seamless redundancy.
- Example calculation of data flow quantities for eight PLCs – without seamless redundancy:
 - $8 \times 512 \times 2 = 8\,192$ streams for C2D communication,
 - $8 \times 64 \times 2 = 1\,024$ streams for C2C communication, and
 - $(8\,192 + 1\,024) \times 2\,000 = 18\,432\,000$ octets data for all streams.

Annex C (informative)

Description of clock control system

C.1 Clock control system introduction

Annex C provides an introductory discussion of a basic clock control system. For more detailed information, see the Bibliography References ITU-T Recommendation G.8260 and ITU-T Series G Supplement 65.

Figure C.1 shows a basic control system model that uses a proportional plus integral (PI) controller. This is meant to be reference model, i.e. it is not meant to specify an implementation. Requirements for the clock control system can be expressed using parameters (e.g. 3 dB bandwidth, gain peaking, frequency response) that are based on this reference model. Any implementation whose parameters are within the requirements is considered to be acceptable. For example, the model of Figure C.1 is expressed in the analog domain (i.e. s-domain), and is shown to be second order. An actual implementation can be digital, and can be higher order, as long as it meets the respective requirements.

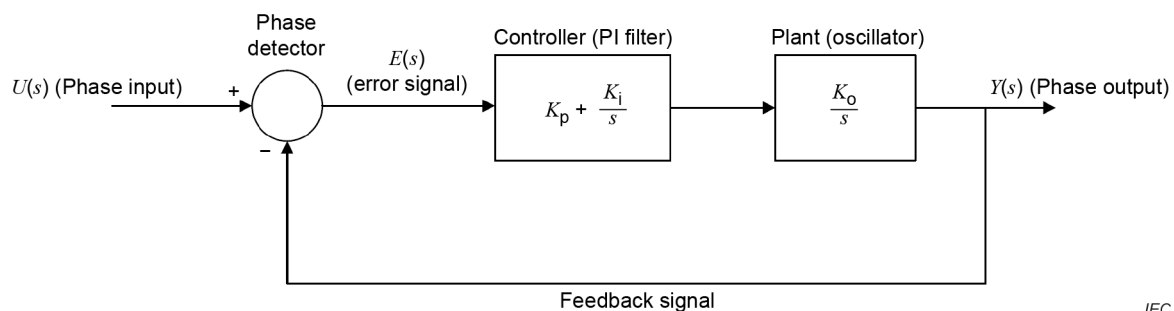


Figure C.1 – Reference model for clock control system

In Figure C.1, the plant, i.e. the entity being controlled, represents the clock oscillator. It is desired that the phase output, $y(t)$, of the oscillator follows the phase input, $u(t)$, as closely as possible (the signals are shown in the frequency domain (i.e. as Laplace Transforms) in Figure C.1; however, they can equivalently be expressed in the time domain, with t representing time). The parameter K_o is the oscillator gain; the oscillator frequency is equal to the oscillator input multiplied by K_o . In some implementations the input signal to the oscillator is a voltage, and the oscillator is referred to as a voltage-controlled oscillator (VCO). However, other implementations are possible, e.g. digital implementations, where the oscillator is a digital controlled oscillator (DCO). Since the input to the oscillator depends on the implementation, it is not labeled in Figure C.1.

The control system of Figure C.1 uses negative feedback to enable the phase output to follow the phase input. The phase detector computes the difference between the input and output signals to produce the error signal $e(t)$. The error signal is then filtered by the PI filter to produce the input to the oscillator. The filter is referred to as a PI filter because its output is the sum of the proportional gain, K_p , multiplied by the error signal and the integral gain, K_i , multiplied by the integral of the error signal. The gains K_o , K_p , and K_i must be chosen such that the performance of the control system is acceptable, i.e. the time-domain behavior of the output with respect to the input is acceptable. However, an alternative set of parameters, which are more convenient, can be defined in terms of K_o , K_p , and K_i ; this is done in Clause C.2.

C.2 Transfer function for control system

From the block diagram of Figure C.1, the input and output are related by:

$$Y(s) = \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right) (U(s) - Y(s)) \quad (\text{C.1})$$

where

$Y(s)$ is the phase output, expressed in the s-domain;

$U(s)$ is the phase input, in the s-domain;

K_p is the proportional gain;

K_i is the integral gain;

K_o is the oscillator gain.

or

$$Y(s) = \frac{\left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)}{1 + \left(K_p + \frac{K_i}{s} \right) \left(\frac{K_o}{s} \right)} U(s) \quad (\text{C.2})$$

This can be simplified by multiplying the numerator and denominator by s^2 to produce:

$$Y(s) = H(s)U(s) \quad (\text{C.3})$$

where the transfer function $H(s)$ is given by:

$$H(s) = \frac{K_p K_o s + K_i K_o}{s^2 + K_p K_o s + K_i K_o} \quad (\text{C.4})$$

In Formula (C.4), the parameter K_o does not appear independently of K_p and K_i ; rather, only the products $K_p K_o$ and $K_i K_o$ appear. The plant and PI filter could have been combined in the model of Figure C.1; this is consistent with the fact that the exact nature of the signal between the PI filter and plant is unimportant in this reference model. The units of $K_p K_o$ are (time)⁻¹ and the units of $K_i K_o$ are (time)⁻². The frequency units are the same as the units of s , e.g. if s has units rad/s, then $K_p K_o$ has units rad/s and $K_i K_o$ has units (rad/s)². The integration operation in the plant results in the transfer function being dimensionless, which is consistent with the fact that the input and output of the control system both have units of phase.

6542 The transfer function can be expressed in an equivalent form by defining the undamped natural
6543 frequency, ω_n , and damping ratio, ζ :

$$H(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (\text{C.5})$$

6544

6545 where

6546 ζ is the damping ratio;6547 ω_n is the undamped natural frequency expressed in rad/s.6548 And, ζ and ω_n are given by:

$$\omega_n = \sqrt{K_i K_o} \quad (\text{C.6})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{K_p}{2} \sqrt{\frac{K_o}{K_i}} \quad (\text{C.7})$$

6549

6550 In the Formula (C.7), the first form shows explicitly that ζ depends only on the products $K_p K_o$
6551 and $K_i K_o$.

6552 C.3 Frequency response for control system

6553 The frequency response is obtained by setting $s = j\omega$ in Formula (C.5) and taking the absolute
6554 value (here j rather than i is used for $\sqrt{-1}$ to avoid confusion with other uses of i), where ω is
6555 the frequency in rad/s. The result is:

$$|H(j\omega)| = \left| \frac{2\zeta\omega_n \omega j + \omega_n^2}{-\omega^2 + \omega_n^2 + 2\zeta\omega_n \omega j} \right| = \left(\frac{4\zeta^2 \omega_n^2 \omega^2 + \omega_n^4}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2 \omega_n^2 \omega^2} \right)^{1/2} \quad (\text{C.8})$$

6556

6557 Dividing the numerator and denominator of Formula (C.7) by ω_n^4 and defining the dimensionless
6558 frequency $x = \omega/\omega_n$ produces:

$$|H(j\omega)| = \left(\frac{4\zeta^2 x^2 + 1}{(1 - x^2)^2 + 4\zeta^2 x^2} \right)^{1/2} \quad (\text{C.9})$$

6559

where

ζ is the damping ratio;

ω is the frequency expressed in rad/s;

$x = \omega/\omega_n$.

Figure C.2 contains plots of frequency response (Formula (C.9)) versus dimensionless frequency x , on a log-log scale, for damping ratio ζ equal to 0,3, 0,5, 0,707, 1,0, 2,0, 3,0, 4,0, and 5,0. It is seen that the frequency response is very close to 1 for values of dimensionless frequency much less than 1 (i.e. for $\omega \ll \omega_n$). The frequency response increases as the frequency approaches the undamped natural frequency (i.e. as dimensionless frequency approaches 1) and reaches a peak for dimensionless frequency slightly less than 1. The frequency response then decreases, eventually having a slope (i.e. roll-off) of 20 dB/decade (i.e. frequency response decreases by a factor of 10 for every factor of 10 increase in x for $x \gg 1$). Figure C.3 shows the detail of frequency response for x in the range 0,1 to 10.

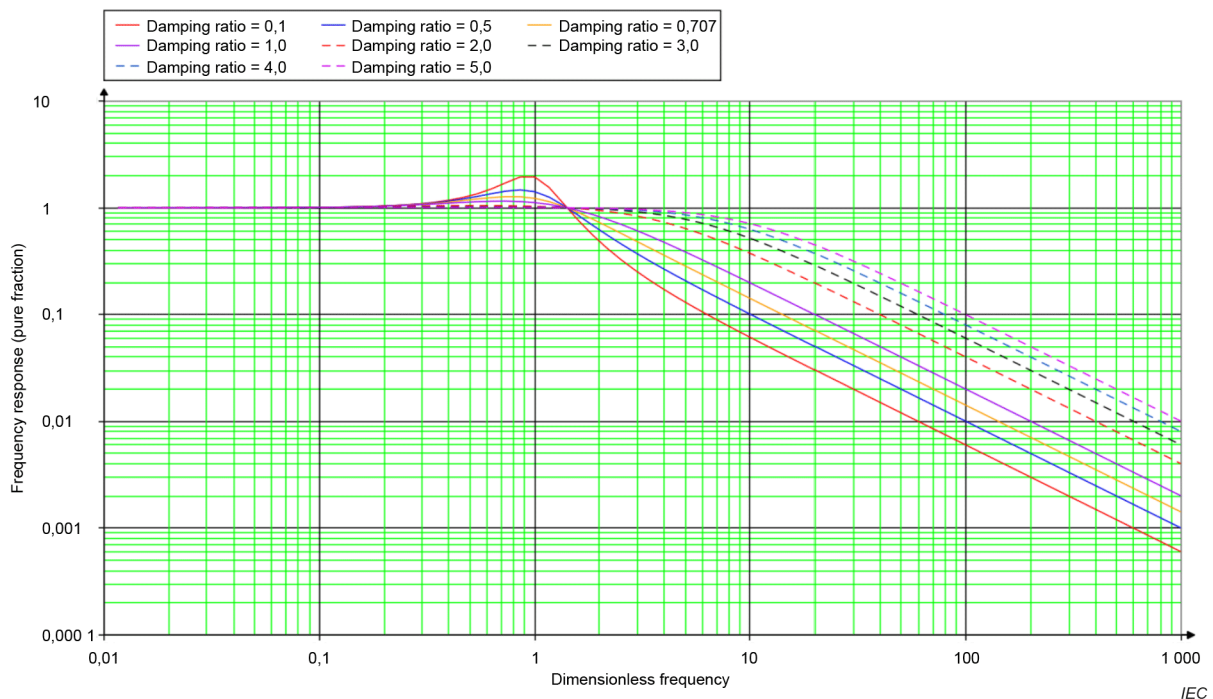


Figure C.2 – Frequency response for the control system of Figure C.1

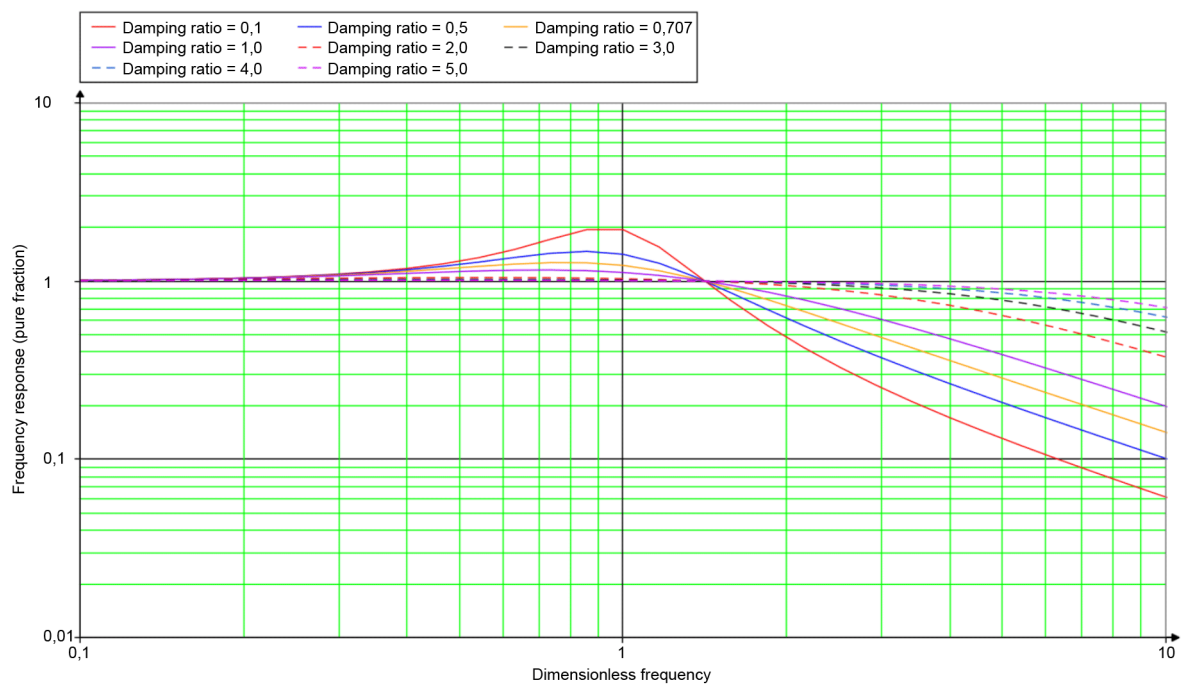


Figure C.3 – Detail of frequency response for the control system of Figure C.1 for dimensionless frequency in the range 0,1 to 10

In addition to undamped natural frequency ω_n and damping ratio ζ , the parameters 3 dB bandwidth and gain peaking are often used when specifying clock performance. The 3 dB bandwidth is defined as the value of frequency for which the frequency response is equal to -3 dB. Since dB is given by 10 multiplied by the logarithm to base 10 of the power ratio, which is 20 multiplied by the logarithm to base 10 of the amplitude ratio, -3 dB corresponds to the value $10^{-3/20}$. The 3 dB bandwidth can be computed by setting Formula (C.8) equal to $10^{-3/20}$ and solving for x in terms of ζ . This is equivalent to setting the quantity in parentheses (i.e. inside the square root) in Formula (C.8) equal to $10^{-3/10}$ and solving for x . Now, $10^{-3/10}$ is approximately equal to 0,501 2, i.e. it is very close to $\frac{1}{2}$. Then the 3dB bandwidth can be obtained by solving the following formula for x in terms of ζ :

$$\frac{4\zeta^2 x^2 + 1}{(1-x^2)^2 + 4\zeta^2 x^2} = \frac{1}{2} \quad (\text{C.10})$$

or

$$x^4 - 2(2\zeta^2 + 1)x^2 - 1 = 0 \quad (\text{C.11})$$

The result is:

$$x = \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (\text{C.12})$$

6592 or

$$\omega_{3\text{dB}} = \omega_n \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2 + 1)^2 + 1} \right]^{1/2} \quad (\text{C.13})$$

6593

6594 The gain peaking is the maximum value of the frequency response, in dB. It is computed by
 6595 differentiating Formula (C.8) with respect to x , setting the result to zero, solving for x , and then
 6596 substituting this value of x into Formula (C.8) to obtain the maximum. The result is:

$$H_p = \left[1 - 2\alpha - 2\alpha^2 + 2\alpha(2\alpha + \alpha^2)^{1/2} \right]^{-1/2} \quad (\text{C.14})$$

6597

6598 where α is related to damping ratio by:

$$\alpha = \frac{1}{4\zeta^2} \quad (\text{C.15})$$

6599

6600 and H_p is the gain peaking expressed as a pure fraction. The gain peaking in dB is equal to
 6601 $20 \cdot \log_{10} H_p$. In some cases, it is necessary to compute damping ratio from gain peaking. The
 6602 result for this is:

$$\alpha = \frac{(1-q)(1+\sqrt{1-q})}{2q} \quad (\text{C.16})$$

6603

6604 where

$$q = \frac{1}{H_p^2} \quad (\text{C.17})$$

6605

6606 Damping ratio is obtained from α using Formula (C.15).

6607 If 3 dB bandwidth and gain peaking are given, damping ratio can be obtained using Formulas
 6608 (C.15), (C.16) and (C.17). Undamped natural frequency can then be obtained using Formula
 6609 (C.13).

6610 Figure C.4 shows gain peaking, expressed as a pure fraction, as a function of damping ratio.
 6611 Figure C.5 shows gain peaking in dB as a function of damping ratio.

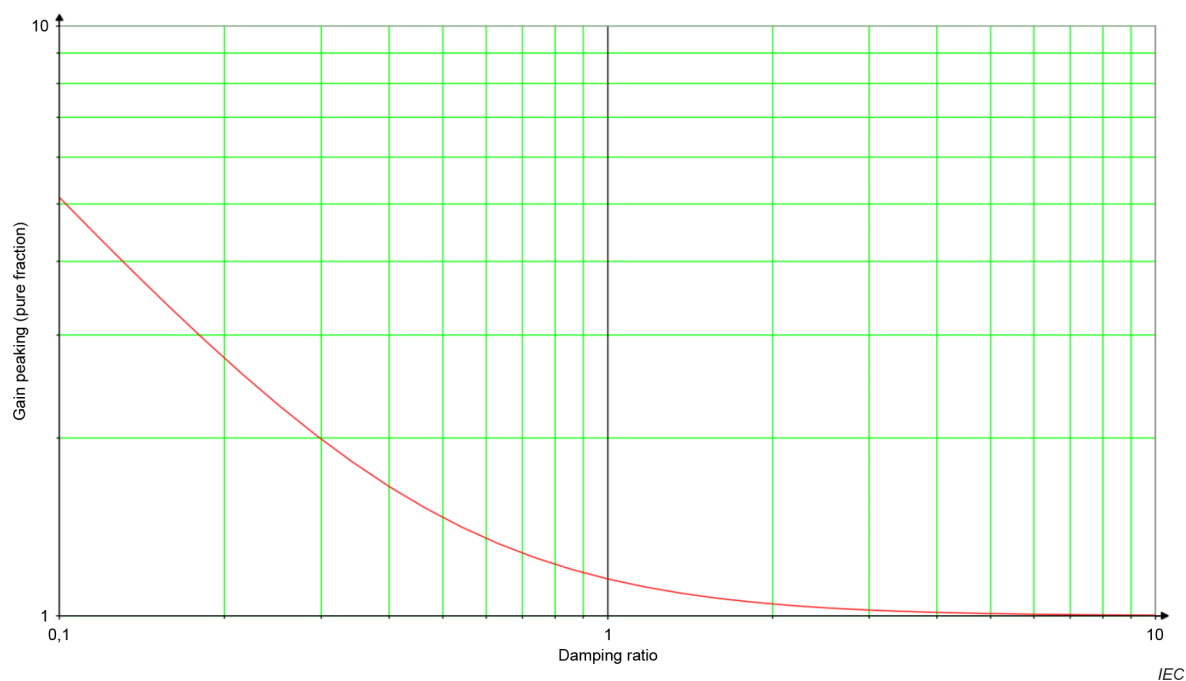


Figure C.4 – Gain peaking (pure fraction) as a function of damping ratio

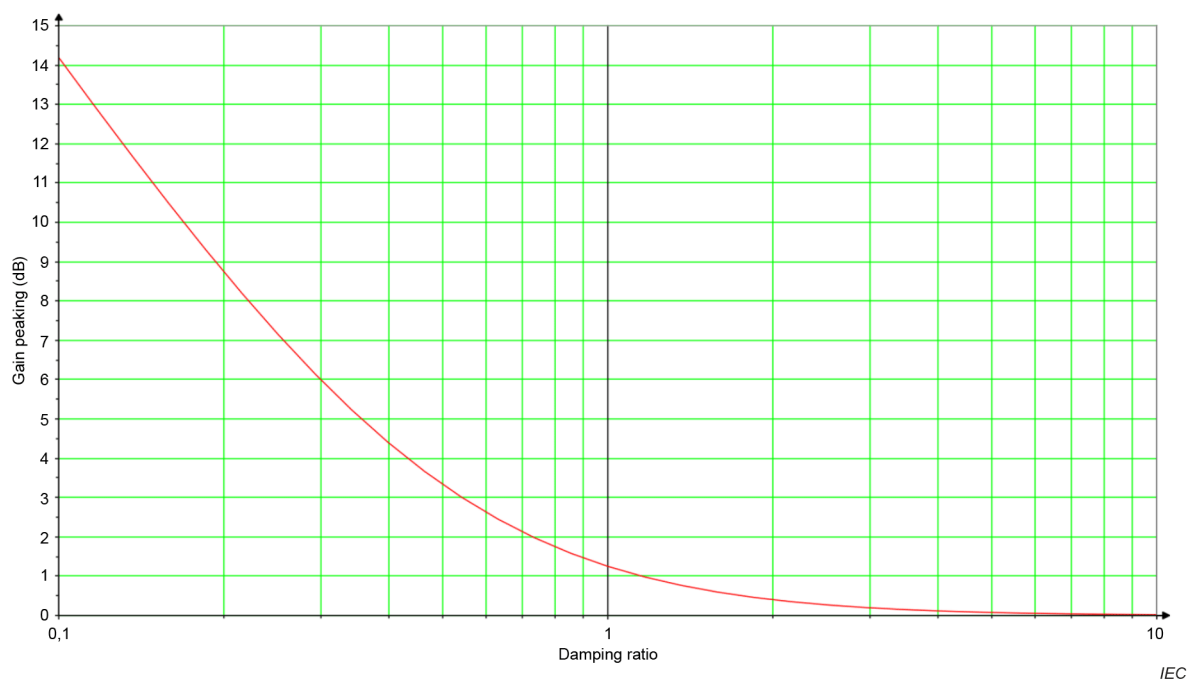


Figure C.5 – Gain peaking in dB as a function of damping ratio

The performance of the clock control system can be described using the frequency response as follows:

- a) Maximum 3 dB bandwidth in Hz,
- b) Maximum gain peaking in dB, and
- c) Frequency response plot (mask) corresponding to (a) and (b) that is not to be exceeded.

6621 C.4 Example

6622 Consider a clock control system with $K_p K_o = 4,23 \text{ rad/s}$ and $K_i K_o = 9,62 (\text{rad/s})^2$. The undamped
6623 natural frequency and damping ratio are:

$$\omega_n = \sqrt{K_i K_o} = \sqrt{9,62 (\text{rad/s})^2} = 3,10 \text{ rad/s} \quad (\text{C.18})$$

$$\zeta = \frac{K_p K_o}{2\sqrt{K_i K_o}} = \frac{4,23 \text{ rad/s}}{2\sqrt{9,62 (\text{rad/s})^2}} = 0,682 \quad (\text{C.19})$$

6624

6625 The gain peaking is obtained from:

$$\alpha = \frac{1}{4(0,682)^2} = 0,537 \quad (\text{C.20})$$

$$H_p (\text{purefraction}) = \left[1 - 2(0,537) - 2(0,537)^2 + 2(0,537) \sqrt{2(0,537) + (0,537)^2} \right]^{\frac{1}{2}} = (\text{C.21})$$

$$H_p (\text{dB}) = 20 \log_{10}(1,29) \text{ dB} = 2,2 \text{ dB} \quad (\text{C.22})$$

6626

6627 The 3dB bandwidth is:

$$f_{3\text{dB}} (\text{Hz}) = \frac{\omega_n}{2\pi} \left[1 + 2\zeta^2 + \sqrt{(1 + 2\zeta^2)^2 + 1} \right]^{1/2} \quad (\text{C.23})$$

$$6628 \quad = \frac{3,10}{2\pi} \left[1 + 2(0,682)^2 + \sqrt{(1 + 2(0,682)^2)^2 + 1} \right]^{1/2} = 1,0 \text{ Hz}$$

6629

6630 The frequency response is shown in Figure C.6.

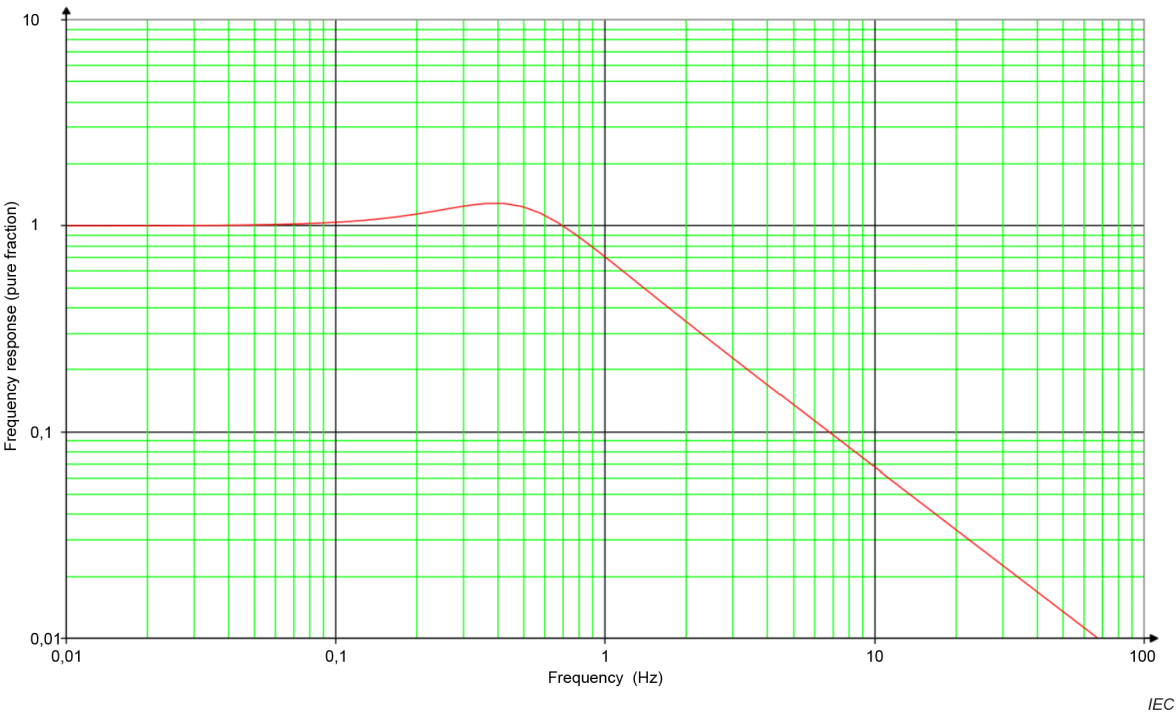


Figure C.6 – Example frequency response

Annex D (informative)

Time synchronization annex

D.1 Overview

Annex D describes how a network of compliant devices can achieve a time synchronization accuracy, at the application level, of $\pm 1 \mu\text{s}$, relative to the Clock Source at the Grandmaster, over 100 network hops. To achieve this, it allocates the overall error budget of 1 000 ns as described in Table D.1.

Table D.1 – Time synchronization error budget

Network Aspect	Error Type	Network-Level Error Budget (ns)
All PTP Instances	Constant time error	200
	Dynamic time error	600
All PTP links	Constant time error	200
	Dynamic time error	

A chain of 1 Grandmaster PTP Instance, 99 PTP Relay Instances and 1 PTP End Instance (100 network hops) that all comply with the normative requirements of 6.2.2, 6.2.3, 6.2.4, and 6.2.5 generates a network-level Time Error at or below the Error Budget for All PTP Instances.

Clause D.2 describes the principles of operation this document assumes.

Clause D.3 provides additional information on specific normative requirements.

The principles of operation include the use of crystal oscillators (XOs) as opposed to more accurate, stable, and costly options such as temperature-compensated crystal oscillators (TCXOs).

Clause D.4 describes a potential approach to testing the normative requirements. It is not a test specification, but rather a high-level overview of one potential approach that can be adopted by a full test specification.

The use of XOs means that some of the normative requirements are difficult or impossible to meet without employing algorithms that track Neighbor Rate Ratio drift and Rate Ratio drift and compensate for consequent errors in calculating Rate Ratio and Correction Field.

Clause D.5 provides examples of algorithms that can be used for this purpose, and which have been shown to enable compliance with the normative requirements.

Implementations that employ TCXOs or other more accurate, stable oscillators can still find some of the normative requirements difficult or impossible to meet without employing algorithms to track and compensate for errors due to clock drift. This is because other PTP Instances that use XOs can still cause the implementation to experience Neighbor Rate Ratio drift, Rate Ratio drift or both.

There is no normative requirement to use the algorithms described in Clause D.5; an implementation can employ alternative algorithms provided the normative requirements are met. Clause D.5 describes the potential risks of deploying a network whose instances employ a mix of different algorithms. It is the responsibility of implementers to mitigate the risks and ensure alternative algorithms deliver the desired network-level performance.

This document does not include normative requirements for PTP Links. Annex D.2.4 describes PTP Link characteristics that influence achieving 1 μ s time synchronization accuracy. It includes some examples using common PTP Link characteristics.

This document's normative requirements regarding instance-level error generation are intended to ensure not just an overall level of dTE generation at each node, but also the performance of drift tracking and error compensation algorithms and the amount of dTE generation due to timestamp error versus clock drift. The algorithms are employed to mitigate errors due to clock drift but cannot mitigate timestamp errors.

D.2 Principles of operation

D.2.1 General

Achieving ± 1 μ s time synchronization accuracy across 100 network hops involves managing the accumulation of errors in the preciseOriginTimestamp plus correctionField and the Rate Ratio as they are passed, via Sync or Follow_Up messages, down the chain of PTP Instances and are then used by the PTP End Instance to keep its ClockTarget in line with the ClockSource at the Grandmaster PTP Instance. The majority of significant errors can ultimately be traced back to one of three sources: timestamp error, clock drift, or path delay asymmetry. The selection of PTP protocol parameters often involves trading off one source of error against the other. This document requires specific PTP protocol configurations, and assumes the use of mechanisms (algorithms), that reduce dTE due to timestamp error but would also – without additional measures – increase dTE due to clock drift to the point where the latter exceeds the allocated error budget. However, this document also assumes additional measures to minimize some sources of dTE due to clock drift, and mechanisms and to track and compensate for errors from other sources to a sufficient degree that the error budget is not exceeded.

The specific protocol configurations and other measures, along with their intended effects, are described in Table D.2.

Table D.2 – Protocol configurations & other measures to achieve dTE budget

Configuration or measure	Description and intended effect(s)
Sync Interval 125 ms	Effects: 1) Calibrates the balance between dTE from timestamp error vs error due to clock drift. Larger intervals lead to less timestamp error and more error due to clock drift. 2) Keeps below acceptable limits the impact of errors in Rate Ratio and Rate Ratio Drift estimation when keeping ClockTarget in line with ClockSource between arrival of Sync messages. Larger intervals increase the impact of any errors.
Drift_Tracking TLV – syncEgressTimestamp	Effect: Enables calculation of NRR using Sync message timestamps, which eliminates error due to NRR clock drift that would otherwise occur between calculation of NRR using Pdelay_Resp messages and use during Sync message processing (i.e. calculation of Rate Ratio and output Correction Field values).

Configuration or measure	Description and intended effect(s)
NRR Smoothing	Description: Algorithm to use timestamps from multiple past Sync messages when calculating NRR. Effect: Reduces the amount of error in the estimate of NRR due to timestamp error while increasing the amount of error due to clock drift.
NRR Drift Tracking & Compensation	Description: Algorithm to use timestamps from multiple past Sync messages to estimate NRR drift rate and then apply compensation to correct for consequent errors in NRR calculation. Effect: Mitigates the effect of errors due to clock drift when calculating and using the estimated NRR.
Drift_Tracking TLV – rateRatioDrift	Description: Carries estimate of Rate Ratio drift rate from one node to the next. Effect: Allows each node to estimate its own Rate Ratio drift rate by combining the incoming Rate Ratio drift rate with the local estimate of NRR drift rate.
RR Drift Compensation	Description: Algorithm that uses the estimate of RR drift rate to compensate for that drift, adjusting the estimated RR over time according to the drift rate. Effect: For PTP Relay Instances, minimizes errors in the Correction Field caused by Rate Ratio drift. For PTP End Instances, a similar approach can reduce errors in keeping ClockTarget in line with ClockSource between arrival of Sync messages, but is outside the scope of this document.
Pdelay Interval Consistency	Description: This document requires tighter control of the interval between Pdelay messages generated at the Grandmaster PTP Instance than the defaults in IEEE Std 802.1AS-2020. Effect: This document requires the use of Sync messages to calculate NRR (see above). However, when a sufficient number of Sync messages are not available, for example on startup or after a reconfiguration, Pdelay_Resp messages can be used instead. In such cases, errors due to clock drift at Relay Instances have a tendency to cancel out. A clock drift that generates a positive error in NRR measurement on receipt of a Pdelay_Resp message generates a negative error in NRR measurement at the next node. The degree of cancellation depends on the consistency of the intervals over which NRR is measured at neighboring nodes. Tighter control of the Pdelay Interval increases the consistency of the measurement interval and thus decreases the amount of error.
Mean Residence Time	Description: This document defines a mean Residence Time requirement, which is significantly lower than the default maximum Residence Time in IEEE Std 802.1AS-2020. Effect: The amount of error in the Correction Field at the PTP End Instance due to clock drift is proportional to the cumulative meanLinkDelay and residenceTime experienced by a Sync message during transit from the Grandmaster PTP Instance to the PTP End Instance. Specifying a lower mean residenceTime reduces this source of error.

D.2.2 Grandmaster PTP Instance implementation

Depending on implementation, a Grandmaster PTP Instance can:

- a) contain a single oscillator used for both Local Clock and Clock Source,
- b) contain separate oscillators for Local Clock and Clock Source, or
- c) contain only an oscillator for Local Clock and accept an external input for Clock Source.

In some cases, a Grandmaster PTP Instance can support more than one mode of operation and transition between them depending on changes in network configuration (see Splitting, Joining and Aligning Time Domains).

In the first case, the rateRatio transmitted by the Grandmaster PTP Instance is one (i.e. 0 ppm) the rateRatioDrift is 0 ppm/s, reflecting the fact there is no difference between the Local Clock and Clock Source frequencies.

In the second and third cases there can be differences between the Local Clock and Clock Source frequencies. Any differences are reflected in the rateRatio and rateRatioDrift fields transmitted by the Grandmaster PTP Instance. This means that the Grandmaster PTP Instance tracks rateRatio over time in order to calculate rateRatioDrift, similarly to PTP Relay Instances and PTP End Instances. The exact implementation can vary.

D.2.3 Splitting, joining and aligning time domains

Modular machines or production cells can allow the splitting and combining of machines if this is required by the production process. When separate, the ClockSources of two machines run separately, each with its own time domain. If both ClockSources are traceable to the same PTP timescale, the difference between the ClockSources can be relatively small. If traceable to different timescales, especially if one or both are ARB timescales, there can be a very large difference between the ClockSources.

When two machines are joined, the first machine's time domain remains unaffected, and it can continue operation without disruption. There are two typical approaches to how the second machine behaves. In the first case, at a time of the end user's choosing, the second machine's time domain ceases to exist, with its PTP Instances becoming part of the first machine's time domain. In the second case, the second machine's time domain is gradually aligned with the first machine's time domain such that control loop cycles are coordinated. In the first case the second machine's time domain is unaffected, and it can continue production even if the machines are accidentally connected, until the end user chooses to join the time domains. In the second case the second machine can continue production while its time domain is being aligned.

D.2.3.1 Joining machines with single time domain

In the first case, where the second machine's time domain ceases to exist, a discontinuity in timing for the second machine's PTP Instances can occur, as they switch to use the first machine's Grandmaster. Some implementations implement measures to limit such timing discontinuities, but these measures are outside the scope of this document. Typically, in this case, the second machine is not operational while it is joined to the first. It resumes operation once its PTP Instances have synchronized with the first machine's Grandmaster.

D.2.3.2 Joining machines with multiple coordinated time domains

In the second case, where the second machine's time domain is gradually aligned with the first machine's time domain, this typically requires both machines to be implementing the same control loop cycle time. The goal is that, once coordinated, each control loop cycle of the first machine is aligned with the start of a control loop cycle of the second machine, even though the two machines maintain separate time domains and there can be a large time difference between their Clock Sources.

6745 In this case, after being joined together, the first machine effectively drives the second
6746 machine's Clock Source faster or slower, during an alignment period, until coordination is
6747 achieved. During the alignment period, this drive from the first machine can result in the second
6748 machine's Clock Source temporarily exceeding the usual normative requirement on range of
6749 fractional frequency offset relative to the nominal frequency of ± 50 ppm. The usual normative
6750 requirement on range of rate of change of fractional frequency offset of ± 1 ppm/s, applicable
6751 when split (i.e. independent) or coordinated (i.e. joined and stable, after the alignment period),
6752 may also be temporarily exceeded. However, if the value stays within the range ± 3 ppm/s, the
6753 network-level performance of 1 μ s time synchronization accuracy can be maintained. For this
6754 reason, this document specifies a separate normative requirement for temporary, externally
6755 driven, rate of rate of change of fractional frequency offset.

6756 Since the second machine experiences no time discontinuities and the network-level
6757 performance is maintained the second machine can, if desired, continue operation during the
6758 alignment period.

6759 Once coordinated, the first machine continues to drive the Clock Source of the second machine
6760 to maintain coordination. In this stable, coordinated mode of operation the normal range of
6761 ± 1 ppm/s is not exceeded.

6762 The mechanism by which the first machine drives the Clock Source of the second machine is
6763 not addressed in this document.

6764 **D.2.3.3 Splitting machines**

6765 In the first case, where the second machine's time domain ceased to exist while joined to the
6766 first, splitting machines means that the second machine must create its own time domain again.
6767 The second machine's Clock Source typically starts at the PTP Grandmaster Instance's last,
6768 best estimate of the first machine's Clock Source. The goal is for no discontinuities in time sync
6769 to occur; however, depending on implementation, it can take some time to before the time
6770 synchronization accuracy of all the second machine's PTP Instances relative to its Grandmaster
6771 can be relied upon. For this reason, it is possible the second machine is not operational during
6772 the split. Hot Standby can be employed to mitigate this transition time, but the details of how to
6773 do so are out of scope for this document.

6774 In the second case, where the second machine maintains its time domain while joined to the
6775 first, splitting machines means that the first machine ceases driving the second machine's Clock
6776 Source to maintain coordination of control loop cycle times. Without this drive, the two time
6777 domains can drift relative to each other resulting in loss of coordination. Time synchronization
6778 performance within the second machine is maintained during the split and the second machine
6779 can, if desired, continue operation throughout the process.

6780 **D.2.4 PTP link characteristics**

6781 A vast majority of time synchronization error due to PTP link characteristics is caused by
6782 asymmetrical path delay in one direction versus the other. The mechanism to measure path
6783 delay assumes the link is symmetrical and cannot detect asymmetry, thus asymmetry causes
6784 an error. The potential maximum asymmetry and thus error typically scales linearly with physical
6785 path length.

6786 The error budget due to PTP link characteristics for an entire network is 200 ns. In any specific
6787 network this budget can be allocated as required with some links allocated a higher budget
6788 (typically longer length) than others.

6789 A typical specified maximum delay skew for Category 6 Ethernet cables is 50 ns per 100 m.
6790 Some Ethernet MAU types such as 100BASE-TX use different pairs for transmit and receive. If
6791 such cables are used, a maximum total cable length between Clock Source and Clock Target
6792 with 99 PTP Relay Instances between them (i.e. 100 network hops) is 400 m. Extending the

6793 cable length beyond 400 m without jeopardizing network-level performance would require the
6794 use of cables with less delay skew or asymmetry compensation for delay skew.

6795 It is possible for the delay skew in one section of cable to cancel all or part of a delay skew in
6796 the opposite direction from prior section but, depending on how cables are manufactured and
6797 deployed, it is feasible for the delay skews of every cable segment between a Grandmaster
6798 PTP Instance and a PTP End Instance to be additive.

6799 **D.3 Notes on normative requirements**

6800 **D.3.1 Oscillator requirements**

6801 Clock drift at the Grandmaster PTP Instance causes greater dTE than the same amount of clock
6802 drift at a PTP Relay Instance or the PTP End Instance. This document therefore requires tighter
6803 limits on maximum fractional frequency offset for an oscillator at the Grandmaster PTP Instance
6804 than at other instances.

6805 This document does not place requirements on operational temperature range or other
6806 environmental factors. The required oscillator behavior is delivered for the operational
6807 conditions across which a device claims it is compliant. These conditions typically include
6808 temperature range but can also include rate of change of ambient temperature, supply voltage
6809 stability, amount of vibration and others.

6810 **D.3.2 Timestamp Granularity Error**

6811 Timestamp Granularity Error (TSGE) is the error in timestamping each incoming and outgoing
6812 message due to the maximum timestamp resolution of which an implementation is capable. It
6813 is typically directly related to an implementation's clock rate.

6814 For example: a clock rate of 125 MHz typically results in a maximum resolution of 8 ns.
6815 Depending on implementation the consequent TSGE range can be –8 ns to 0 ns, 0 ns to 8 ns,
6816 or anything in between. In some implementations, offsets are applied to ensure the average
6817 TSGE is 0 ns with, assuming uniform error distribution, a range of –4 ns to +4 ns

6818 Similarly, a clock rate of 500 MHz results in a maximum resolution of 2 ns; a consequent TSGE
6819 range between –2 ns to 0 ns and 0 ns to 2 ns; and, if a suitable offset is applied to ensure a
6820 TSGE average of 0 ns, a range of –1 ns to +1 ns.

6821 A minimum resolution of 8 ns, i.e. minimum clock rate of 125 MHz is assumed. It is further
6822 assumed that TSGE for the sum of the preciseOriginTimestamp and followUpCorrectionField at
6823 the Grandmaster PTP Instance (see IEEE Std 802.1AS-2020 as amended by IEEE Std
6824 802.1ASdm-2024, 10.2.9.2.1) has an average of 0 ns and that the TSGE averages for other
6825 timestamps are stable and consistent across all a PTP Instance's ports. No assumption
6826 regarding the value of the TSGE average for these other timestamps is necessary as they are
6827 always used to measure intervals such that any stable, consistent offset cancels out.

6828 **D.3.3 Dynamic Timestamp Error**

6829 Dynamic Timestamp Error (DTSE) is the, effectively random, error in timestamping each
6830 incoming and outgoing event message due to an implementation's inherent inaccuracies,
6831 excluding TSGE. It is assumed to vary between a minimum of –6 ns and a maximum of + 6 ns
6832 with an average of 0 ns. Lower levels of DTSE are better.

6833 If an implementation timestamps an incoming or outgoing message at a point other than as
6834 specified in IEEE Std. 802.3-2022 as amended by IEEE Std. 802.3cx-2023 and IEEE Std
6835 802.3de-2022, any variability in delay between that point and the delay point specified in 802.3
6836 translates to DTSE. Some common implementations were not designed to limit this variability.
6837 If care is not taken to avoid implementations with high variability, the assumed DTSE range is

6838 easily exceeded. Such implementations can find some of the normative requirements difficult
6839 or impossible to meet.

6840 D.3.4 Grandmaster PTP Instance error generation

6841 Table 12 sets normative requirements for error generation at a Grandmaster PTP Instance that
6842 ensure the relevant fields in the Sync and Follow_Up messages it transmits are sufficiently
6843 accurate to deliver the network-level performance. Table D.3 describes how the normative
6844 requirements align with major sources of error.

6845 **Table D.3 – Main sources of error affecting normative requirements for Grandmaster**
6846 **PTP Instance**

Item	Normative requirement	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Working Clock at Grandmaster (acting as a Clock Source)	Timestamp error relative to Clock Source plus accuracy measuring any internal delay between generation of the preciseOriginTimestamp and Sync message transmission.
2	rateRatio vs Direct measurement of Rate Ratio of Clock Source vs Local Clock	Accuracy of internal mechanism to measure Rate Ratio of Clock Source vs. Local Clock, potentially including algorithms that track RateRatioDrift and modify Rate Ratio accordingly ^a
3	syncEgressTimestamp vs Direct measurement of Local Clock	Timestamp error relative to Local Clock
4	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	Timestamp error relative to Local Clock
^a Only applicable if Clock Source and Local Clock are not locked to the same frequency by the implementation. If they are locked, then rateRatio is 0 ppm and rateRatioDrift is 0 ppm/s.		

6847

6848 D.3.5 PTP Relay Instance error generation

6849 Table 13 sets normative requirements for error generation at a PTP Relay Instance that ensure
6850 the relevant fields in the Sync and Follow_Up messages it transmits as part of Sync processing
6851 are sufficiently accurate to deliver the network-level time sync performance. The requirements
6852 include the ability to mitigate errors in rateRatio and rateRatio drift that would otherwise occur
6853 due to clock drift at the current PTP Relay Instance, an adjacent PTP Relay Instance, or the
6854 Grandmaster PTP Instance. Table D.4 describes how the normative requirements align with
6855 major sources of error.

6856 **Table D.4 – Main sources of error affecting normative requirements for PTP Relay**
6857 **Instance**

Item	Normative requirement	Clock drifts	Main sources of error
1	preciseOriginTimestamp + correctionField vs Direct measurement of Clock Source at Grandmaster PTP Instance	None	Timestamp errors relative to Local Clock when measuring Residence Time, i.e. Sync message ingress and egress. Accuracy of meanLinkDelay measurement. Errors in Rate Ratio used when translating Residence Time measured in terms of Local Clock to Residence time in terms of Clock Source, although these are typically orders of magnitude smaller than those from Timestamp Errors.

Item	Normative requirement	Clock drifts	Main sources of error
2	rateRatio vs Direct measurement of Rate Ratio of Clock Source vs Local Clock	None	Timestamp error affecting measurement of NRR when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two requirements.
3		Clock Source (RR Drift)	Accuracy of measurement of NRR when there is no NRR Drift (as above). Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation.
4		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation Accuracy of calculation of rateRatio, including algorithms for RR Drift tracking & error compensation. Combined with test 3 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
5	rateRatioDrift vs Direct measurement of Rate Ratio Drift of Clock Source vs Local Clock	None	Timestamp error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two requirements.
6		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
7		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 6 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
8	syncEgressTimestamp vs Direct measurement of Local Clock	None	Timestamp error relative to Local Clock
9	(responseOriginTimestamp – requestReceiptTimestamp) vs Direct measurement of Local Clock interval between reception of Pdelay_Req and transmission of Pdelay_Resp	None	Timestamp error relative to Local Clock
10	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm

Item	Normative requirement	Clock drifts	Main sources of error
			mitigates the impact of any such single Pdelay measurement.

6858

6859 D.3.6 PTP End Instance error generation

6860 Table 14 sets normative requirements for error generation at a PTP End Instance that ensure
6861 the ClockTarget it generates from incoming Sync and Follow_Up messages is sufficiently
6862 accurate to deliver the network-level time sync performance. Table D.5 describes how the
6863 normative requirements align with major sources of error.

6864 **Table D.5 – Main sources of error affecting normative requirements for PTP End**
6865 **Instance**

Item	Normative requirement	Clock drifts	Main sources of error
1	ClockTarget vs ClockSource	None	Timestamp error affecting measurement of NRR Drift when there is no NRR Drift. The effect should be low. This normative requirement is a baseline for the next two tests.
2		Clock Source (RR Drift)	Accuracy of measurement of NRR Drift when there is no NRR Drift (as above). Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation.
3		Clock Source and Local Clock at previous PTP Instance (RR Drift & NRR drift)	Accuracy of measurement of NRR Drift when there is NRR Drift, including algorithms for NRR Drift tracking & error compensation. Accuracy of calculation of rateRatioDrift, including algorithms for RR Drift tracking & error compensation. Combined with test 2 this effectively requires a level of performance regarding NRR Drift tracking & error compensation, whether the source of the NRR drift is the Local Clock of the current PTP Instance or the previous PTP Instance.
4	Mean Link Delay	None	Timestamp error affecting measurement of meanLinkDelay. Each Pdelay measurement can be subject to relatively large errors from both the local and neighboring node. This normative requirement ensures a suitable algorithm mitigates the impact of any such single Pdelay measurement.

6866

6867 D.3.7 Explanation for the asymmetric normative requirements for the allowable range 6868 of dTE in Table 14, rows 2 and 3

6869 The PTP End Instance requirements for dTE in Table 14, rows 2 and 3, are asymmetric, i.e.,
6870 the allowable range in both cases is -145 ns to +15 ns. These tests are performed with the
6871 fractional frequency offset of the Working Clock (acting as the ClockSource) at the Grandmaster
6872 PTP Instance increasing at the rate of 1 ppm/s. dTE is measured at the output of the PTP End
6873 Instance clock control system after steady-state is reached, i.e., after any transients due to
6874 initiating the 1 ppm/s frequency drift rate have decayed. The clock control system is a second-
6875 order filter, with bandwidth and gain peaking requirements given in Table 11. Table 11 also
6876 indicates that the filter has a minimum roll-off of 20 dB/decade. The worst-case dTE response
6877 can be analyzed using the second-order filter model described in Annex C. The transfer function
6878 for this filter is given by Eq. (C.5). The example in C.4 shows that the damping ratio, ζ , for a

filter whose gain peaking is 2.2 dB, i.e., the maximum allowed in Table 11, is 0.682 (see Eq. (C.19)). The allowable range for filter 3dB bandwidth in Table 11.

is 0.9 Hz – 1.0 Hz. The example in C.4 indicates that the undamped natural frequency, ω_n corresponding to the maximum bandwidth of 1.0 Hz is 3.10 rad/s (see Eq. (C.18)).

It is well-known that if a frequency drift is applied to a linear, second-order filter with 20 dB/decade roll-off, the steady-state output has a time offset. This means that if the input to the PTP End Instance from the ClockSource (see Figure D.5) has a frequency drift of 1 ppm/s, the measured dTE output of the ClockTarget has a steady-state offset. Any errors due to other sources (e.g., timestamp granularity, dynamic timestamp error, clock noise) are in addition to this. If these other errors are symmetric about zero, the addition of the steady-state offset results in an asymmetric range of dTE.

The requirement in Table 14 is on the Working Clock (acting as ClockTarget) at the PTP End Instance minus the Working Clock (acting as ClockSource) at the Grandmaster. This difference, which is the error between the input and output and is denoted by $E(s)$ in the s-domain, is given by:

$$\begin{aligned} E(s) &= Y(s) - U(s) \\ &= H(s)U(s) - U(s) \\ &= (H(s) - 1)U(s) \\ &= H_e(s)U(s) \end{aligned} \quad (D.1)$$

6894

where $Y(s)$ and $U(s)$ are the Laplace transforms of the output and input, respectively, $H(s)$ is the transfer function between the input and output and is given by Eq. (C.5), and $H_e(s)$ is the transfer function between the input and error. Using Eq. (C.5), the error transfer function is given by:

$$H_e(s) = -\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (D.2)$$

The phase drift corresponding to an $A = 1$ ppm/s = 1000 ns/s² frequency drift is $0.5At^2$, where t is the time in seconds and the phase drift is in ns if A is in ns/s². The Laplace transform of this waveform is $U(s) = A/s^3$. The steady-state value of the filter output, $y_{\text{steady-state}}$, due to this drift is obtained from the final value theorem:

$$y_{\text{steady-state}} = \lim_{s \rightarrow 0} sH(s)U(s) = \lim_{s \rightarrow 0} s \cdot \left(-\frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \cdot \frac{A}{s^3} = -\frac{A}{\omega_n^2} \quad (D.3)$$

Since the 3dB bandwidth is proportional to the undamped natural frequency (see Eq. (C.23)), Eq. (D.32) indicates that the worst case, i.e., largest steady-state error occurs for the minimum 3dB bandwidth, i.e., 0.9 Hz in Table 11. In the example of C.4, the undamped natural frequency of 3.10 rad/s and damping ratio 0,682 correspond to a 3dB bandwidth of 1,0 Hz. Then, due to the direct proportionality between 3dB bandwidth and undamped natural frequency (for the same damping ratio), a 3dB bandwidth of 0.9 Hz corresponds to an undamped natural frequency of:

$$\omega_n = \left(\frac{0.9 \text{ Hz}}{1.0 \text{ Hz}} \right) (3.10 \text{ rad/s}) = 2.79 \text{ rad/s} \quad (D.4)$$

Then, using Eq. (D.32), the steady-state error corresponding to a frequency drift of 1 ppm/s is:

$$\gamma_{\text{steady-state}} = -\frac{1000 \text{ ns/s}}{(2.79 \text{ rad/s})^2} = -128 \text{ ns} \quad (\text{D.5})$$

6910 Then, assuming the error without the filter is 15 ns as given in Table 14, row 1, the lower end
6911 of the range of dTE when the filter is present is:

$$\text{dTE} = -128 \text{ ns} - 15 \text{ ns} = -143 \text{ ns} \approx -145 \text{ ns} \quad (\text{D.6})$$

6912 **D.4 Approach to testing normative requirements**

6913 **D.4.1 General**

6914 This document does not specify tests to ensure conformance with the requirements. However,
6915 it is important that the requirements are, in principle, testable. Clause D.4 describes, at a high
6916 level, approaches a test specification can take to testing conformance with some of the
6917 requirements related to time synchronization.

6918 It is assumed that test equipment can precisely measure the output of the ClockSource (at a
6919 Grandmaster PTP Instance), ClockTarget (at a PTP End Instance) and Local Clock (at any PTP
6920 Instance) to ensure conformance with frequency offset and frequency offset drift requirements.
6921 This might be via a Pulse per Second (PPS) plus Time-of-Day information or another
6922 mechanism.

6923 It is also assumed that test equipment can generate sequences of PTP messages with precise
6924 timing and content (for testing PTP Relay Instances and PTP End Instances) and receive, log,
6925 and process sequences of PTP messages with precise timing measurement (e.g. of message
6926 arrival).

6927 **D.4.2 Testing Grandmaster PTP Instance**

6928 Figure D.1 illustrates an approach to testing the three normative requirements discussed in
6929 D.3.4.

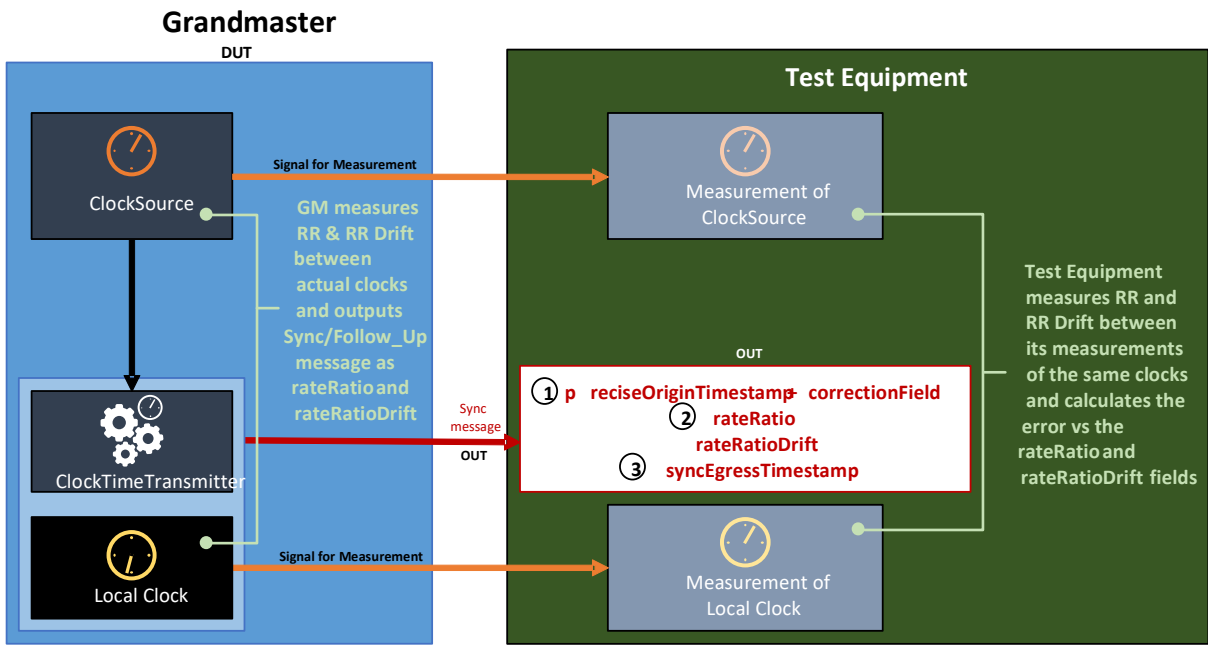


Figure D.1 – Approach to testing normative requirements for Grandmaster PTP Instance

The test equipment can calculate the time the Sync message is output at the DUT by subtracting the link delay from the measured arrival time at the test equipment.

For test 1, the test equipment compares the value of the preciseOriginTimestamp + correctionField against its measurement of the ClockSource.

For test 2, the test equipment compares the value in the rateRatio field with its calculation of the equivalent value based on its measurement of the ClockSource.

For test 3, the test equipment compares the value of the syncEgressTimestamp against its measurement of the Local Clock.

D.4.3 Testing PTP Relay Instance

Figure D.2 illustrates an approach to testing normative requirements 1, 2, 5 and 8 discussed in D.3.5.

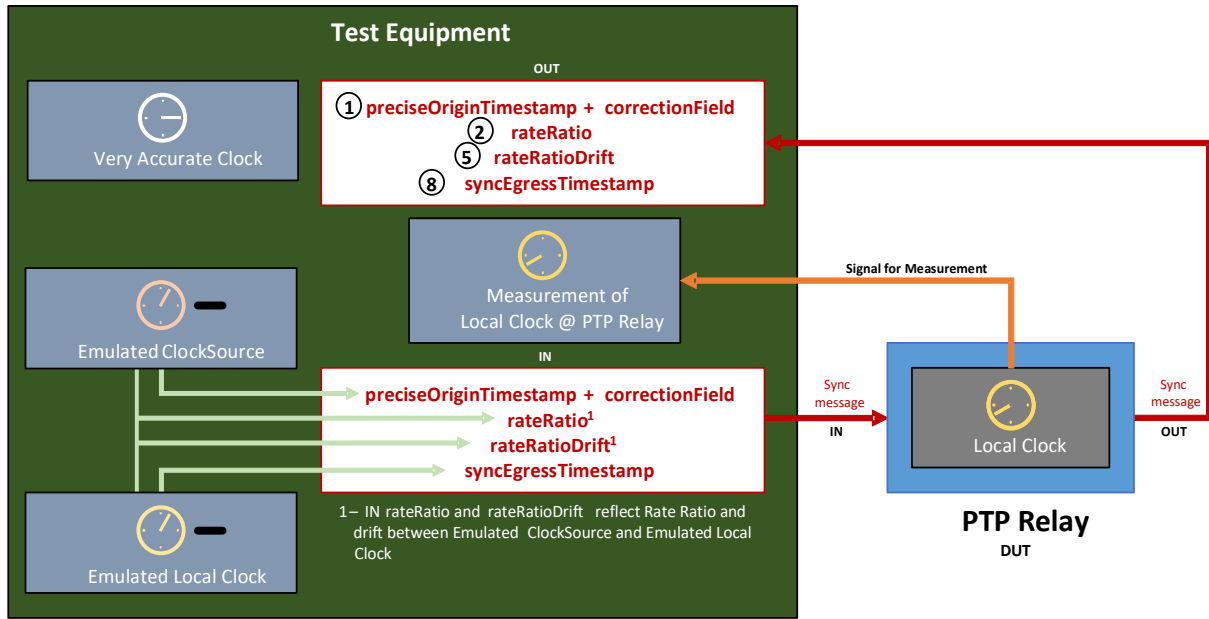


Figure D.2 – Approach to testing normative requirements for PTP Relay Instance - 1

The test equipment can compare the DUT's output Sync message to the expected result given the measurement of the Local Clock and the timing of the input PTP timing message transmission and output PTP timing message reception.

For these four tests, the Emulated ClockSource and Emulated Local Clock are stable and in sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input Follow_Up information TLV, rateRatio is 0 ppm, and in the input Drift_Tracking TLV rateRatioDrift is 0 ppm/s. If the Local Clock of the PTP Relay Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

The test equipment can calculate the time the output Sync message is output at the DUT by subtracting the link delay from the measured arrival time at the test equipment.

For test 1, the test equipment can compare the increase in the value of the correctionField to the measured meanLinkDelay (from the test equipment to the DUT) plus residenceTime. The test equipment accounts for the additional delay between the PTP Relay Instance's transmission of the input PTP timing message and its reception by the test equipment.

For tests 2 and 5, the test equipment can compare the rateRatio and rateRatioDrift fields in the output PTP timing message with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

For test 8, the test equipment can compare syncEgressTimestamp value in the output PTP timing message with its measurement of the Local Clock.

Figure D.3 illustrates an approach to testing normative requirements 3 and 6 discussed in D.3.5.

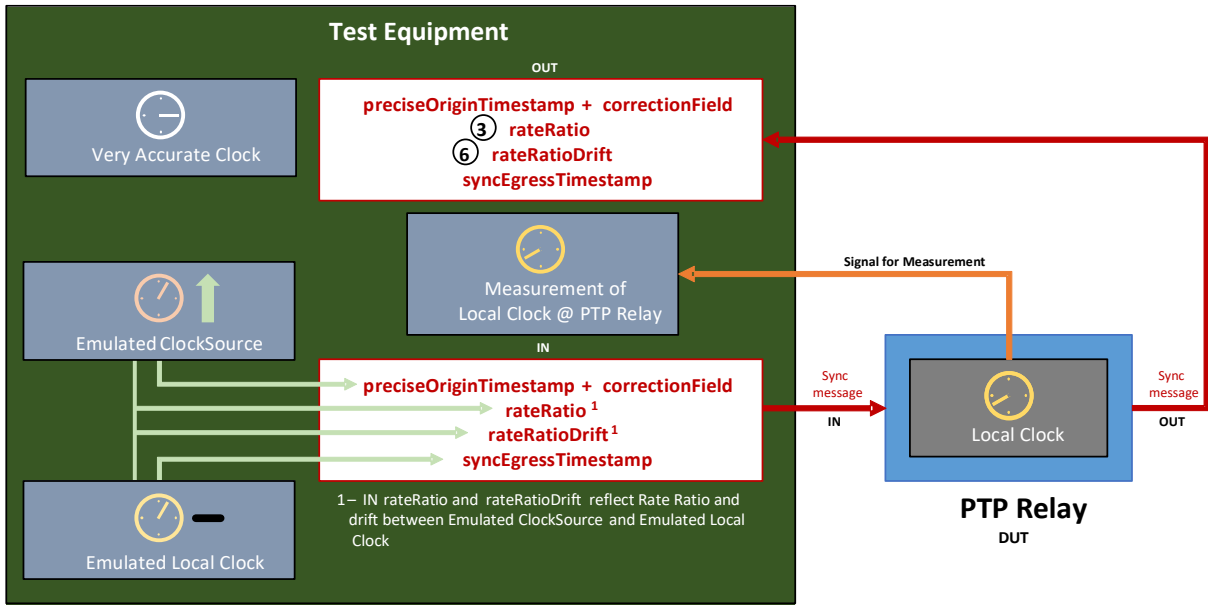


Figure D.3 – Approach to testing normative requirements for PTP Relay Instance – 2

For these two tests, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and in the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures a NRR of 0 ppm and a NRR Drift of 0 ppm/s.

For tests 3 and 6, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

Figure D.4 illustrates an approach to testing normative requirements 4 and 7 discussed in D.3.5.

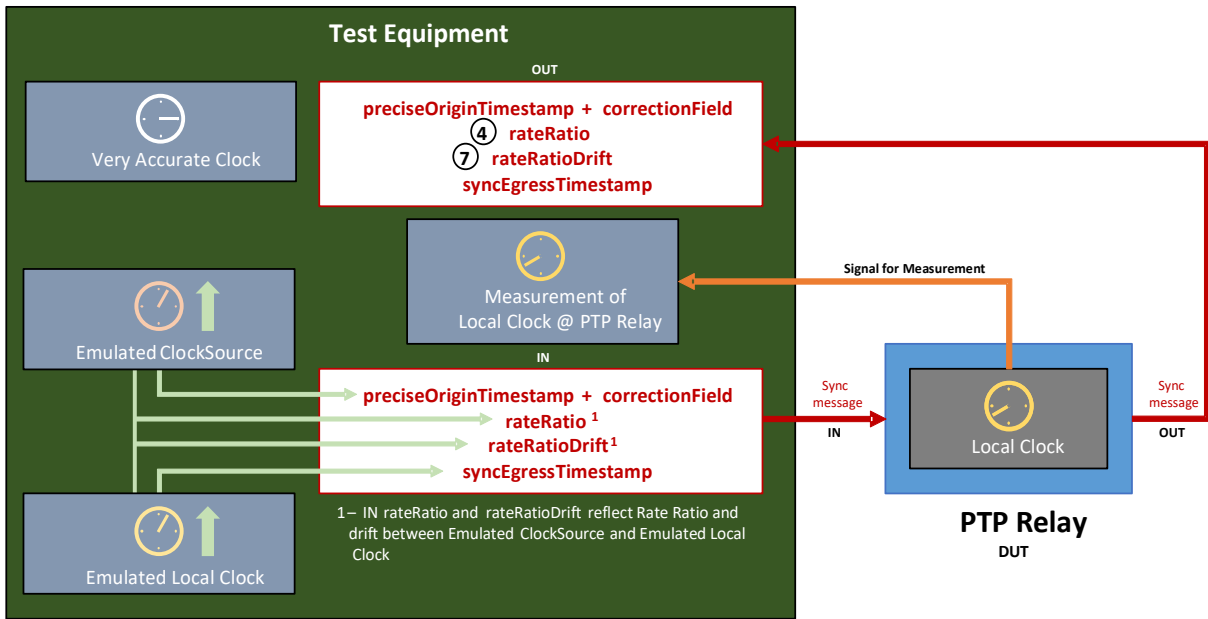


Figure D.4 – Approach to testing normative requirements for PTP Relay Instance – 3

For these two tests, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output Drift_Tracking TLV the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP Relay Instance is stable, the NRR it measures increases over time and the NRR Drift it measures maintains a matching positive value.

For tests 4 and 7, the test equipment can compare the rateRatio and rateRatioDrift fields in the output Follow_Up information TLV and Drift_Tracking TLV respectively with the equivalent calculated values between the measured Local Clock and the Emulated ClockSource.

D.4.4 Testing PTP End Instance

Figure D.5 illustrates an approach to testing the three normative requirements discussed in D.3.6.

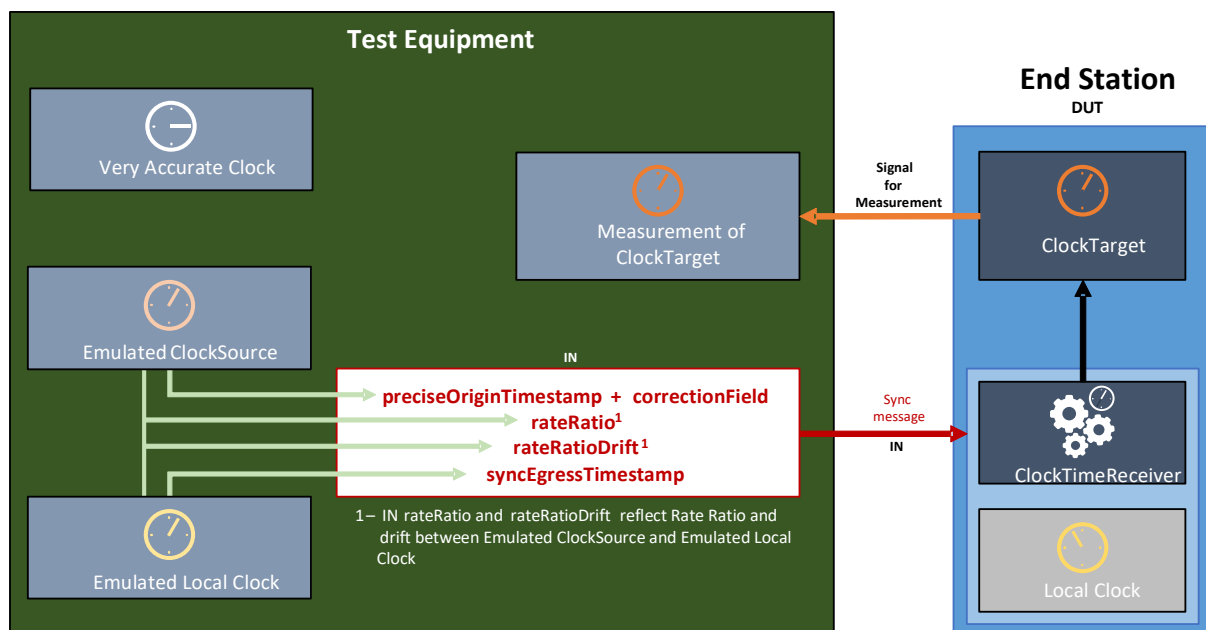


Figure D.5 – Approach to testing normative requirements for PTP End Instance

The test equipment can compare its measurement of the DUT's ClockTarget to the Emulated ClockSource. It accounts for the additional delay between its transmission of the input Sync message and the reception of the message by the DUT.

For test 1, the Emulated ClockSource and Emulated Local Clock are stable and in sync. In practice, both can be equal to the test equipment's Very Accurate Clock. In the input Follow_Up information TLV, rateRatio is 0 ppm, and in the input Drift_Tracking TLV, rateRatioDrift is 0 ppm/s. If the Local Clock of the PTP End Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

For test 2, the fractional frequency offset of the Emulated ClockSource is increasing at a defined ppm/s rate relative to the Very Accurate Clock. The Emulated Local Clock is stable; in practice, it can be equal to the test equipment's Very Accurate Clock. In the output Follow_Up information TLV, the rateRatio field increases over time, and the output Drift_Tracking TLV, the rateRatioDrift field maintains a matching positive value. If the Local Clock of the PTP Relay Instance is also stable, it measures NRR of 0 ppm and NRR Drift of 0 ppm/s.

7008 For test 3, the fractional frequency offsets of the Emulated ClockSource and the Emulated Local
7009 Clock are equal and increasing at a defined ppm/s rate relative to the Very Accurate Clock. In
7010 the output Follow_Up information TLV, the rateRatio field is 0 ppm, and in the output output
7011 Drift_Tracking TLV, the rateRatioDrift field is 0 ppm/s. If the Local Clock of the PTP Relay
7012 Instance is stable, the NRR it measures increases over time and the NRR Drift it measures
7013 maintains a matching positive value.

7014 **D.4.5 Testing meanLinkDelay Accuracy**

7015 D.3.5 and D.3.6 both discuss a normative requirement to test the accuracy of meanLinkDelay
7016 measurements for PTP Relay Instances and PTP End Instances respectively. This can be
7017 achieved by using a cable of known length and asymmetry to connect the test equipment to the
7018 DUT. The test equipment then responds to the DUT's Pdelay_Req messages with Pdelay_Resp
7019 messages, allowing the DUT to estimate the path delay.

7020 The DUT's estimate of the path delay can be accessed via the meanLinkDelay global variable
7021 (see IEEE 802.1AS-2020, Clause 14.8.8).

7022 The error in a single path delay measurement via a Pdelay_Req / Pdelay_Resp exchange can
7023 be large relative to the normative requirement, but an estimate with significantly less error can
7024 be generated by considering more than one measurement, hence the "mean" in
7025 "meanLinkDelay". Two consequences of this are noteworthy.

7026 Firstly, since errors can be introduced by Timestamp Errors at both the DUT and a neighboring
7027 node, and the DUT is responsible for mitigating both, the test equipment simulates typical errors
7028 on its timestamps for receiving Pdelay_Req and transmitting Pdelay_Resp messages to provide
7029 a complete test.

7030 Secondly, since the meanLinkDelay measurement can take some time to stabilize based on
7031 multiple Pdelay_Req / Pdelay_Resp exchanges, leaving sufficient time for these multiple
7032 exchanges before assessing meanLinkDelay accuracy results in a more accurate test.

7033 **D.5 Example algorithms**

7034 **D.5.1 General**

7035 This document does not place normative requirements on the use of specific algorithms.
7036 However, the normative requirements assume the use of algorithms to reduce the effect of
7037 errors in meanLinkDelay and to track clock drift and compensate for consequent errors. PTP
7038 Instances that do not implement algorithms can find it difficult or impossible to meet the
7039 normative requirements.

7040 D.5 provides examples of algorithms that can be used for:

- 7041 • tracking NRR drift;
- 7042 • correcting for errors in measured NRR (mNRR) due to NRR drift;
- 7043 • calculating RR drift;
- 7044 • correcting for errors in measured RR (mRR) due to RR drift;
- 7045 • reducing the effect of errors in meanLinkDelay.

7046 **D.5.2 Algorithm for tracking NRR drift**

7047 For measured NRR, measured RR, and meanLinkDelay, an example for how startup behavior
7048 can be handled is provided.

7049 NRR Drift Tracking and Error Correction is carried out for each network hop, i.e. at every node
7050 other than the Grandmaster. It is based on pairs of timestamps with each pair associated with
7051 a Sync message transmitted from the previous node ($n-1$) to the current node (n).

- 7052 • t_{s1outP} – Timestamp of the Sync message egress from the previous node ($n-1$), timestamped
7053 by that node's Local Clock. Unit: ns.
- 7054 • t_{s2in} – Timestamp of the Sync message ingress to the current node (n), timestamped by that
7055 node's Local Clock. Unit: ns.

7056 All timestamps are affected by Timestamp Errors.

7057 The algorithm uses information from the 32 most recent Sync messages. However, a node only
7058 keeps track of the 9 most recent pairs of timestamps from the most recent (x) to the 9th most
7059 recent ($x-8$) Sync message. The algorithm generates one measurement of NRR using the prior
7060 2 s of Sync message data (on average, based on a nominal Sync Interval of 125 ms), and a
7061 second measure based on the 2 s of Sync message data prior to that. It then uses the difference
7062 in the two measurements over the interval between the effective measurement points to
7063 calculate the NRR drift rate.

7064 On arrival of a new Sync message (x), or Follow_Up in the case of two-step time transport, a
7065 node executes a NRR calculation:

$$NRRcalc(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-8)}{t_{s2in}(x) - t_{s2in}(x-8)} - 1 \right) \times 10^6 \quad (D.7)$$

7066

7067 where

7068 $NRRcalc$ is the calculated Neighbor Rate Ratio, expressed in ppm;

7069 x is the most recent Sync message;

7070 t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$),
7071 timestamped by that node's Local Clock, expressed in ns;

7072 t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped
7073 by that node's Local Clock, expressed in ns.

7074 with an associated effective measurement point:

$$NRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-8)}{2} \quad (D.8)$$

7075

7076 where

7077 $NRRcalcT$ is the effective measurement point, expressed in ns;

7078 t_{s1outP} is the timestamp of the Sync message egress from the previous node ($n-1$),
7079 timestamped by that node's Local Clock, expressed in ns;

7080 t_{s2in} is the Timestamp of the Sync message ingress to the current node (n), timestamped
7081 by that node's Local Clock, expressed in ns.

7082 A node keeps track of the 24 most recent NRR calculations and effective measurement points,
7083 from the most recent (x) to the 24th most recent ($x-23$).

7084 After of a new most-recent NRR calculation, a node calculates an NRR drift rate:

$$NRR_{averageA} = \sum_{i=x-7}^x \frac{mNRR_{calc}(i)}{8} \quad (D.9)$$

7085

7086 where

7087 $NRR_{averageA}$ is the average of the 8 most recent Neighbor Rate Ratio calculations, expressed
7088 in ppm.

$$NRR_{averageB} = \sum_{i=x-23}^{x-16} \frac{mNRR_{calc}(i)}{8} \quad (D.10)$$

7089

7090 where

7091 $NRR_{averageB}$ is the average of the 8 least recent Neighbor Rate Ratio calculations, expressed
7092 in ppm.

$$NRR_{driftInterval} = \sum_{i=x-7}^x \frac{mNRR_{calcT}(i)}{8} - \sum_{i=x-23}^{x-16} \frac{mNRR_{calcT}(i)}{8} \quad (D.11)$$

7093

7094 where

7095 $NRR_{driftInterval}$ is the period across which Neighbor Rate Ratio drift is measured, expressed
7096 in ns.

$$NRR_{driftRate}(n) = \left(\frac{NRR_{averageA} - NRR_{averageB}}{NRR_{driftInterval}} \right) \times 10^9 \quad (D.12)$$

7097

7098 where

7099 $NRR_{driftRate}(n)$ is the the NRR drift rate for the current Node n , expressed in ppm/s.

7100 **D.5.3 Algorithm to compensate for errors in measured NRR due to clock drift**

7101 **D.5.3.1 General**

7102 The algorithm to measure NRR uses data from the previous 1 s of Sync message data,
7103 combined with the NRR drift estimate from the previous step. This smaller amount of data (vs
7104 that used for either of the NRR measurements in the previous step) is employed as it improves
7105 responsiveness to sudden changes in NRR drift with minimal loss of accuracy.

7106 On arrival of a Sync message (x), or Follow_Up in the case of two-step time transport, a node
7107 executes a NRR calculation:

$$mNRR_{calc}(x) = \left(\frac{t_{s1outP}(x) - t_{s1outP}(x-4)}{t_{s2in}(x) - t_{s2in}(x-4)} - 1 \right) \times 10^6 \quad (D.13)$$

7108

7109 with an associated effective measurement point:

$$mNRRcalcT(x) = \frac{t_{s2in}(x) + t_{s2in}(x-4)}{2} \quad (D.14)$$

7110

7111 A node keeps track of the 4 most recent mNRR calculations and effective measurement points,
 7112 from the most recent (x) to the 4th most recent ($x-3$). The mNRR calculations use information
 7113 from the 5 most recent Sync messages, but the node is already keeping track of information
 7114 from the 9 most recent Sync messages for the NRR drift tracking algorithm.

7115 The node then calculates an error corrected measured NRR value.

7116 For $i = x$ to $(x-3)$

$$mNRRcorrected(i) = mNRRcalc(i) + \left(NRRdriftRate(n) \times \frac{(t_{s2in}(x) - mNRRcalcT(i))}{10^9} \right) \quad (D.15)$$

7117

7118 where

7119 $mNRRcorrected(i)$ is the error-corrected measured NRR value at message i , expressed in ppm.

$$mNRR(n) = \sum_{i=x-3}^x \frac{mNRRcorrected(i)}{4} \quad (D.16)$$

7120

7121 where

7122 $mNRR$ is the error-corrected measured NRR value, expressed in ppm.

7123 The result is a measured NRR value, error-corrected to the time when the most recent Sync
 7124 message was received.

7125 **D.5.3.2 Measured NRR algorithm – Startup behavior**

7126 NRR is used when calculating meanLinkDelay and output Sync/Follow_Up message fields. The
 7127 first NRR drift calculation is only be available after receipt of 32 Sync/Follow_Up messages, i.e.
 7128 after approximately 4 seconds of operation given the 125 ms Sync Interval. During this time
 7129 meanLinkDelay and output Sync/Follow_Up messages fields have not been calculated, so an
 7130 alternative must be used, even if it cannot deliver the same assurances regarding network-level
 7131 performance.

7132 If measured NRR from Sync/Follow_Up message information is unavailable but equivalent
 7133 information from Pdelay_Resp messages is available, it may be substituted for Sync/Follow_Up
 7134 message information. However, measuring NRR using Pdelay_Resp messages is vulnerable to
 7135 additional error due to clock drift between the time NRR is measured, on receipt of the latest
 7136 Pdelay_Resp message, and use of the measurement during Sync message processing. This is
 7137 the reason using Sync/Follow_Up message information is preferable. It also means that a switch
 7138 to using Sync/Follow_Up message information as soon as possible is desirable. It is technically
 7139 possible to calculate a NRR using a combination of Pdelay_Resp and Sync messages but this
 7140 can be risky due to the potential for very short intervals between messages and resulting high
 7141 error due to timestamp errors, so it not recommended.

7142 It is the responsibility of implementers to decide whether and when to use Pdelay_Resp
 7143 message information and when to switch to using Sync/Follow_Up message information. The
 7144 normative requirements in this document can be achieved after 32 Sync/Follow_Up messages
 7145 have been received and assume the use of the algorithms in D.5.2 and D.5.3, or more effective
 7146 algorithms. Implementations that continue to use Pdelay_Resp message information to
 7147 calculate NRR after 32 messages have been received can find some of the normative
 7148 requirements difficult or impossible to meet.

7149 The following describes potential startup behavior applicable to either Sync/Follow_Up or
 7150 Pdelay_Resp message information.

- 7151 a) At least two messages must be received before calculating a NRR value.
- 7152 b) Prior to two messages being received, $NRR = 1$ (i.e. 0 ppm) should be used.
- 7153 c) Once two messages have been received, NRR should be calculated using the formula:

$$2^{\text{nd}} \text{ message: } mNRR = \left(\left(\frac{t_3(x) - t_3(x-1)}{t_4(x) - t_4(x-1)} \right) - 1 \right) \times 10^6 \quad (\text{D.17})$$

7154
 7155 where

7156 $t_3(i)$ is the timestamp of the i^{th} Pdelay_Resp message on egress from the previous node ($n-1$),
 7157 timestamped by that node's Local Clock, expressed in ns;

7158 $t_4(i)$ is the timestamp of the Pdelay_Resp message on ingress to the current node (n),
 7159 timestamped by that node's Local Clock, expressed in ns.

- 7160 d) When three to four messages have been received, NRR should be calculated using the
 7161 following formula:

$$3^{\text{rd}} \text{ message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-2)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-2)} \right) - 1 \right) \times 10^6 \quad (\text{D.18})$$

$$4^{\text{th}} \text{ message: } mNRR = \left(\left(\frac{t_{1\text{outP}}(x) - t_{1\text{outP}}(x-3)}{t_{2\text{in}}(x) - t_{2\text{in}}(x-3)} \right) - 1 \right) \times 10^6 \quad (\text{D.19})$$

7162

- 7163 e) On arrival of the 5th Sync/Follow_Up message the first mNRRcalc and mNRRcalcT
 7164 calculations can take place and should be used for NRR:

$$mNRRcalc(x) = \left(\frac{t_{s1\text{outP}}(x) - t_{s1\text{outP}}(x-4)}{t_{s2\text{in}}(x) - t_{s2\text{in}}(x-4)} - 1 \right) \times 10^6 \quad (\text{D.20})$$

$$mNRRcalcT(x) = \frac{t_{s2\text{in}}(x) + t_{s2\text{in}}(x-4)}{2} \quad (\text{D.21})$$

$$5^{\text{th}} \text{ Sync message: } mNRR = mNRRcalc(x) \quad (\text{D.22})$$

7165

7166 f) As the 6th, 7th and 8th messages arrive an average can be taken and used for NRR, so:

$$6^{\text{th}} \text{ message: } mNRR = \sum_{i=x-1}^x \frac{mNRR_{calc}(i)}{2} \quad (\text{D.23})$$

$$7^{\text{th}} \text{ message: } mNRR = \sum_{i=x-2}^x \frac{mNRR_{calc}(i)}{3} \quad (\text{D.24})$$

$$8^{\text{th}} \text{ message: } mNRR = \sum_{i=x-3}^x \frac{mNRR_{calc}(i)}{4} \quad (\text{D.25})$$

7167 g) For the 9th to the 31st message, the Formula (D.19) can be used.

7168 h) Once the 32nd message arrives, the regular formulas with NRR drift tracking and error
7169 correction can be used.

7170 **D.5.4 Algorithm for tracking RR Drift**

7171 A Sync or Follow_Up message carries the rateRatio field, which informs each node of the
7172 previous node's estimate of its (the previous node's) Rate Ratio. This document also requires
7173 support for the Drift_Tracking TLV, which carries the rateRatioDrift field and informs each node
7174 of the previous node's estimate of its (the previous node's) Rate Ratio Drift.

7175 If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock
7176 (at the Grandmaster PTP Instance) are linked such that the two are always operating at the
7177 same frequency, the rateRatio field received by the first node (Node 1) is always 0 ppm and the
7178 rateRatioDrift field is always 0 ppm/s. Thus, at Node 1, RR equals NRR, RR Drift equals NRR
7179 Drift, and therefore D.5.3 and D.5.2 describe how to calculate RR and RR Drift at Node 1.

7180 If the implementation of the Grandmaster PTP Instance means the ClockSource and Local Clock
7181 (at the Grandmaster PTP Instance) can operate at different frequencies, the implementation
7182 populates the rateRatio and rateRatioDrift field with values reflecting those differences.

7183 In either case all PTP Instances, other than the Grandmaster PTP Instance, calculate an
7184 estimate of the local Rate Ratio Drift when the latest Sync/Follow_Up Message is received,
7185 based on the received rateRatioDrift field and the local measure of NRR Drift. The Rate Ratio
7186 Drift Rate from the previous node is in ppm/s relative to the timebase of its Local Clock (i.e. the
7187 "s" in "ppm/s"). For highest precision, this can be converted to the timebase of the current
7188 node's Local Clock.

$$rateRatioDrift(n) = \frac{rateRatioDrift(n-1)}{\left(1 + \frac{mNRR(n)}{10^6}\right)} + NRRdriftRate(n) \quad (\text{D.26})$$

7189
7190 where

7191 *rateRatioDrift* is the Rate Ratio drift rate, expressed in ppm/s.

7192 However, given that adding ppm/s already lacks the precision of multiplying actual ratios, this
7193 simplification delivers similarly accurate results.

$$rateRatioDrift(n) = rateRatioDrift(n-1) + NRRdriftRate(n) \quad (D.27)$$

7194

7195 D.5.5 Algorithm to compensate for errors in measured RR due to clock drift

7196 On receipt of a Sync or Follow_Up message, all PTP Relay Instances estimate a measured RR
 7197 ($mRR(n)$) based on the received rateRatio field ($mRR(n-1)$) and the local measure of NRR
 7198 ($mNRR(n)$). An $mRR(n)$ value is used to translate the sum of meanLinkDelay and residenceTime
 7199 from Local Clock timebase into Grandmaster timebase. An $mRR(n)$ value is also passed in the
 7200 transmitted Sync or Follow_Up message's rateRatio field to the next node. Errors in these
 7201 estimates due to clock drift can be reduced by taking account of RR Drift. Since the optimal
 7202 point in time for each estimate is different, the amount of applicable RR Drift is different, and
 7203 hence the estimates are different.

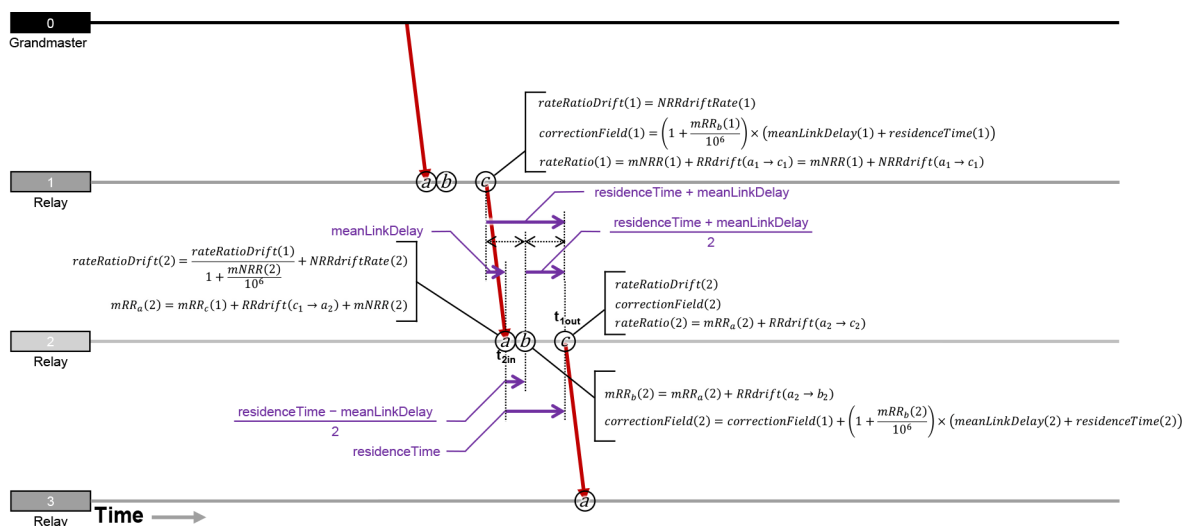
7204 (For discussion of how different Grandmaster PTP Implementations affect the behavior of a
 7205 PTP Relay Instance at Node 1 – or not – see D.5.4.)

7206 A PTP End Instance is similar in that it estimates $mRR(n)$ on receipt of a Sync message,
 7207 subsequently uses an $mRR(n)$ value, and errors in the latter due to clock drift can be mitigated
 7208 by taking account of RR Drift. However, unlike a PTP Relay Instance, the $mRR(n)$ value is used
 7209 to keep the ClockTarget in line with the ClockSource and transmitting a rateRatio field to a
 7210 subsequent node is unnecessary.

7211 For a PTP Relay Instance there are three points in time of interest:

- 7212 • Point a: receipt of the Sync Message by the current node (Node n),
- 7213 • Point b: mid-point between transmission of the Sync message by the previous node (Node
 7214 $n-1$) and transmission of the consequent Sync message by the current node (Node n), and
- 7215 • Point c: transmission of the Sync Message by the current node (Node n).

7216 Figure D.6 illustrates these points and the associated calculations.



7217

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7218 **Figure D.6 – RR Drift tracking and error compensation calculations –**
 7219 **PTP Relay Instance**

7220 The estimate of RR when the Sync message arrives can be calculated as follows:

$$mRR_a(n) = rateRatio(n-1) + RRdrift(c_{n-1} \rightarrow a_n) + mNRR(n). \quad (D.28)$$

$$7221 = rateRatio(n-1) + \left(rateRatioDrift(n-1) \times \left(1 + \frac{mNRR(n)}{10^6} \right) \times meanLinkDelay(n) \right) + mNRR(n)$$

7222 where

7223 mRR_a is the estimate of RR when the Sync message arrives at node n , expressed in ppm;

7224 $RRdrift(c_{n-1} \rightarrow a_n)$ is the amount $rateRatio(n-1)$ drifts between transmission of the Sync
7225 message at Node $n-1$ and reception at Node n , expressed in ppm/s.

7226 $RRdrift(c_{n-1} \rightarrow a_n)$ is equivalent to $rateRatioDrift(n-1)$ multiplied by meanLinkDelay but,
7227 since meanLinkDelay is measured in terms of Node n 's Local Clock and rateRatioDrift is in
7228 terms of Node $n-1$'s Local Clock the former should be multiplied by the NRR at Node n for the
7229 highest accuracy.

7230 However, given that adding ppm/s already lacks the precision of multiplying actual ratios, the
7231 following simplification delivers similarly accurate results:

$$mRR_a(n) = rateRatio(n-1) + (rateRatioDrift(n-1) \times meanLinkDelay(n)) + mNRR(n) \quad (D.29)$$

7232

7233 Once the time when Node n transmits the consequent Sync message is known, the
7234 correctionField value can be calculated.

$$mRR_b(n) = mRR_a(n) + RRdrift_n(a \rightarrow b) \quad (D.30)$$

$$7235 = mRR_a(n) + \left(rateRatioDrift(n) \times \frac{residenceTime(n) - meanLinkDelay(n)}{2} \right)$$

7236 where

7237 $mRR_b(n)$ is the estimate of RR when Node n transmits the consequent Sync message,
7238 expressed in ppm.

7239 The correctionField is calculated as follows:

$$correctionField(n) = correctionField(n-1) + \left(1 + \frac{mRR_b(n)}{10^6} \right) \quad (D.31)$$

$$\times (meanLinkDelay(n) + residenceTime(n))$$

7240

7241 where

7242 $correctionField(n)$ is the value of the correction field transmitted by node n , expressed in
7243 ns.

7244

7245 The rateRatio field is calculated as follows:

$$7246 \quad \begin{aligned} rateRatio(n) &= mRR_a(n) + RRdrift_n(a \rightarrow c) \\ &= mRR_a(n) + (RRdriftRate(n) \times residenceTime(n)) \end{aligned} \quad (D.32)$$

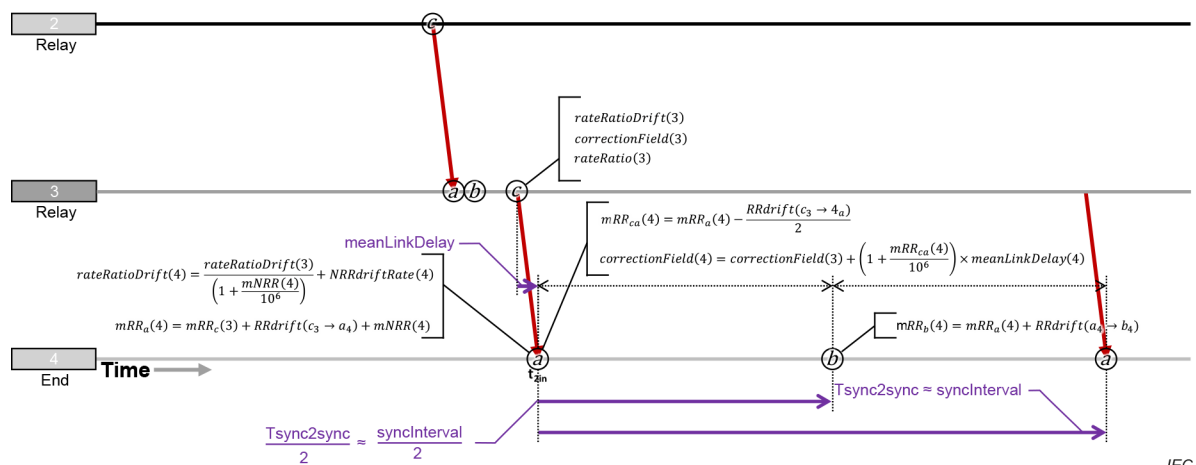
7247

7248 where

7249 $rateRatio(n)$ is the value of the rateRatio field transmitted by node n , expressed in ppm.

7250 D.5.6 Algorithm to compensate for errors in measured RR due to clock drift at PTP 7251 End Instance

7252 Figure D.7 illustrates a possible approach to applying similar RR drift tracking and error
7253 compensation at a PTP End Instance.



7254

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7255 **Figure D.7 – RR Drift tracking and error compensation calculations –**
7256 **PTP End Instance**

7257 The initial calculations for rateRatioDrift and mRR_a are exactly the same as for a PTP Relay
7258 Instance. Instead of using RR to translate meanLinkDelay + residenceTime from the Local Clock
7259 timebase to the Grandmaster timebase – as is done at a PTP Relay Instance – a PTP End
7260 Instance uses mRR to translate meanLinkDelay to the Grandmaster timebase (there is no
7261 residenceTime at an End Instance), adding the result to the incoming correctionField to obtain
7262 an estimate of the ClockSource at the time the Sync message arrives, then uses mRR to keep
7263 its ClockTarget in line with the ClockSource until arrival of the next Sync message. The optimal
7264 mRR value for translating meanLinkDelay is halfway between meanLinkDelay's transmission
7265 (at c_{n-1} , i.e. point C at the previous node) and reception (at a_n , i.e. point A at the current node);
7266 in the formulas below, this value is referred to as mRR_{ca} .

$$\begin{aligned}
mRR_{ca}(n) &= mRR_a(n) - \frac{RRdrift(c_{n-1} \rightarrow a_n)}{2} \\
&= mRR_a(n) - \left(rateRatioDrift(n) \times \frac{meanLinkDelay(n)}{2} \right)
\end{aligned} \tag{D.33}$$

7268

7269 where

7270 $mRR_{ca}(n)$ is the estimate of RR calculated based upon one-half of the meanLinkDelay,
 7271 expressed in ppm.

7272 The correction field at the PTP End Instance is given by:

$$correctionField(n) = correctionField(n-1) + mRR_{ca}(n) \times meanlinkDelay(n) \tag{D.34}$$

7273

7274 The optimal value of mRR for keeping the ClockTarget in line with the Clock Source is mRR_b ,
 7275 where Point B is halfway between the most recently received Sync message and the next Sync
 7276 message.

7277 Of course, the exact interval until the next Sync message's arrival (Tsync2sync in Figure D.7)
 7278 can't be known before it happens, but the Rate Ratio value is required as soon as possible after
 7279 arrival of the most recent Sync message. The solution is to use the nominal value of the interval,
 7280 i.e. syncInterval, which is 125 ms.

7281

$$mRR_b(n) = mRR_a(n) + RRdrift_n(a \rightarrow b)$$

7282

$$\begin{aligned}
&= mRR_a(n) + \left(rateRatioDrift(n) \times \frac{syncInterval}{2} \right) \\
&= mRR_a(n) + (rateRatioDrift(n) \times 0,0625)
\end{aligned} \tag{D.35}$$

7283

7284 where

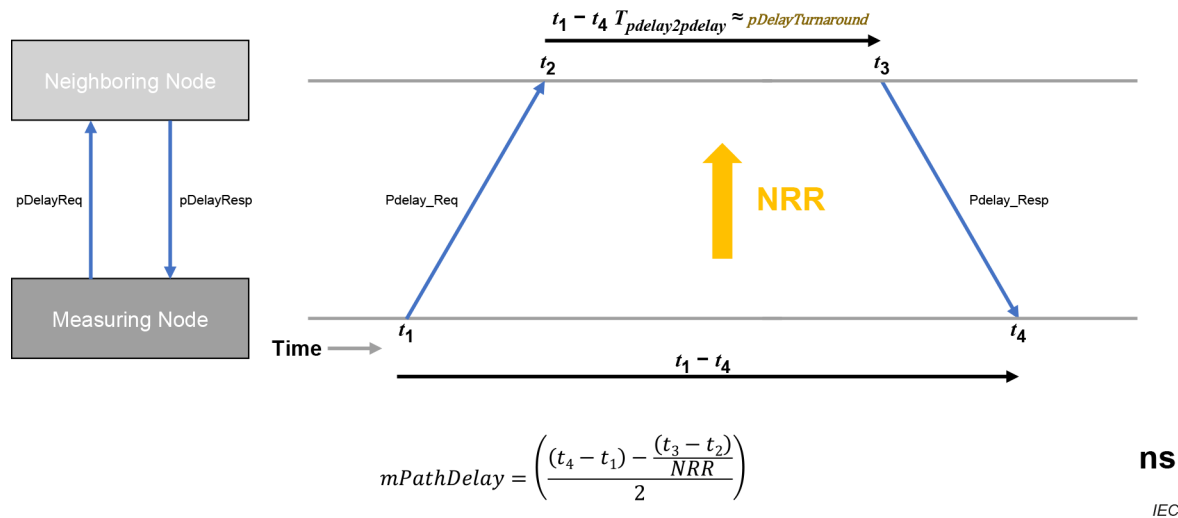
7285 *syncInterval* is the nominal value of the interval between sync messaged, expressed in ms.

7286 It is also possible to use more complex algorithms that repeatedly or continuously adjust the
 7287 mRR value between Sync messages, but such an approach is not addressed in this document.

7288 **D.5.7 Mean Link Delay averaging**

7289 The actual Path Delay from one node to the next – for a wired connection – is very stable and
 7290 errors measuring it due to Timestamp Error average to zero. Thus, taking a long average or
 7291 applying a low-pass filter with a low bandwidth is an effective way to reduce error in
 7292 meanLinkDelay. Care is taken during system startup or after any other initialization of the
 7293 algorithm, to quickly converge on a stable value.

7294 The basic Pdelay calculation, used by the Common Mean Link Delay service, remains the same.
 7295 Figure D.8 illustrates it.



7297 **Figure D.8 – Signals and timestamps to measure path delay**

7298 Following each Pdelay_Req – Pdelay_Resp exchange, the measured path delay (mPathDelay)
 7299 is calculated.

7300 For the x^{th} message after initialization

$$mPathDelay(x) = \frac{(t_4 - t_1) - \frac{(t_3 - t_2)}{NRR}}{2} \quad (\text{D.36})$$

7301

7302 where

7303 $mPathDelay$ is the measured path delay between the measuring node and the neighboring
 7304 node expressed in ns;

7305 t_1 is a measurement point as defined in Figure D.8, expressed in ns;

7306 t_2 is a measurement point as defined in Figure D.8, expressed in ns;

7307 t_3 is a measurement point as defined in Figure D.8, expressed in ns;

7308 t_4 is a measurement point as defined in Figure D.8, expressed in ns.

7309 The meanLinkDelay is then updated via an IIR (Infinite Impulse Response) filter. For the first
 7310 measurement, the filter is initialized:

$$meanLinkDelay(x) = mPathDelay(x) \quad (\text{D.37})$$

7311

7312 For the next few minutes after initialization (when $x < 1\,000$) the filter is in startup mode. It then
 7313 transitions to steady-state mode.

7314 If $x < 1\,000$ then $f = x$ else $f = 1\,000$

$$meanLinkDelay(x) = \frac{(meanLinkDelay(x-1) \times (f-1)) + mPathDelay(x)}{f} \quad (D.38)$$

7315 For example:

$$meanLinkDelay(100) = \frac{(meanLinkDelay(99) \times 99) + pDelay(x)}{100} \quad (D.39)$$

7316

$$meanLinkDelay(5\ 836) = \frac{(meanLinkDelay(5\ 835) \times 999) + pDelay(x)}{1\ 000} \quad (D.40)$$

7317

7318 It is possible to automatically reinitialize the algorithm if an *mPathDelay* value, or series of
7319 values, deviates too much from the *meanLinkDelay*, but the details are outside the scope of this
7320 document.

7321 The behavior of timestamp error means that, for shorter actual link delays, *mPathDelay* can be
7322 a negative value. It can seem tempting to reject negative values, since a negative delay is
7323 impossible. However, at a device level, including negative values of *mPathDelay* in the input to
7324 the IRR filter results in a more accurate filter output, i.e. *meanLinkDelay* value; values lower
7325 than the actual delay, even when negative, are balanced by values higher than the actual delay.

7326 Similarly, at a network level, using negative values of *meanLinkDelay*, i.e. the output of the IIR
7327 filter, results in a more accurate *correctionField* calculation at the PTP End Instance when there
7328 are many networking hops between it and the Grandmaster PTP Instance.

Annex E

(normative)

60802 device info TLV definition

E.1 60802 device info TLV

E.1.1 General

In the TLV definition that follows, any fields that are labelled as “Reserved” are transmitted as zero and ignored on receipt.

The 60802 device info TLV is a mandatory TLV that indicates a device’s configuration capabilities.

Figure E.1 shows the 60802 device info TLV.

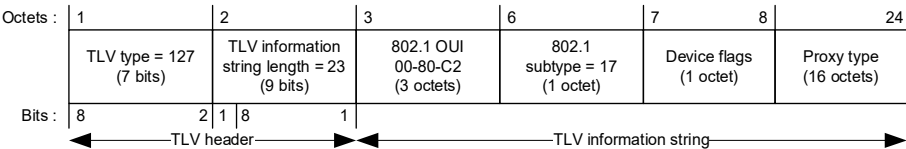


Figure E.1 – 60802 device info TLV

E.1.2 Device flags

The Device flags field shall contain a bit map indicating the device configuration capabilities as defined in Table E.1.

Table E.1 – Device flags

Bit	Value/meaning	Function
0	1 = Proxy required 0 = no Proxy required	Proxy required
1	1 = supported 0 = not supported	NETCONF configuration supported
2-7	(set to zero)	Reserved for future standardization

E.1.3 Proxy type

The proxy type field is conditionally mandatory if the device flags indicate that the device requires a proxy. It indicates the proxy type that is required to configure the device. The format is a human readable string, that is specific for a specific proxy and its associated devices.

Annex F (informative)

MAC and PHY delay

F.1 Accumulated delays in industrial automation

F.1.1 General

MAC and PHY delays affect the performance of industrial communication. These delays impact forwarding delay, the accuracy of time-triggered transmission, and synchronization accuracy. Factors contributing to these delays include:

- the clock system between the MAC application, the MAC, the Media Independent Interface (MII), and the PHY,
- the processing of the frame, and
- the transmission of bits from the MAC application over MAC, MII and PHY to the MDI.

These delays consist of two components:

- the propagation delay, and
- the propagation delay variance (often called jitter).

F.1.2 Forwarding delay

Linear topologies, as commonly used in industrial communication, result in cumulative MAC and PHY delays. With 64 hops, there are 130 MAC and PHY delays (64 hops x 2 + 2x end stations) in end-to-end transmission.

For example, a frame sent from the end station at the beginning of a linear topology (64 Hops) to an end station at the end of the line is delayed by 26 μ s / 65 μ s / 130 μ s solely due to a PHY delay of 200 ns / 500 ns / 1 μ s. This means that the lower the PHY delay, the shorter the achievable update interval and thus, the roundtrip delay.

F.1.3 Time-triggered transmission

The reference point for time-triggered transmission is at the MDI. However, the frames are handed over from the transmission control to the MAC in a time-controlled manner. To transmit a frame at the correct time at the reference point, the MAC and PHY delay are compensated by the transmission control.

For example, the magnitude of the propagation delay reduces the computation time available for the application to calculate new process values, as the frames are handed over from the transmission control to the MAC earlier so they can be sent at the correct time at the reference point. Additionally, the frames are also received later by the application, which further reduces the computation time (see 4.1).

The magnitude of the propagation delay variance affects the safety margin of the windows in TAS and is a consideration when configuring PSFP and scheduled traffic. Additionally, it increases the required TAS window size in linear topologies to ensure that all planned frames can pass through their assigned TAS windows.

F.1.4 Synchronization

The reference point for synchronization is also at the MDI. The actual point at which the transmitted or received synchronization frames are timestamped depends on the implementation and can be in the PHY, on the MII, or behind the MAC. Depending on the

implementation, part or all the MAC and PHY delay is compensated to determine the time at which the synchronization frame was at the reference point.

For example, during synchronization, the mean link delay and residence time are determined. The propagation delay variance may be different for the transmission of pdelay frames and synchronization frames. This results in an uncompensated delay error for the sync frames, which accumulates in linear topologies.

F.1.5 Recommended delays

Ethernet hardware used for industrial communication typically considers the requirements for MAC and MII propagation delay and propagation delay variance, minimizing these values during development. PHYs, if not integrated, are appropriately developed or selected.

Table F.1 shows the recommended values for PHYs in industrial communication.

Table F.1 – Recommended PHY delay

MAU Type	Recommended Propagation delay	Recommended Propagation delay variance
10BASE-T1L	maximum < 5 µs minimum < 500 ns	maximum < 16 ns minimum < 4 ns
100BASE-TX, 100BASE-FX, 1000BASE-SX, and 1000BASE-T	maximum < 1 µs minimum < 500 ns	maximum < 16 ns minimum < 4 ns

7406

7407

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